ملخص مادة الحاكمات المنطقية المبرمجة

عمل الطالب عبدالله شاويش

فريق سبكتروم

Chapter 1

PLC stands for Programmable Logic Controller. It's a type of industrial digital computer used to automate electromechanical processes like controlling machinery on factory assembly lines, amusement rides, or lighting fixtures. PLCs are designed for real-time use and often used in situations where reliability, high availability, and ruggedness are critical.

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ما الهدف من تعلم PLC:

هنالك العديد من المتحكمات المبرمجة التي تستخدم في مجالات متخصصة مثل المصانع والبنوك والفنادق وأنظمة الاتصالات وكل بيئة يتم العمل بها بمتحكم دقيق قادر على العمل في هذه البيئة و في هذه المادة يتم شرح المتحكم الدقيق المتخصص في العمل في بيئة المصانع

والمتحكم الدقيق هو حاسوب مصغر يحتوي رام وميموري وCPU و قابل للبرمجة بلغة برمجة معينة وفي حالة plc يتم برمجته من خلال 3 لغات برمجة اهمهم ladder diagram

بسواعدنا نبنيها

advantages of PLCs

1. Flexibility: Easy modifications and updates without rewiring, ideal for dynamic environments.
2. Reliability: Designed for industrial use, withstand harsh conditions, operate continuously.
3. Scalability: Easily scale up or down with additional modules.
4. Real-Time Operation: Crucial for immediate actions, ensures precise responses.
5. Integration: Interfaces with various input/output signals, seamless control.
6. Diagnostics: Extensive diagnostic capabilities for monitoring and troubleshooting.
7. Programming: Supported by intuitive environments, with simulation tools for verification.

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8. Cost Effectiveness: Eliminates complex wiring, reduces installation time, cost-efficient.

9. Communication Abilities: Supports various protocols, facilitates data exchange and remote monitoring.

10. Safety Features: Plays a vital role in ensuring safe operation, with built-in safety functions.

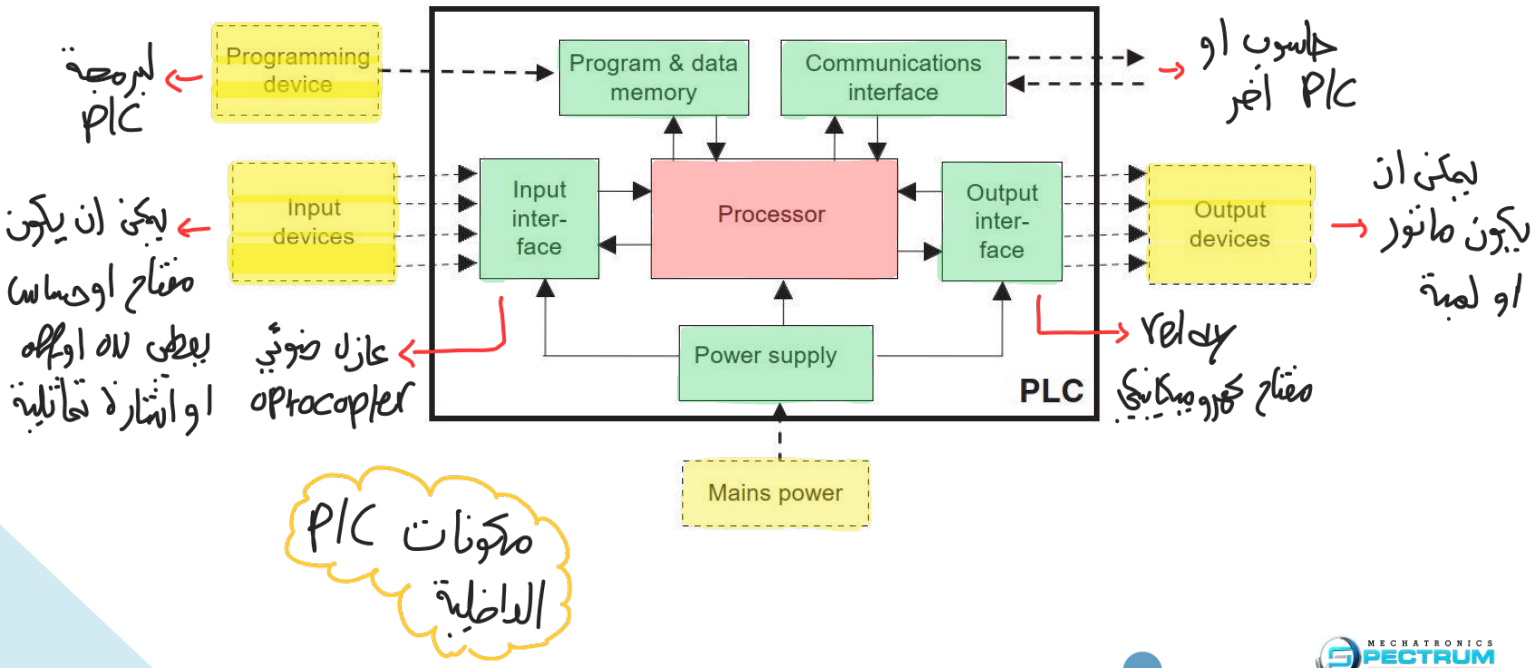
Parts of a PLC

A. central processing unit (CPU)

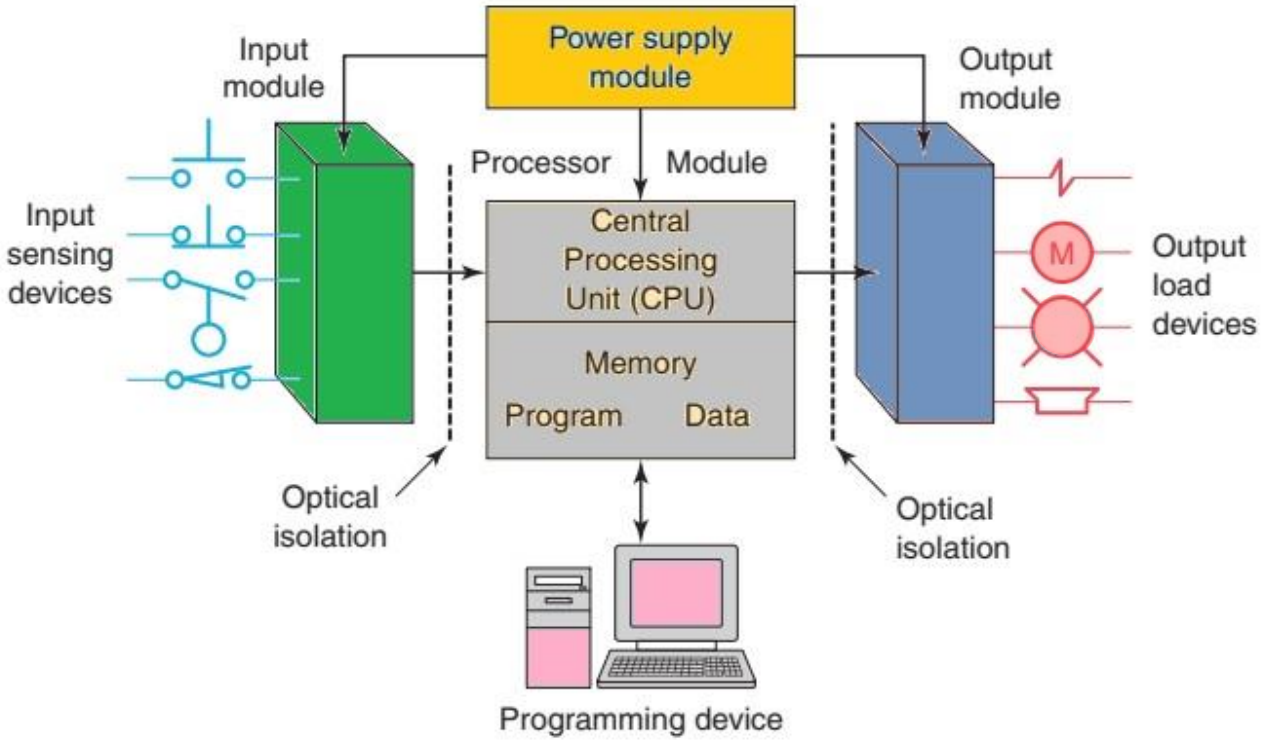
B. input/output (I/O) section

C. The power supply

D. The programming device



PLC system



يمتلك plc وحدات ادخال ووحدات اخراج ووحدة معالجة مركزية ووحدة تزويد الطاقة يزوده بتيار DC بقيمة 24 فولت

تعمل وحدة الادخال على اخذ قراءات الحساسات ومعالجتها عبر وحدة المعالجة وبناءا على الكود البرمجي يتم تعديل المخرجات مثل تشغيل ماتور او اطفاءه

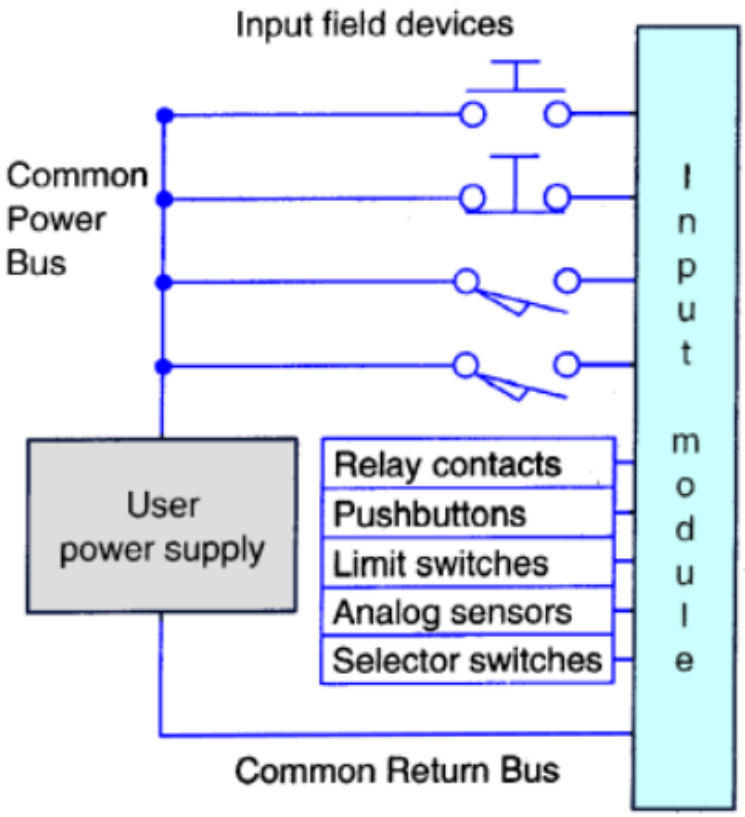
ويمتلك plc عدد مداخل ومخارج محدود ويتم تزويدهم من خلال موديول خارجي ان لزم

Inputs

مداخل plc يمكن ان تكون حساسات حرارة او ضغط او لون او مسافة او مياه او تقاربية اكتشاف الاجسام او ممكن ان تكون مفاتيح عادية بكل انوعها

هذه الحساسات او المفاتيح ترسل إشارات يمكن ان تكون إشارات تماثلية analog او ممكن ان تكون إشارات رقمية digital حسب نوع الحساس اما المفاتيح ترسل فقط إشارات رقمية اما on او off

يتم اخذ إشارات هذه الحساسات والمفاتيح لوحدة المعالجة ليتم اتخاذ امر معين على output



يمكن للحساسات ان تغطي PLC 220 فولت او 24 فولت في حالة استعارة تيار معين

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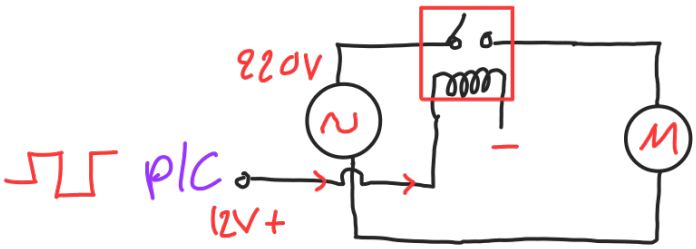
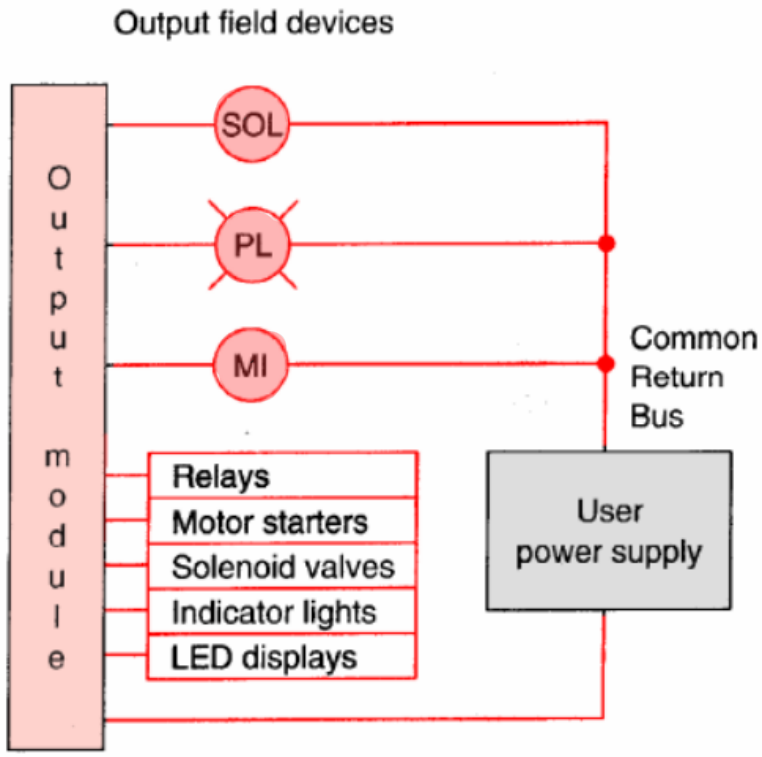
Outputs

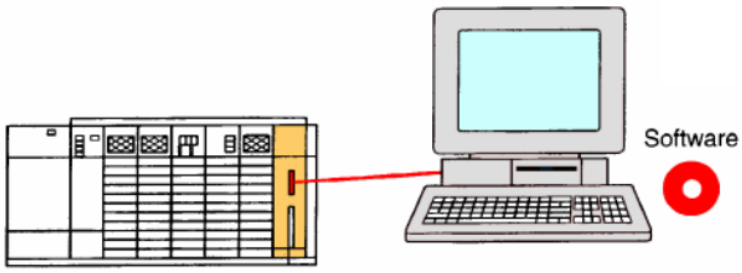


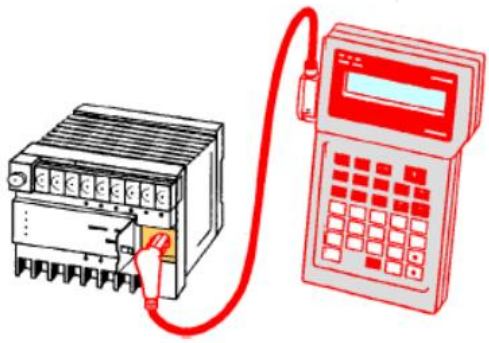
يحتوي plc على مخارجه مرحلات (ريليه) وهو مفتاح كهربائي ميكانيكي يتم التحكم به عبر وحدة المعالجة المركزية لتشغيل احمال تعمل على فولتيات عالية و تيارات عالية جدا مثل المواتير والمصابيح والسخانات ويمكن لل plc اخراج إشارات analog او digital حسب الحاجة



programming

يتم برمجة plc عن طريق الحاسوب او عن طريق hand held unit



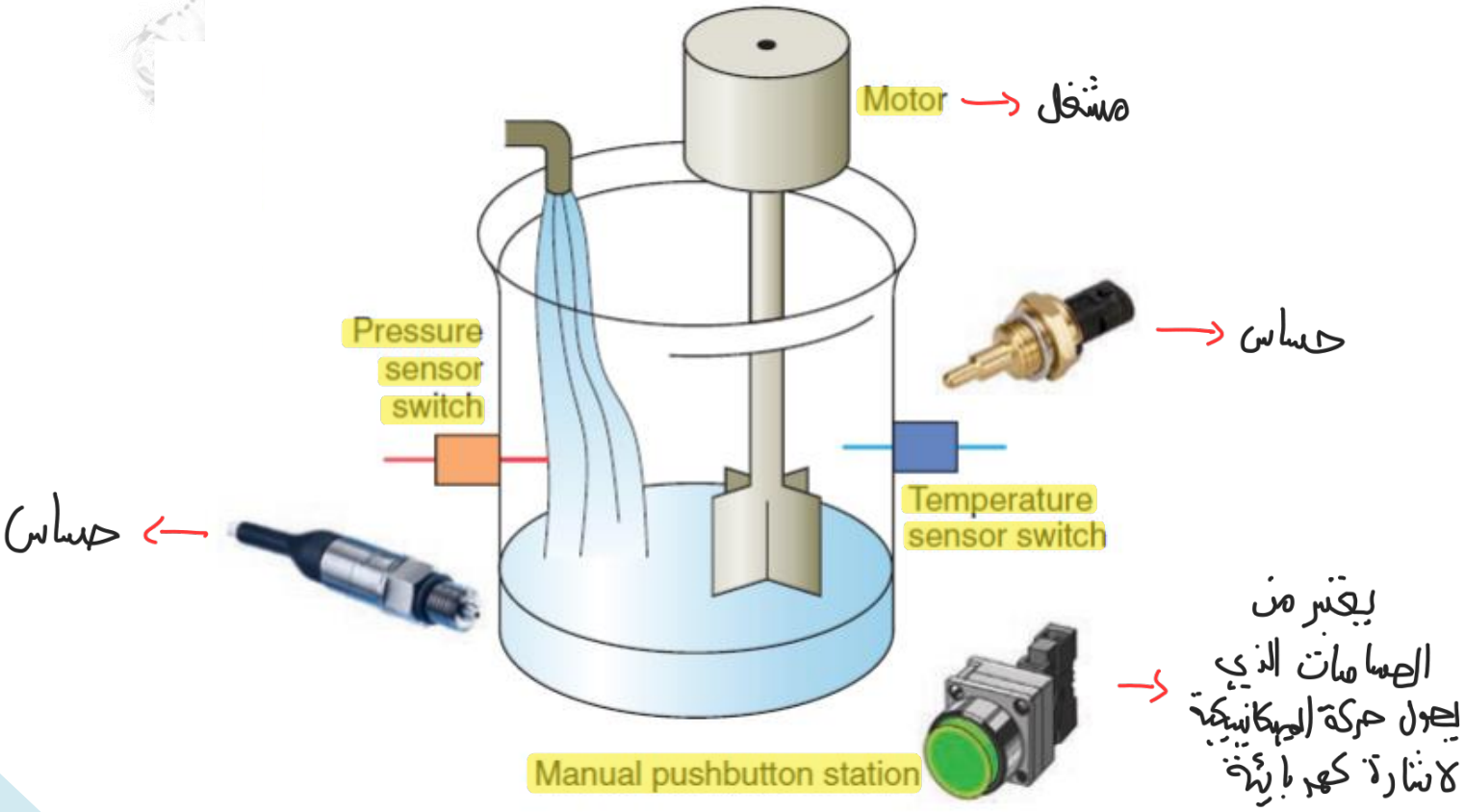


Principles of Operation

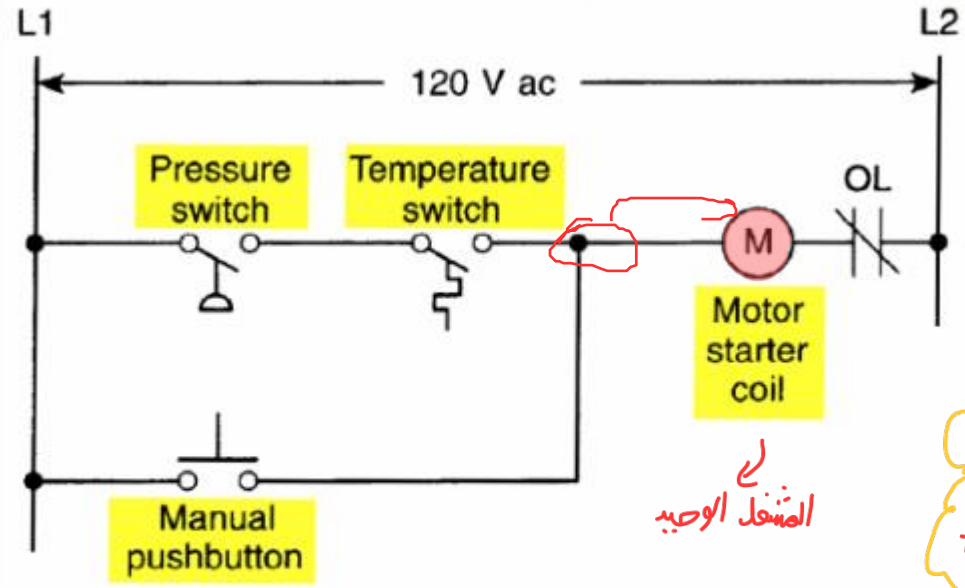
1.Mixer process control problem example

Here a mixer motor is to be used to automatically stir the liquid in a vat when the temperature and pressure reach preset values. In addition, direct manual operation of the motor is provided by means of a separate pushbutton station.

The process is monitored with temperature and pressure sensor switches that close their respective contacts when conditions reach their preset values



Process Control Relay Ladder Diagram



مكونات Ladde digram الرئيسية حساسات + مشغلات

المشغل الوحيد

Motor starter coil is energized when both the pressure and temperature switches are closed or when the manual pushbutton is pressed

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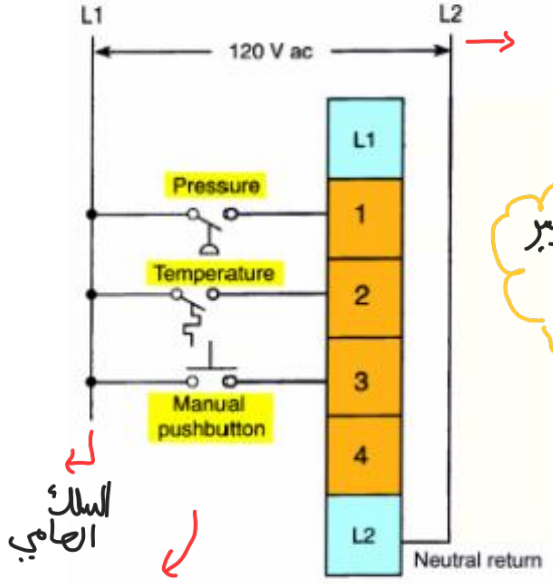
PLC Input Module Connections

- The same input field devices are used
- These devices are wired to the input module according to the manufacturer's labeling scheme



السلك البارد

ملاحظة: البارد يسير من الخط العامي للخط البارد

السلك العامي

الحساسات و المفاتيح

مداخل PLC

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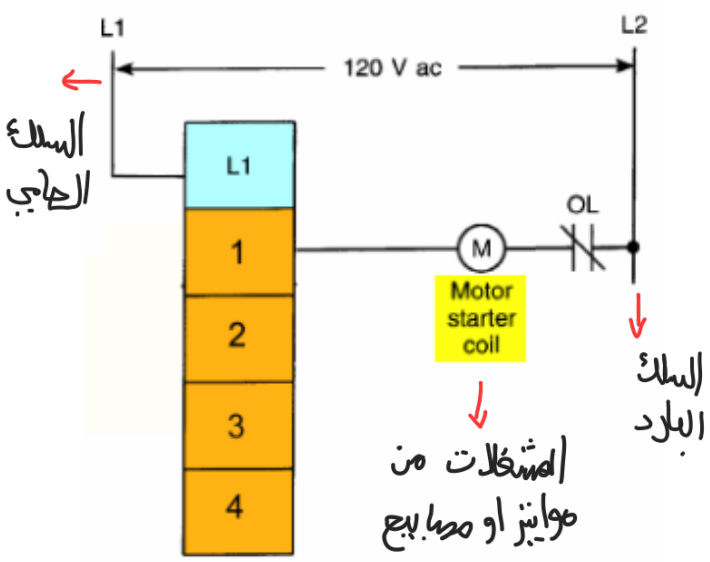
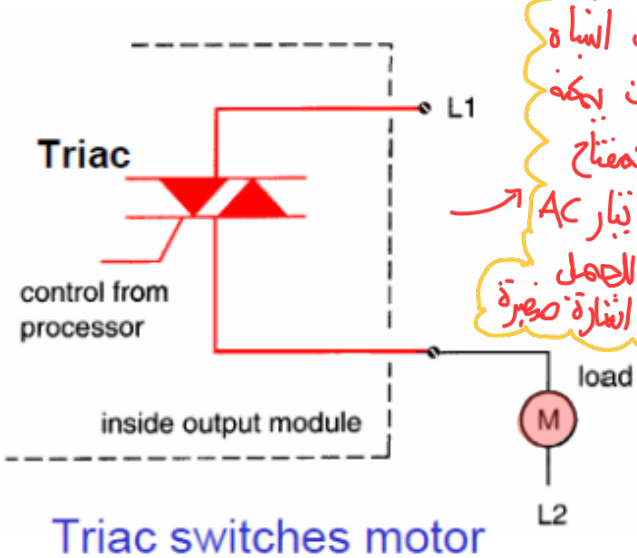
PLC Output Module Connections

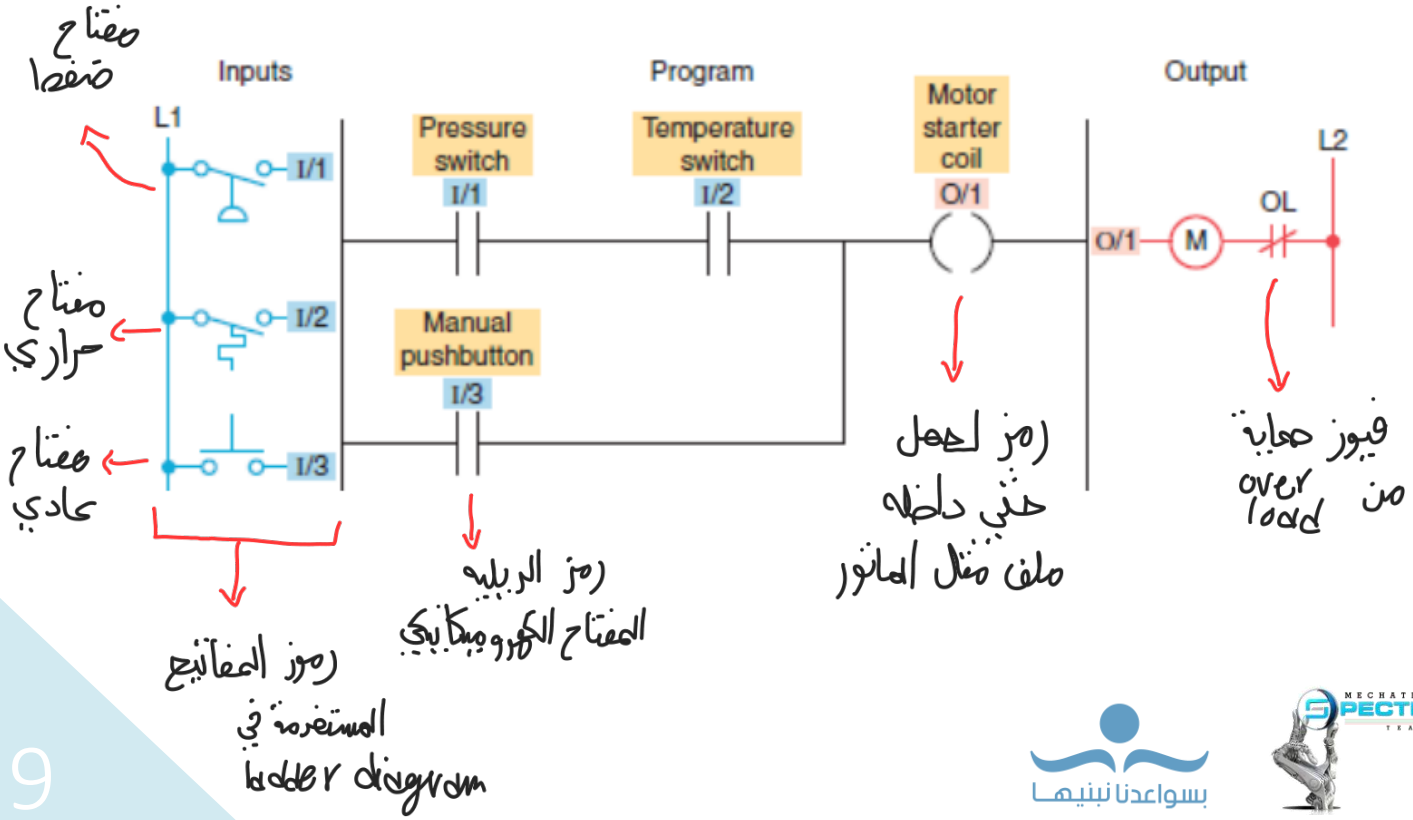
Same output field device is used and wired to the output module



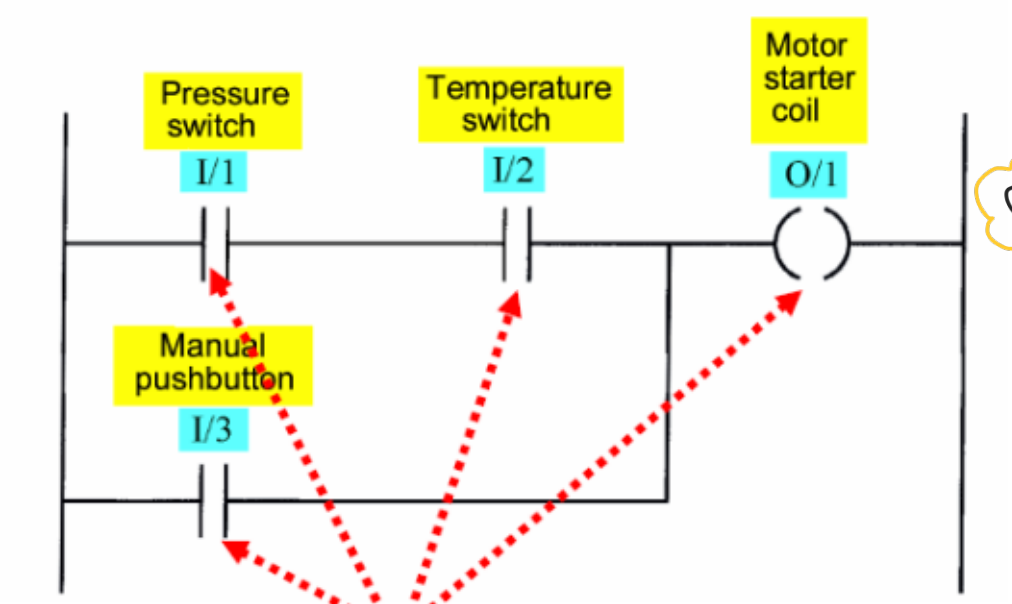


Triac switches motor ON and OFF in accordance with the control signal from the processor

Ladder diagram



PLC Ladder Logic Program



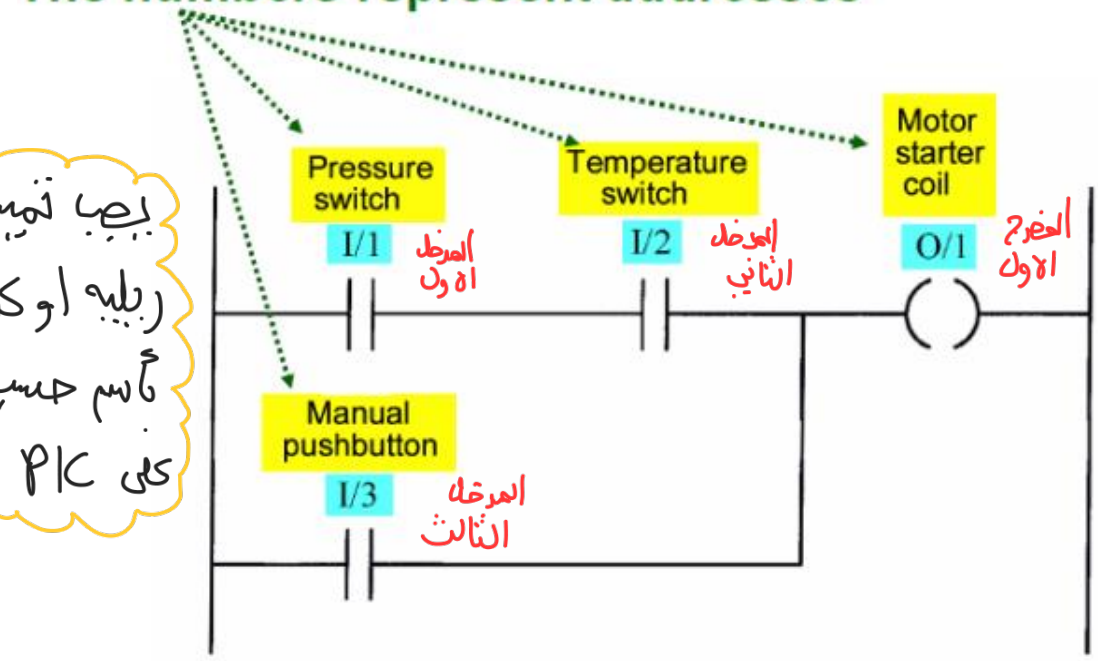
كما نرى
نمثل كل حساس
برمز الزبليه

The *symbols* represent *instructions*

T E A M

PLC Ladder Logic Program

The *numbers* represent *addresses*



يجب تمييز كل
زبليه او كل حساس
بأسم حسب موقعه
على PLC

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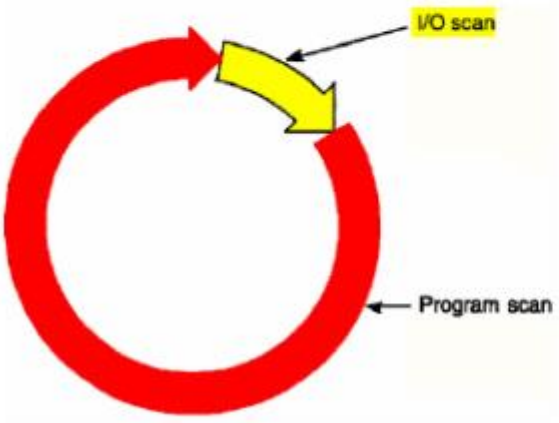
PLC Operating Cycle

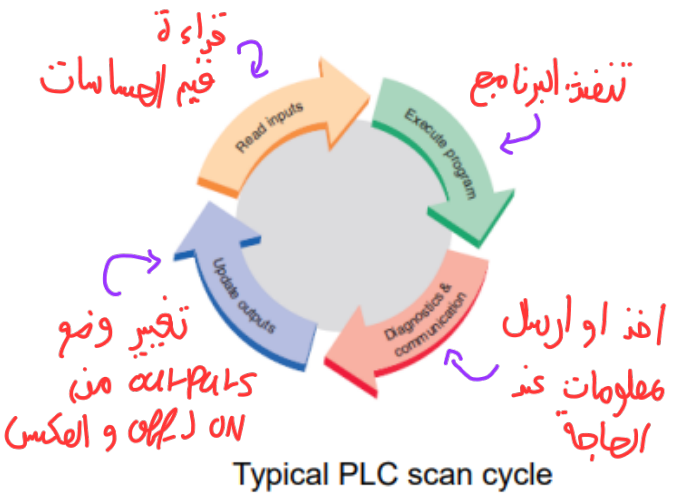
During each operating cycle, the controller examines the status of input devices, executes the user program, and changes outputs accordingly



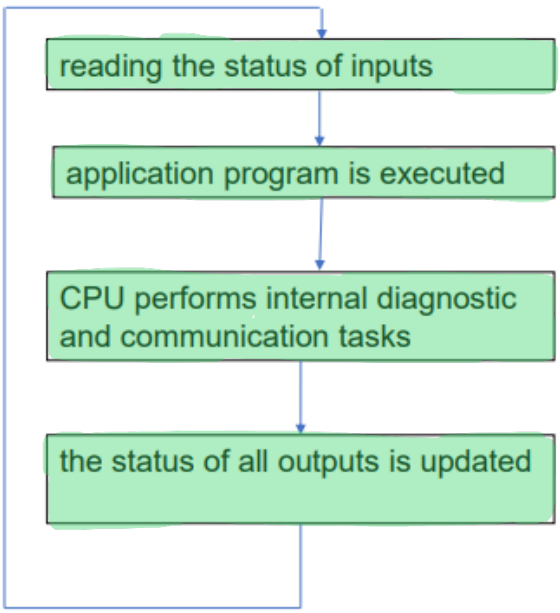
The completion of one cycle of this sequence is called a *scan*. The scan time, the time required for one full cycle, provides a measure of the speed of response of the PLC

T E A M

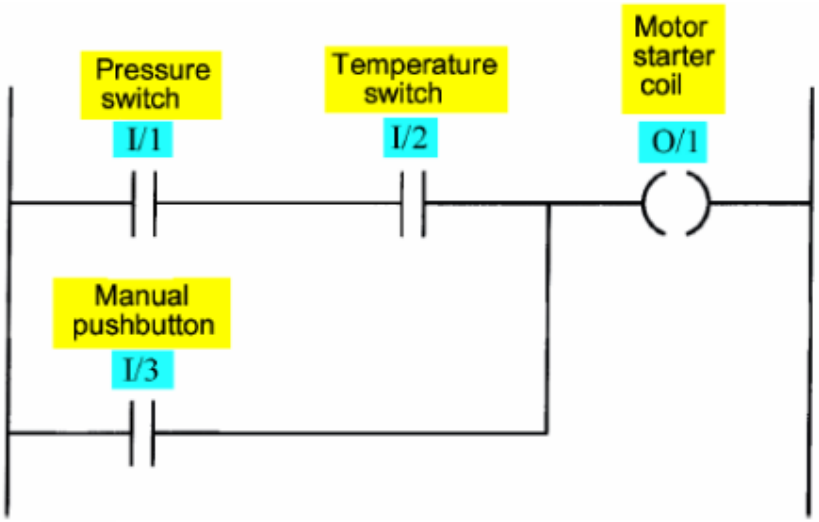
PLC program is executed



Typical PLC scan cycle



PLC Operating Cycle



نعطيه ON لتمرير التيار

normaly open

normaly close

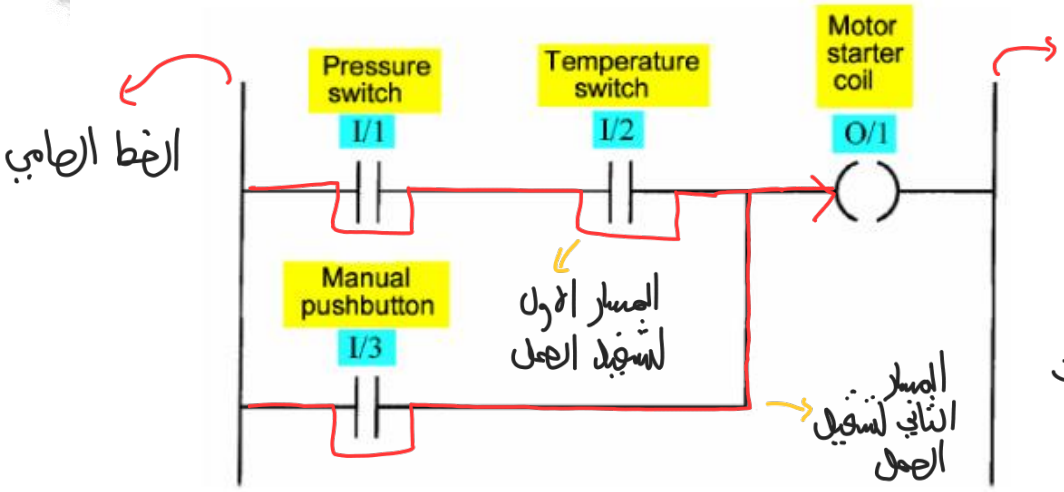
نعطيه off لتمرير التيار

Each ⊣⊢ can be though of as a set of normally open contacts

The ( ) can be considered to represent a coil that, when energized, will close a set of contacts.

T E A M

PLC Operating Cycle



ملاحظة: حتى يعمل المحرك يحتاج خط حامي وبارد والخط الطاقي مقطوع بسبب وجود حساسات ومفاتيح

Coil O/1 is energized when contacts I/1 and I/2 are closed or when contact I/3 is closed. Either of these conditions provides a continuous path from left to right across the rung that includes the coil.

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The RUN operation for the process

control scheme for the example

It can be described by the following sequence of events:

1. The pressure switch, temperature switch, and pushbutton

inputs are examined and their status is recorded in the

controller's memory.

2. Closed contact is recorded in memory as logic 1 and an open

contact as logic 0.

3. The ladder diagram is evaluated, with each internal contact

given an OPEN or CLOSED status according to its recorded 1

or 0 state

4. When the states of the input contacts provide logic continuity from left to

right across the rung, the output coil memory location is given a logic 1 value

and the output module interface contacts will close.

5. When there is no logic continuity of the program rung, the output coil

memory location is set to logic 0 and the output module interface contacts will

be open.

6. The completion of one cycle of this sequence by the controller is called a

scan. The scan time, the time required for one full cycle, provides a measure

of the speed of response of the PLC.

7. Generally, the output memory location is updated during the scan but the

actual output is not updated until the end of the program scan during the I/O

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2.Installation

If Allen-Bradley Pico controller is used, it equipped with 8 inputs

and 4 outputs is used to control and monitor the process.

Installation can be summarized as follows:

• Fused power lines, of the specified voltage type and level, are

connected to the controller's L1 and L2 terminals.

• The pressure switch, temperature switch, and pushbutton field

input devices are hardwired between L1 and controller input

terminals I1, I2, and I3, respectively.

• The motor starter coil connects directly to L2 and in series with

Q1 relay output contacts to L1.

• The ladder logic program is entered using the front keypad and

LCD display.

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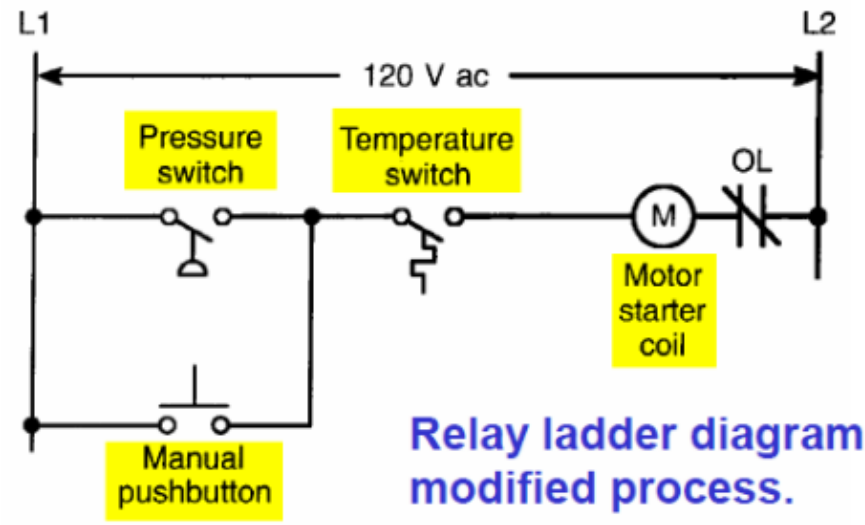
Modifying A PLC Program

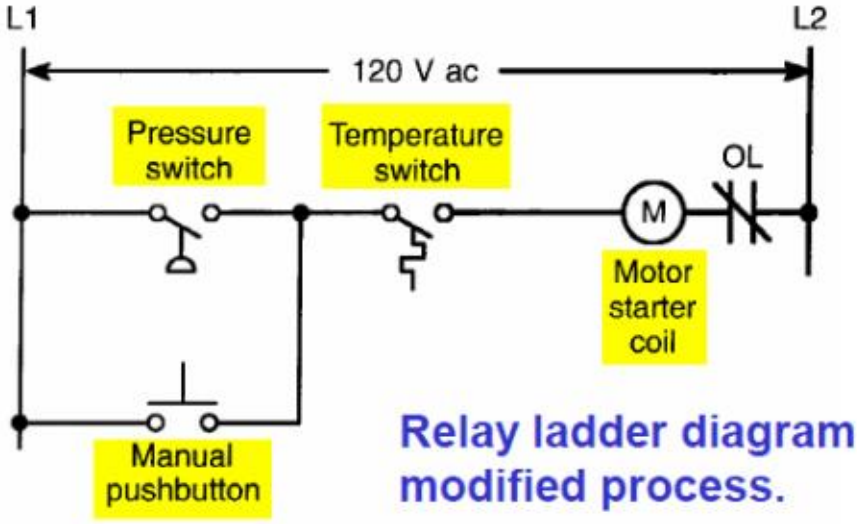


Relay ladder diagram for modified process.

The change requires that the manual pushbutton control should be permitted to operate at any pressure but not unless the specified temperature setting has been reached.

T E A M

Modifying A PLC Program

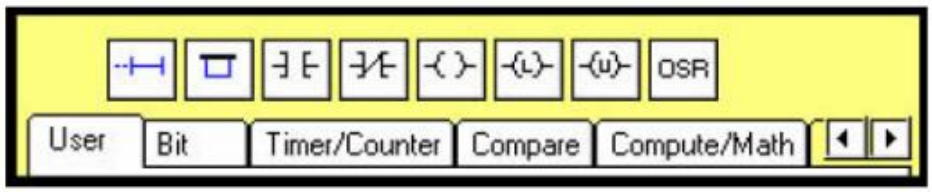


Relay ladder diagram for modified process.

If a relay system were used, it would require some rewiring of the system, as shown, to achieve the desired change.

PLC instruction list

The instruction set for a particular PLC type lists the different types of instructions supported.



الرموز داخل البرنامج البرمجية

An instruction is a command that will cause a PLC to perform a certain predetermined operation.

Typical PLC Instructions

XIC (Examine ON) . . . . . Examine a bit for an ON condition
XIO (Examine OFF) . . . . . Examine a bit for an OFF condition
OTE (Output Energize) . . . . Turn ON a bit (non retentive)
OTL (Output Latch) . . . . . Latch a bit (retentive)
OTU (Output Unlatch) . . . . Unlatch a bit (retentive)
TOF (Timer Off-Delay) . . . . . Turn an output ON or OFF after its rung has been OFF a preset time interval
TON (Timer On-Delay) . . . . Turn an output ON or OFF after its rung has been ON for a preset time interval
CTD (Count Down) . . . . . Use a software counter to count down from a specified value
CTU (Count Up) . . . . . Use a software counter to count up to a specified value

Types of PLC application

1.A single ended or stand-alone PLC

application

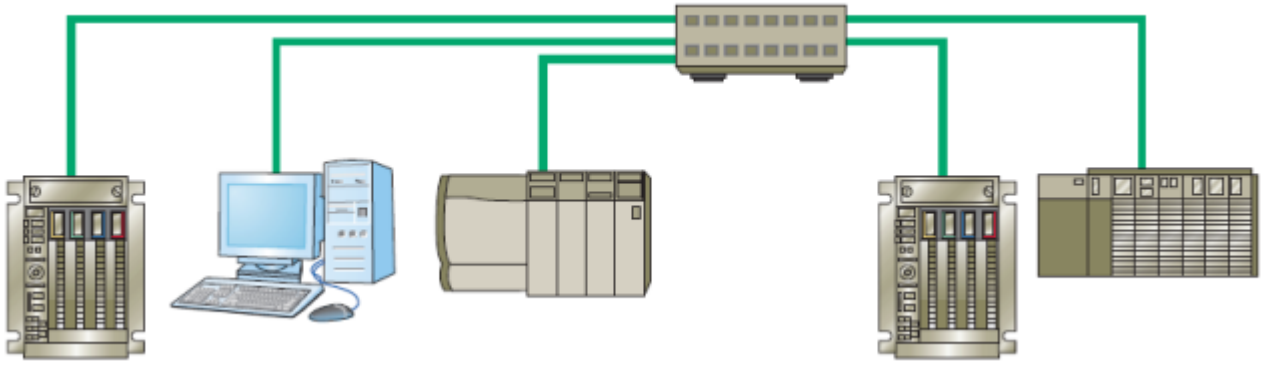
- It involves one PLC controlling one process
- It would not be used for communicating with other computers or PLCs.

2.A multitask PLC application

- It involves one PLC controlling several processes.

3.Control management PLC application

It involves one PLC controlling several others



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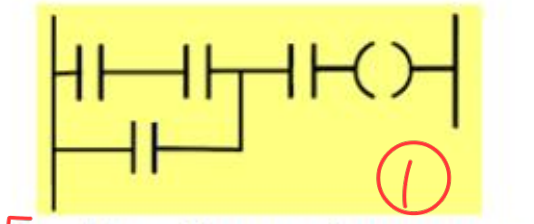
Chapter 2

PLC programmable language

The term *PLC programming language* refers to the method by which the user communicates information to the PLC.

The three most common language structures are: ladder diagram language, Boolean language, and functional chart.

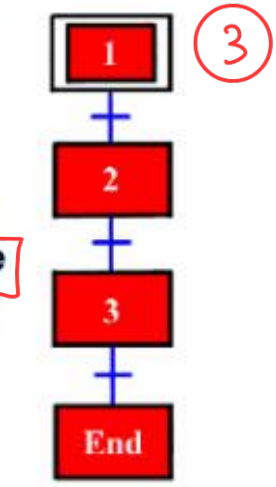


Ladder diagram language

START	PB1
AND	CR1
OR	LS1
AND NOT	CR2
OUT	SOL

Boolean equation: $Y = [(AB) + C]\,\overline{D}$

Boolean language



Functional chart

Processor Memory Organization

The memory of a PLC is organized by types.

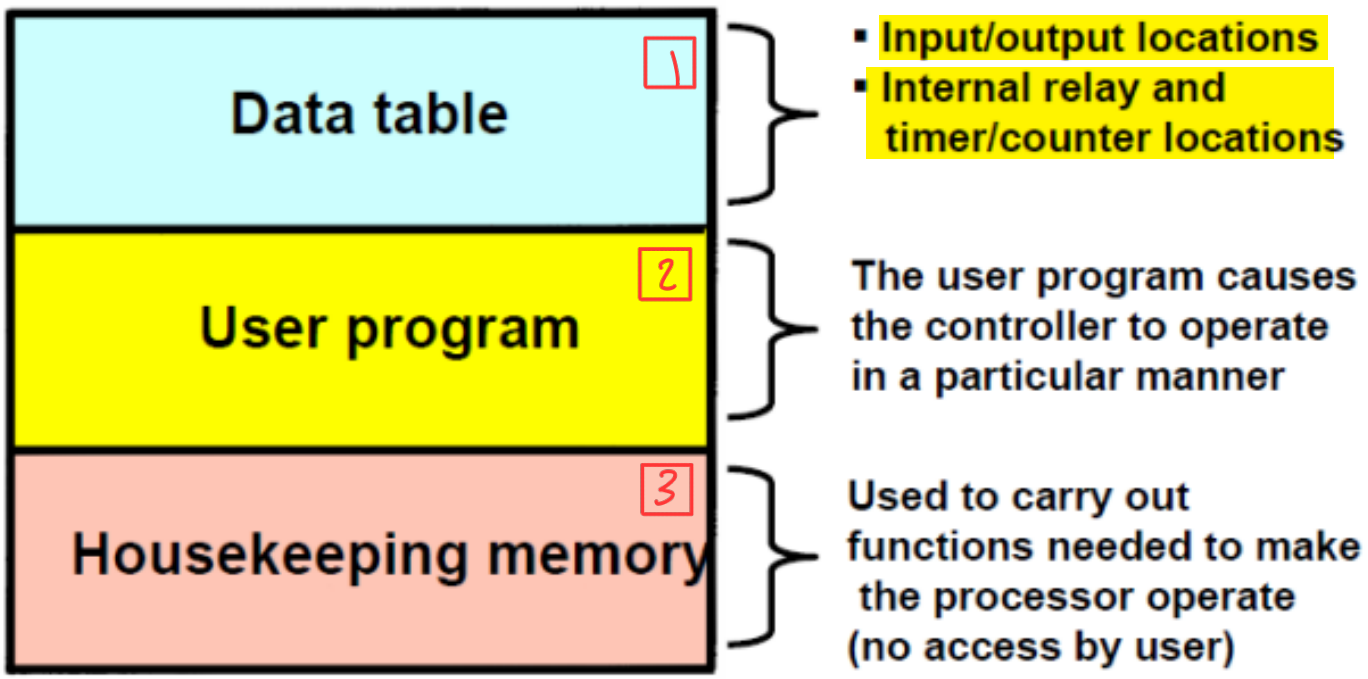
The memory space can be divided into two broad categories:

program and data memory.

Advanced ladder logic functions allow controllers to perform calculations, make decisions and do other complex tasks. Timers and counters are examples of ladder logic functions. They are more complex than basic inputs contacts and output coils and rely heavily upon data stored in the memory of the PLC.

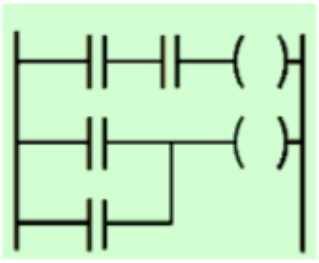
Memory Map

A memory map can be used to show how memory is organized in a PLC.

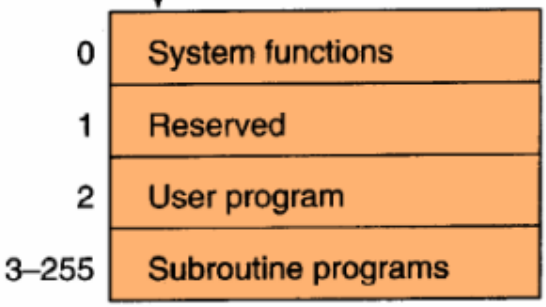


Program file

The user program will account for most of the memory of a PLC system.



مكان الذاكرة الذي يخزن البرنامج المعد من المبرمج

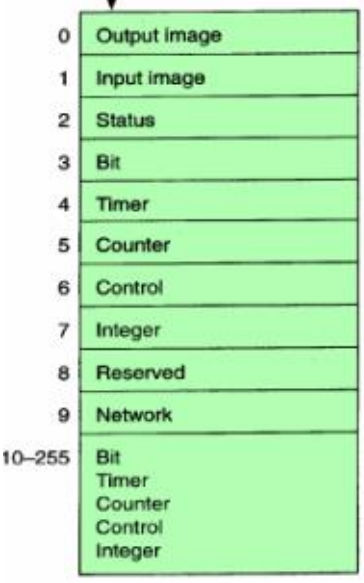


- Program files contain the logic controlling machine operation.
- This logic consists of instructions that are programmed in a ladder logic format.

Program file organization for SLC-500 controller.

Data file

Data file organization for SLC-500 controller.



The data file portion of memory stores input and output status, processor status, the status of various bits and numerical data.

معلومات موجودة مسبقاً من المصنع تمثل اماكن منافذ الادخال والاخراج

بسواعدنا نبنيها
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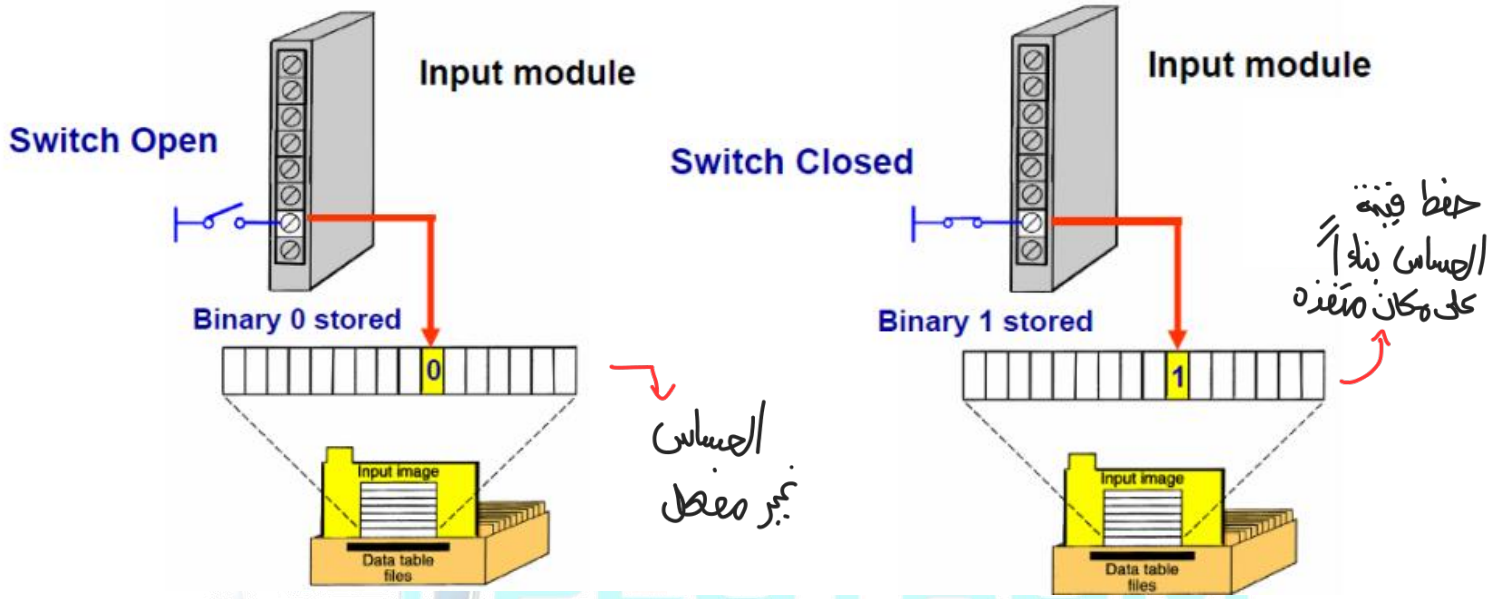
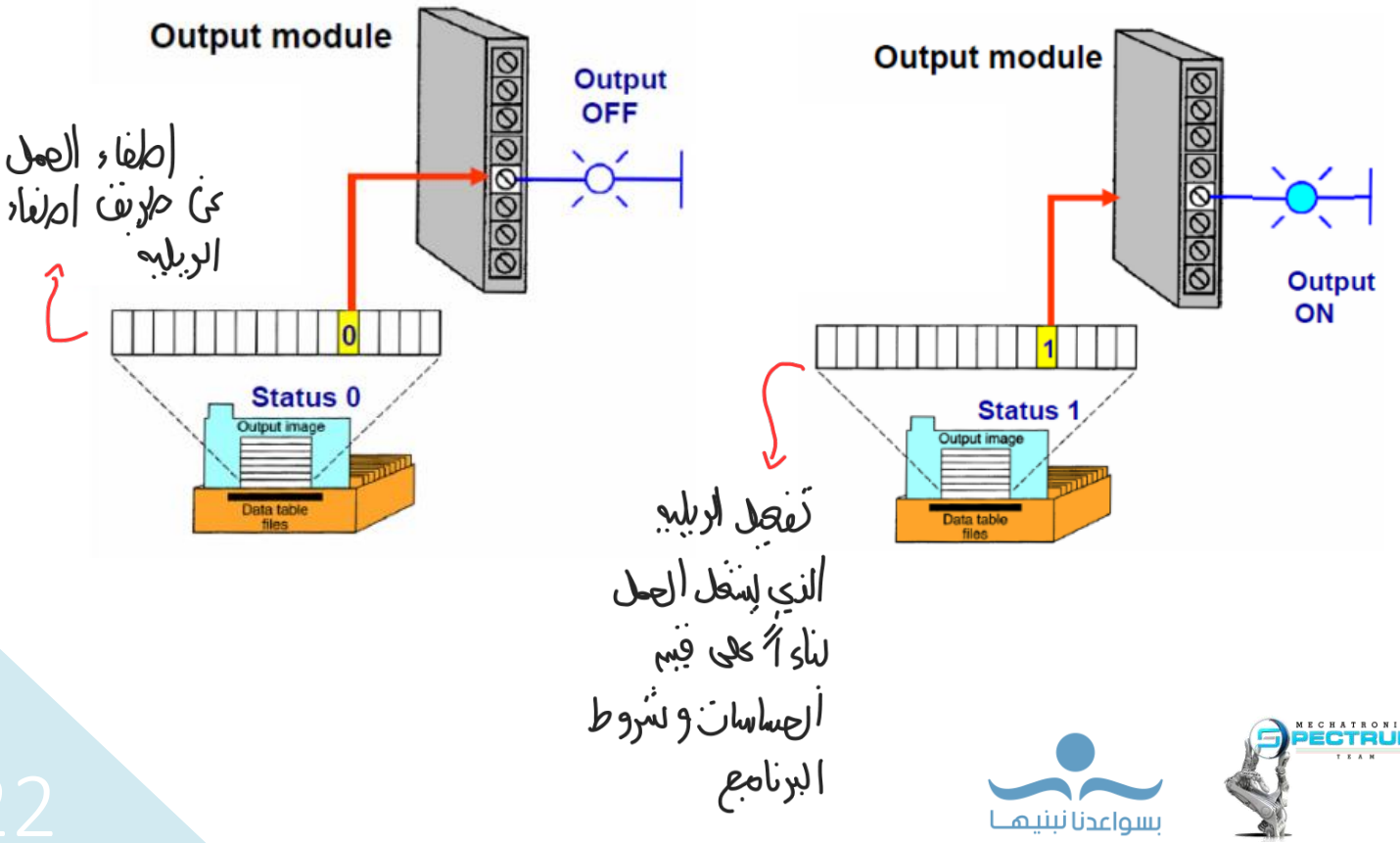
Input Table file operation

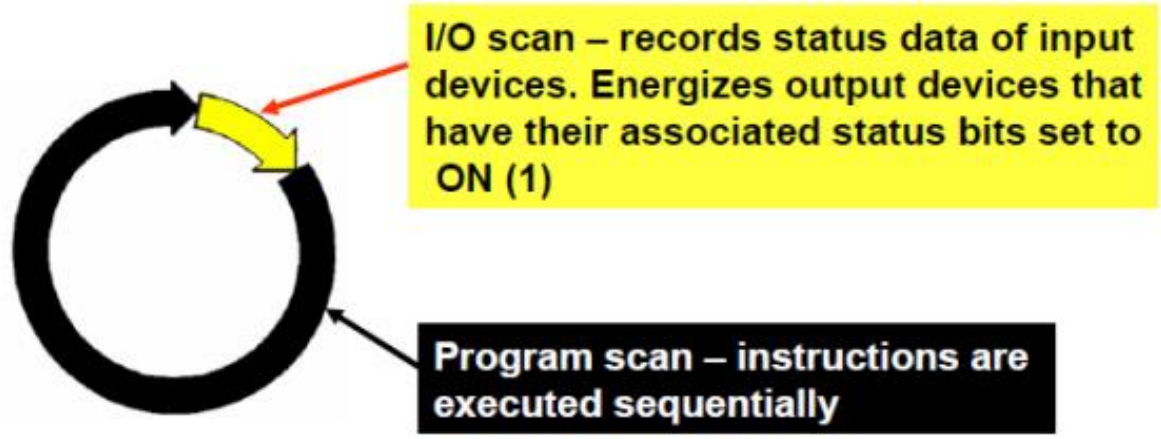


Output able file operation

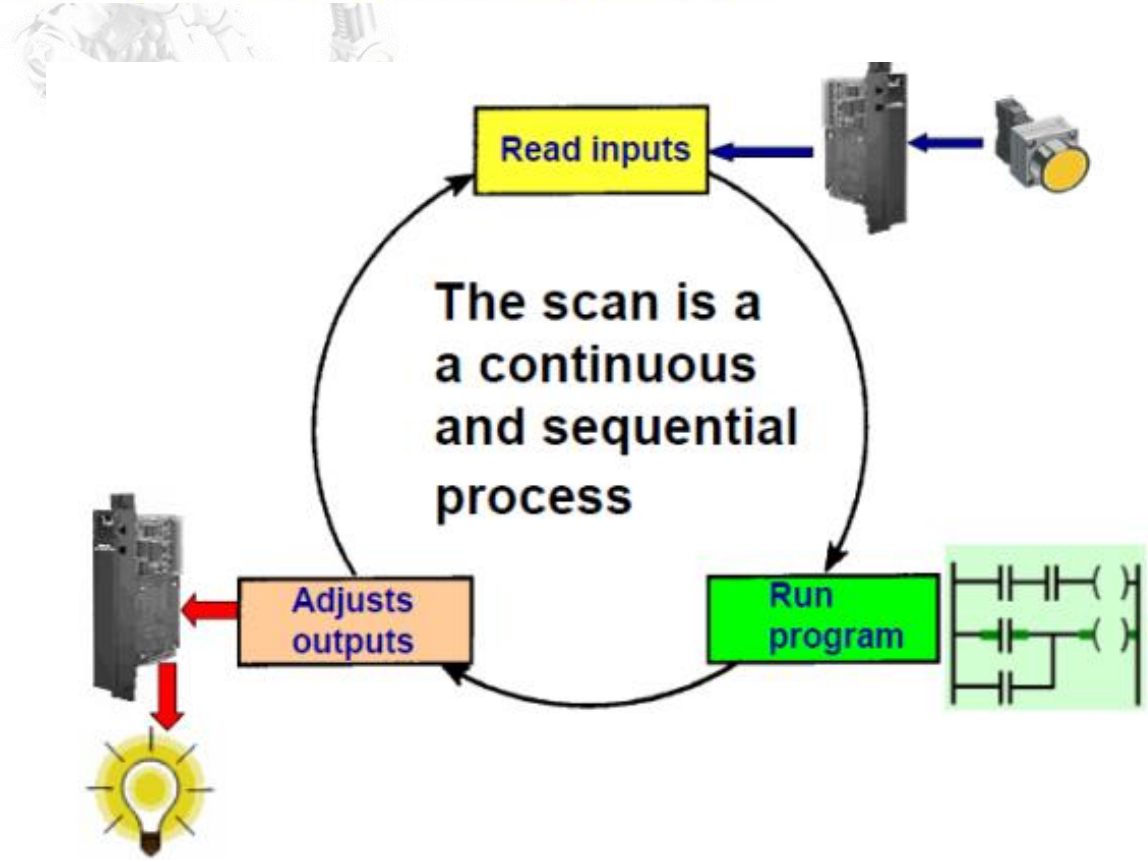


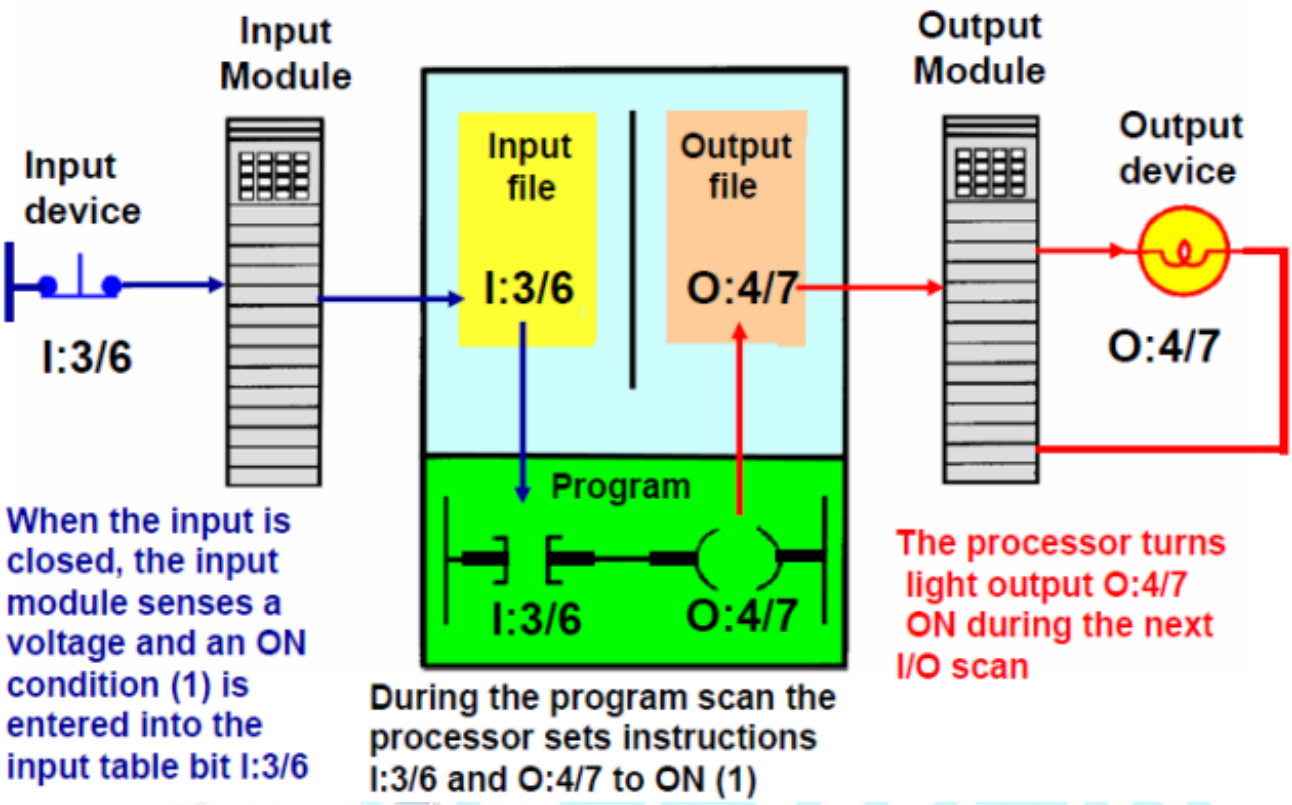
Program Scan

During each operating cycle, the processor reads all inputs, takes these values, and energizes or de-energizes the outputs according to the user program. This process is known as a *scan*.



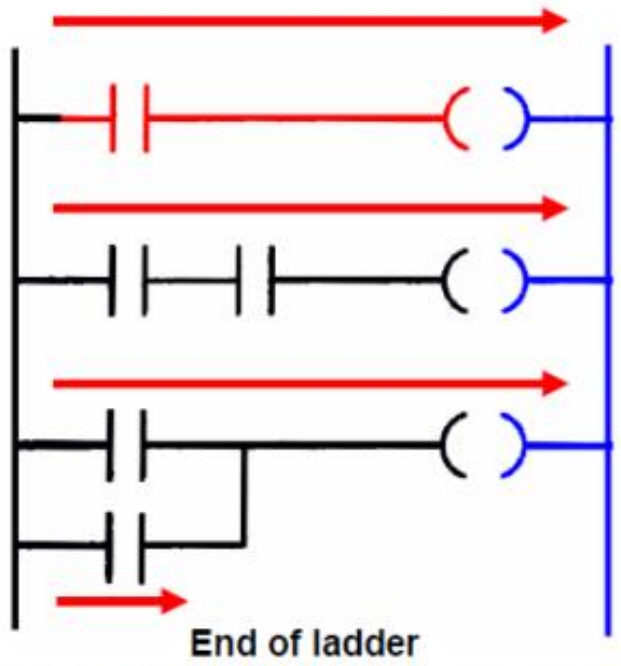
Because the inputs can change at any time, the PLC must carry on this process continuously.





T E A M

Scan patterns



مبدأ عمل PLC داخلياً:

أولاً قراءة كل قيم الحساسات

ثانياً تنفيذ البرنامج

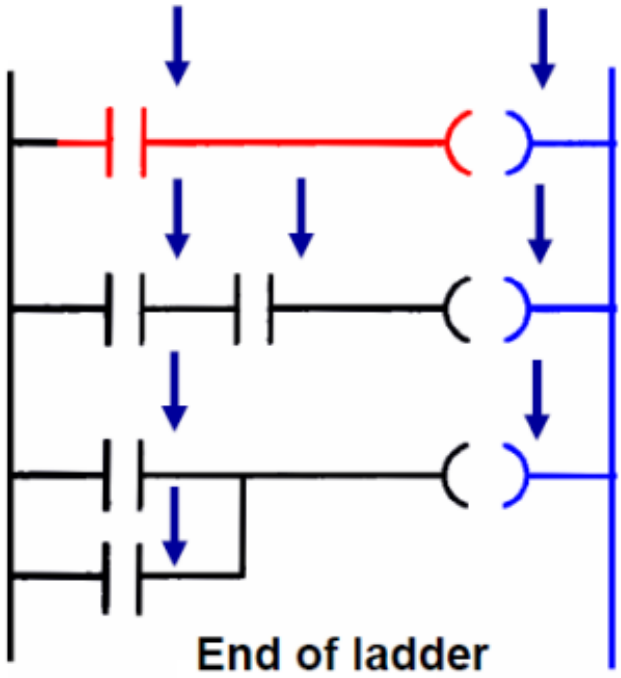
ثالثاً تعديل وضع الـ output بناءً على قيم الحساسات و البرنامج

In addition to the program itself, the scan time is also dependent on the clock frequency of the processor!

بسواعدنا نبنيها
MECHATRONICS SPECTRUM TEAM

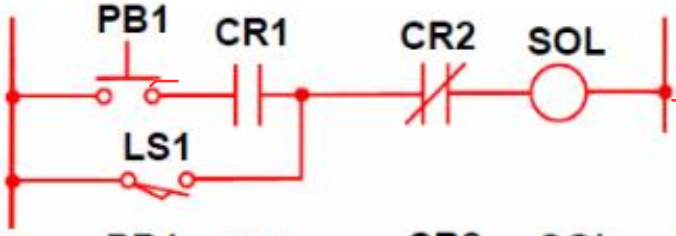
Vertical Scanning Order

The processor examines input and output instructions from the first command, vertically, column by column and page by page. Pages are executed in sequence.

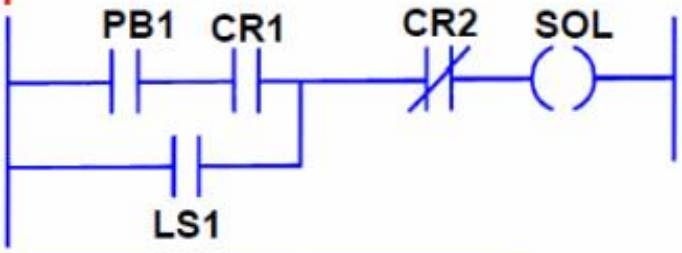


Misunderstanding the way the PLC scans can cause programming bugs!

Comparing program language



Relay Schematic → وضع رموز الحساسات



Equivalent ladder diagram language → استبدال رموز الحساسات ب رليهات

START	PB1
AND	CR1
OR	LS1
AND NOT	CR2
OUT	SOL

Boolean equation: $Y = [(AB) + C]\,\overline{D}$

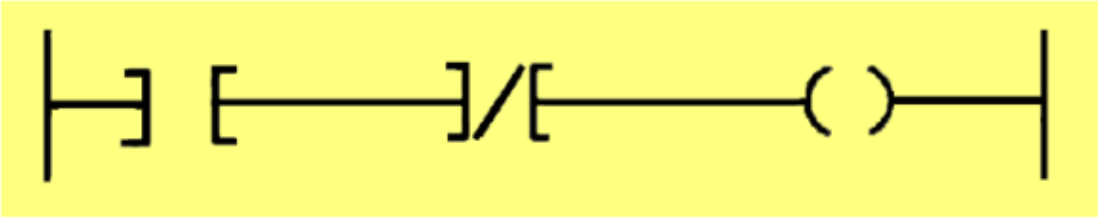
Equivalent Boolean language → لغة برمجة اخرى للـ PLC

بسواعدنا نبنيها
MECHATRONICS SPECTRUM TEAM

Relay type instruction

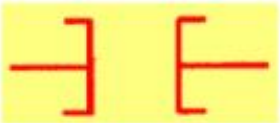
The ladder diagram language is basically a *symbolic* set of instructions used to create the controller program.

These ladder instructions symbols are arranged to obtain the desired control logic.



XIC instruction

Symbol

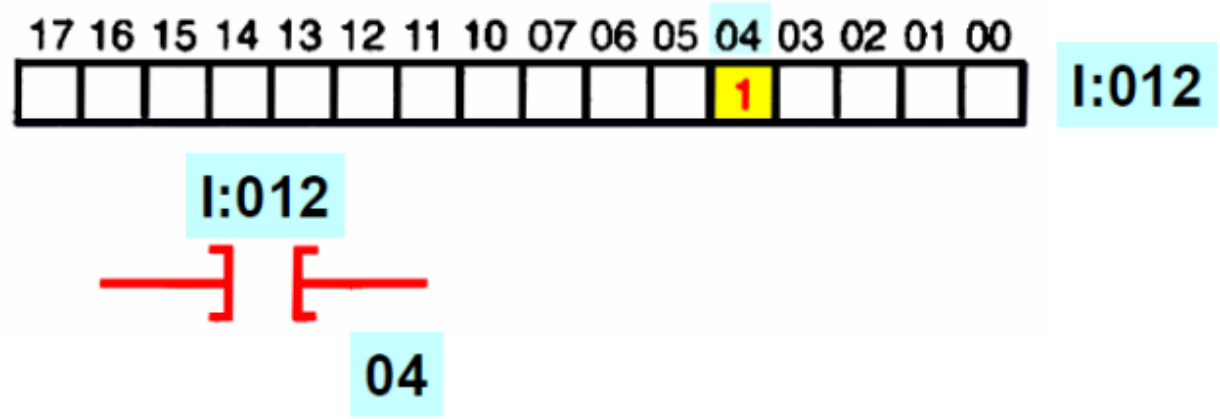


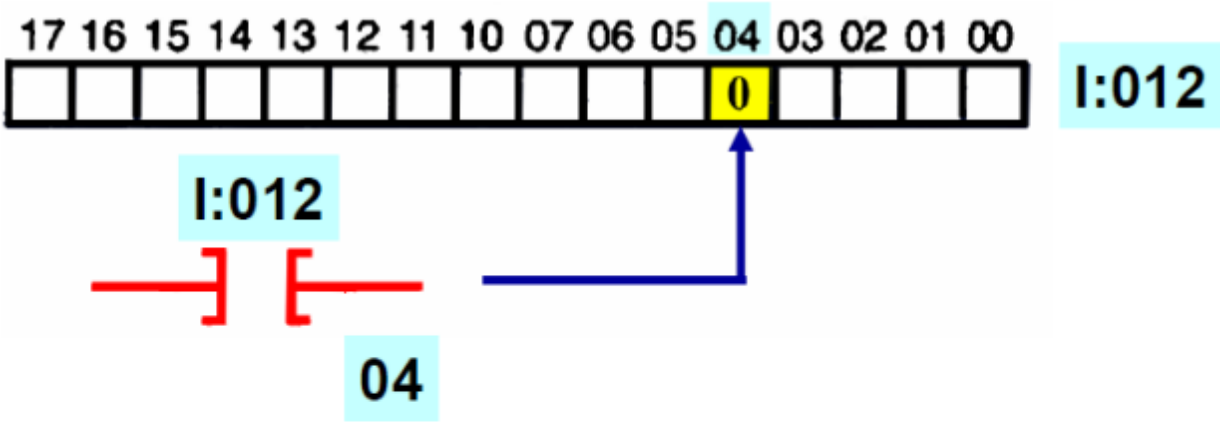
Analogous to the normally open relay contact. For this instruction we ask the processor to EXAMINE IF (the contact is) CLOSED (XIC)

Typically represents any input. Can be a switch or pushbutton, a contact from a connected output, or a contact from an internal output.

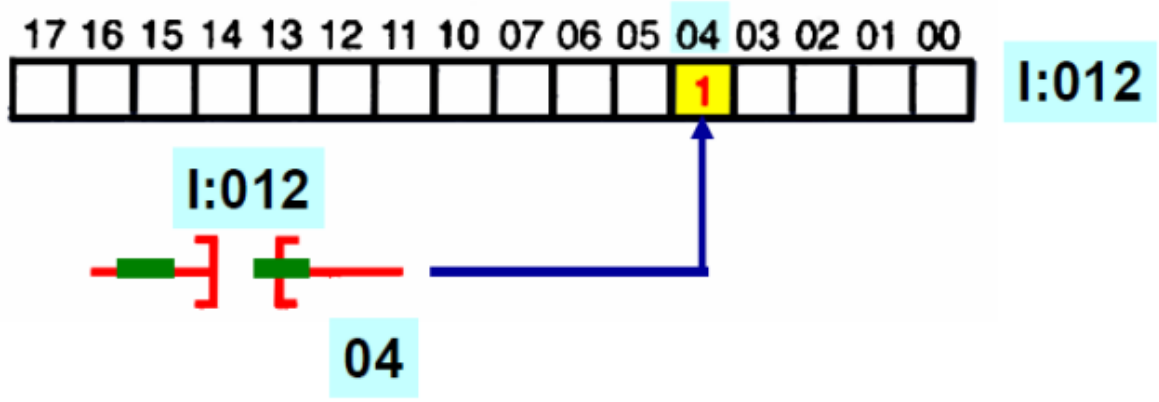
Has a bit-level address which is examined for an ON condition.

The status bit will be either 1 (ON) or 0 (OFF).





If the status bit is 0 (OFF), then the instruction is FALSE.



If the status bit is 1 (ON), then the instruction is TRUE.

XIO instruction

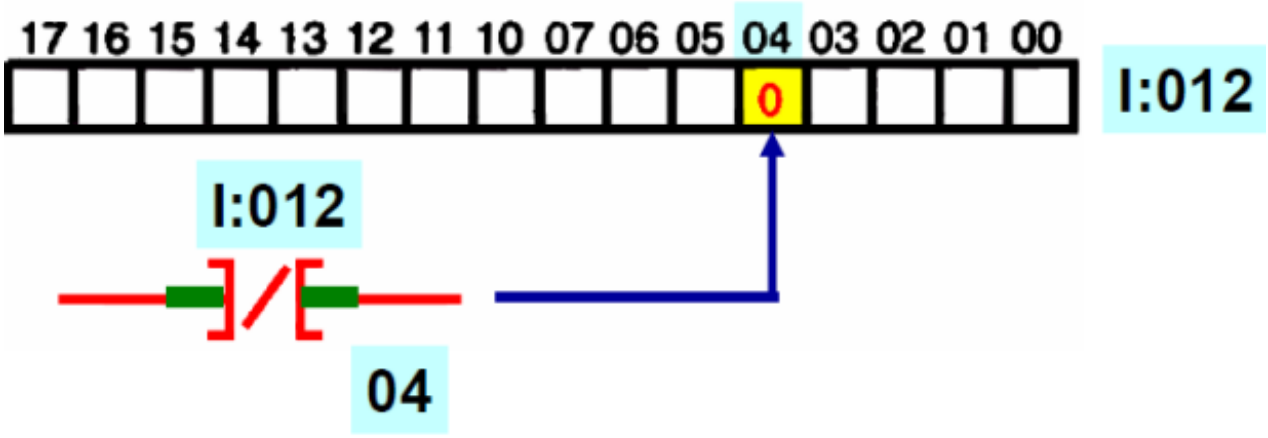
Symbol

Analogous to the normally closed relay contact. For this instruction we ask the processor to EXAMINE IF (the contact is) OPEN (XIO).

Typically represents any input. Can be a switch or pushbutton, a contact from a connected output, or a contact from an internal output.

Has a bit-level address which is examined for an OFF condition.

The status bit will be either 1 (ON) or 0 (OFF).



If the status bit is 0 (OFF), then the instruction is TRUE.

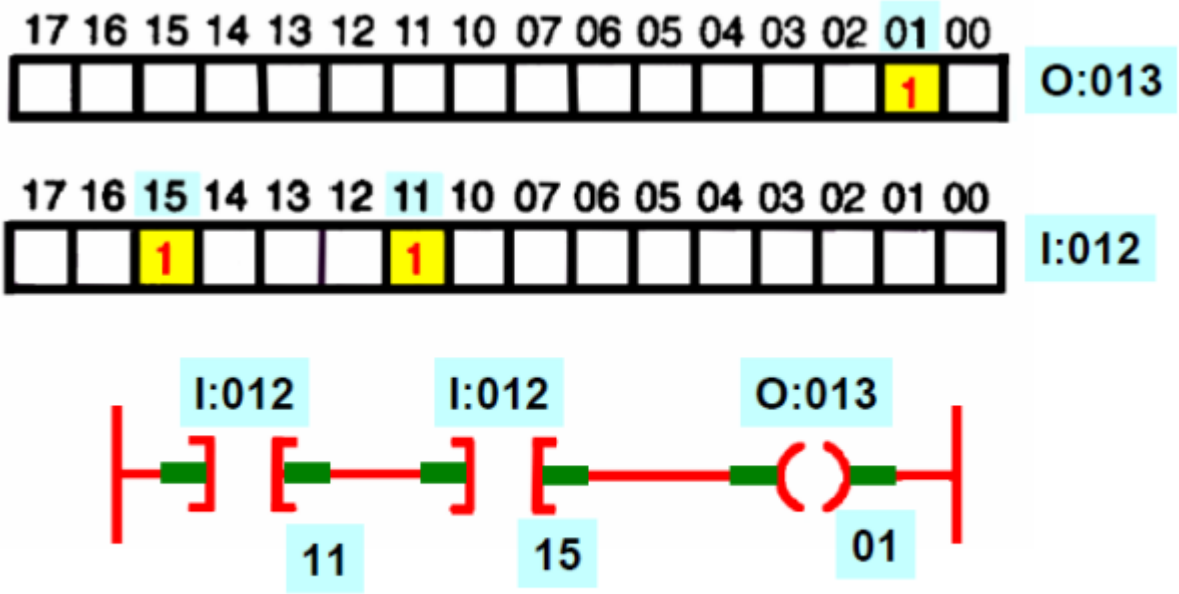
OTE instruction

Symbol

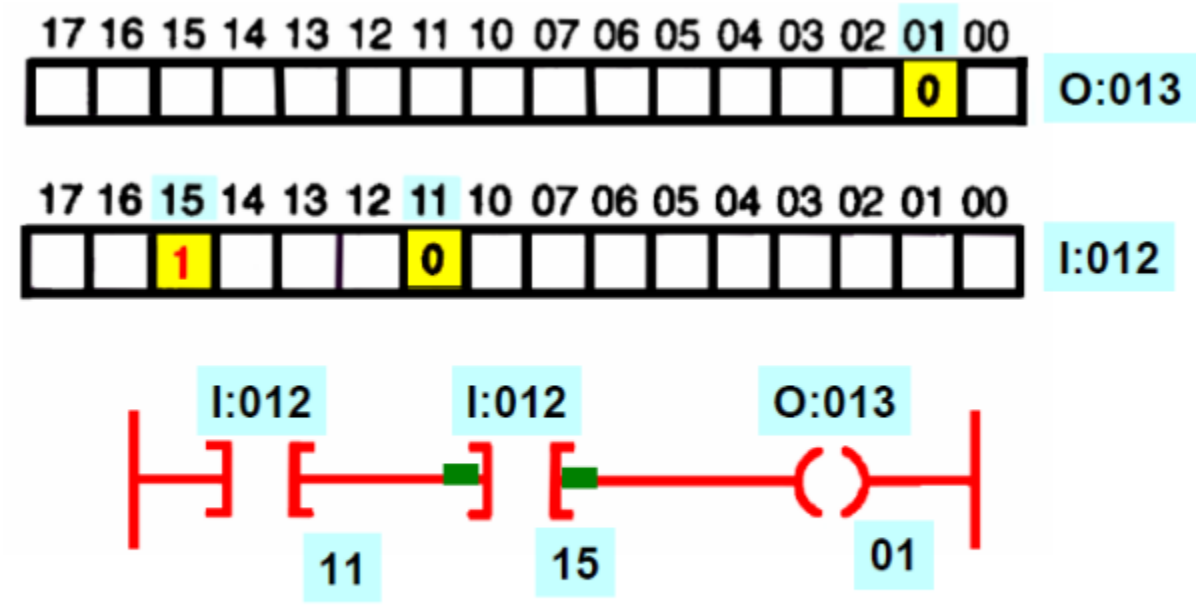
Analogous to the relay coil. The processor makes this instruction true (analogous to energizing a coil) when there is path of true XIC and XIO instructions in the rung.

Typically represents any output that is controlled by some combination of input logic. Can be a connected device or an internal output (internal relay).

If any left-to-right path of input conditions is TRUE, the output is energized (turned ON).

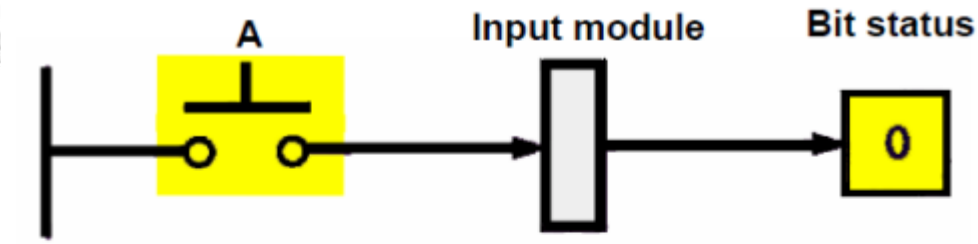


OUTPUT ENERGIZE instruction - TRUE

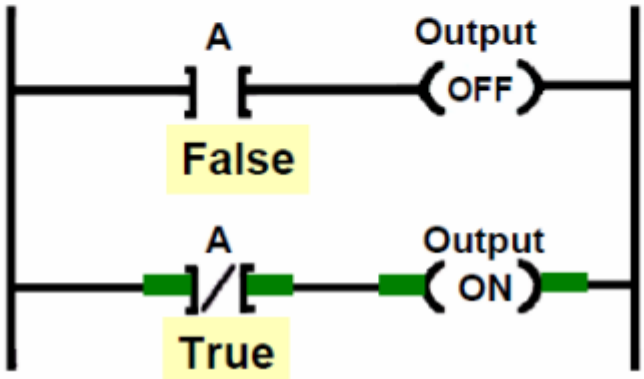


OUTPUT ENERGIZE instruction - FALSE

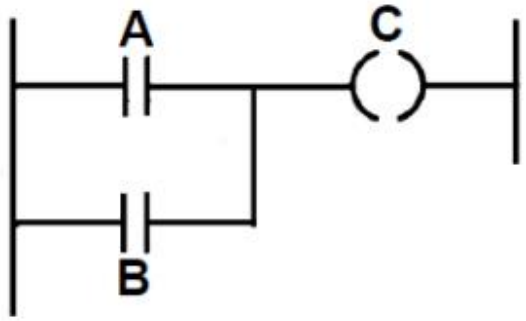
Status Bit example



Button not actuated

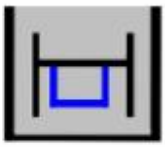


Parallel Input



Branch instructions are used to create parallel paths of input condition instructions. If at least one of these parallel branches forms a true logic path, the logic is enabled.

Add A Branch

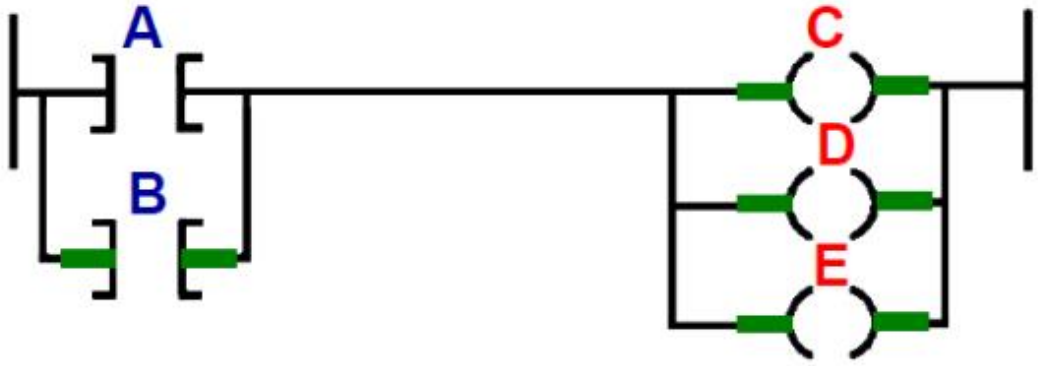


Click this icon on the instruction toolbar to place a branch in your ladder logic. If your cursor is on an instruction, the branch is placed immediately to the right of the instruction. If your cursor is on the rung number, the branch is placed on the rung.

معناها ان الحمل C يمكن ان يعمل من الحساس A او B

Parallel Output

معناها ان الحساس A او B يمكن ان يشغل الثلاث احمال C و D و E معاً

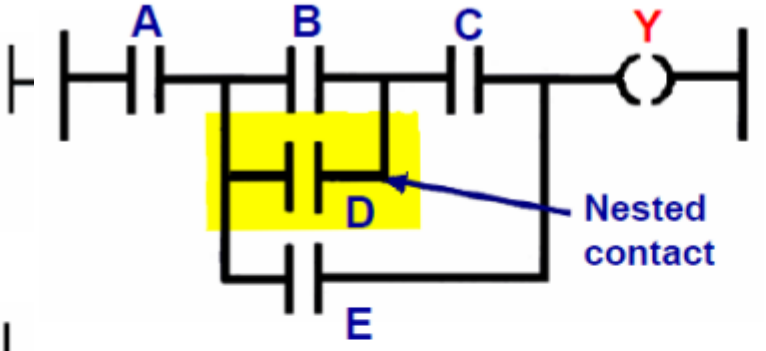


On most PLC models, branches can be established at both the input and output portion of the rung.

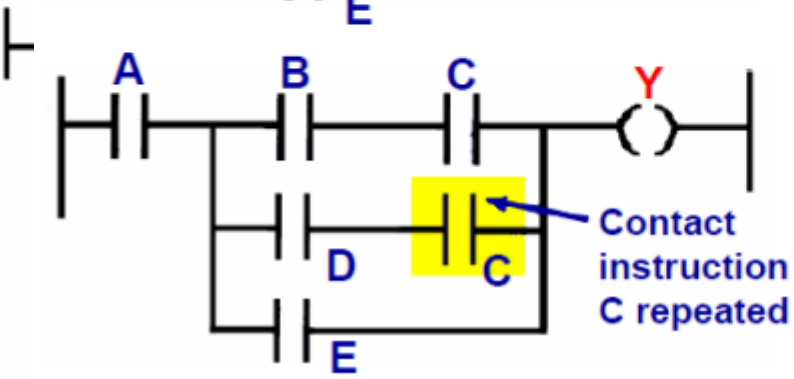
With output branching, you can program parallel outputs on a rung to allow a true logic path to control multiple outputs.

بسواعدنا نبنيها
MECHATRONICS SPECTRUM TEAM

Nested Connect program



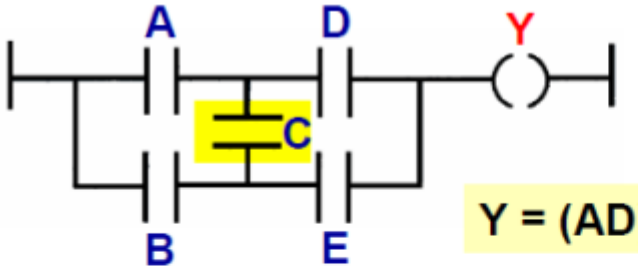
On some PLC models, the programming of a nested branch circuit cannot be done directly



Reprogrammed to obtain the required logic.

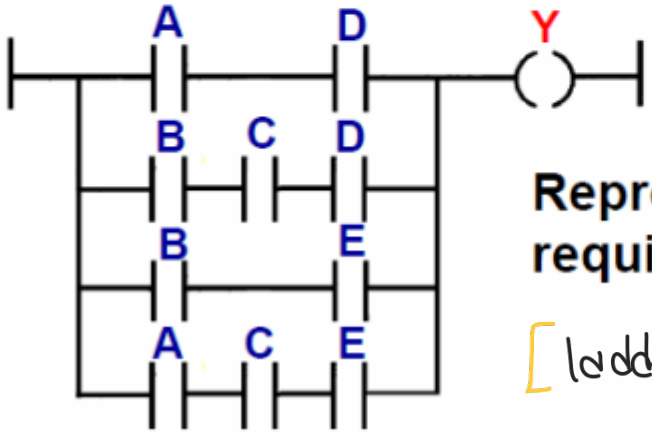
[تبسيط شكل ladder]

Simplifying the ladder Diagram



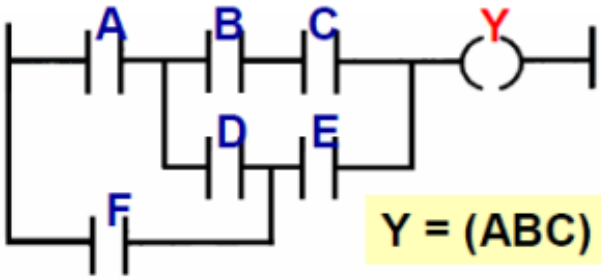
Original program

Y = (AD) + (BCD) + (BE) + (ACE)



Reprogrammed to obtain the required logic

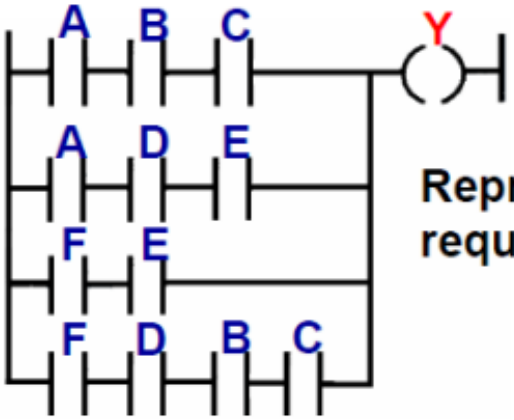
[تبسيط شكل ladder]



Original program

$$Y = (ABC) + (ADE) + (FE) + (FDBC)$$

[تبسيط شكل ladder]



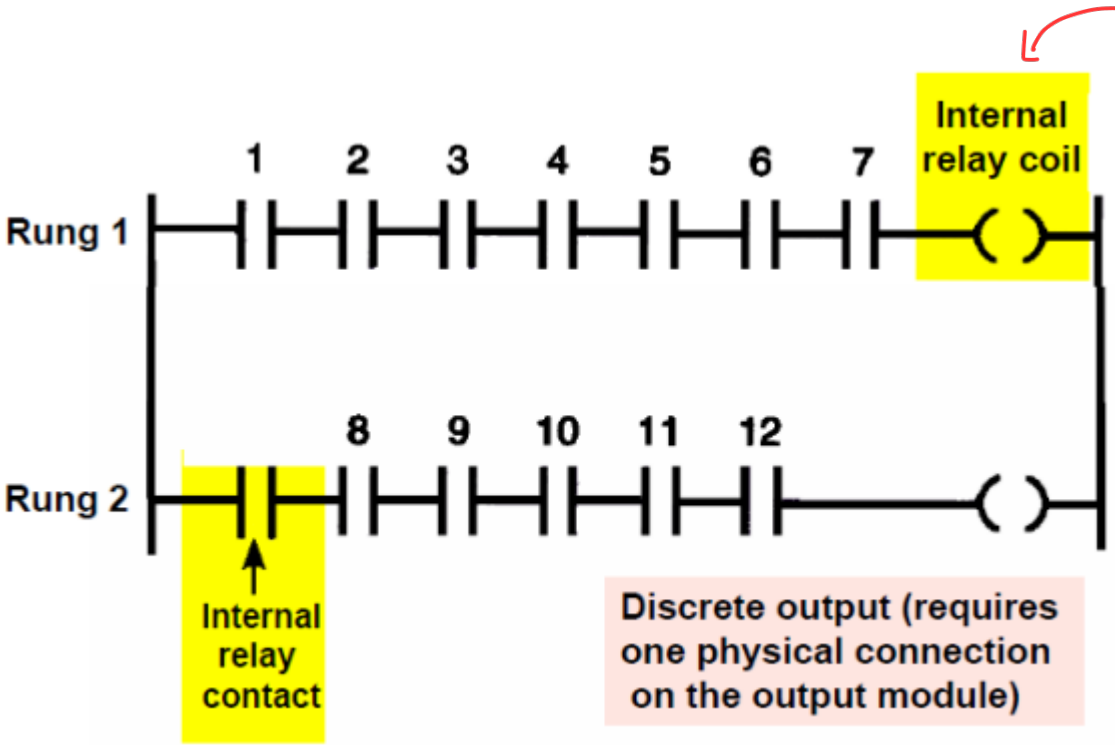
Reprogrammed to obtain the required logic

Internal control relay

The internal output operates just as any other output that is controlled by programmed logic; however, the output is used strictly for internal purposes.

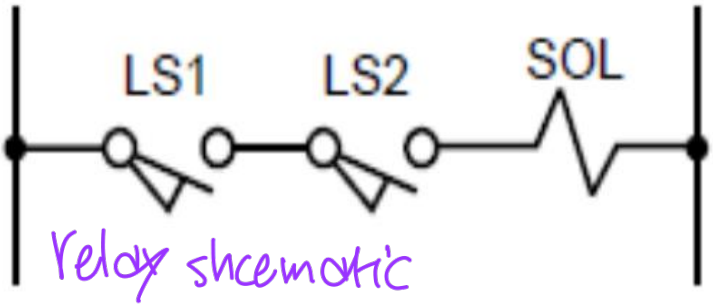
The internal output does not directly control an output device.

The advantage of using internal outputs is that there are many situations where an output instruction is required in a program, but no physical connection to a field device is needed. Their use in this type of instance can minimize output card requirements.



يستخدم بلمارة لعمل latch البرنامج

Examples of ladder diagram

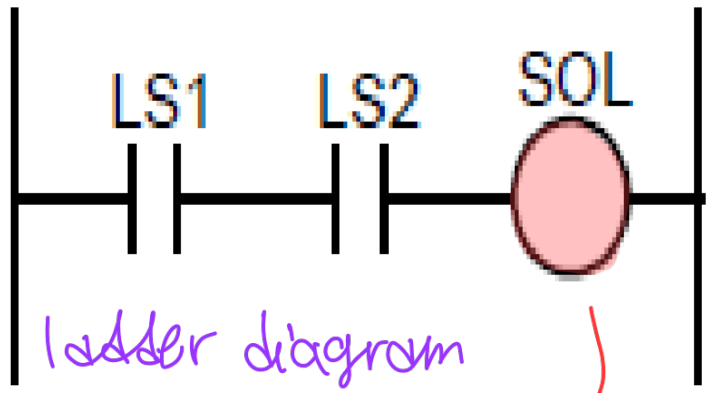


Relay shcematic

مبدأ الحل : سيعمل السخان عند تفعيل حساس LS1 و LS2 معاً لتشغيل الماء في التنك

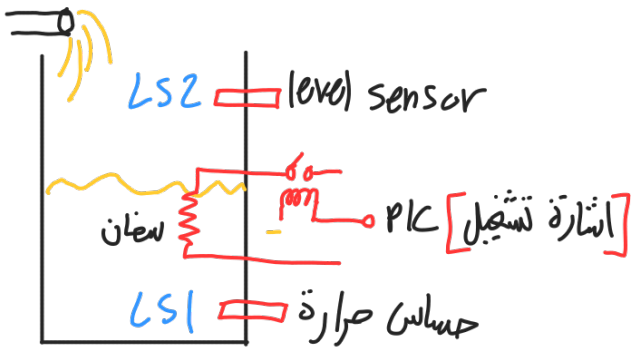
AND gate

تحويل

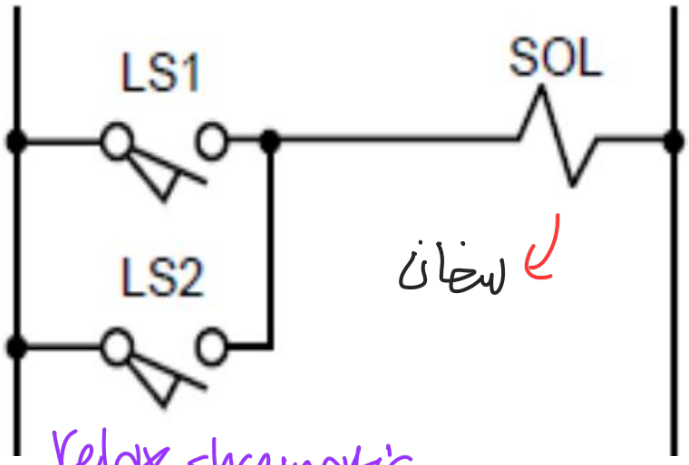


ladder diagram

سخان



بسواعدنا نبنيها
MECHATRONICS SPECTRUM TEAM

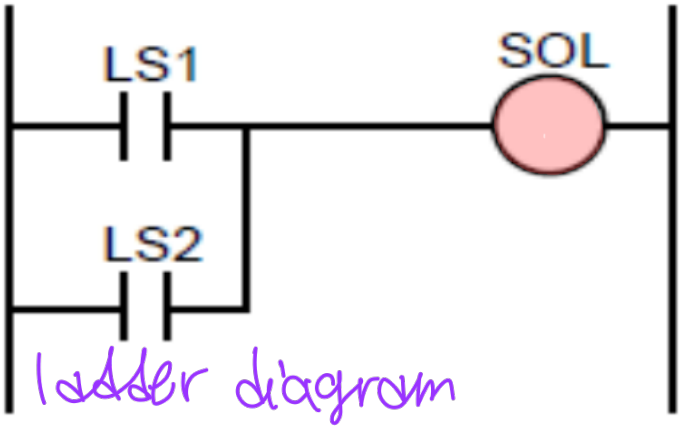


Relay shcematic

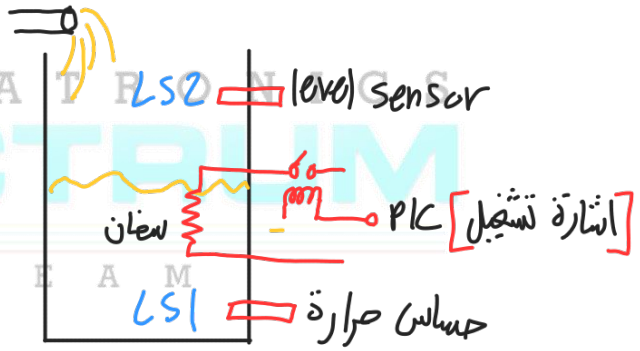
مبدأ العمل : سيتم تشغيل السخان لتسخين الماء عند تفعيل الحساس LS1 او LS2

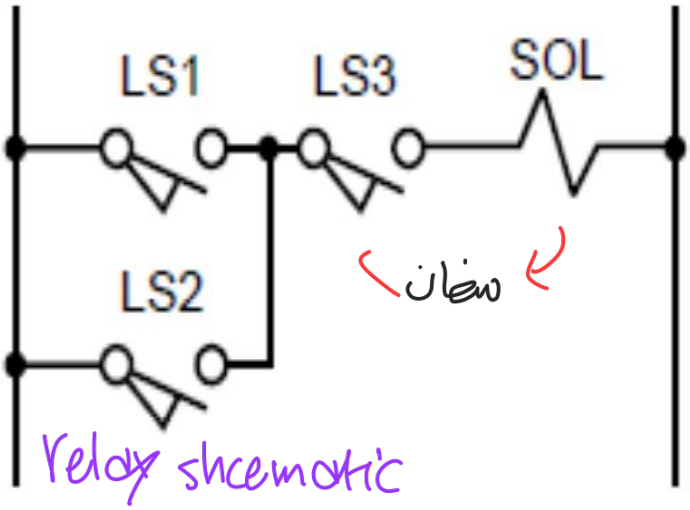
OR gate

تحويل



ladder diagram





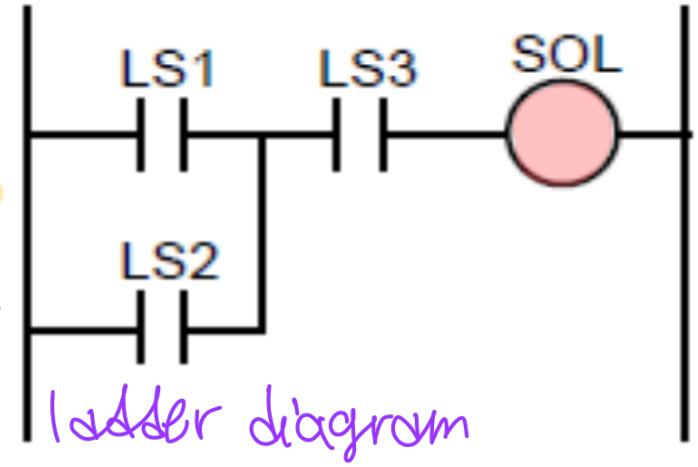
Relay shcematic

مبدأ العمل : سيتم تشغيل السخان عند تفعيل الحساس LS1 او الحساس LS2 مع الحساس LS3

OR gate + AND gate

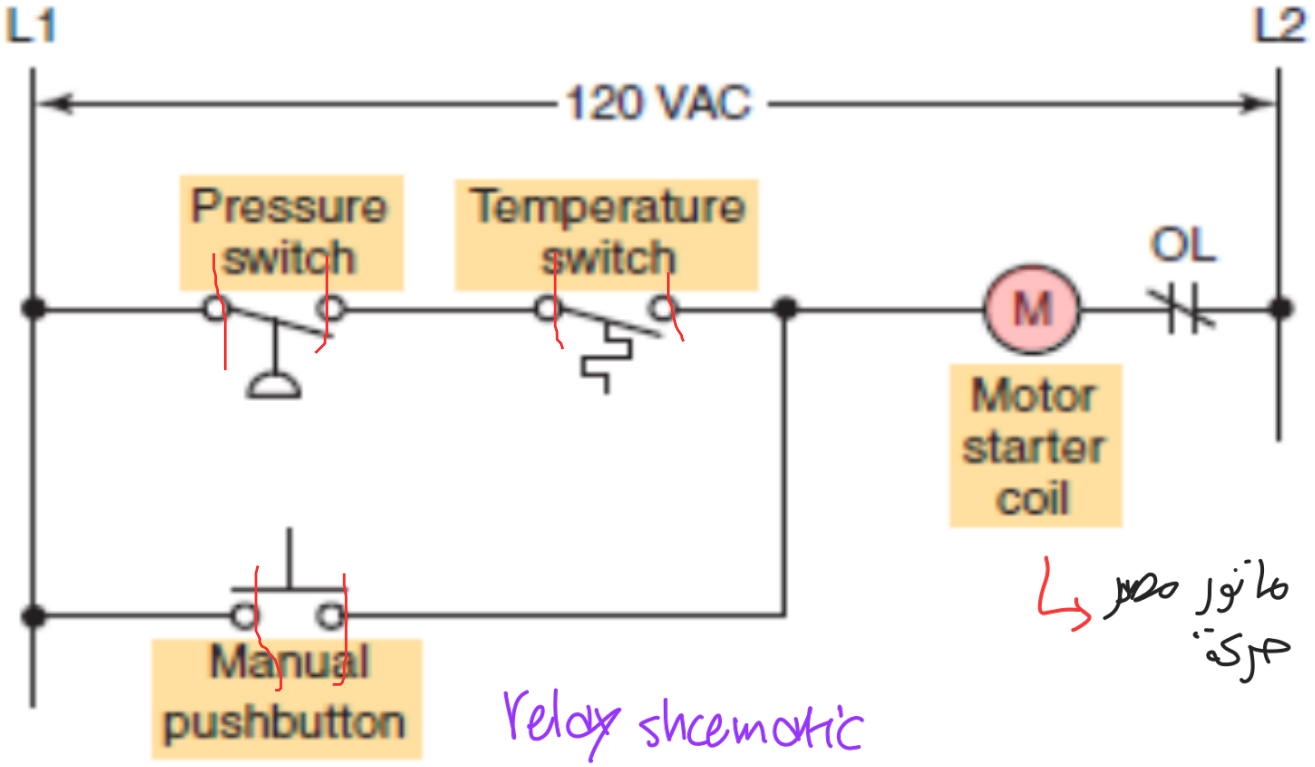
بسواعدنا نبنيها
MECHATRONICS SPECTRUM TEAM

تم تحويل رمز
الحساسات و الخانتج
لرموز الريلاي



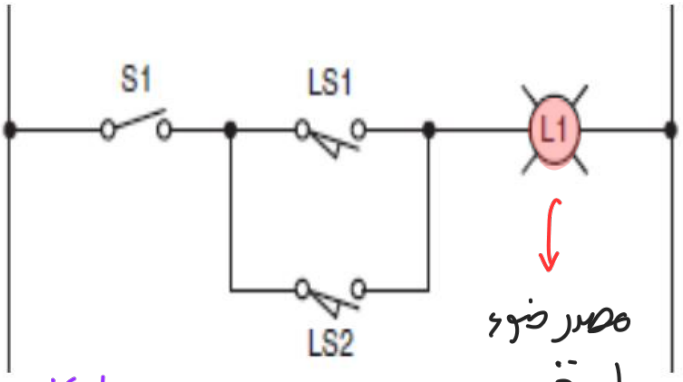
تحويل

ladder diagram



ماتور مصدر حركة

Relay shcematic

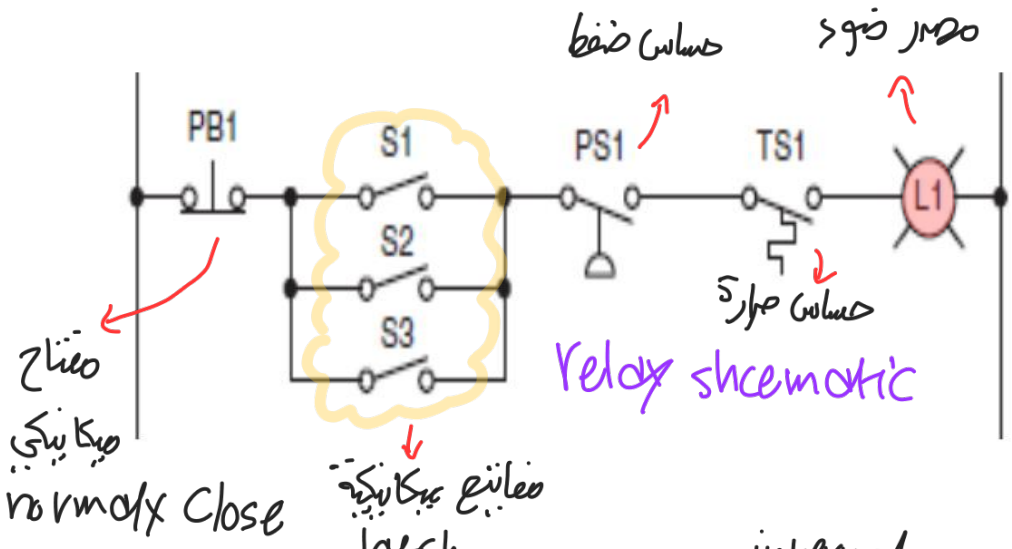


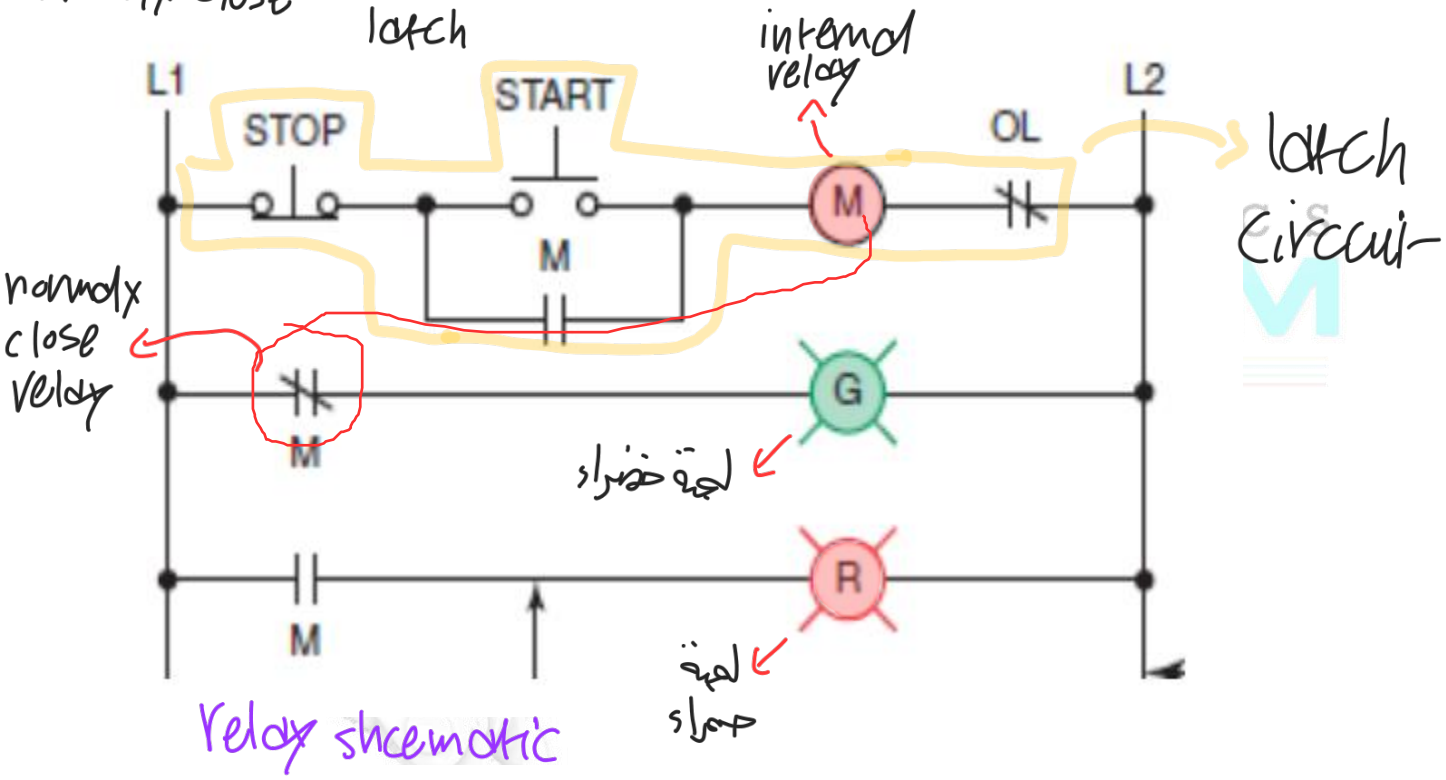
مصدر ضوء لمبة

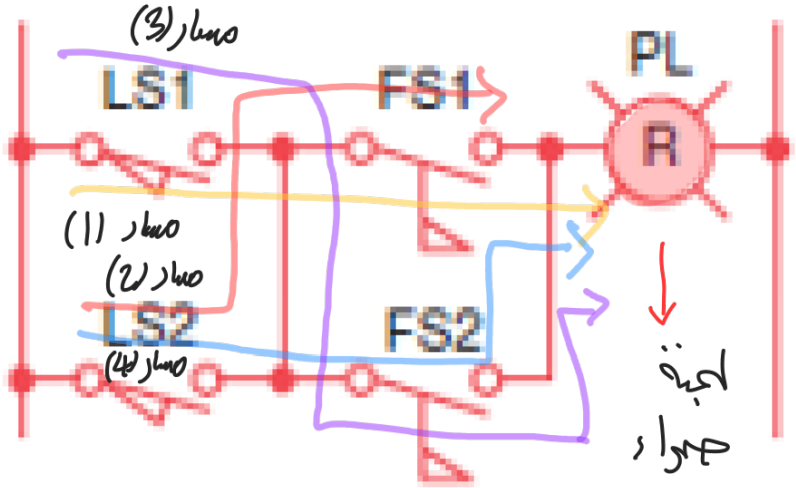
Relay shcematic

مبدأ العمل: سيتم تشغيل المصباح عند تفعيل الحساس او المفتاح S1 مع LS1 او مع LS2

AND gate + OR gate

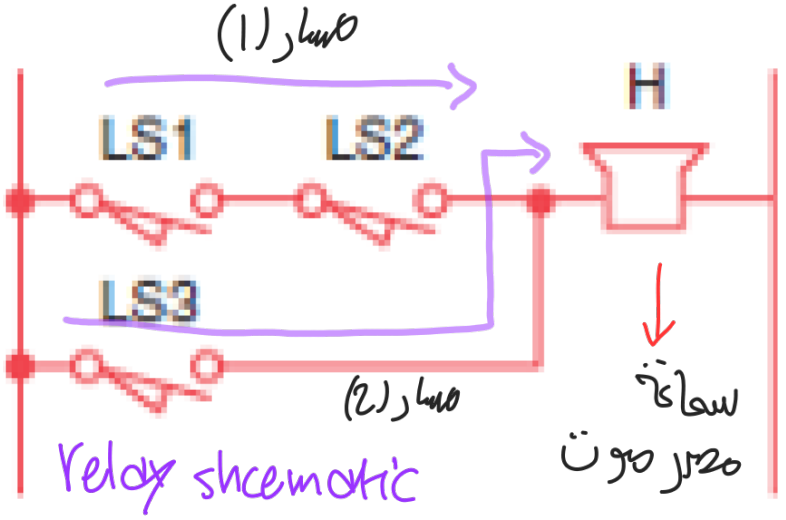




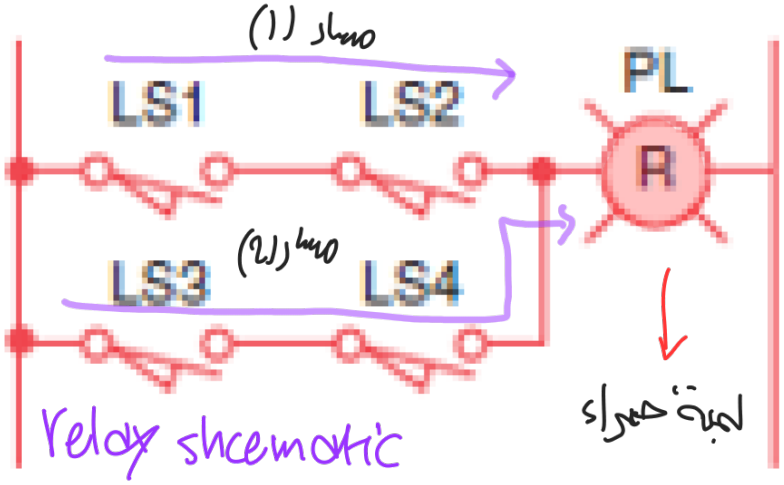


هناك اربع مسارات لتفعيل المصباح 4 شروط

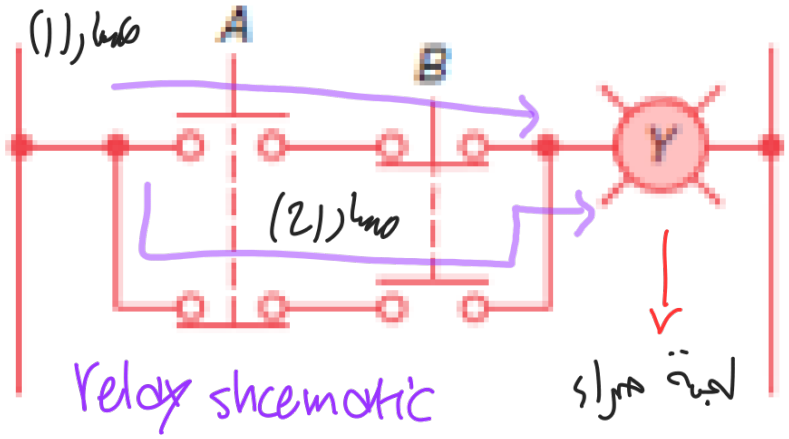
بسواعدنا نبنيها
MECHATRONICS SPECTRUM TEAM



أمثلة مسارين لتشغيل السماعة



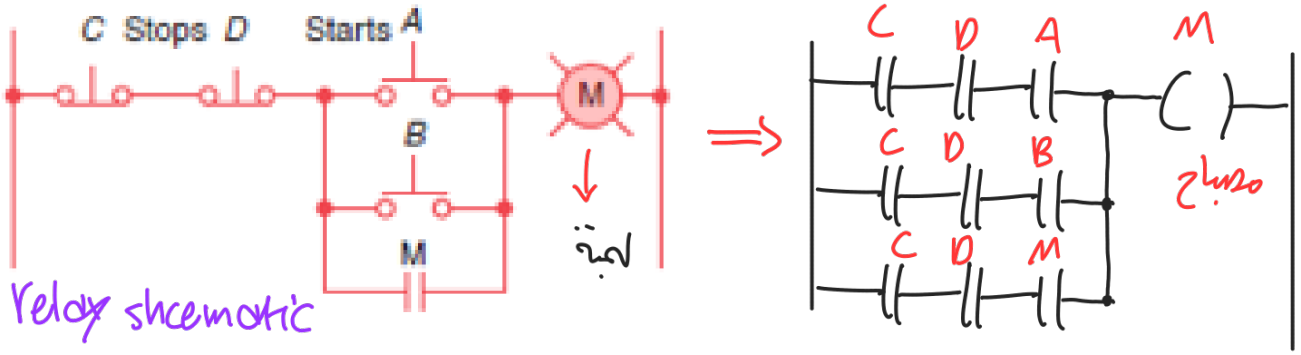
أمثلة مسارين لتشغيل السماعة



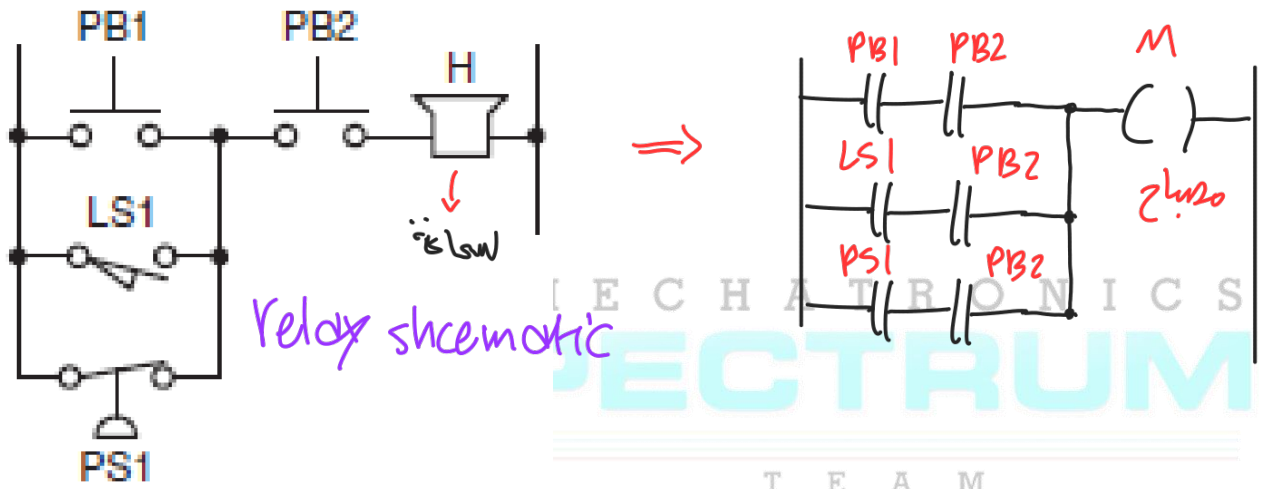
أمثلة مسارين لتشغيل السماعة

TRONICS
TRUM
A M

بسواعدنا نبنيها
MECHATRONICS SPECTRUM TEAM

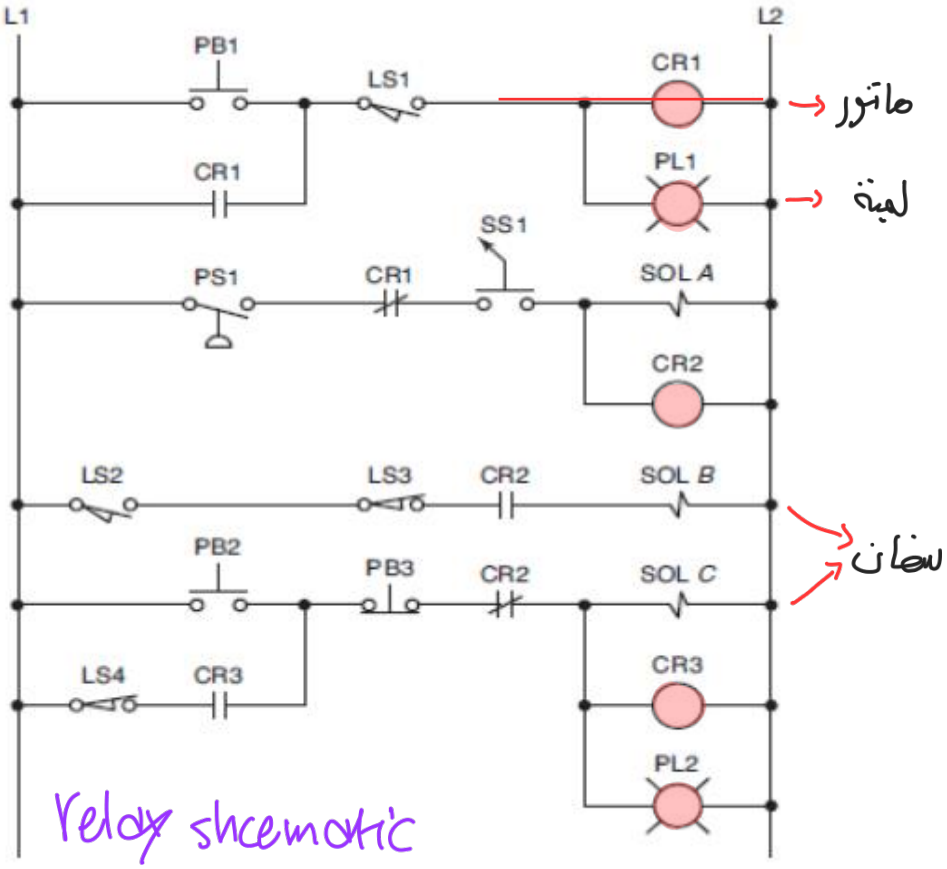


Relay shcematic



Relay shcematic

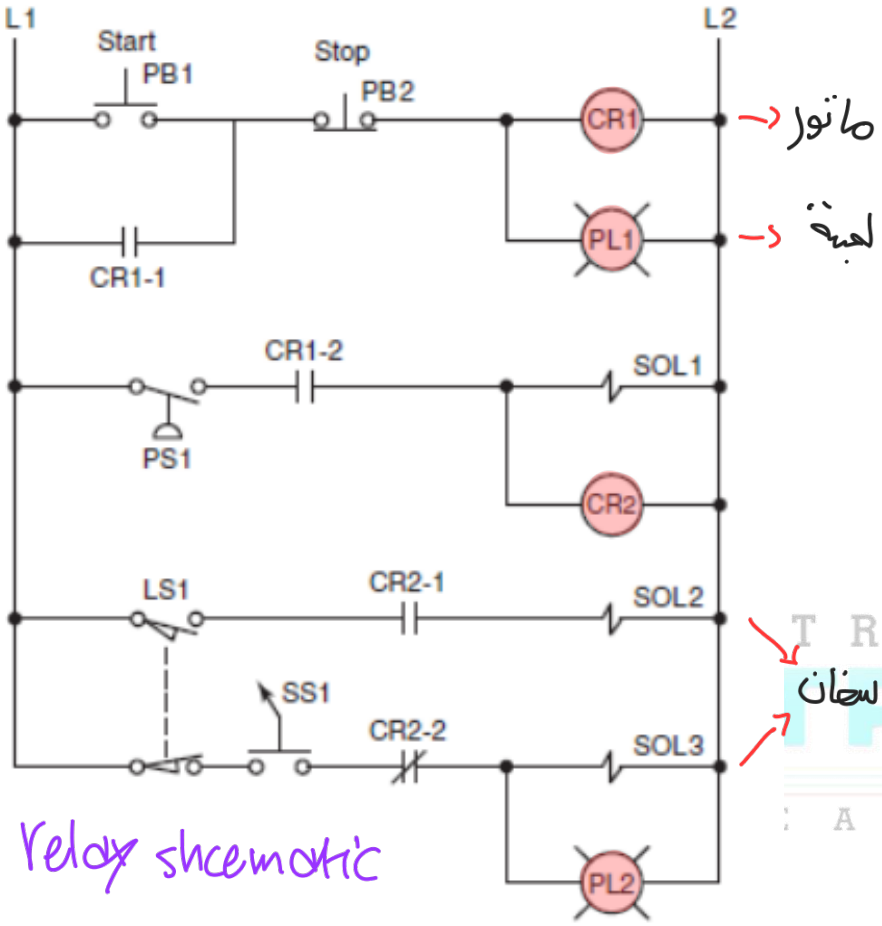
MECHATRONICS SPECTRUM TEAM



Relay shcematic

ملاحظة: يمنع تكرار output في ladder أكثر من rung

بسواعدنا نبنيها
MECHATRONICS SPECTRUM TEAM



Relay shcematic

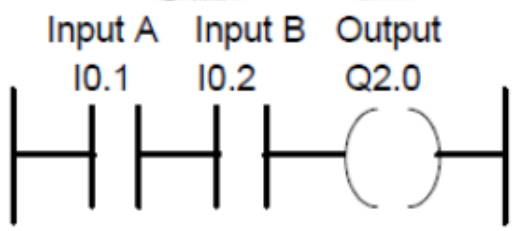
انواع المدخلات على PLC
- حساس مسافة
- مفتاح
- حساس حرارة
- حساس مياه
- حساس وزن
- حساس الوان

انواع الاحمال على PLC
- مصدر ضوء مثل لمبة
- مصدر صوت مثل سماعة
- مصدر حركة مثل ماتور
- مصدر حرارة مثل سخان

Chapter 3

IEC 1131-3 Operators	Mitsubishi	OMRON	Siemens/ Telemecanique	Operation	Ladder diagram
LD	LD	LD	LD	Load operand into result register	Start a rung with open contacts
LDN	LDI	LD NOT	LDN	Load negative operand into result register	Start a rung with closed contacts
AND	AND	AND	A	Boolean AND	A series element with open contacts
ANDN	ANI	AND NOT	AN	Boolean AND with negative operand	A series element with closed contacts
OR	OR	OR	O	Boolean OR	A parallel element with open contacts
ORN	ORI	OR NOT	ON	Boolean OR with negative operand	A parallel element with closed contacts
ST	OUT	OUT	=	Store result register into operand	An output

T E A M



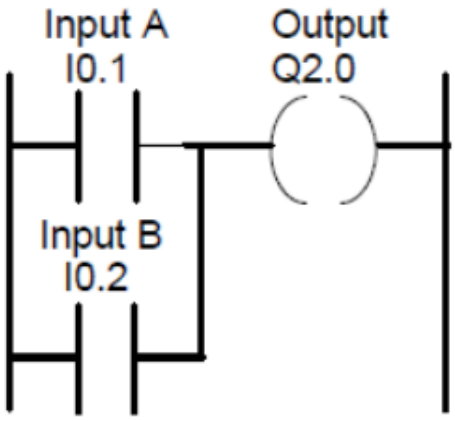
rung

```
LD    I0.1
A     I0.2
=     Q2.0
```

AND

output

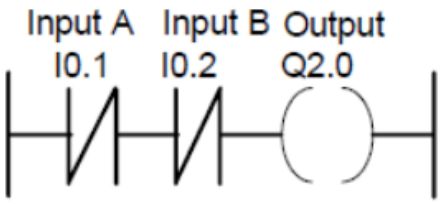
تعويل instruction list



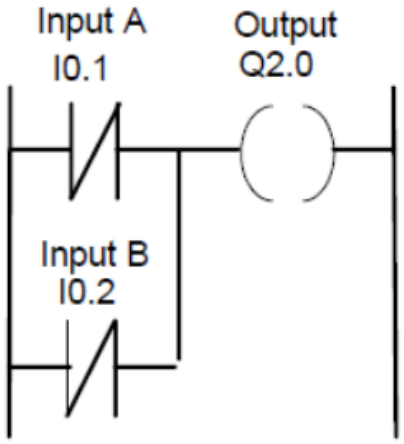

```
LD    I0.1
O     I0.2
=     Q2.0
```

OR

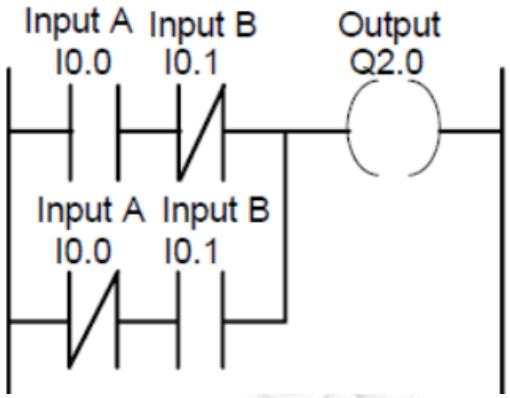
بسواعدنا نبنيها
MECHATRONICS SPECTRUM TEAM



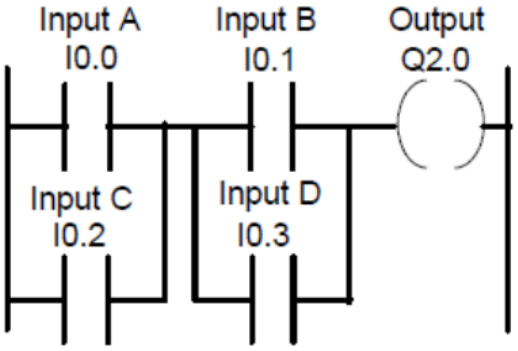

```
LDN     I0.1
AN      I0.2
=       Q2.0
```




```
LDN     I0.1
ON      I0.2
=       Q2.0
```



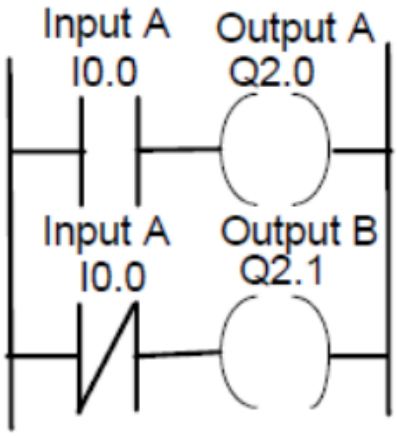

```
LD      I0.0
AN      I0.1
LDN     I0.0
A       I0.1
OLD
=       Q2.0
```



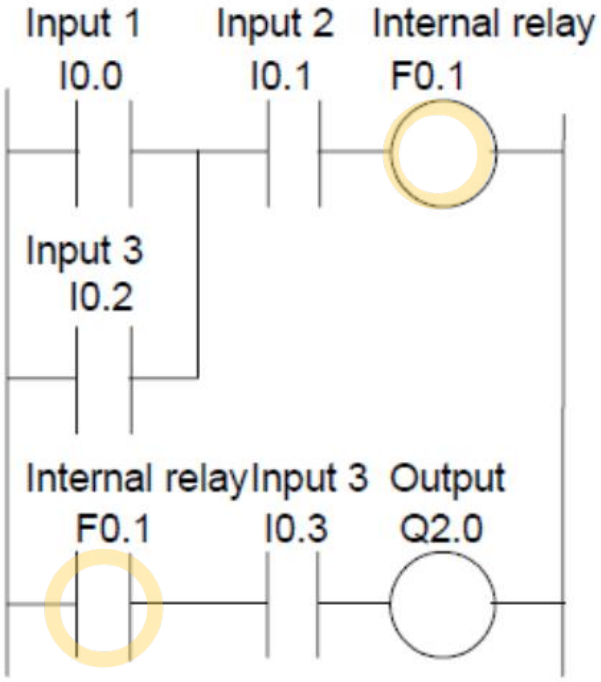

```
LD      I0.0
O       I0.2
LD      I0.1
O       I0.3
ALD
=       Q2.0
```

I => input // I0.1 - I0.9 // I1.0 - I1.9 // I2.0 - I2.9

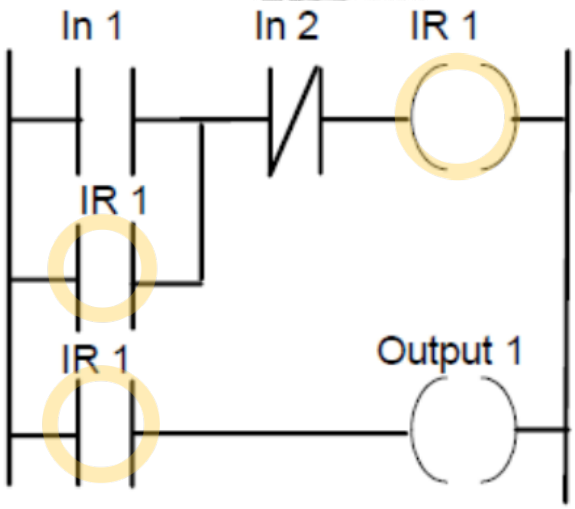
Q => output // Q0.1 - Q0.9 // Q1.0 - Q1.9 // Q2.0 - Q2.9



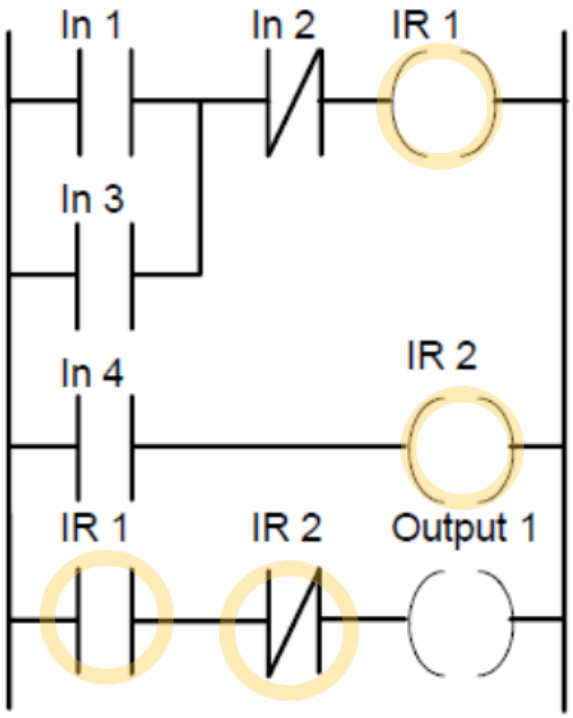

```
LD      I0.0
=       Q2.0
LDN     I0.0
=       Q2.1
```



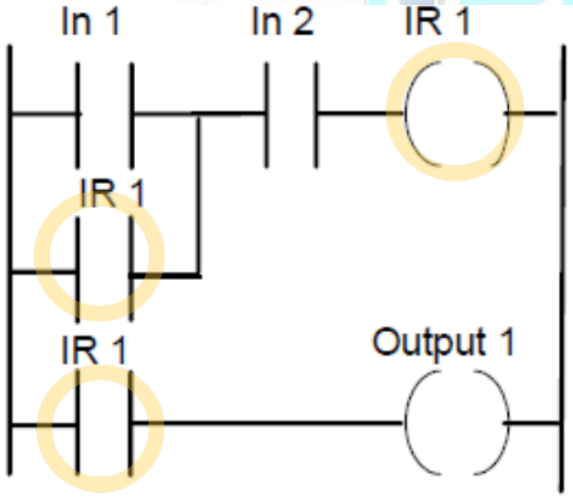

```
LD      I0.0
O       I0.2
A       I0.1
=       F0.1
LD      F0.1
A       I0.3
=       Q2.0
```




```
LD  IN1
O   IR1
AN  IN2
=   IR1
LD  IR
=   OUTPUT1
```




```
LD    IN1
O     IN3
AN    IN2
=     IR1
LD    IN4
=     IR2
LD    IR1
AN    IR2
=     output 1
```

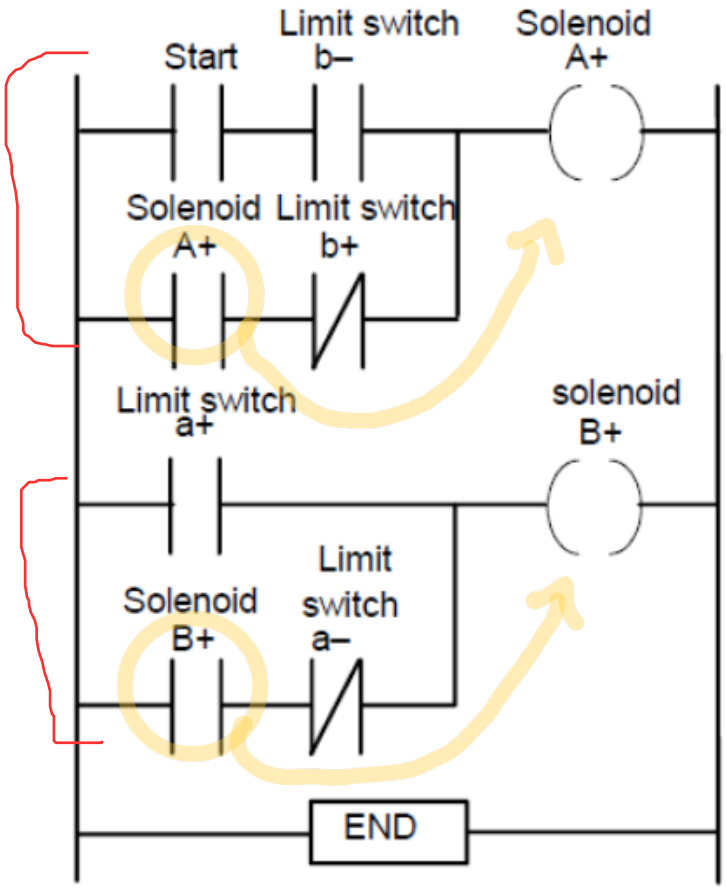



```
LD    IN1
O     IR1
A     IN2
=     IR1
LD    IR1
=     output 1
```

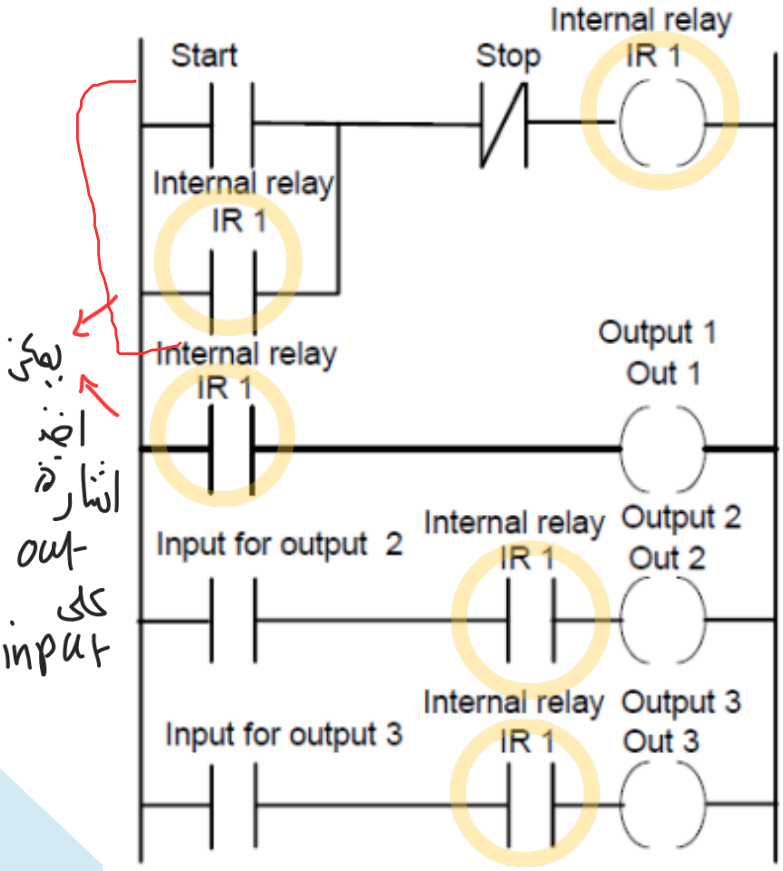
تعني ان اشارة output مرتبطة في inputs

CHATRONICS SPECTRUM TEAM

بسواعدنا نبنيها



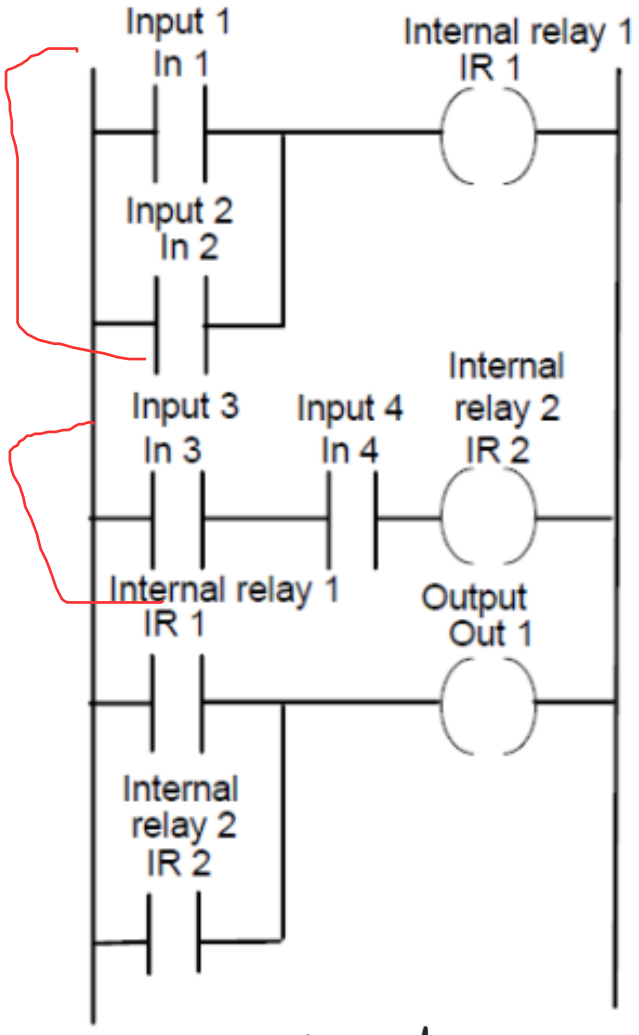

```
LD      start
A       b-
LD      A+
AN      b+
OLD     ...
=       A+      (out تابع لها)
LD      a+
LD      B+
AN      a-
OLD     ...
=       B+      (out- تابع لها)
```



يعني اخذ اشارة out- كـ input

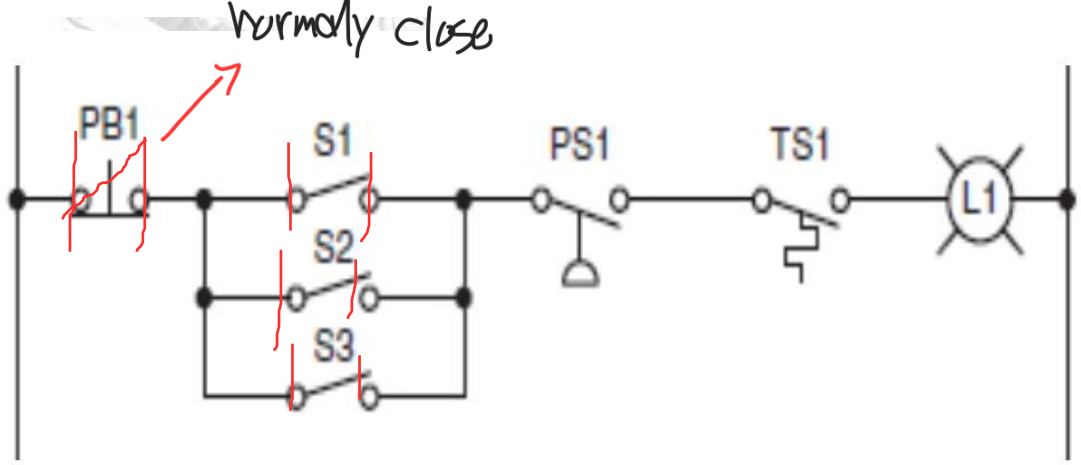
```
LD      start
O       IR1
AN      stop
=       IR1     (تابع لـ)
LD      IR1
=       out 1   (تابع لـ)
LD      input for output 2
A       IR1     (تابع لـ)
=       out 2
LD      input for output 3
A       IR1     (تابع لـ)
=       out 3
```

MECHATRONICS SPECTRUM TEAM
بسواعدنا نبنيها

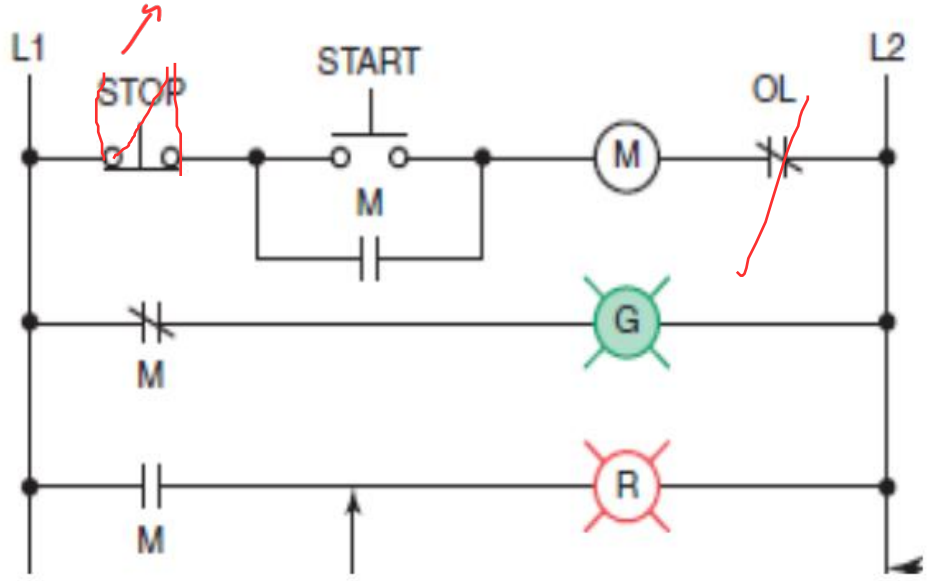



```
LD   in1
O    in2
=    IR1
LD   in3
A    in4
=    IR2
LD   IR1
O    IR2
=    out1
```

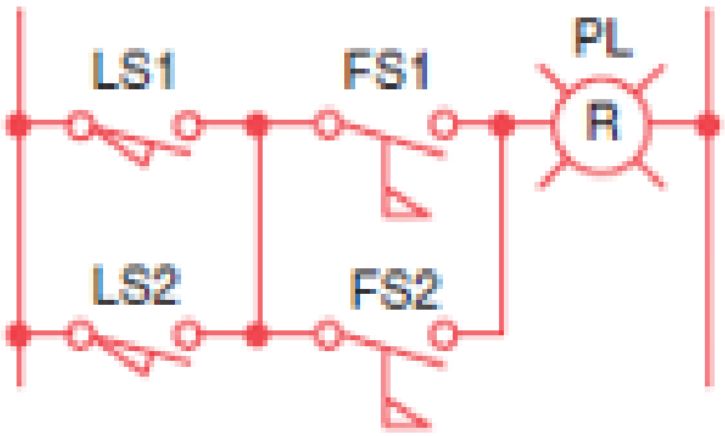
CHATRONICS SPECTRUM TEAM



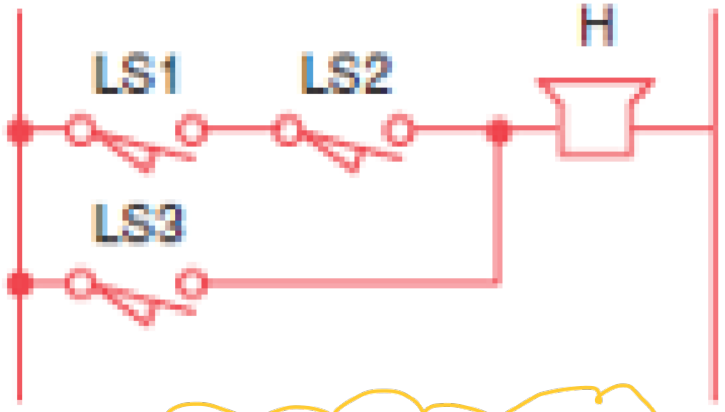

```
LDN  PB1          LO   S3          ALD  ...
LD   S1           LD   PS1         =    L1
O    S2           A    TS1
```




```
LDN   STOP
LD    START
O     M
ALD   ----
=     M
LDN   M
=     G
LD    M
=     R
```




```
LD   LS1
O    LS2
LD   FS1
O    FS2
ALD  ---
=    R
```

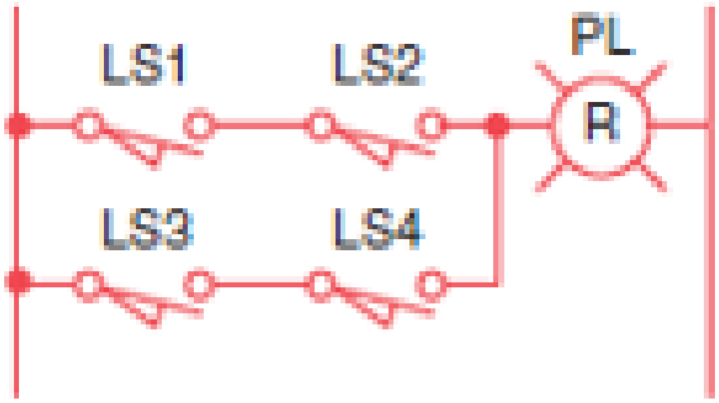



```
LD   LS1
A    LS2
LD   LS3
OLD  ---
=    H
```

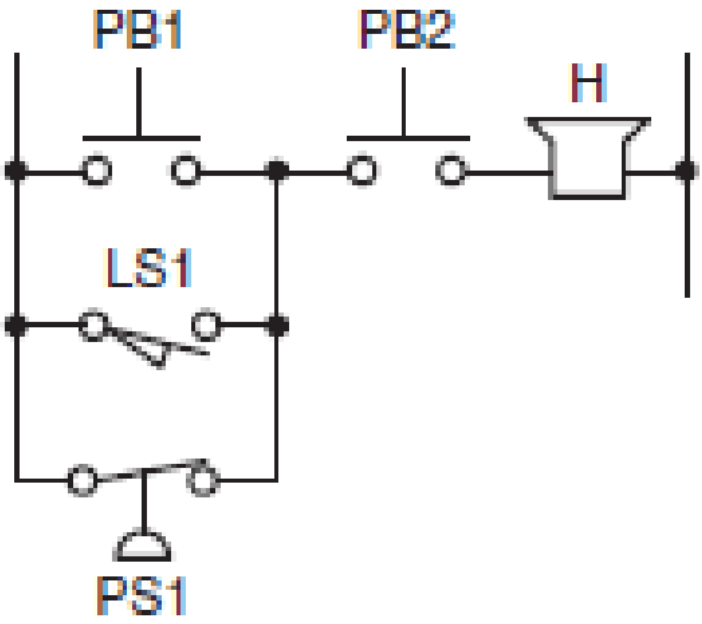
OLD ⇒ يربط جميع rung التي قبله بـ OR gate

ALD ⇒ يربط جميع rung التي قبله بـ AND gate

MECHATRONICS SPECTRUM TEAM
بسواعدنا نبنيها

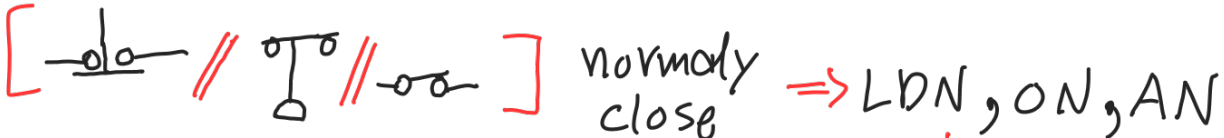



```
LD   LS1
A    LS2
LD   LS3
A    LS4
OLD  ....
=    R
```



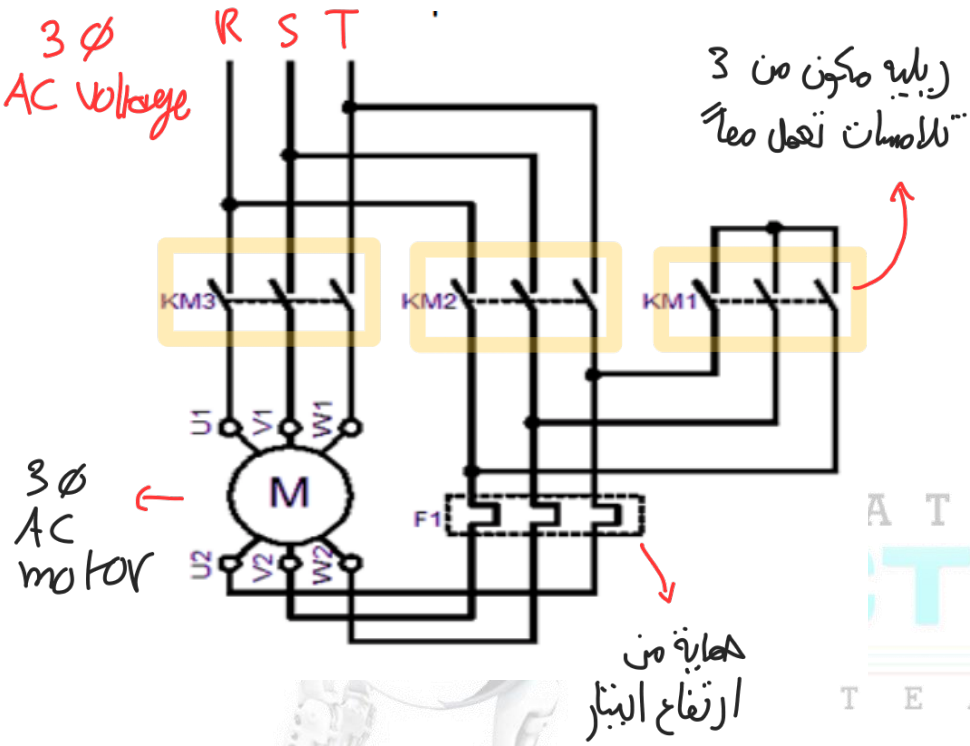

```
LD   PB1
O    LS1
ON   PS1
LD   PB2
ALD  ...
=    H
```

MECHATRONICS SPECTRUM TEAM



بسواعدنا نبنيها

Star-delta 3 phase motor control



نستخدم هذه الطريقة في
التحكم في المانور لتقليل
تيار البدء المسحوب المانور
و بسبب اعطاء المانور
سرعة اكبر

خطوات التحكم :

1- عند بداية عمل المانور
نحول للتوصيلة Y بسبب تقليل
تيار البدء تشغيل المانور

2- ثم نحول للتوصيلة Δ
لزيادة سرعة المانور على
عكس توصيلة Y التي تعطي
سرعات اقل

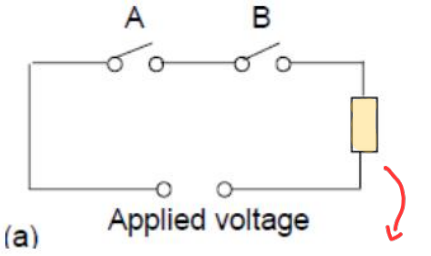
[سؤال سنوات]

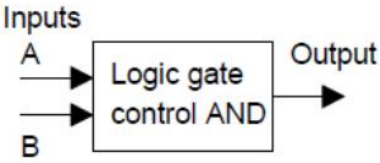
```
LD   km3
O    latch relay
AN   stop
=    latch relay
LD   latch relay
AN   km2
=    km1
LD   latch relay
A    km1
=    km2
```

Chapter 4

Logic functions

AND

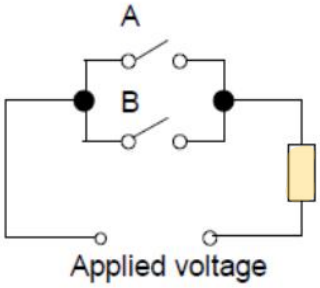


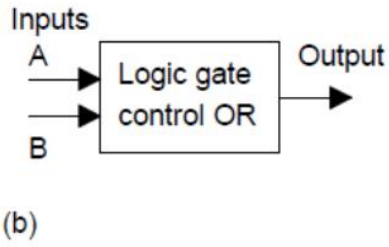


Inputs		Output
A	B	
0	0	0
0	1	0
1	0	0
1	1	1

Truth table

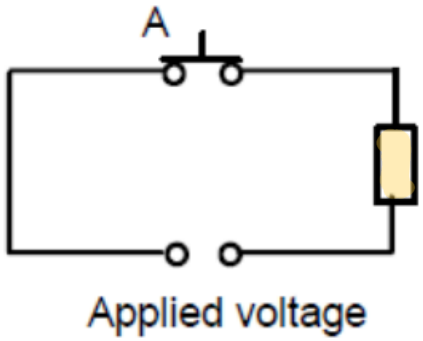
OR





Inputs		Output
A	B	
0	0	0
0	1	1
1	0	1
1	1	1

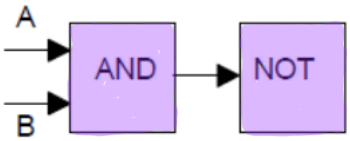
NOT



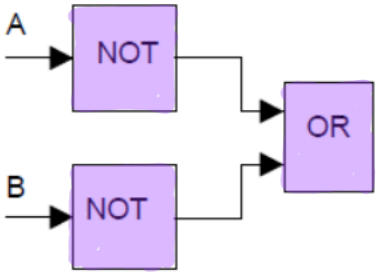
Input A	Output
0	1
1	0

MECHATRONICS SPECTRUM TEAM

NAND



(a)

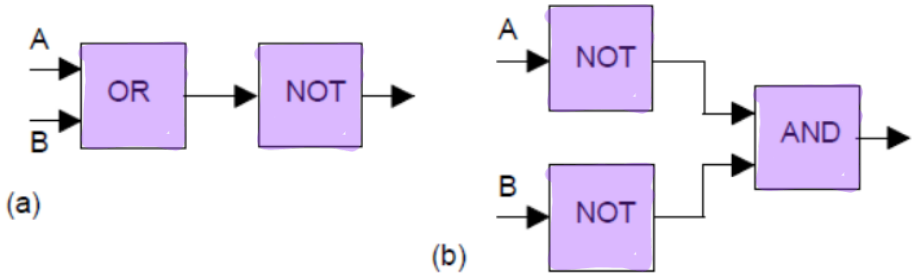


(b)

Inputs		Output
A	B	
0	0	1
0	1	1
1	0	1
1	1	0

بسواعدنا نبنيها

NOR

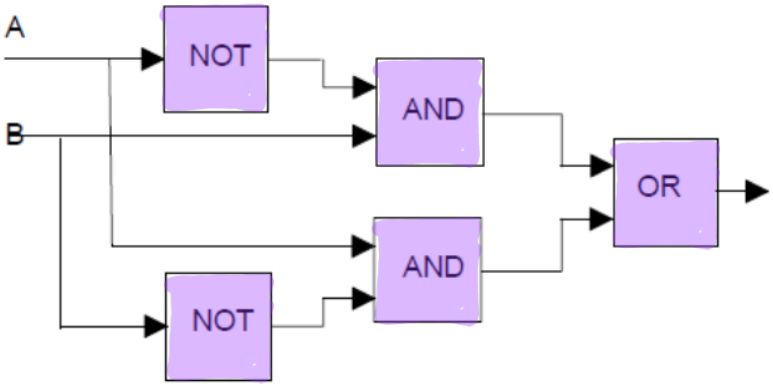


Inputs		Output
A	B	
0	0	1
0	1	0
1	0	0
1	1	0

المتشابه (١)

MECHATRONICS SPECTRUM TEAM

XOR

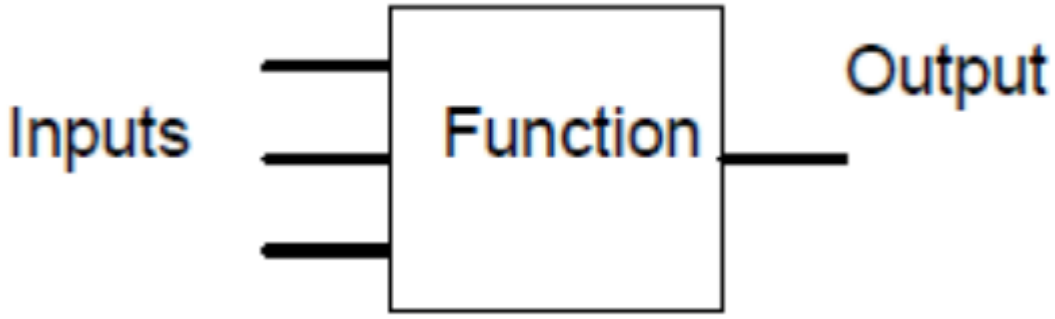


Inputs		Output
A	B	
0	0	0
0	1	1
1	0	1
1	1	0

المختلف (١)

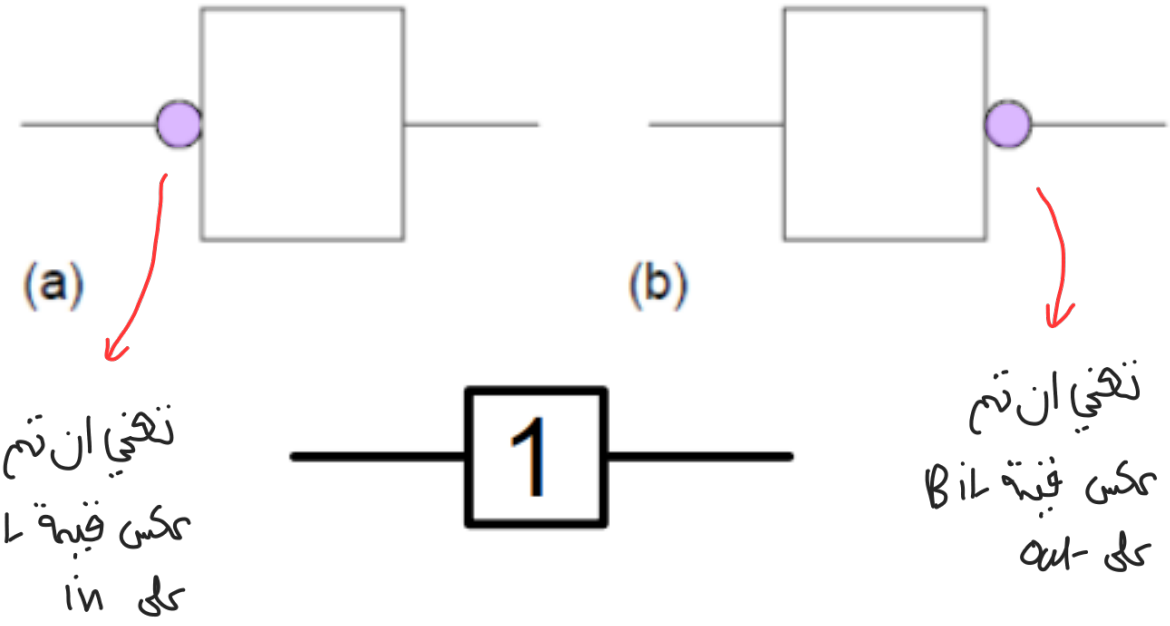
بسواعدنا نبنيها
MECHATRONICS SPECTRUM TEAM

Logic gates Function blocks

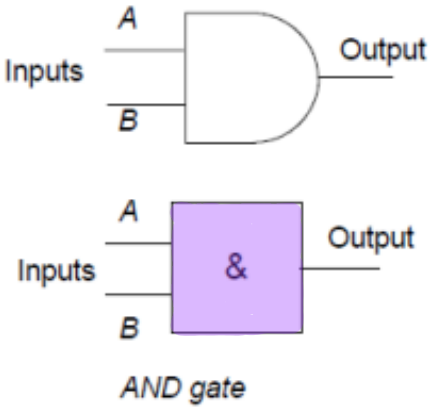


MECHATRONICS SPECTRUM

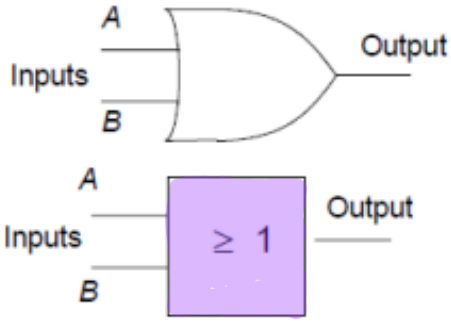
(a) Negated input, (b) negated output



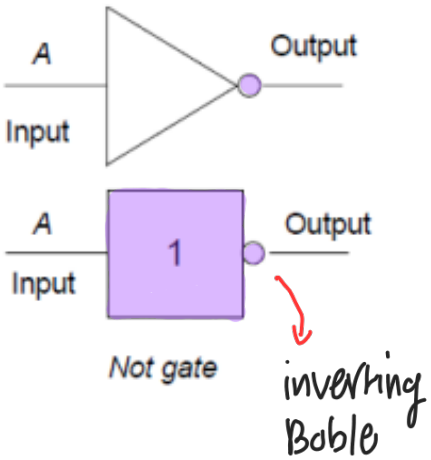
بسواعدنا نبنيها
MECHATRONICS SPECTRUM TEAM



AND gate

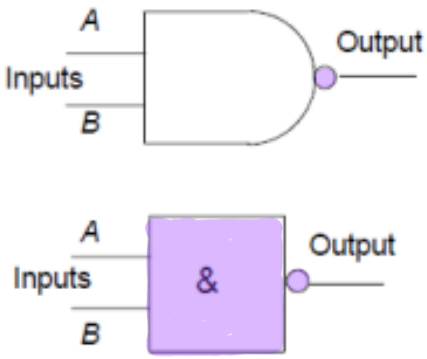


OR gate

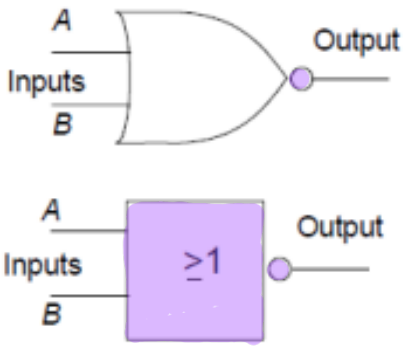


Not gate

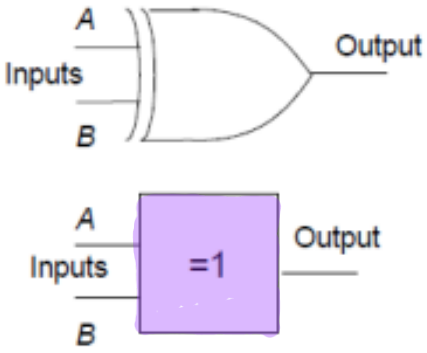
> = 1



NAND gate



NOR Gate

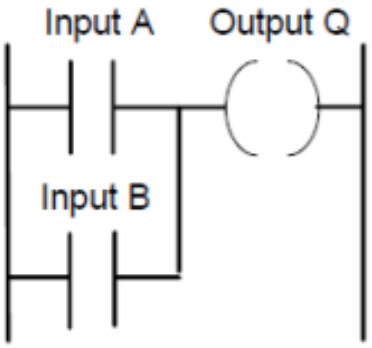


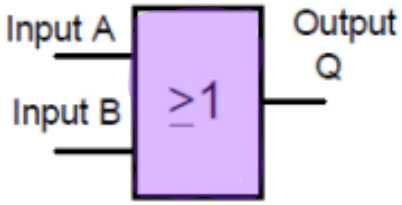
XOR Gate

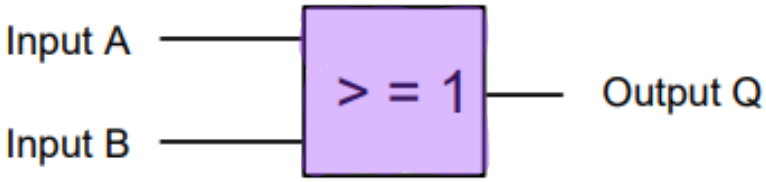
> = 1

Ladder diagram and equivalent functional

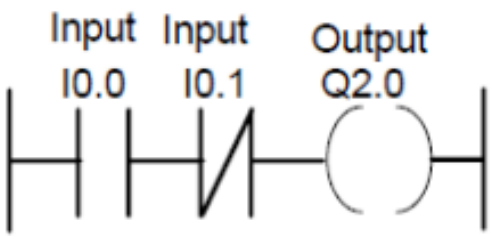
OR

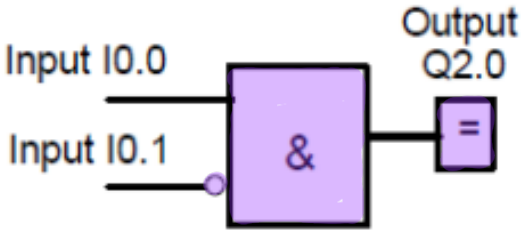




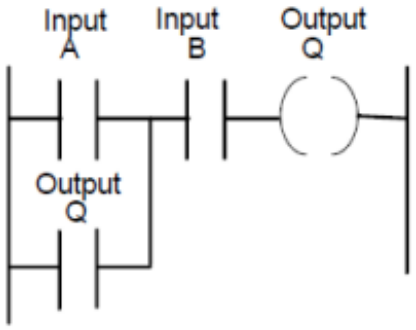


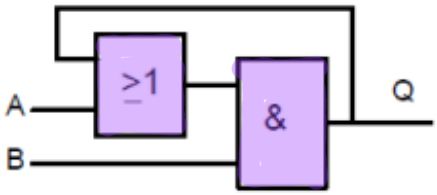
AND

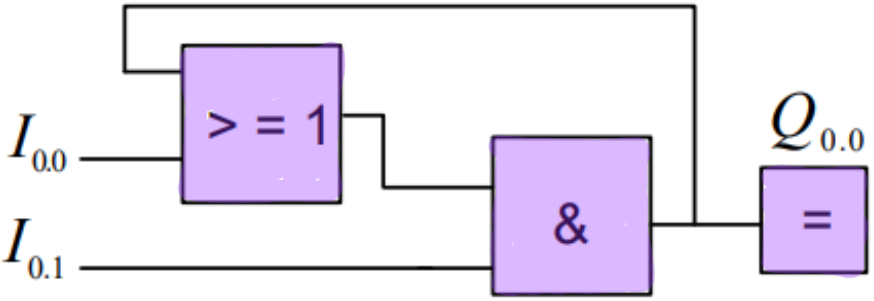


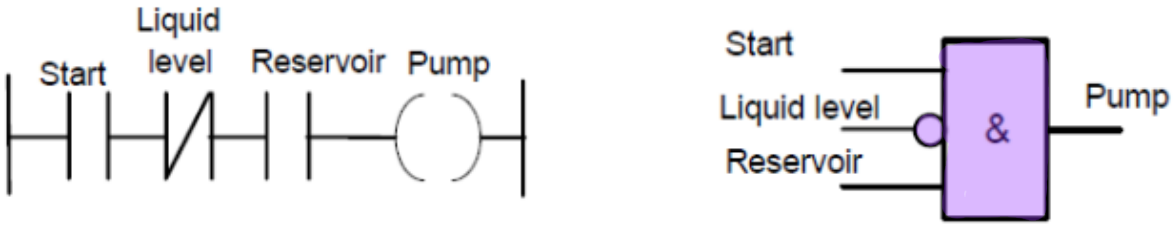


Examples

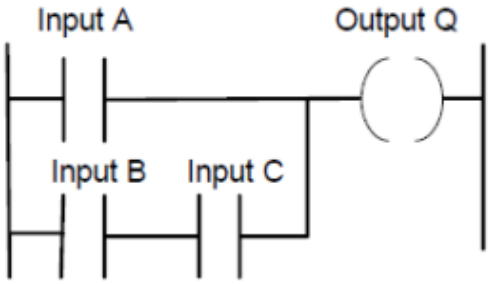

Input
A
Input
B
Output
Q
Output
Q

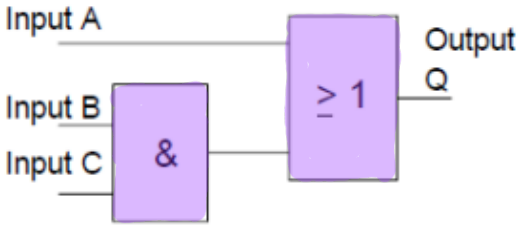

≥1
A
B
&
Q

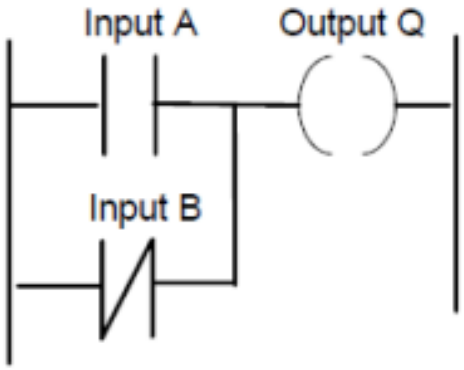

$I_{0.0}$
$I_{0.1}$
> = 1
&
$Q_{0.0}$
=

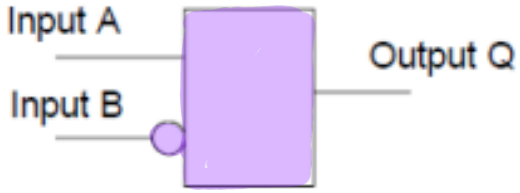

Start
Liquid
level
Reservoir
Pump

Start
Liquid level
Reservoir
&
Pump

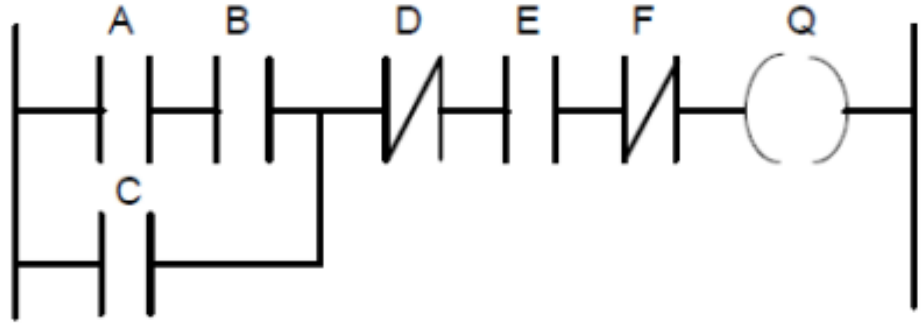

Input A
Output Q
Input B
Input C

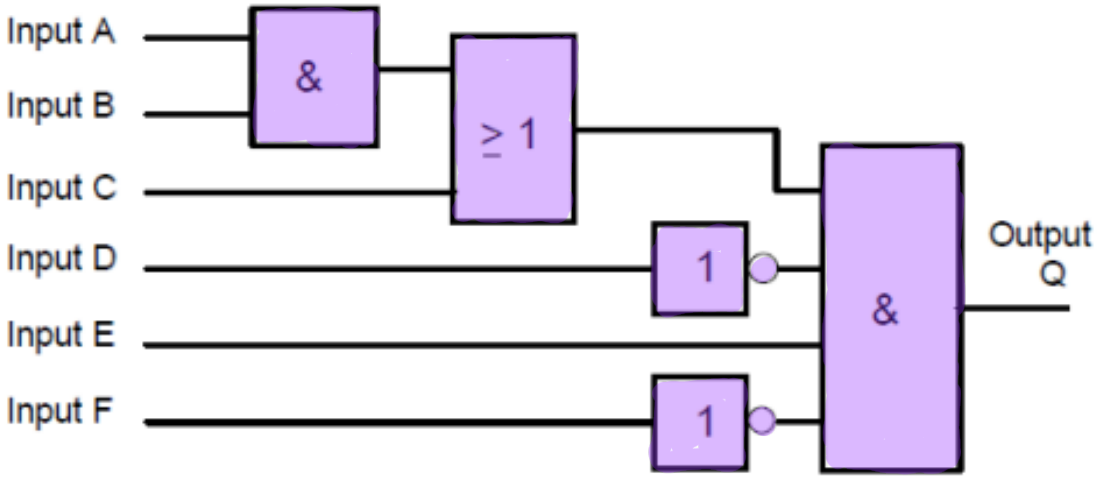

Input A
Input B
Input C
&
≥ 1
Output
Q



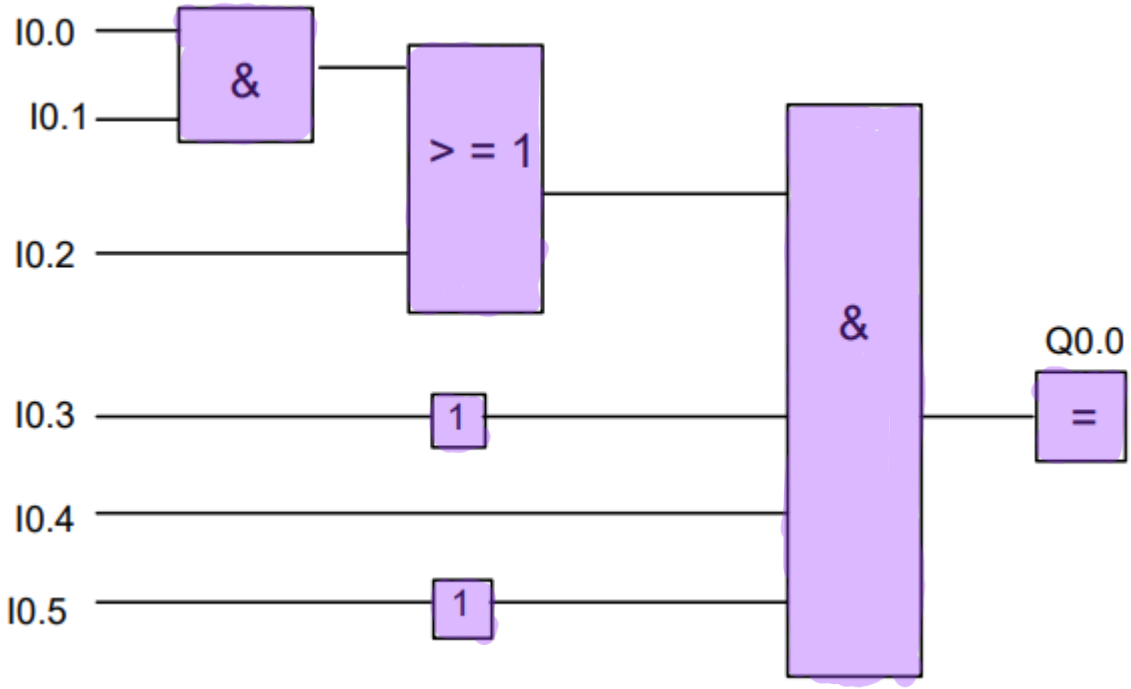


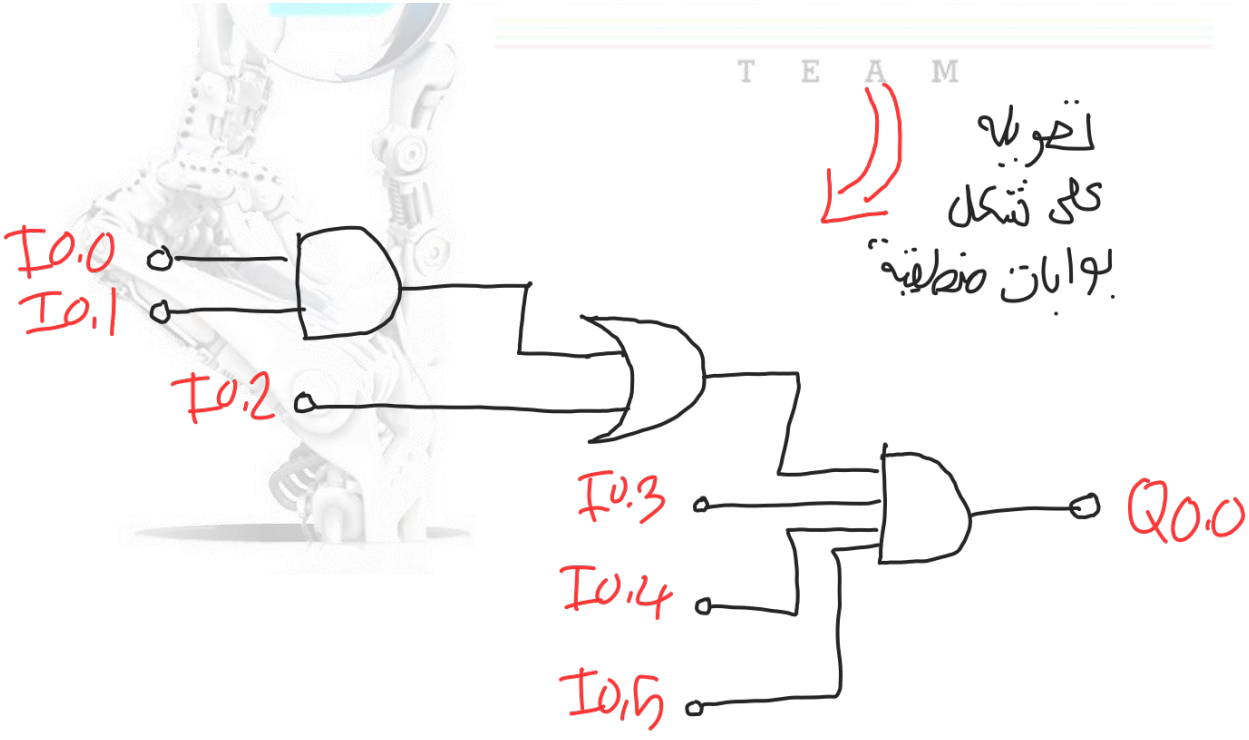
MECHATRONICS SPECTRUM





بسواعدنا نبنيها
MECHATRONICS SPECTRUM TEAM





تصويبة
على تشكل
بوابات منطقية

Mat-Comp-Logic Instructions

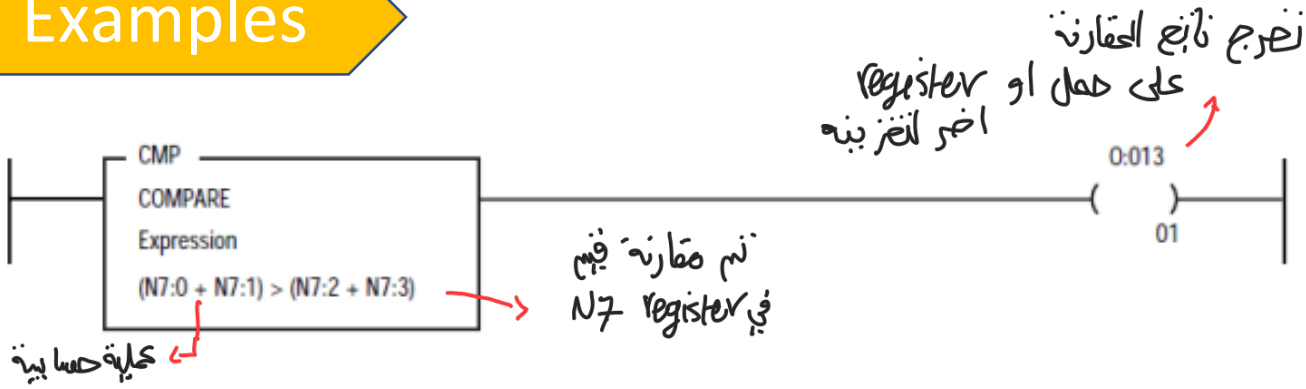
Compare Instructions

=	equal to	if A = B, then ...
<>	not equal to	if A <> B, then ...
==	exactly equal	if A == B, then ...
<	less than	if A < B, then ...
<=	less than or equal to	if A <= B, then ...
>	greater than	if A > B, then ...
>=	greater than or equal to	if A >= B, then ...

Compare Instructions

+	add	A+B
–	subtract	A-B
*	multiply	A*B
/	divide	A/B
SQR	square root	SQR I0:0

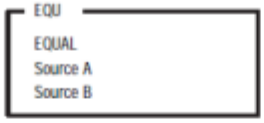
Examples



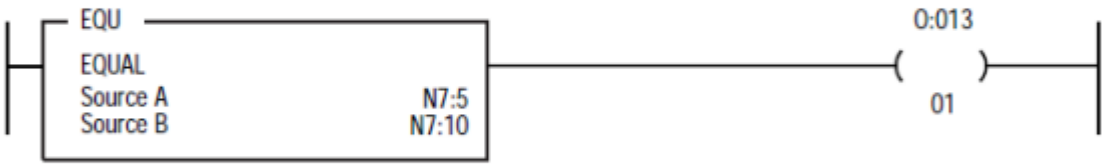
- The CMP instruction tells the processor: if the sum of the values in N7:0 and N7:1 is greater than the sum of the values in N7:2 and N7:3, set output bit O:013/01.

MECHATRONICS SPECTRUM TEAM

Equal to (EQU)

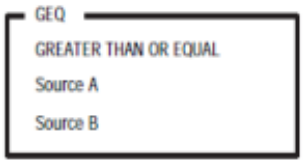


Example:

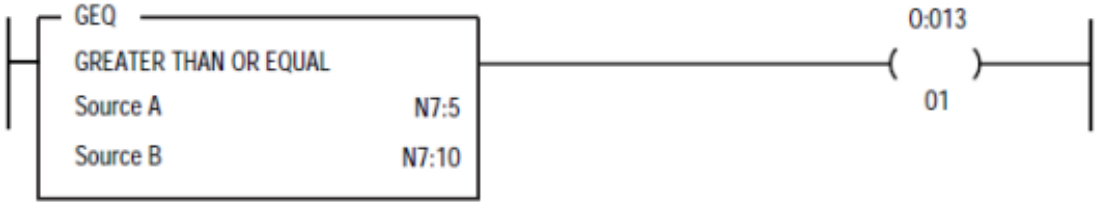


If the value in N7:5 is equal to the value in N7:10, set output bit O:013/01.

Description: Use the GEQ instruction to test whether one value (Source A) is greater than or equal to another value (Source B). Source A and Source B can be values or addresses that contain values.



Example:

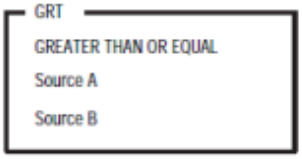


If the value in N7:5 is greater than or equal to the value in N7:10, set output bit O:013/01.

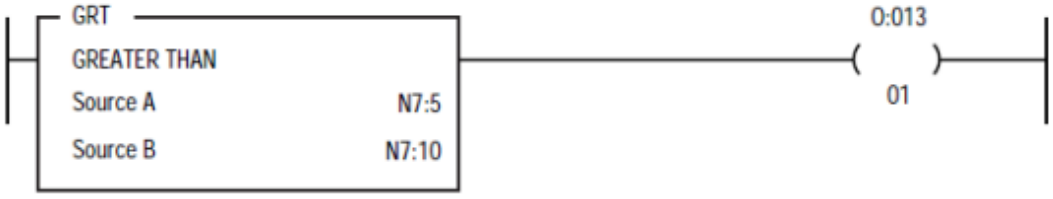
SPECTRUM TEAM

* يجب حفظ اختصار كل تعليمة

Description: Use the GRT instruction to test whether one value (Source A) is greater than another value (Source B). Source A and Source B can either be values or addresses that contain values.



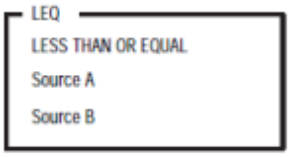
Example:



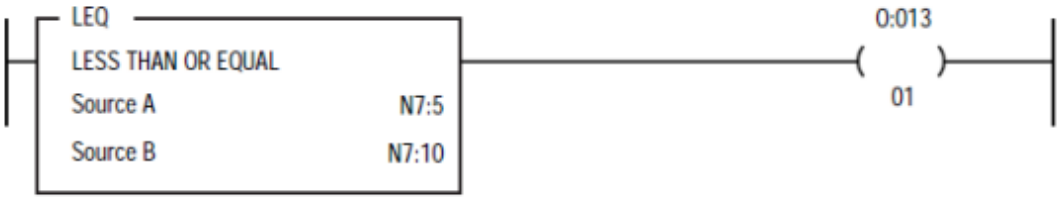
If the value in N7:5 is greater than the value in N7:10, set output bit O:013/01.

بسواعدنا نبنيها
MECHATRONICS SPECTRUM TEAM

Description: Use the LEQ instruction to test whether one value (Source A) is less than or equal to another value (Source B). Source A and Source B can either be values or addresses that contain values.



Example:

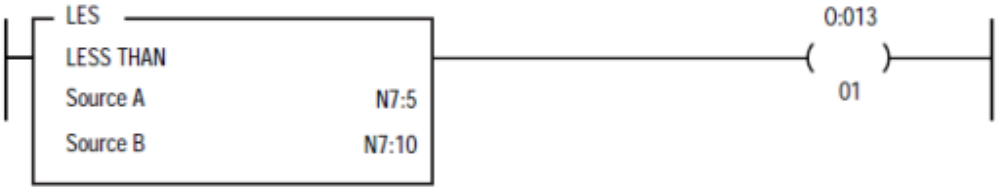


If the value in N7:5 is less than or equal to the value in N7:10, set output bit O:013/01.

MECHATRONICS SPECTRUM TEAM

Description: Use the LES instruction to test whether one value (Source A) is less than another value (Source B). Source A and Source B can be values or addresses that contain values.

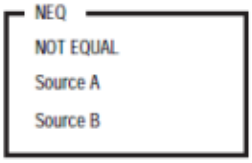
Example:



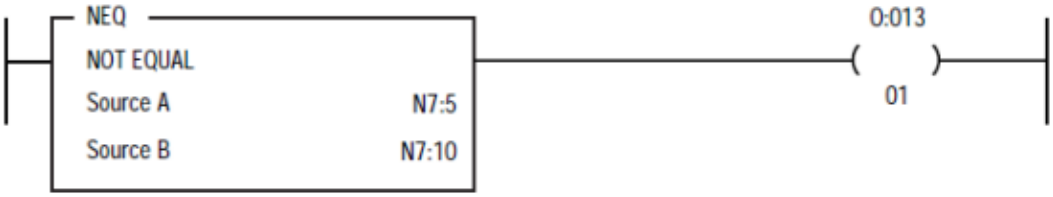
If the value in N7:5 is less than the value in N7:10, set output bit O:013/01.

بسواعدنا نبنيها
MECHATRONICS SPECTRUM TEAM

Description: Use the NEQ instruction to test whether two values are not equal. Source A and Source B can be values or addresses.



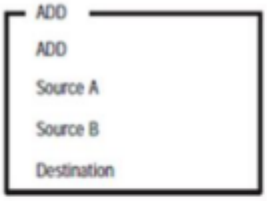
Example:



If the value in N7:5 is not equal to the value in N7:10, set output bit O:013/01.

SPECTRUM TEAM

Description: Use the ADD instruction to add one value (Source A) to another value (Source B) and place the result in the destination. Source A and Source B can either be values or addresses that contain values.



تعديه موقع لتضربن
و فيم العملية

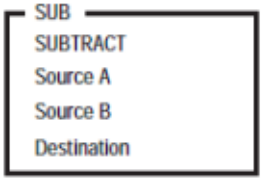
Example:

If input word 12, bit 10 is set, add the value in N7:3 to the value in N7:4 and store the result in N7:20.

نلاحظ ان: في عمليات AND و OR و العمليات الحسابية يوجد in و out بدون

بسواعدنا نبنيها
MECHATRONICS SPECTRUM TEAM

Description: Use the SUB instruction to subtract one value (Source B) from another value (Source A) and place the result in the destination. Source A and Source B can be values or addresses that contain values.



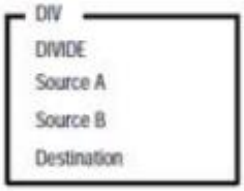
Example:

اذا كان هذا حساس او مفتاح
وتم تفعيله يتم البدء في العملية الحسابية

If input word 12, bit 10 is set, subtract the value in N7:4 from the value in N7:3 and store the result in N7:20.

SPECTRUM
T E A M

Description: Use the DIV instruction to divide one value (Source A) by another value (Source B) and place the result in the Destination. Source A and Source B can either be values or addresses that contain values.

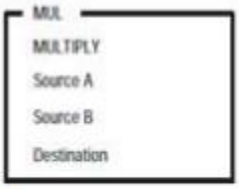


Example:

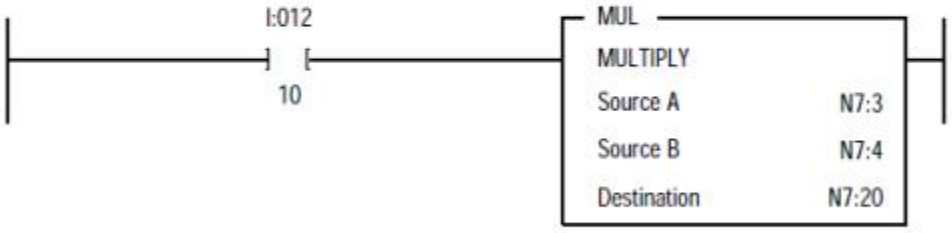
If input word 12, bit 10 is set, divide the value in N7:3 by the value in N7:4 and store the result in N7:20.

بسواعدنا نبنيها
MECHATRONICS SPECTRUM TEAM

Description: Use the MUL instruction to multiply one value (Source A) by another value (Source B) and place the result in the destination. Source A and Source B can be values or addresses.



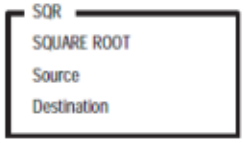
Example:



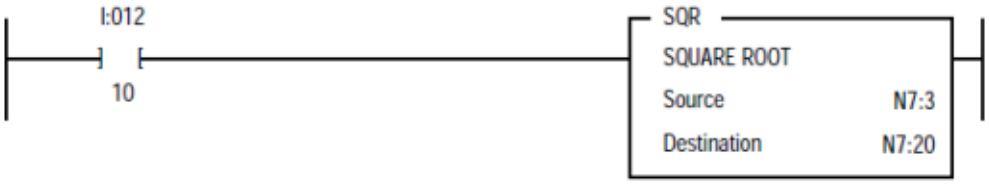
If input word 12, bit 10 is set, multiply the value in N7:3 by the value in N7:4 and store the result in N7:20.

SPECTRUM
TEAM

Description: Use the SQR instruction to take the square root of a value and store the result in the destination. The source can be a value or an address. If the source value is negative, the processor takes its absolute value and performs the square root function.



Example:



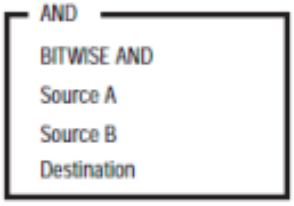
If input word 12, bit 10 is set, take the square root of the value in N7:3 and store the result in N7:20.

بسواعدنا نبنيها
MECHATRONICS SPECTRUM TEAM

Logical Instructions AND, NOT, OR, XOR

AND Operation (AND)

Description: Use the AND instruction to perform an AND operation using the bits in the two source addresses.

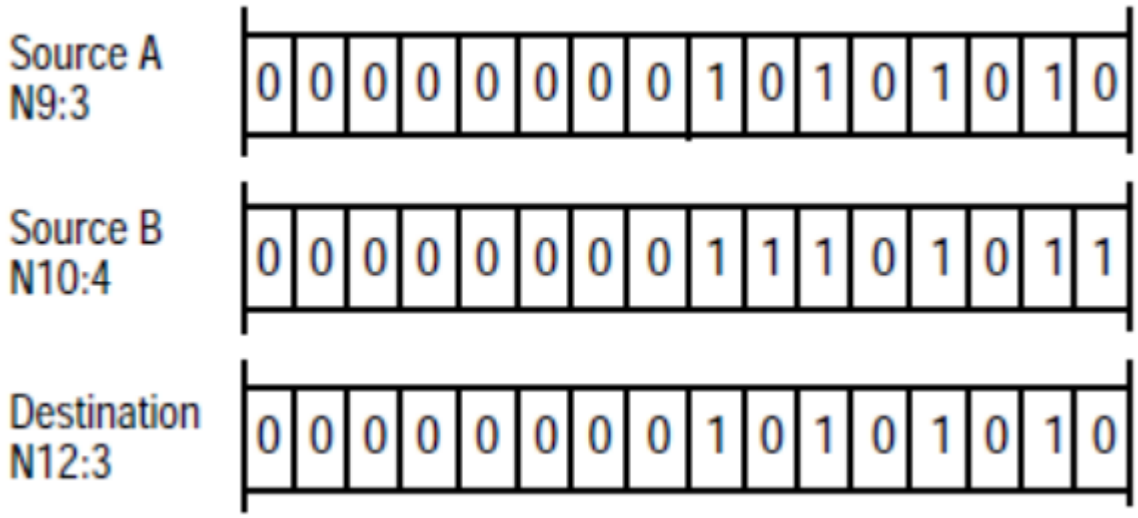


Source A	Source B	Result
0	0	0
1	0	0
0	1	0
1	1	1

T E A M

Example

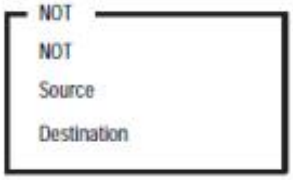
If input word 12, bit 10 is set, the processor performs an AND operation on N9:3 and N10:4 and stores the result in N12:3.



registers

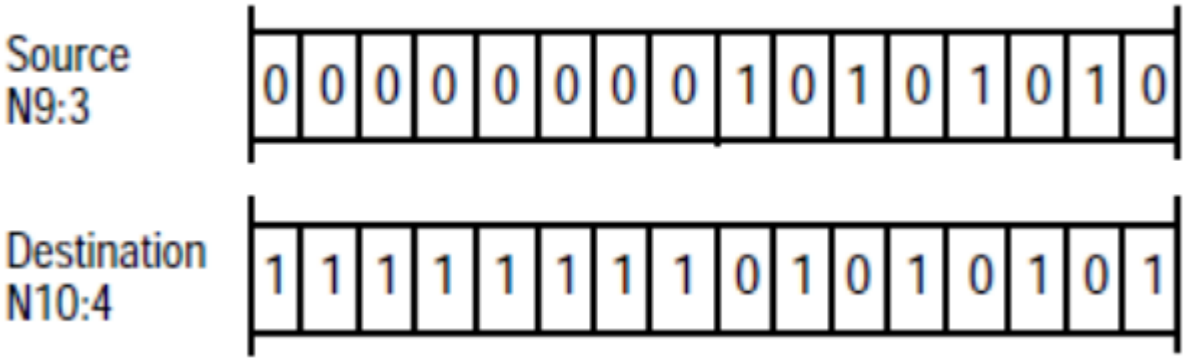
NOT Operation (NOT)

Description: Use the NOT instruction to perform a NOT operation using the bits in the source address. This operation is also know as a bit inversion.

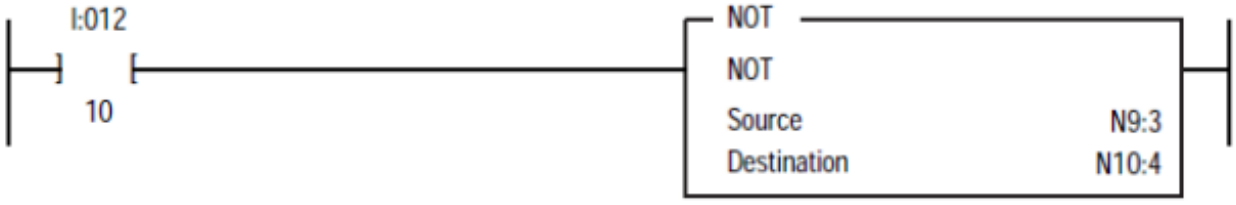


Truth Table for a NOT Operation

Source	Result
0	1
1	0



Example

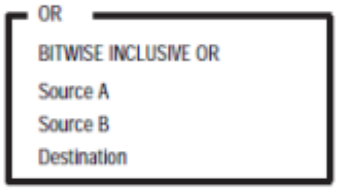


If input word 12, bit 10 is set, the processor performs a NOT operation on N9:3 and stores the result in N10:4

SPECTRUM
TEAM

OR Operation (OR)

Description: Use the OR instruction to perform an OR operation using the bits in the two sources (constants or addresses).



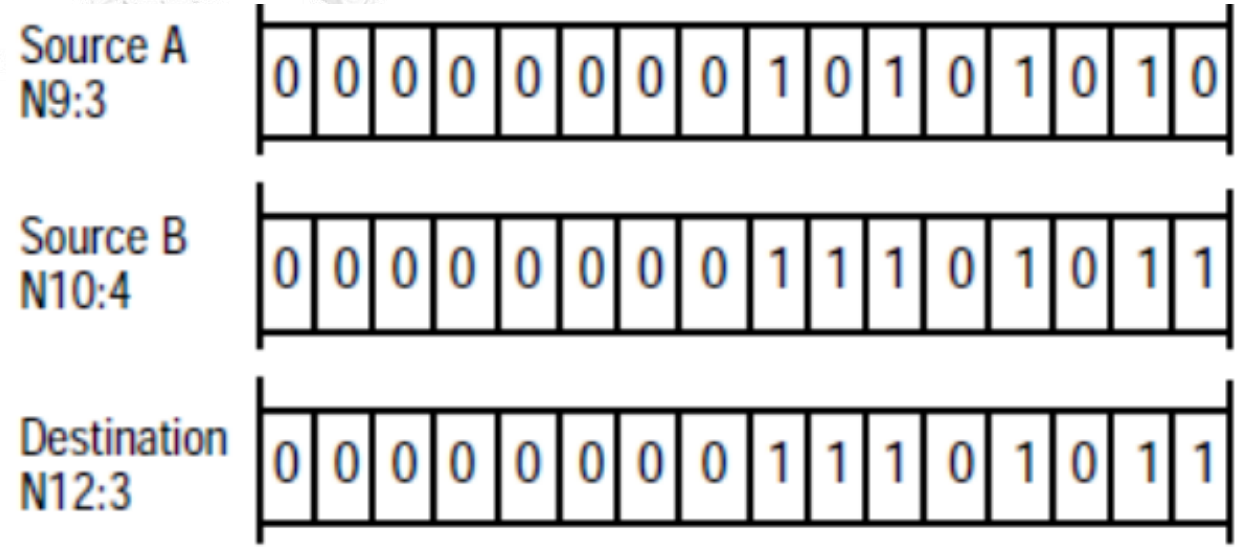
Truth Table for an OR Operation

Source A	Source B	Result
0	0	0
1	0	1
0	1	1
1	1	1

بسواعدنا نبنيها
MECHATRONICS SPECTRUM TEAM

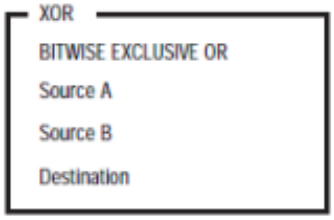
Example

If input word 12, bit 10 is set, the processor performs an OR operation on N9:3 and N10:4 and stores the result in N12:3.



Exclusive OR Operation (XOR)

Description: Use the XOR instruction to perform an exclusive OR operation using the bits in the two sources (constants or addresses).

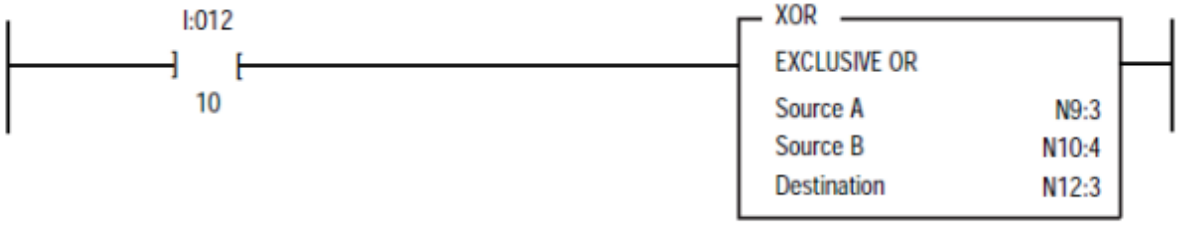


Truth Table for an XOR Operation

Source A	Source B	Result
0	0	0
1	0	1
0	1	1
1	1	0

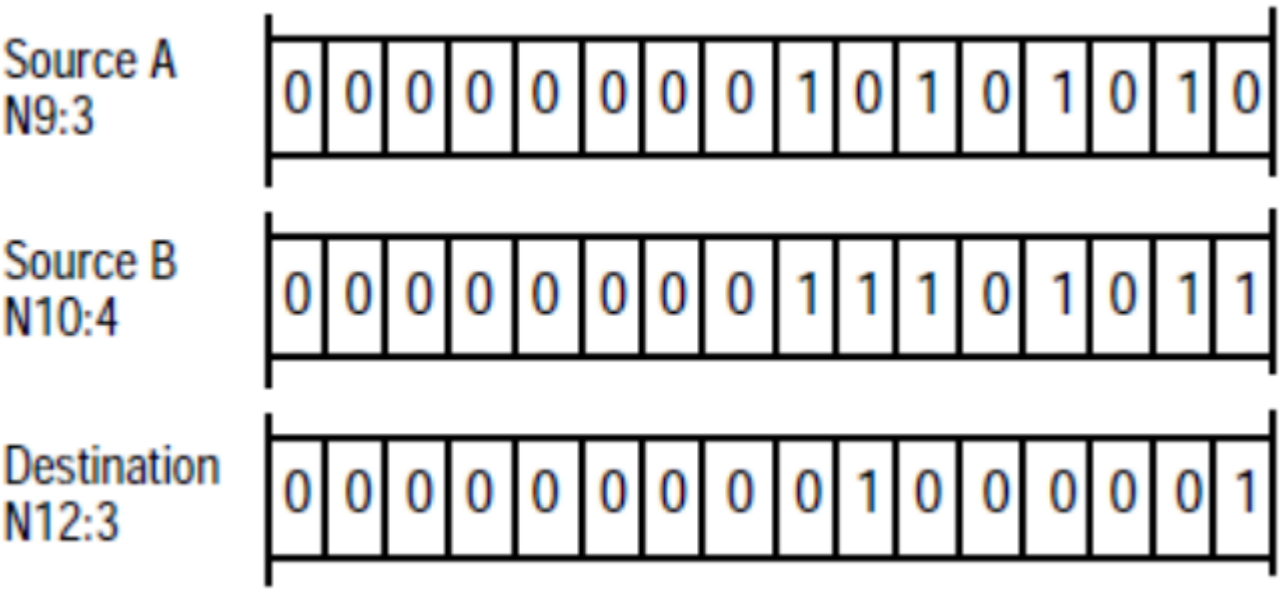
SPECTRUM
TEAM

Example

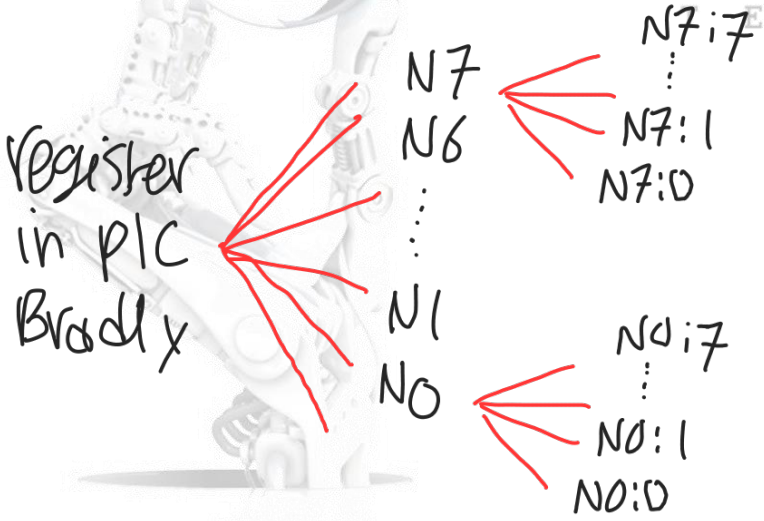


If input word 12, bit 10 is set, the processor performs an XOR operation on N9:3 and N10:4 and stores the result in N12:3.

بسواعدنا نبنيها
MECHATRONICS SPECTRUM TEAM

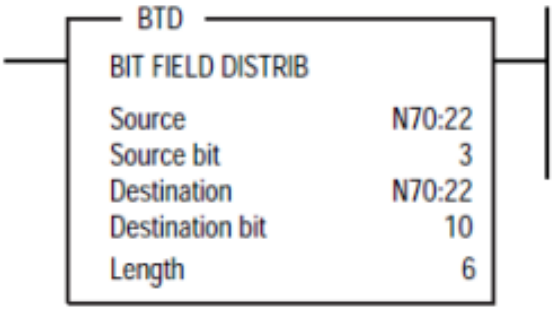


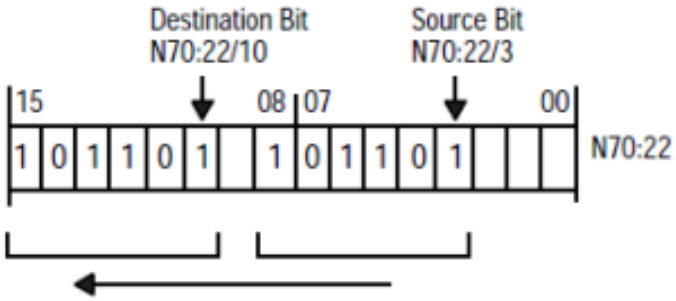
MECHATRONICS SPECTRUM TEAM



Move Instructions

Moving Bits Within a Word

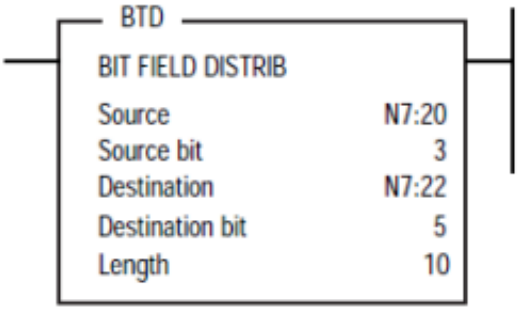


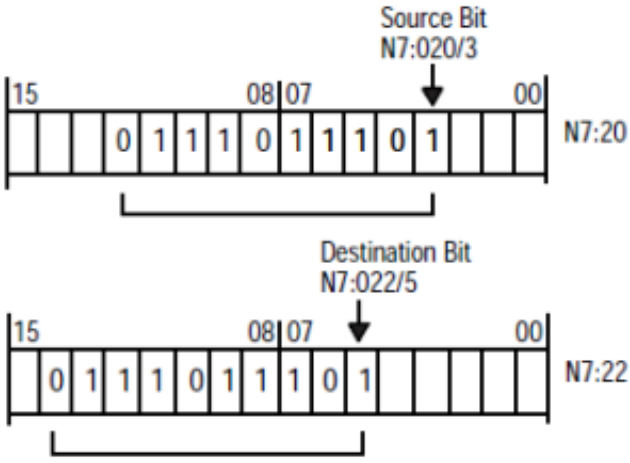


نقل 6 Bits من
register

MECHATRONICS SPECTRUM TEAM

Moving Bits Between Words



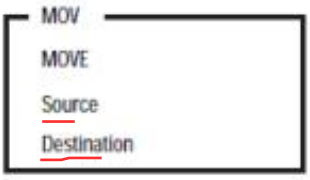


نقل 10 Bits من
register معين

بسواعدنا نبنيها
MECHATRONICS SPECTRUM TEAM

Move (MOV)

Description: The MOV instruction is an output instruction that copies the Source address to a Destination. As long as the rung remains true, the instruction moves the data each scan.



copy الى
paste
register

MECHATRONICS SPECTRUM

To program this instruction, you must provide the processor with the following:

Parameter:	Definition:
source	is a program constant or data address from which the instruction reads an image of the value. You can also use a symbol, as long as the symbol name is more than 1 character. The source remains unchanged.
destination	the data address to which the instruction writes the result of the operation. The instruction writes over any data stored at the destination.

بسواعدنا نبنيها
MECHATRONICS SPECTRUM TEAM

Jump (JMP) and Label (LBL)

Description: Use JMP and LBL instructions in pairs to skip portions of the ladder program.

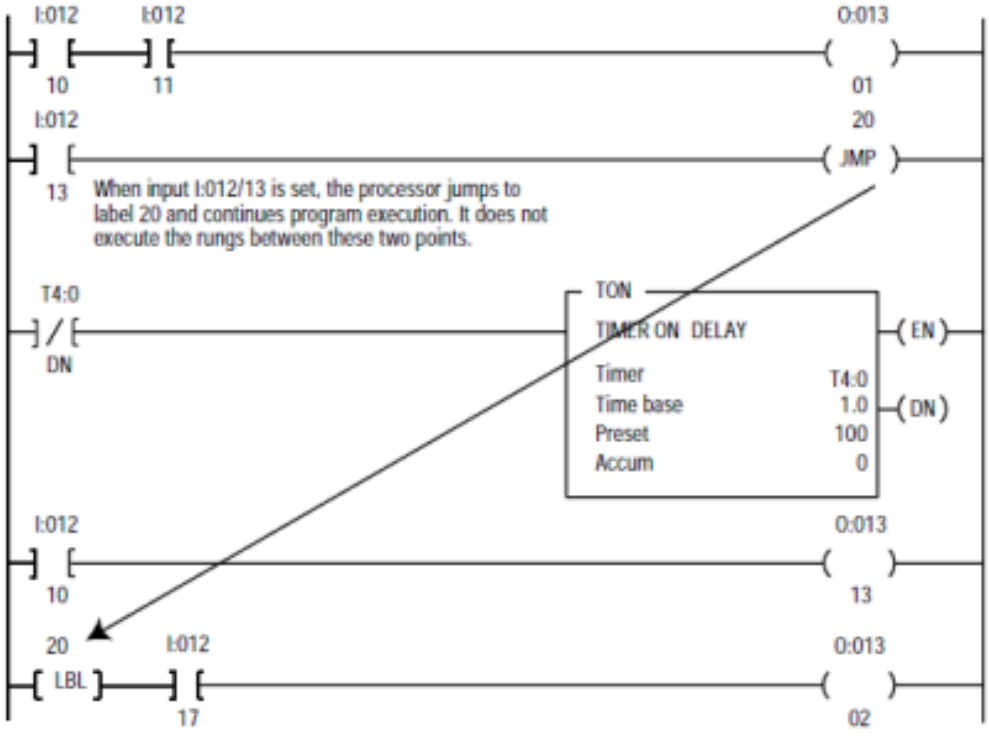
—(JMP)—

|— LBL —|

If the Jump Rung Is:	Then the Processor:
True	Skips from the JMP rung to the LBL rung and continues executing the program. You can jump forward or backward.
False	Ignores the JMP instruction

Jumping forward to a label saves program scan time by omitting a program segment until needed. Jumping backward lets the processor repeat iterations through a program segment until its logic is complete.

JMP and LBL Example: When the rung containing the JMP instruction goes true, the processor jumps over successive rungs until it reaches the rung that contains the LBL instruction with the same number. The processor resumes executing at the LBL rung.



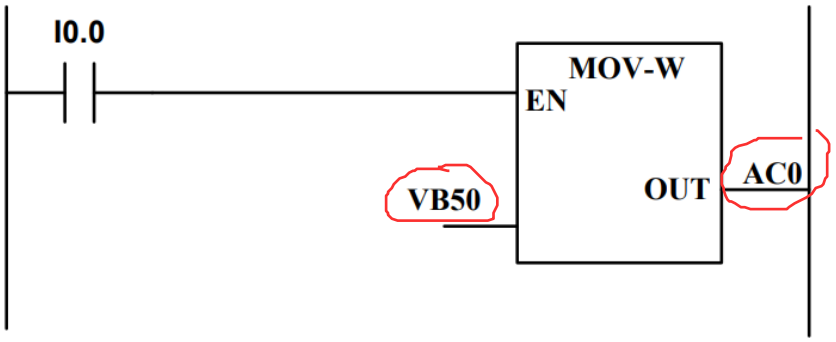
اذا تم تفعيل هذا الحساس او المفتاح بيتم تفعيل امر Jum الذي يجعل الكود يقفز لـ Veng مطلوب لبستخدم في حالات الطوارئ

بسواعدنا نبنيها
MECHATRONICS SPECTRUM TEAM

Math-Comp-Logic-Siemens

MOV

كل ما بستق بحط PLC برادي

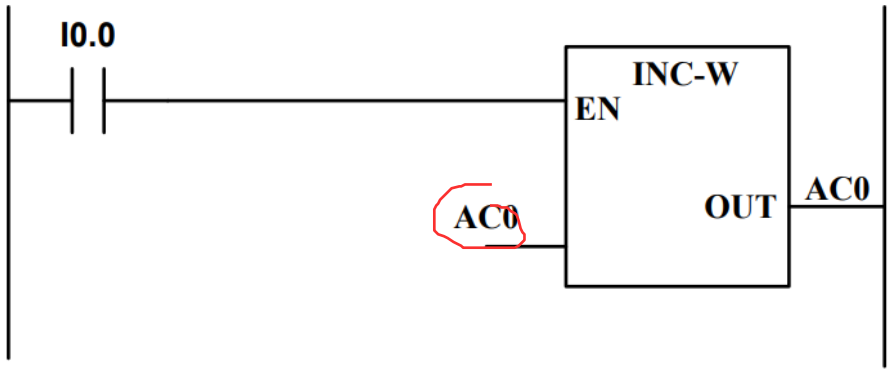



```
LD      I0.0
MOV-W   VB50, AC0
MEND
```

in out

SPECTRUM TEAM

INCREMENT (INC)



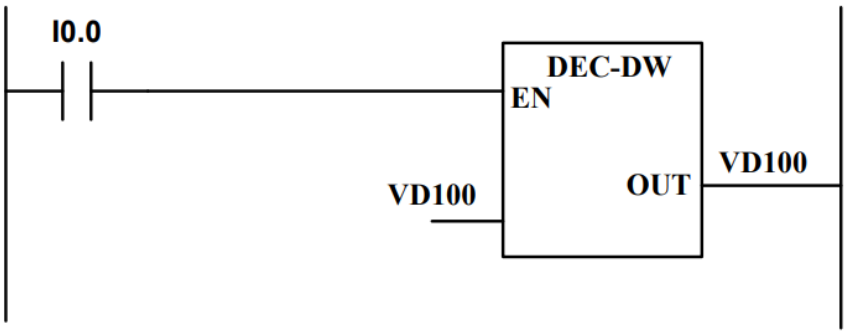

```
LD      I0.0
INC-W   AC0
MEND
```

in + out

او register الاصل نفس الشيء

بسواعدنا نبنيها
MECHATRONICS SPECTRUM TEAM

DECREMENT (DEC)

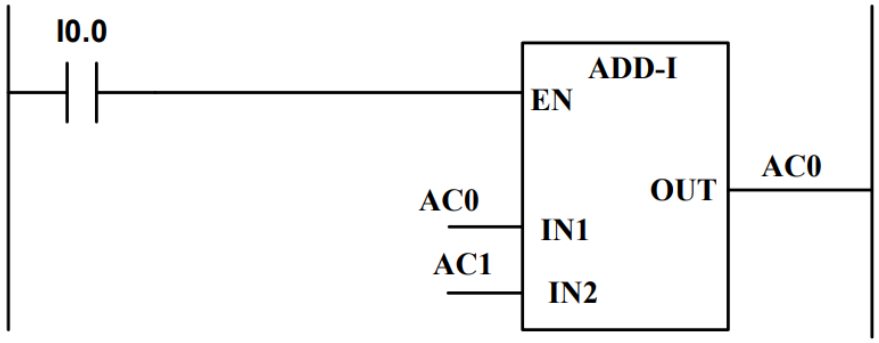



```
LD        I0.0
DEC-DW    VD100
MEND
```

مكرر

MECHATRONICS SPECTRUM TEAM

ADD

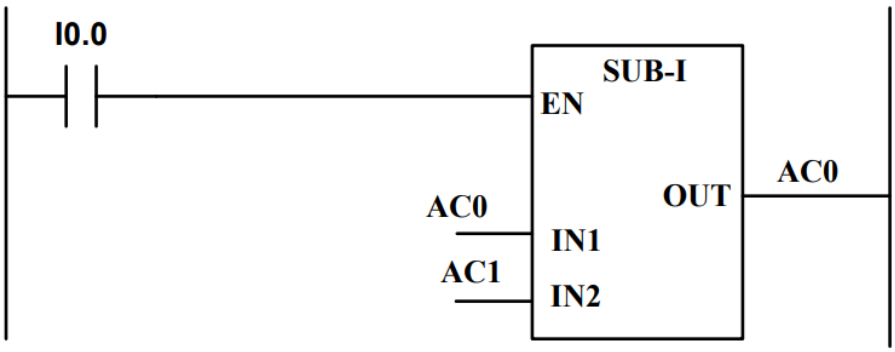



```
LD        I0.0
ADD-I     AC1,AC0
MEND
```

لم نكرر AC0 بسبب تكراره على in و out

بسواعدنا نبنيها

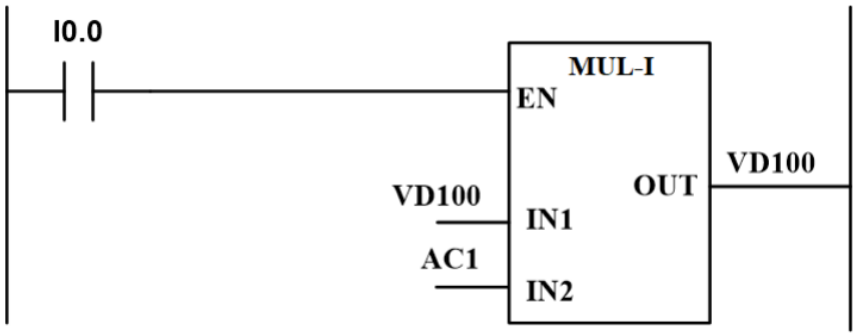
SUB




```
LD      I0.0
SUB-I   AC1,AC0
MEND
```

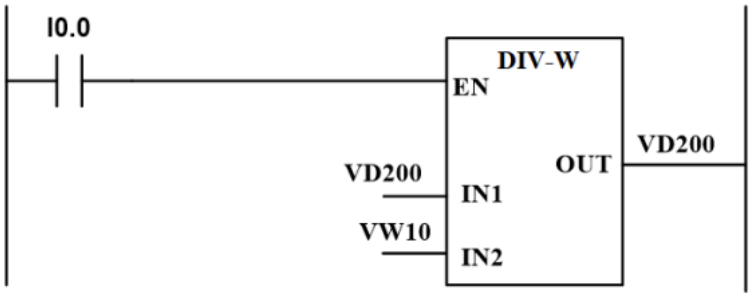
MECHATRONICS SPECTRUM TEAM

MUL




```
LD      I0.0
MUL-I   AC1,VD100
MEND
```

DIV

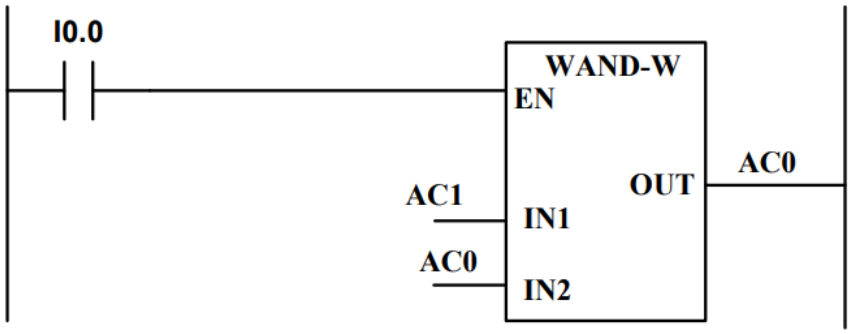



```
LD      I0.0
DIV-W   VW10,VD200
MEND
```

مكرر

MECHATRONICS SPECTRUM TEAM

AND

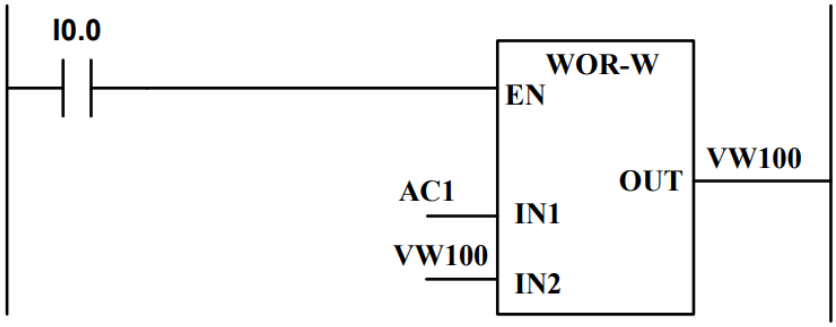



```
LD      I0.0
AND-W   AC1,AC0
MEND
```

مكرر

بسواعدنا نبنيها

OR

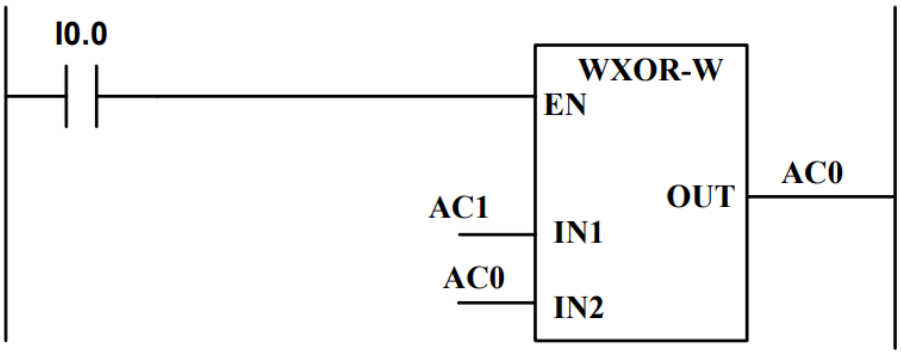



```
LD      I0.0
OR-W    AC1,VW100
MEND
```

مكرر

MECHATRONICS SPECTRUM TEAM

XOR



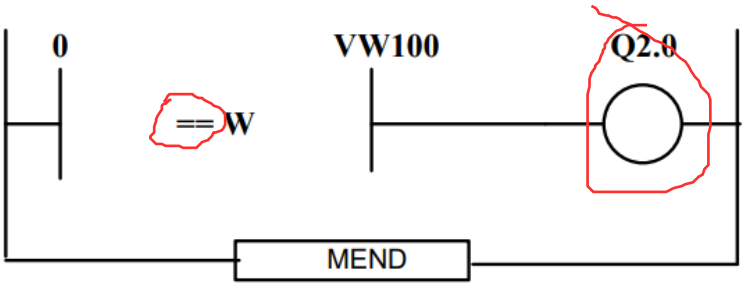

```
LD      I0.0
XOR-W   AC1,AC0
MEND
```

مكرر

بسواعدنا نبنيها

Compare Instruction

ضروري قبل الرقم
نضع #

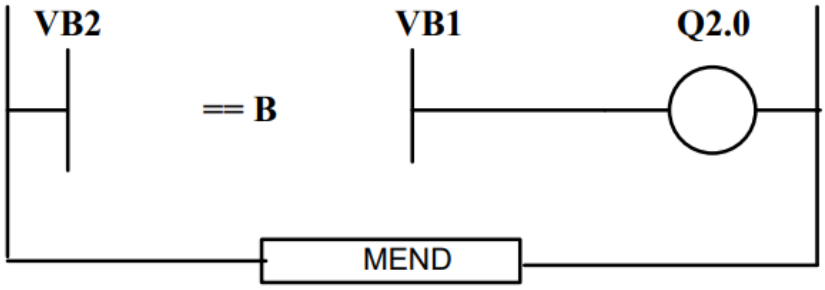



```
LD W==  #0,VW100
=       Q2.0
MEND
```

W Word

MECHATRONICS SPECTRUM TEAM

نعكس عكانهم
==



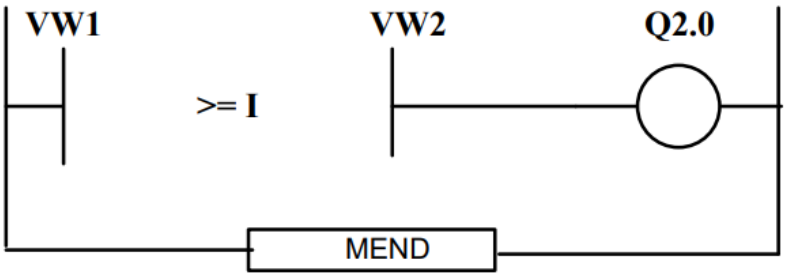

```
LD B==  VB2,VB1
=       Q2.0
MEND
```

B Byte

يمكن ان تكون
قيم Register قيم
حساس حرارة او حساس
وزن او مسافة

نقارن بين قيم في Register2
او قيمة Register و رقم

بسواعدنا نبنيها
MECHATRONICS SPECTRUM TEAM

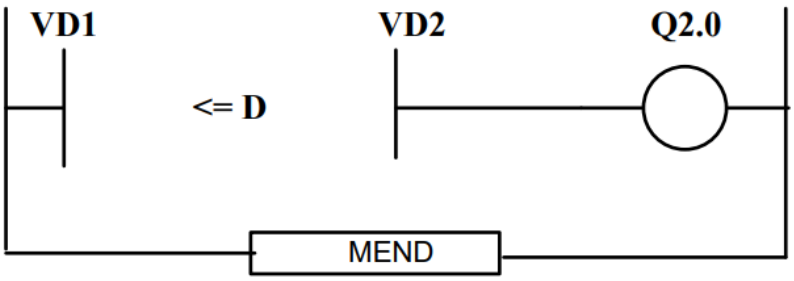


نعكس عكانهم
>=

```
LD I>=    VW1,VW2
=         Q2.0
MEND
```

I Integer

MECHATRONICS SPECTRUM TEAM



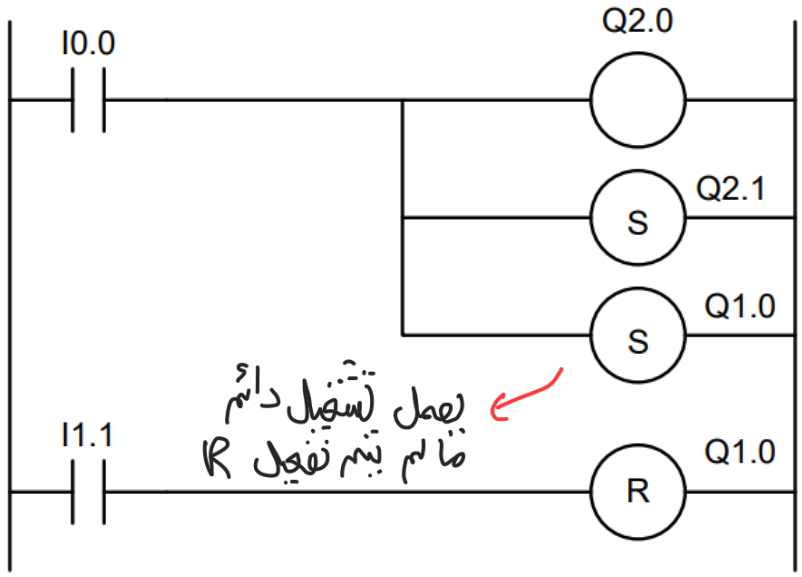
نعكس عكانهم
<=

```
LD  D <=    VD1,VD2
=           Q2.0
MEND
```

D Double Word

بسواعدنا نبنيها

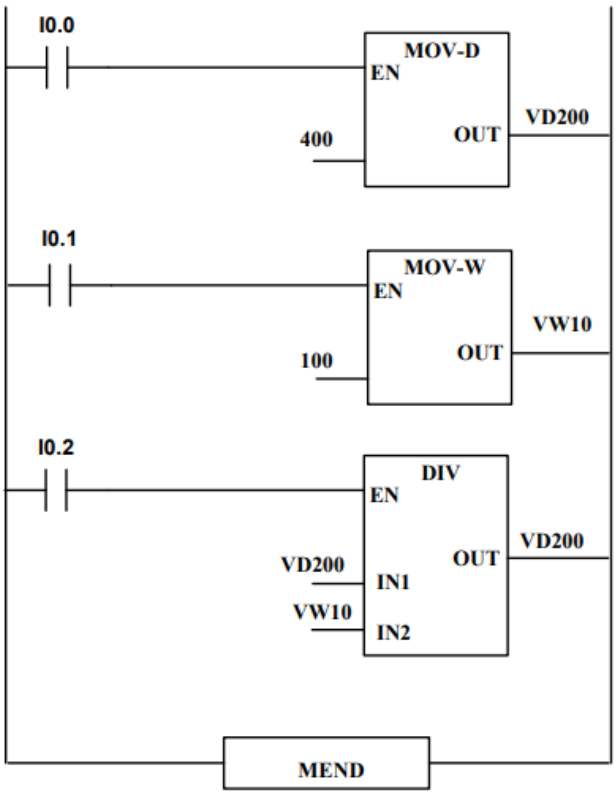
Set (S) and Rest (R)



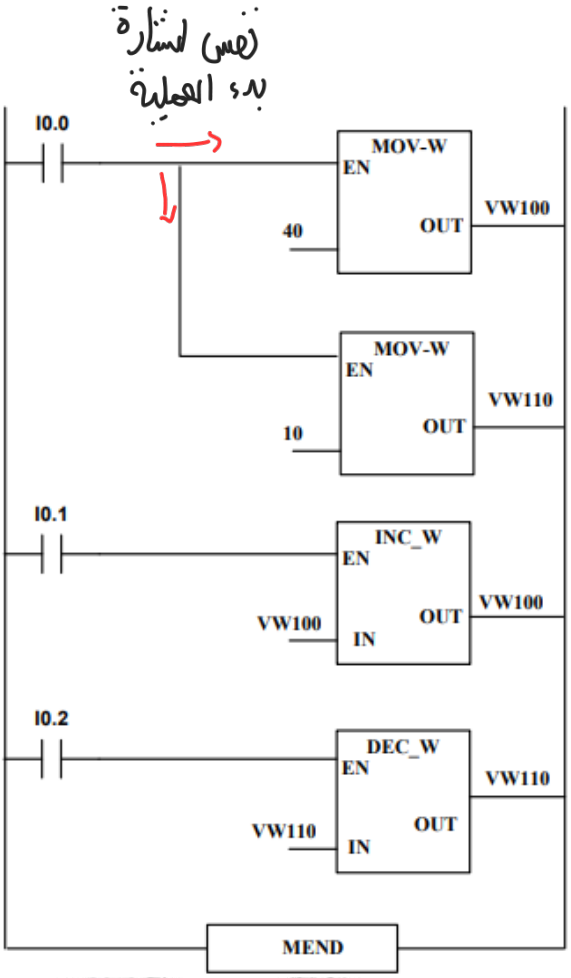

```
LD      I0.0
=       Q2.0
S       Q2.1
S       Q1.0
LD      I1.1
R       Q1.0
MEND
```

T E A M

Examples

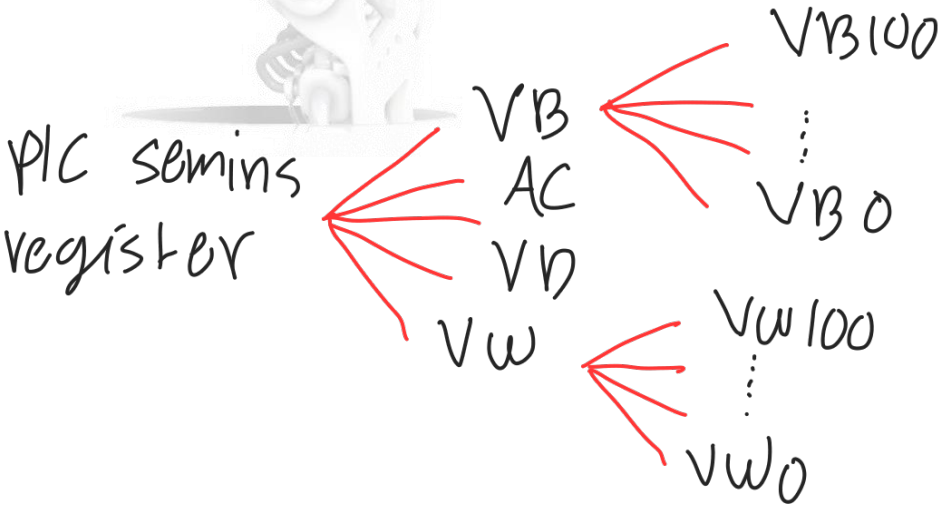



```
LD      I0.0
MOV-D   #400,VD200
LD      I0.1
MOV-W    # 100,VW10
LD      I0.2
DIV     VW10,VD200
MEND
```




```
LD      I0.0
MOV-W   #40,VW100
MOV-W   # 10,VW110
LD      I0.1
INC-W   VW100
LD      I0.2
DEC-W   VW110
MEND
```

مكرر



بسواعدنا نبنيها
MECHATRONICS SPECTRUM TEAM

Chapter 5

PLC Examples

Generate Outputs Based on Equations

Description

The following calculation will be made when input I:1/0 is true.

$$X = 2^{Y}\sqrt{(1+\sin Y)}$$

If the result x is between 1 and 10 then the Output 0 will be turned on. The value of x will be output as an Analog voltage. Perform this operation in PLC using Ladder Diagram programming language.

Solution:

- Perform tasks individually as defined below.
- Assign integer number addresses to store the 16/32 bit data into registers.
- Use this address in the mathematical instructions as Source and Destination.
- Use (X to power Y) instruction to find the input 2 to power y.

- Use SIN instruction to find $\sin y$. Add this with 1 by using ADD instruction.

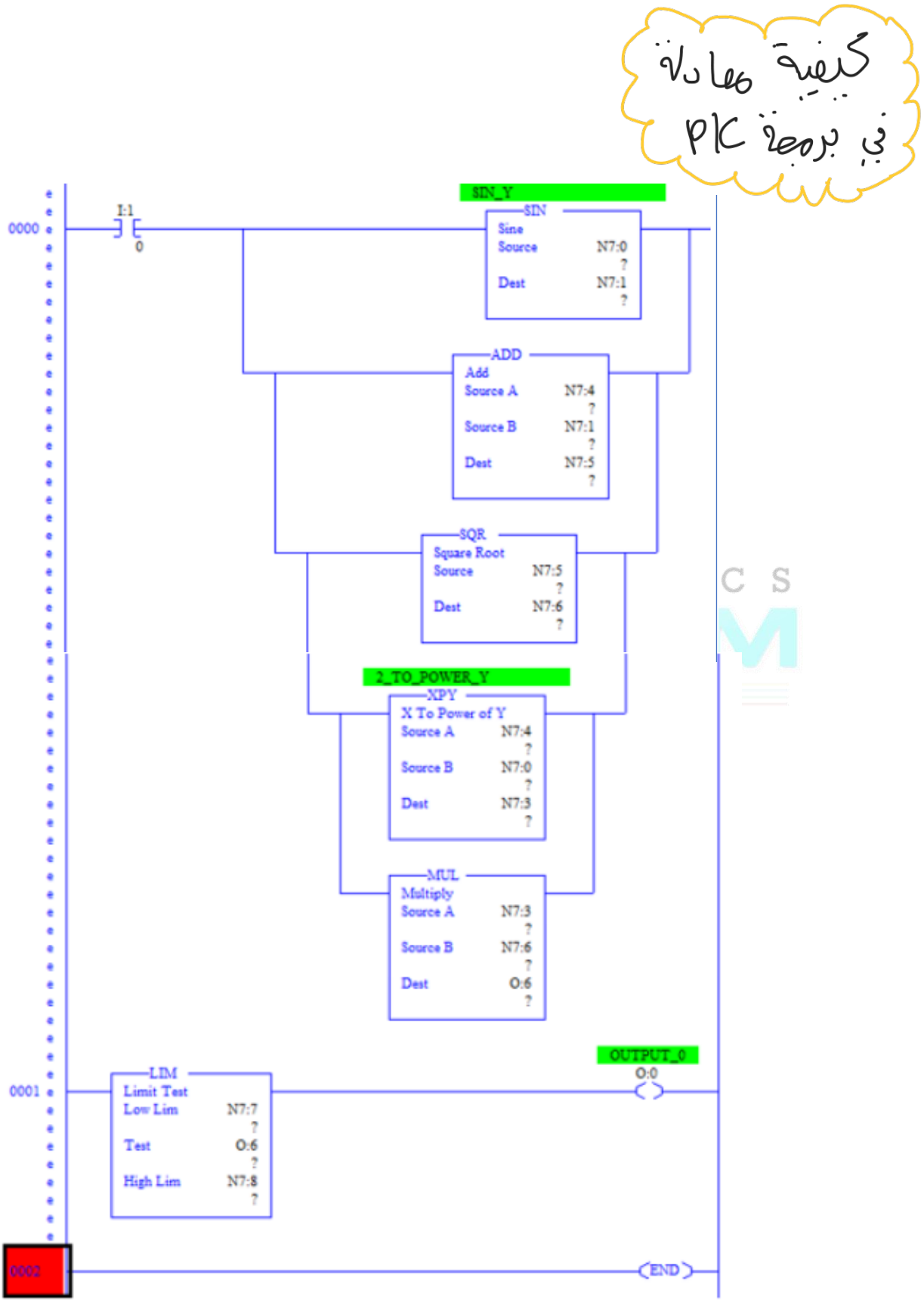
- Square Root this $1 + \sin y$ by using SQR instruction and multiply the

output of 2^y and SQR $\sqrt{1 + \sin y}$.

- Compare this data with the lower and upper limit and control output.

List of Inputs and Outputs

```
I:1/0   = Start Input                                  (Input)
N7:0 = Y input to SIN function                         (Input)
N7:1 = Sin Y operation data                            (Input)
N7:4 = Numeric Value "1" (To perform 1+Sin(y))         (Input)
N7:5 = Output of addition 1+Sin(Y)                     (Input)
N7:6 = Square Root of 1+Sin(Y)                         (Input)
N7:4 = Numeric Value "2" (To perform 2^y)              (Input)
N7:3 = Output of X=(2^y)Square Root of 1+Sin(Y)        (Input)
N7:7 = Lower Limit (01)                                (Input)
N7:8 = Higher Limit (10)                               (Input)
O:6  = Output x, multiplication of N7:6 & N7:3         (Output)
O:0  = Output 0                                        (Output)
```


كيفية معادلة
في برمجة PLC
SIN_Y
SIN
Sine
Source N7:0
Dest N7:1
ADD
Add
Source A N7:4
Source B N7:1
Dest N7:5
SQR
Square Root
Source N7:5
Dest N7:6
2_TO_POWER_Y
XPY
X To Power of Y
Source A N7:4
Source B N7:0
Dest N7:3
MUL
Multiply
Source A N7:3
Source B N7:6
Dest O:6
OUTPUT_0
O:0
LIM
Limit Test
Low Lim N7:7
Test O:6
High Lim N7:8
I:1
0
0000
0001
0002
END

Description

- As we can see, step-by-step procedure is done.
- The output of individual operations are first obtained.
- The output of all these data are stored into Integer Registers.
- Operation starts from the smallest function as shown in the RUNG0000 that first Sin(Y) (N7:0) is obtained, then Sin Y is added with the integer 1.
- The output of this 1+Sin Y is then square rooted using SQR instruction and the result is stored in the register N7:6.
- Then 2^y is performed. This is obtained by using X to power Y instruction where X is N7:4 (digit 2) and Y is N7:0 (y).
- The output of this both are Multiplied to finally obtain the output of equation X (O:6).
- LIM instruction is then used to compare the output of the equation with its lower limit and upper limit. Accordingly, the output is generated.
- If O:6 is between 1 to 10, output O:0 energizes turning ON Output 1.

Drilling Process

Devise a ladder program that could be used to operate the following task.

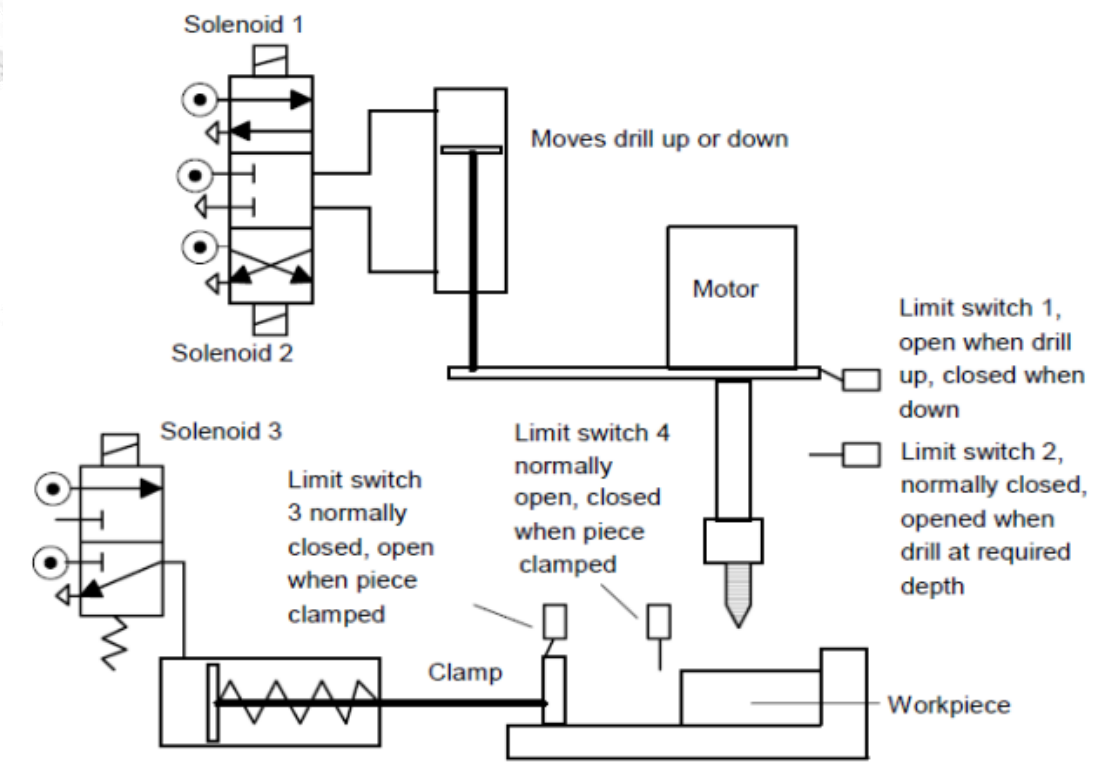
1- Start the drill motor and the pump for the air pressure for the pneumatic valves.

2- Clamp the workpiece.

3- Lower the drill and drilling started to the required depth.

4- Then retract the drill and unclamp the workpiece.

The task is shown in the following figure



MECHATRONICS SPECTRUM TEAM

I0.0 Motor Start P.B

I0.1 Motor Stop P.B

I0.2 Pump Start P.B

I0.3 Pump Stop P.B

I0.4 Limit switch 3

I0.5 Limit Switch 4

I0.6 Limit Switch 2

I0.7 Limit Switch 1

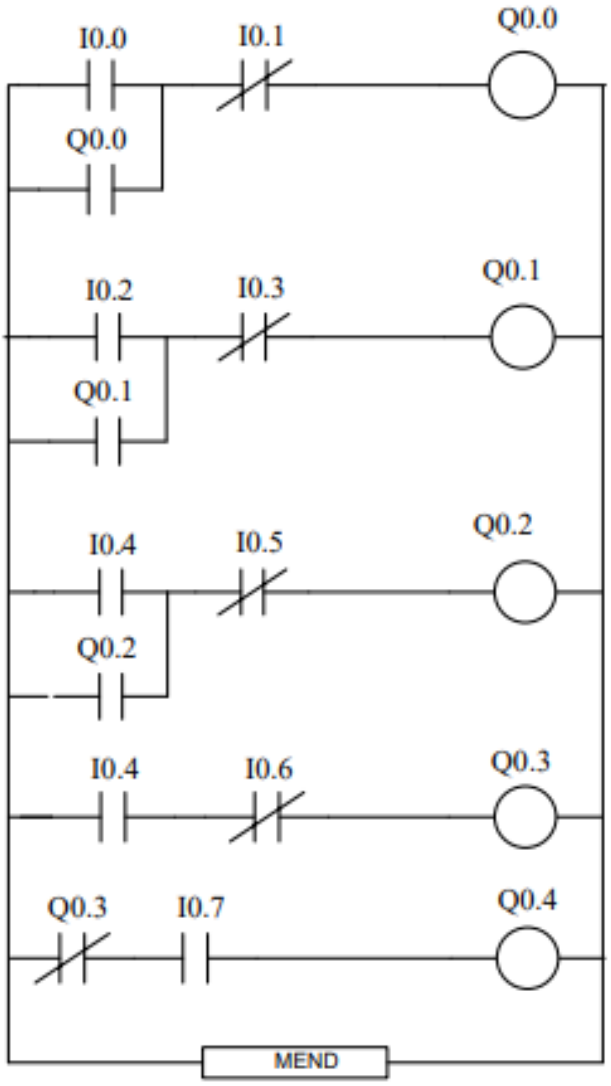
Q0.0 Motor

Q0.1 Pump

Q0.2 Solenoid 3

Q0.3 Solenoid 1

Q0.4 Solenoid 2



Heating and Mixing of Products

Description

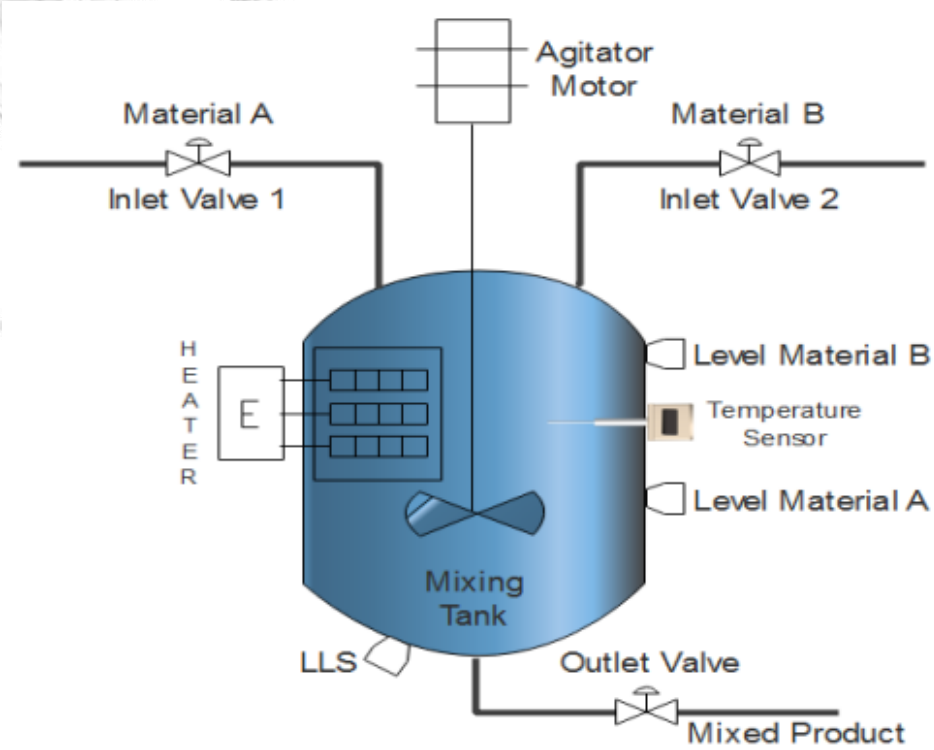
Material A and Material B are collected in a tank.

These products are to be mixed and heated till it reaches set point

temperature value.

Implement automation of this system in PLC using Ladder Diagram

programming language and instruction list for an S7-200 Siemens PLC



Solution

- To detect level of Material A and Material B, two separate level switches

are used.

- And to detect low level, one more level switch is used at the bottom of the

tank.

- These give output in digital terms that is when corresponding levels are

detected.

- To control level of this system, Single Acting Piston valve can be used

which has two states, either fully open or fully close.

- To control mixing, motor is used .

- Heater is installed inside the tank and temperature sensor is used to detect

the temperature of liquid in the tank.

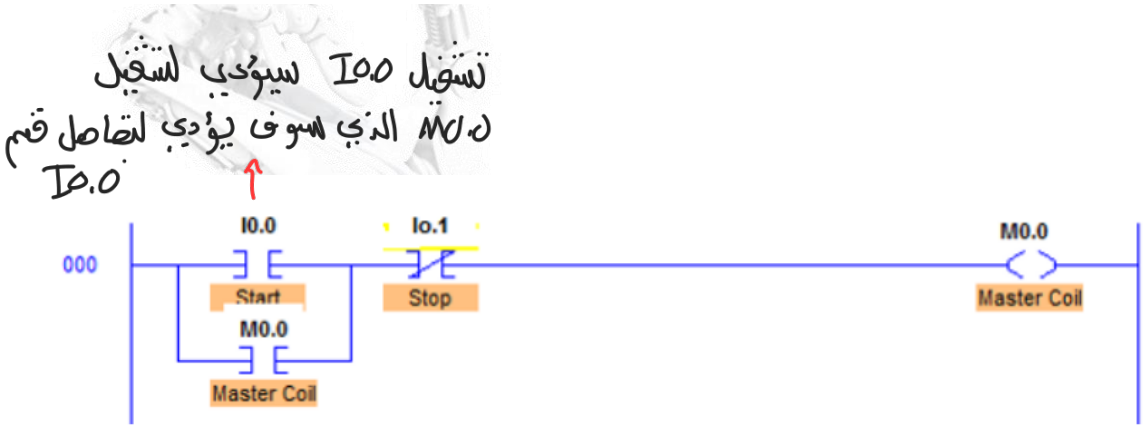
- This mixer is mixed until set point temperature is achieved.

- Outlet valve is then operated to drain the mixed product.

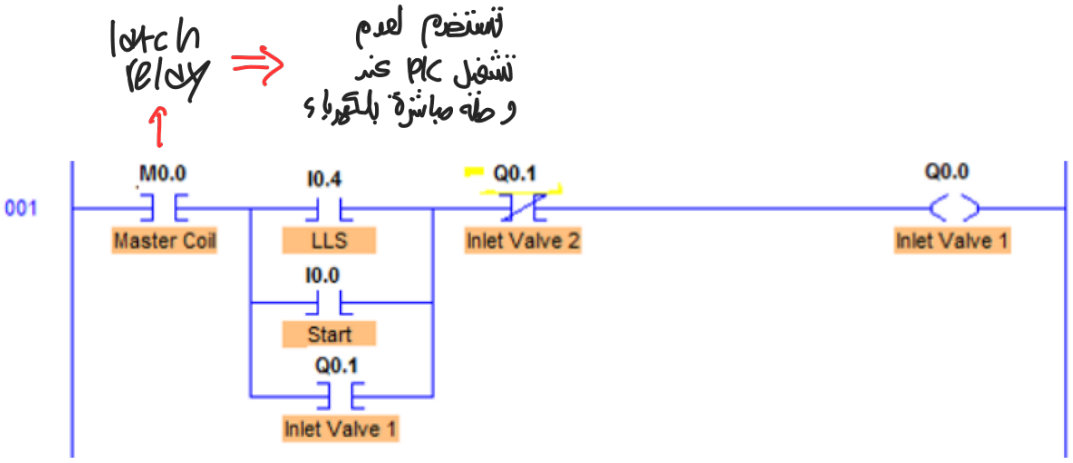
MECHATRONICS SPECTRUM TEAM

List of Inputs and Outputs

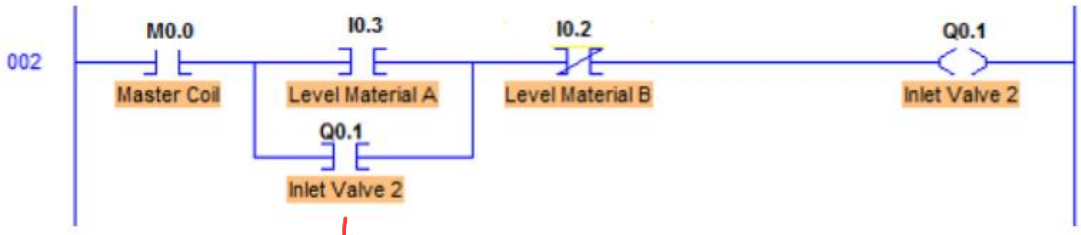
I0.0	Start	(Input)
I0.1	Stop	(Input)
M0.0	Master Coil	(Internal)
Io.2	Level of Material B	(Input)
I0.3	Level of Material A	(Input)
I0.4	Low Level Switch (empty tank)	(Input)
I0.5	Temperature detector	(Input)
Q0.0	Inlet Valve 1 (Material A Feed)	(Output)
Q0.1	Inlet Valve 2 (Material B Feed)	(Output)
Q0.2	Agitator Motor (Mixing)	(Output)
Q0.3	Heater	(Output)
Q0.4	Outlet valve to drain the product	(Output)



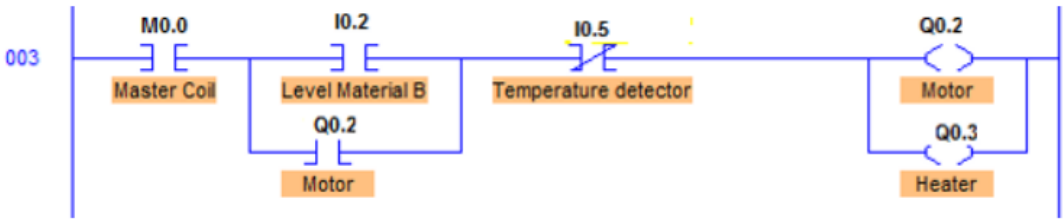
- RUNG000 contain master start/stop with address of Start PB I0.0 and Stop PB I0.1 and internal relay.



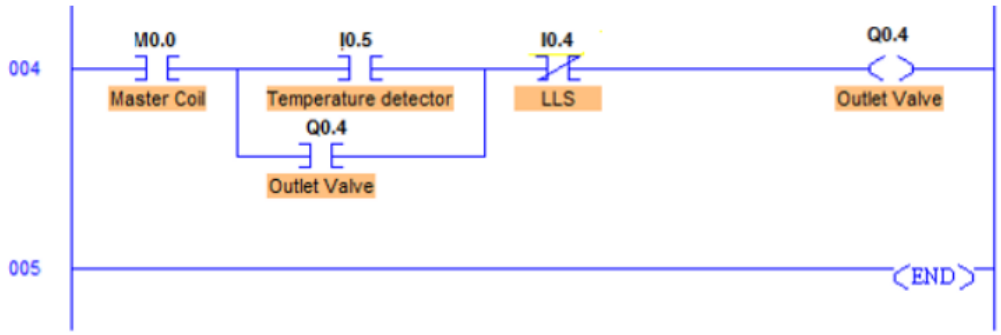
- RUNG001 is to operate Inlet Valve of Material A with address Q0.0. It is operated when Low Level of the tank is detected by Level Low Switch I0.4. And it is closed when Inlet Valve of material B starts that is when Level Material A is detected by a switch with address I0.3.
- Start PB is also connected in parallel with the LLS, it is done so that when LLS or level of Material A is not detected, Inlet Valve is operated by Start PB.



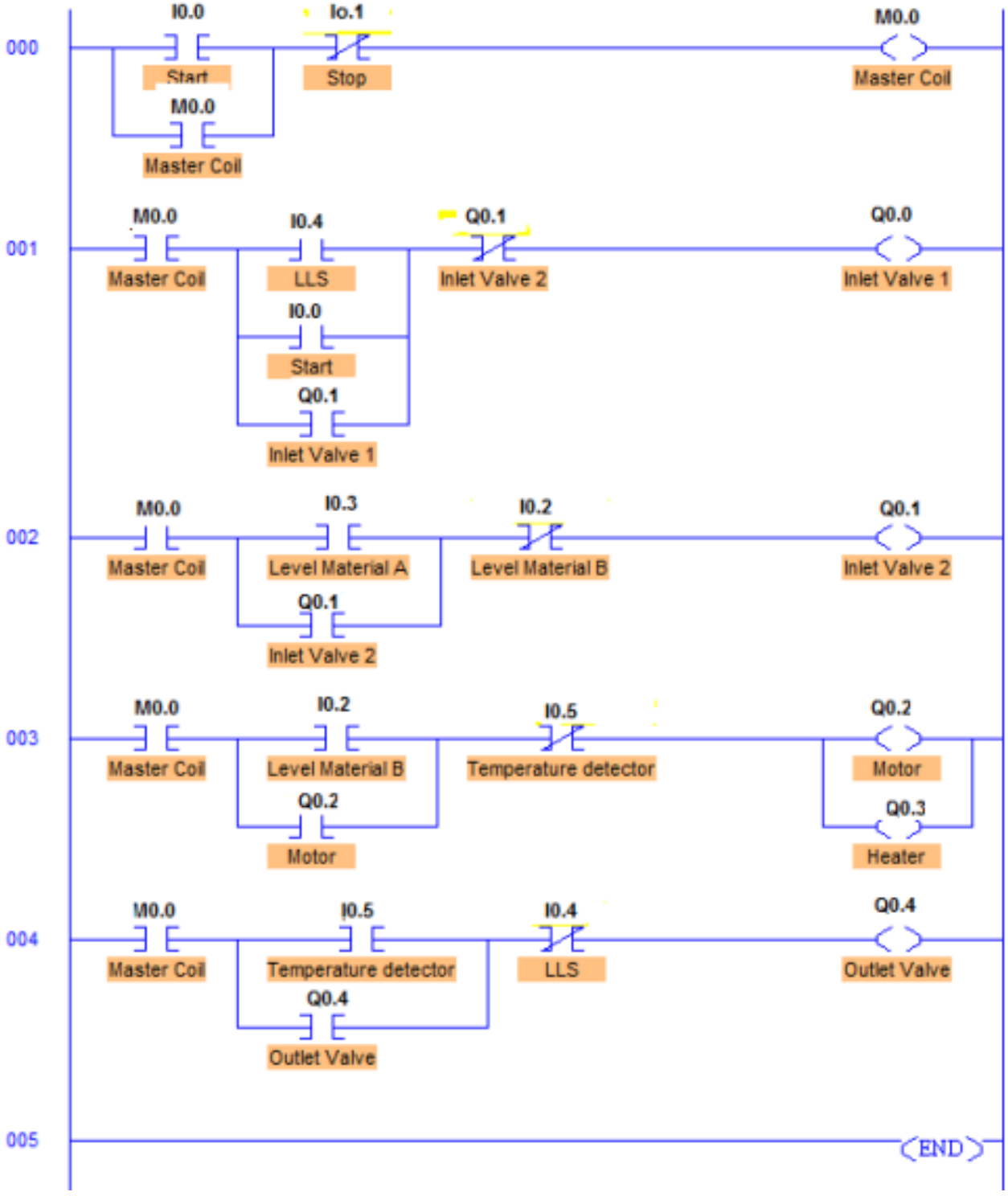
- RUNG002 is to operate Inlet Valve of Material B with address Q0.2. It is opened when Material A is filled to its desired level (Level Material A). In other words, Valve of Material B is opened when Level of Material A is detected by I0.3 and it is closed when Level of Material B is detected or Tank is full.



- RUNG003 operates Motor and Heater with address Q0.2 and Q0.3 respectively.
- When the tank is full with Material A and B, Level High (Level Material B) is detected. Motor is connected to the address Q0.2 and Heater to Q0.3. Motor and Heater are run when I0.2 is set to 1 that is when tank is full.



- RUNG004 is for Outlet Valve with address Q0.3. It is operated when the entire mixing and heating process is completed that is when temperature set point is reached. And this valve is closed when LLS is again detected.


000
I0.0
Start
M0.0
Master Coil
Io.1
Stop
M0.0
Master Coil
001
M0.0
Master Coil
I0.4
LLS
I0.0
Start
Q0.1
Inlet Valve 1
Q0.1
Inlet Valve 2
Q0.0
Inlet Valve 1
002
M0.0
Master Coil
I0.3
Level Material A
Q0.1
Inlet Valve 2
I0.2
Level Material B
Q0.1
Inlet Valve 2
003
M0.0
Master Coil
I0.2
Level Material B
Q0.2
Motor
I0.5
Temperature detector
Q0.2
Motor
Q0.3
Heater
004
M0.0
Master Coil
I0.5
Temperature detector
Q0.4
Outlet Valve
I0.4
LLS
Q0.4
Outlet Valve
005
END

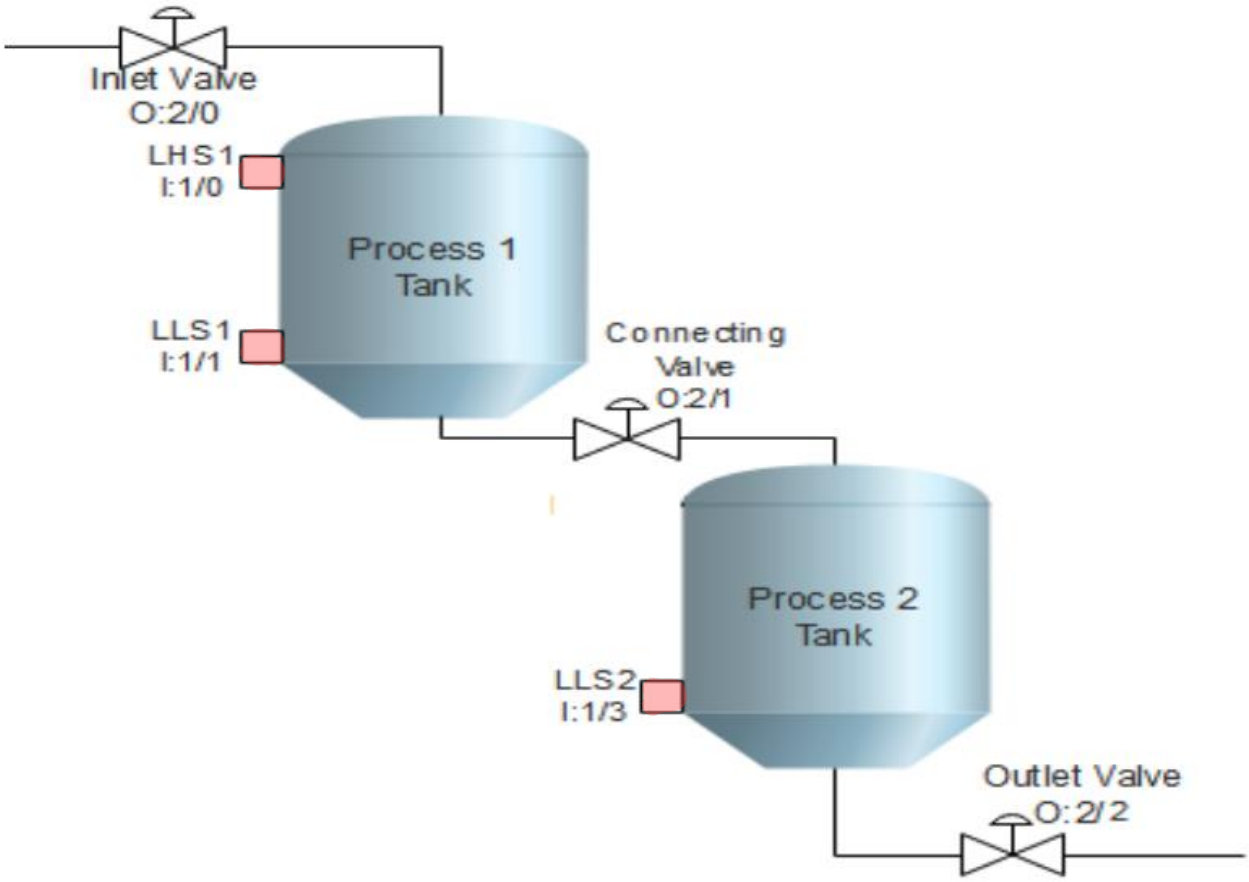
Control Level of Two Tanks

Description

-Two simultaneous processes are to be performed in two separate tanks which are connected through a valve.

-Process 1 takes place in the 1st tank and Process 2 takes in the 2nd tank.

-Control level of these tanks in PLC using Ladder Diagram programming language.



Solution

- Mount two level switches in the first tank and two switches in

the second tanks. Both switches of tanks detect High and Low

level of the tanks.

- Install inlet valve to control inlet process flow and outlet valve to

control outlet process flow.

- Install one more connecting valve to control draining of process

material from Process 1 Tank to Process 2 Tank.

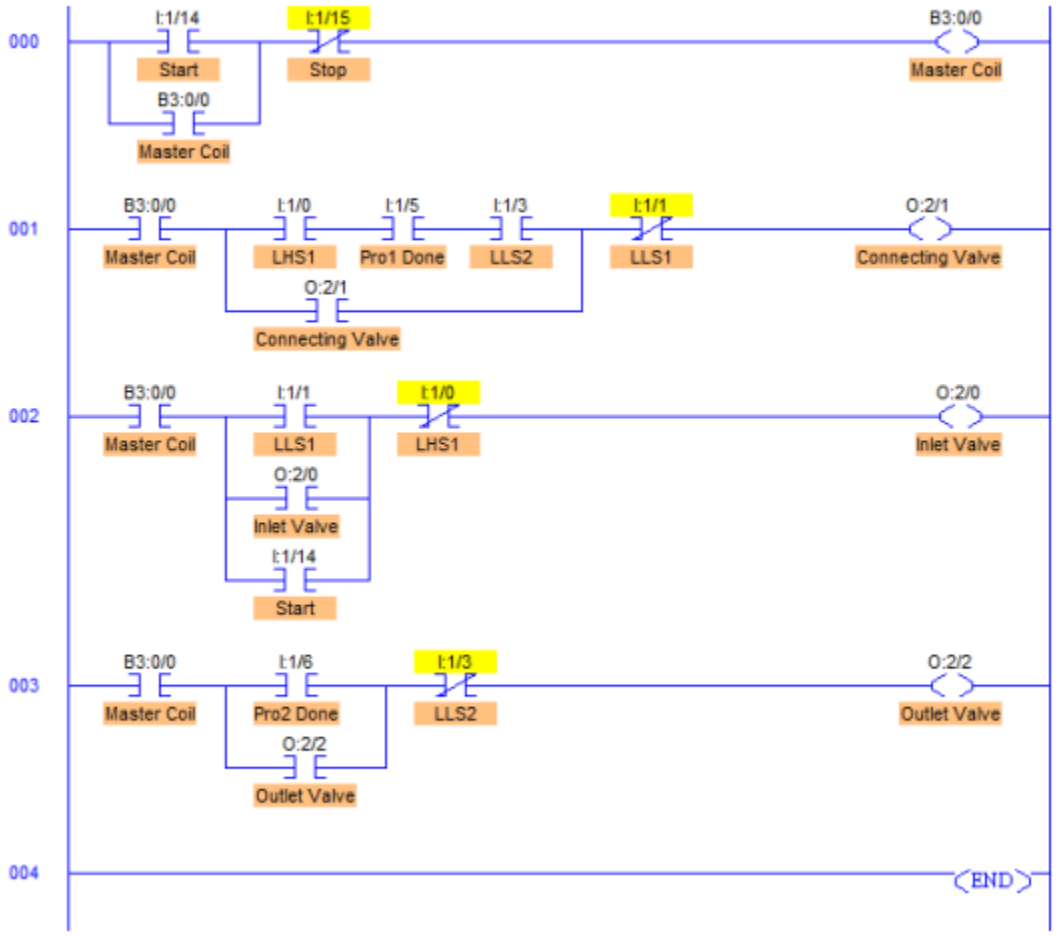
- Provide interlocking to prevent from malfunctioning or

overflowing.

MECHATRONICS SPECTRUM TEAM

List of inputs or Outputs

I:1/0= Level High Switch-Tank 1, LHS1 (Input)
I:1/1= Level Low Switch-Tank 1, LLS1 (Input)
I:1/3= Level Low Switch-Tank 2, LLS2 (Input)
O:2/0 = Inlet Valve (Output)
O:2/1 = Tank Connecting Valve (Output)
O:2/2 = Outlet Valve (Output)
I:1/14 = Start I0.0 (Input)
I:1/15 = Stop (Input)
B3:0/0 = Master Coil Bit (Bit)



Description

- RUNG000 is a master start/stop rung to Start and Stop the entire process.

- RUNG001 is for controlling Connecting Valve output O:2/1. It is opened when LHS1 I:1/0, Pro1 Done I:1/5 and LLS2 I:1/3 are detected.

And it is closed when LLS1 I:1/1 is detected, or in other words, when Process 1 Tank is empty.

- Important thing to note here is that if Tanks'size/shape were not same and any one XIC contact from O:2/1, I:1/0 or I:1/5 was not connected, there would have been a chance of malfunctioning/overflowing.

- RUNG002 is for controlling Inlet valve O:2/0. It is allows the inlet flow by opening Inlet Valve whenever Low Level switch of Process1 Tank LLS1 with address I:1/0 is detected.

- RUNG003 is for controlling Outlet Valve with address O:2/2. It allows the process material to flow out when Pro1 Done I:1/6 is detected.

- Consider any set point of Process1 and Process2 as I:1/5 and I:1/6.

Jump to Other Process

Description

Implement Jumping from one process to another process in PLC using Ladder Diagram programing language.

Solution

- This type of operation is most widely used in hazardous area where some action must be taken immediately despite of the status of currently running process.
- Use JMP instruction to perform this task.
- This Jump instruction does not differ from any Microcontroller

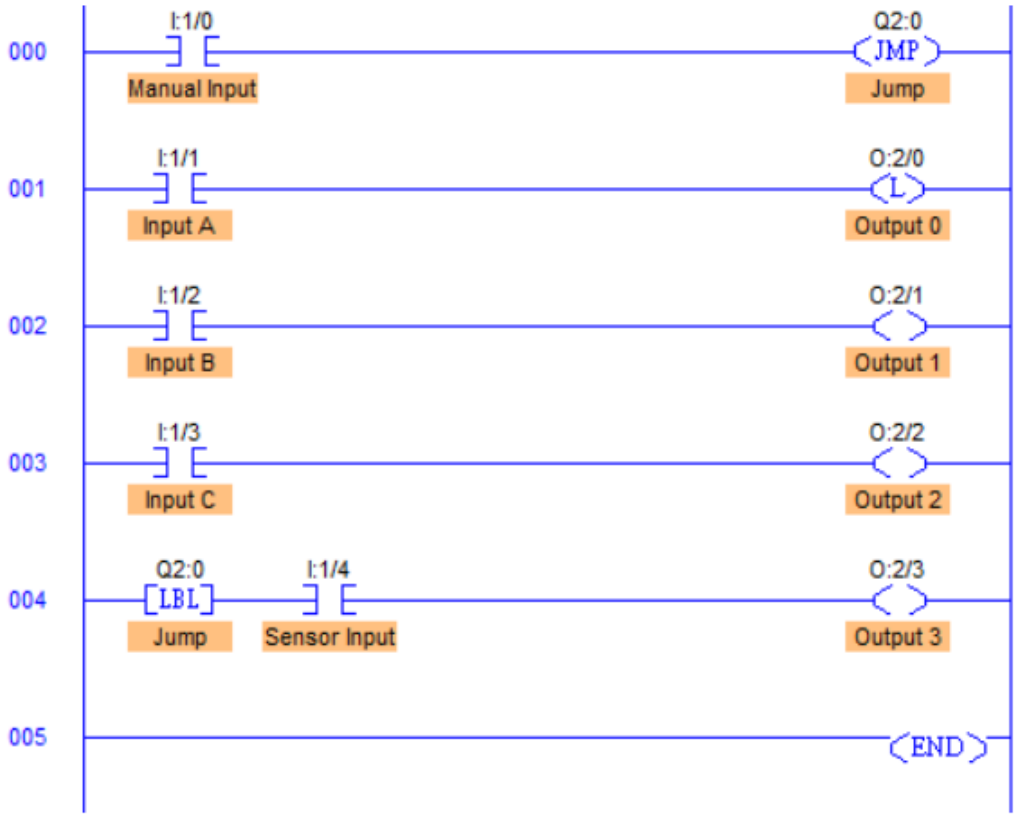
Jump instruction, this Jump instruction also must be used with LBL (Label) instruction.

- Simply explaining, when JMP is activated, all the outputs between JMP and LBL are disabled until this is active. By saying ‘all outputs’, it does not include latch –(L)– outputs.

MECHATRONICS SPECTRUM TEAM

List of Inputs and Outputs

I:1/0	= Input to Jump to Label	(Input)
I:1/1	= Input A	(Input)
I:1/2	= Input B	(Input)
I:1/3	= Input C	(Input)
I:1/4	= Sensor Input	(Input)
O:2/0	= Output 0	(Output)
O:2/1	= Output 1	(Output)
O:2/2	= Output 2	(Output)
O:2/3	= Output 3	(Output)
Q2:0	= Jump (JMP and LBL)	(I/O)



Program Description

- RUNG000 comprises of a Manual input which may be a toggle

button or sensor output by which operation is performed.

- Until I:1/0 is pressed, JMP does not energize, that means program is being scanned rung by rung and it does not affect program or process at all.

- When I:1/0 is pressed, program is directly jumped to the LBL and scans RUNG004.

- If Sensor Input I:1/4 is true, it energizes O:2/3 turning ON Output 3.

- As long as JMP is energized, all the output between JMP and LBL rungs are skipped and Input I:1/1 to I:1/3 do not affect O:2/0 to O:2/2 at all.

- Important thing to note here is that if Latch Output is used between JMP-LBL rungs and is active then it is not unlatched when JMP energizes.

MECHATRONICS SPECTRUM TEAM

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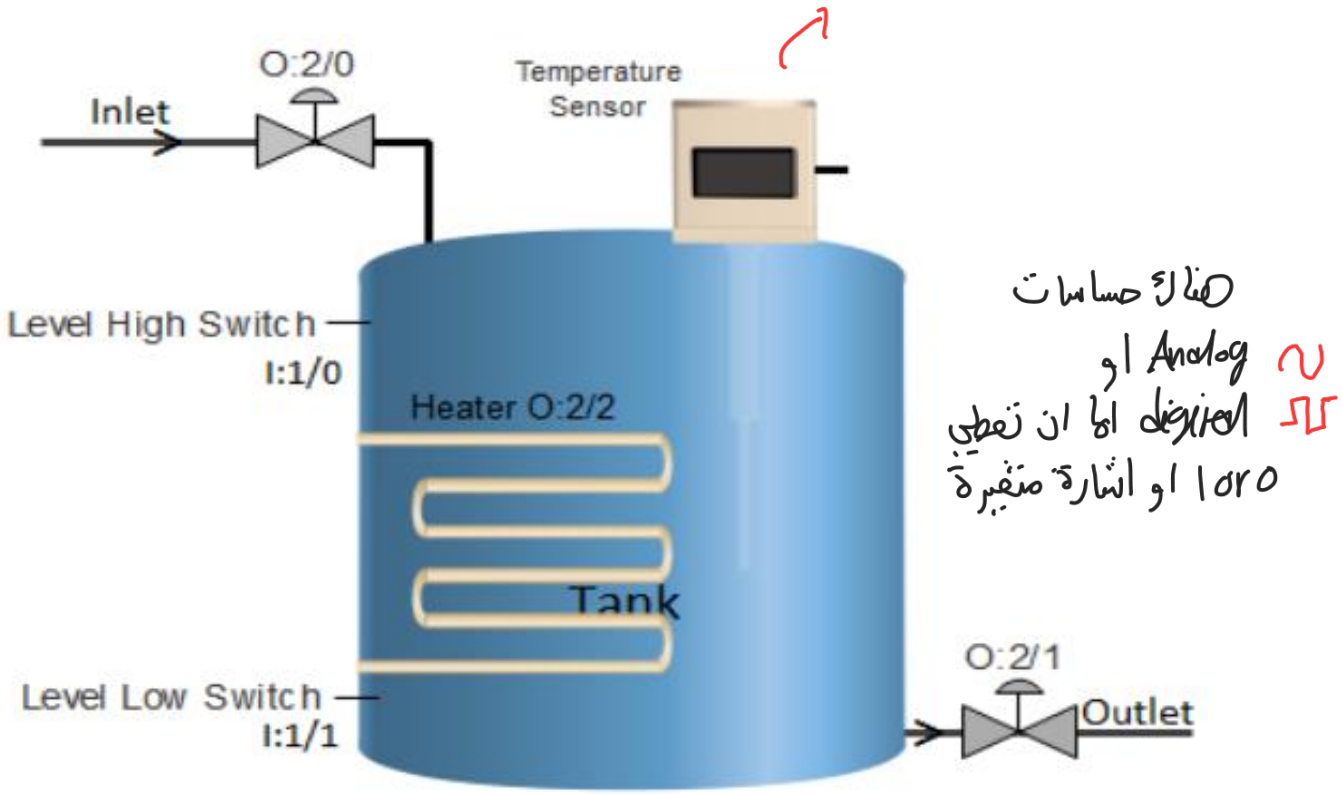
Heating Liquid in a Tank using electric

Description

-Controlling heating of Liquid in a tank using heater.

-Implement this process in PLC using Ladder Diagram programming language.



Solution

- To detect high and low level of liquid in the tank, two level switches are used

which gives output in digital terms, that is when corresponding levels are

detected, it gives output high otherwise remain low.

- To control level of this system, Single Acting Piston valve can be used which

has two states, either fully open or fully close and to heat the liquid, heater is used.

- Low Level Switch is mounted at the bottom of the tank and Level High

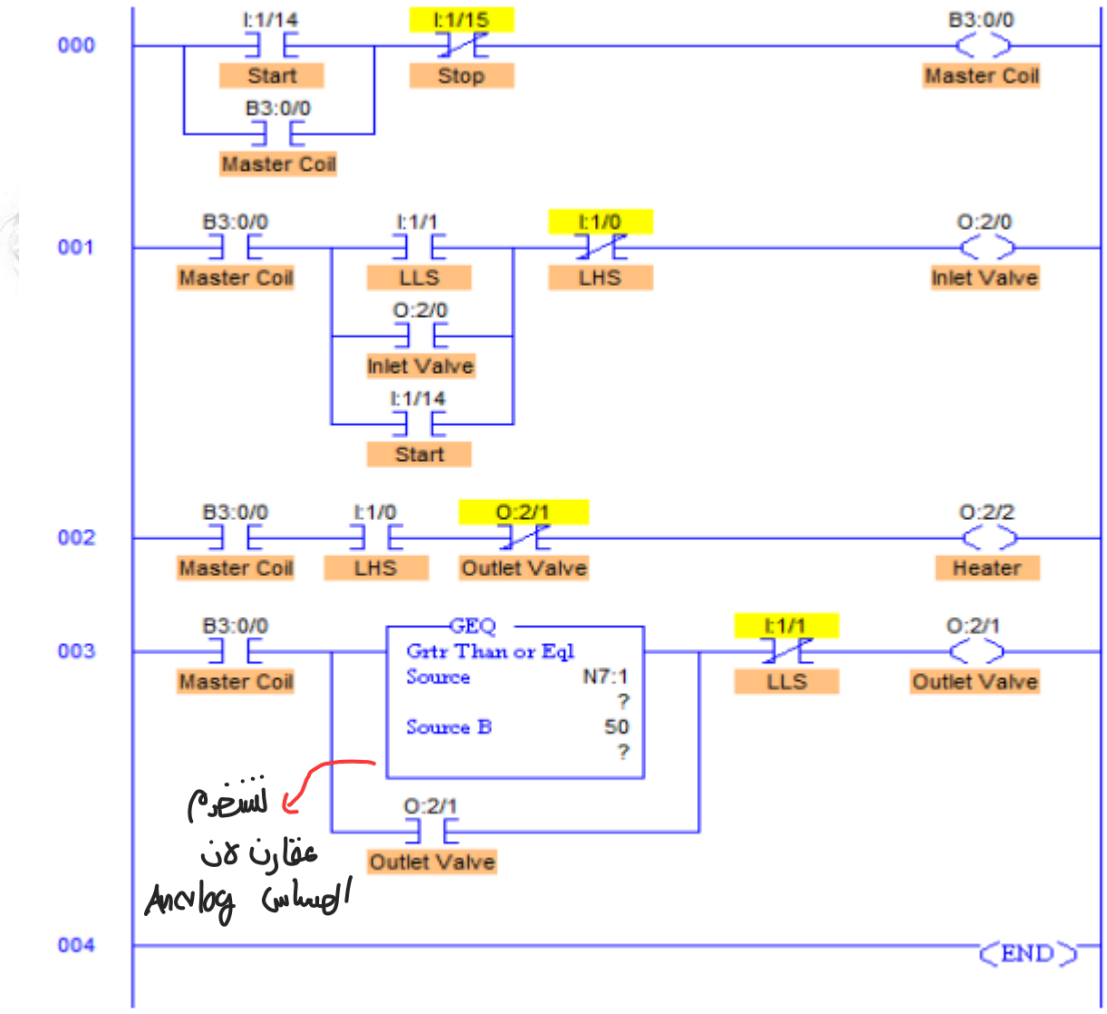
switch is mounted at the side-upper most position.

- Heater is installed inside the tank and temperature sensor may be used to detect the temperature of liquid in the tank.

MECHATRONICS SPECTRUM TEAM

List of Inputs and Outputs

I:1/0	= Level High Switch	(Input)
I:1/1	= Level Low Switch	(Input)
O:2/0	= Inlet Valve	(Output)
O:2/1	= Outlet Valve	(Output)
O:2/2	= Heater	(Output)
I:1/14	= Start	(Input)
I:1/15	= Stop	(Input)
N7:1	= Temperature Data	(Register)



Description

- RUNG000 is simply for latching a coil and master start-stop buttons.

- Similarly in RUNG001, it works exactly same. It is energized

when Level Low is detected. The only difference in RUNG001 is that extra I:1/14 contact in parallel with LLS.

- In RUNG002, heater coil O:2/2 energizes when LHS I:1/0 is detected and heating is started. This is done until temperature

reaches 50°C, O:2/1 is energized de-energizing O:2/2 which in turn stops heating.

- RUNG003 is to control the outlet valve through O:2/1. This is done when Temperature is above 50°C. When temperature is greater than 50°C, outlet valve starts draining the liquid. XIO of Level Low

Switch is connected in series so that when Level Low is detected, it goes true closing the outlet valve.

-Suppose when the system is started and the tank is partially filled,

neither LHS nor LLS is detected, in this case heater output does not

get energized and heating is not controlled.

-To eliminate this error, I:1/14 (Start) is connected in parallel to LLS

I:1/1 contact. This checks if LHS (I:1/0) is detected or not. If LHS is

not detected, then it opens the inlet valve until LHS is detected and heating is started

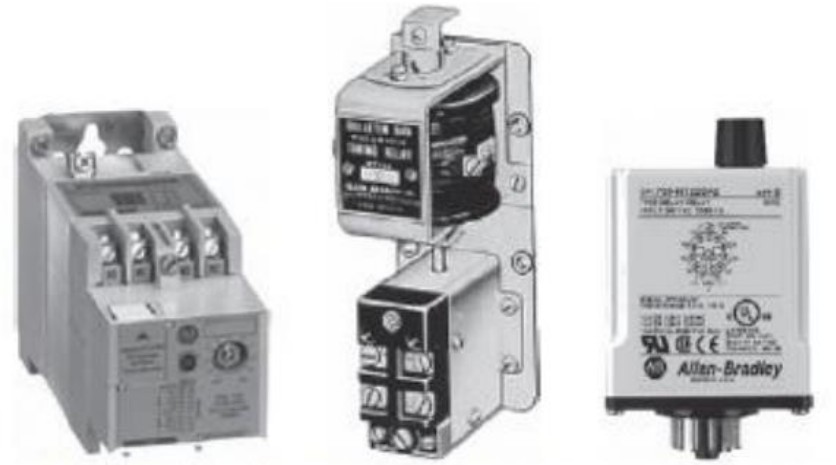
MECHATRONICS SPECTRUM TEAM

Chapter 6

Timers

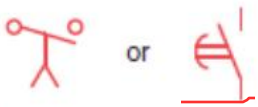
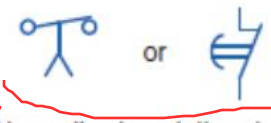
In many control tasks there is a need to control time. For example, a motor or a pump might need to be controlled to operate for a particular interval of time, or perhaps be switched on after some time interval. PLCs thus have timers as built-in devices. Timers count fractions of seconds or seconds using the internal CPU clock. This lecture shows how such timers can be programmed to carry out control tasks.

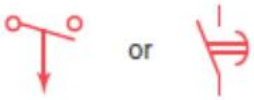
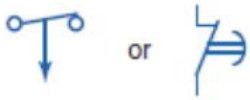
Mechanical Timing



Solid-state timing relay | Pneumatic timing relay | Plug-in timing relay

Timed contact symbols

On-delay symbols	
	
Normally open, timed closed contact (NOTC).	Normally closed, timed open contact (NCTO).
Contact is open when relay coil is de-energized.	Contact is closed when relay coil is de-energized.
When relay is energized, there is a time delay in closing.	When relay is energized, there is a time delay in opening.

Off-delay symbols	
	
Normally open, timed open contact (NOTO).	Normally closed, timed closed contact (NCTC).
Contact is normally open when relay coil is de-energized.	Contact is normally closed when relay coil is de-energized.
When relay coil is energized, contact closes instantly.	When relay coil is energized, contact opens instantly.
When relay coil is de-energized, there is a time delay before the contact opens.	When relay coil is de-energized, there is a time delay before the contact closes.

T E A M

Types of timers

- PLC manufacturers differ on how timers should be programmed and how they can be considered.

A common approach is to consider timers to behave like relays with coils which when energized result in the closure or opening of contacts after some preset time.

The timer is thus treated as an output for a rung with control being exercised over pairs of contacts.

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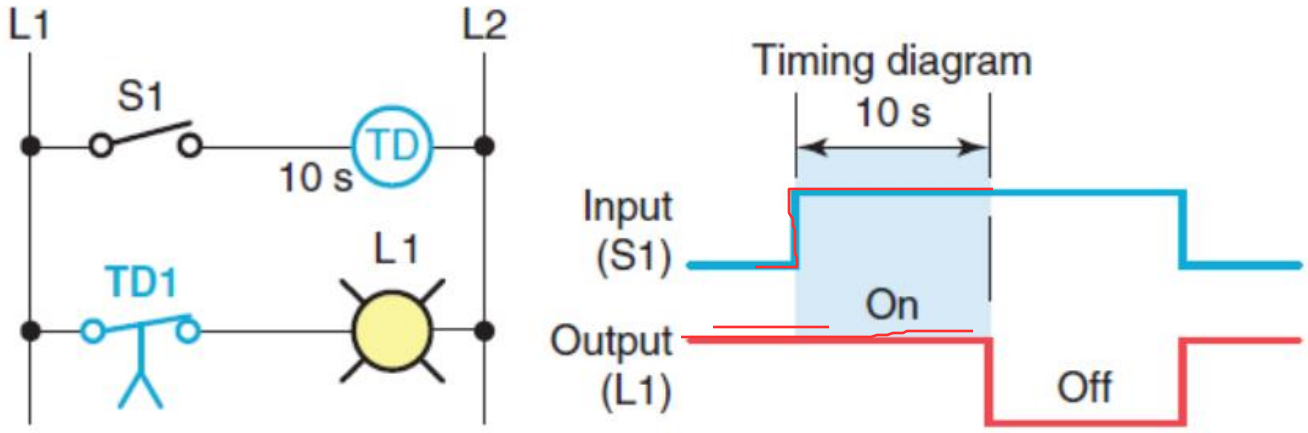
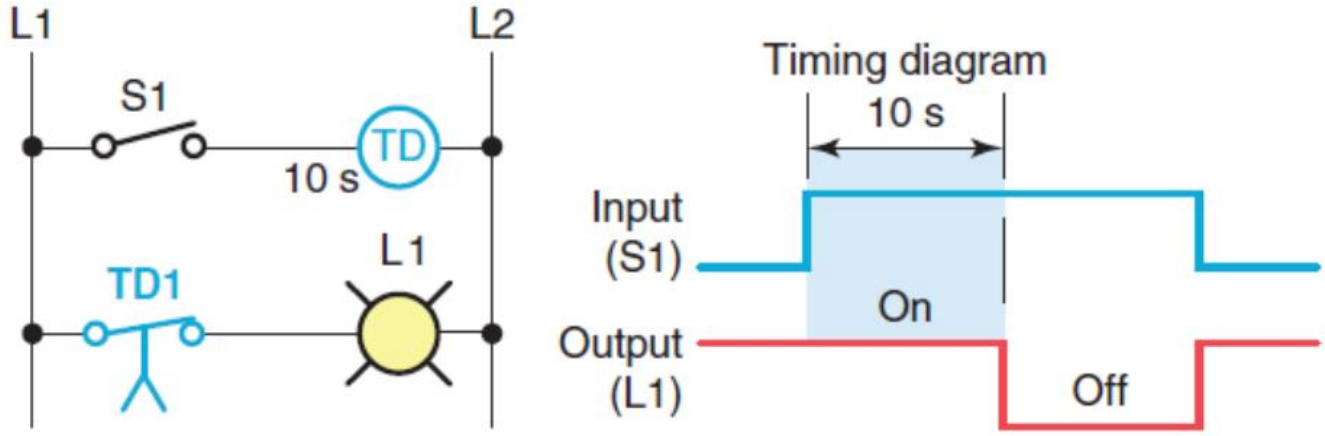
MECHATRONICS SPECTRUM TEAM

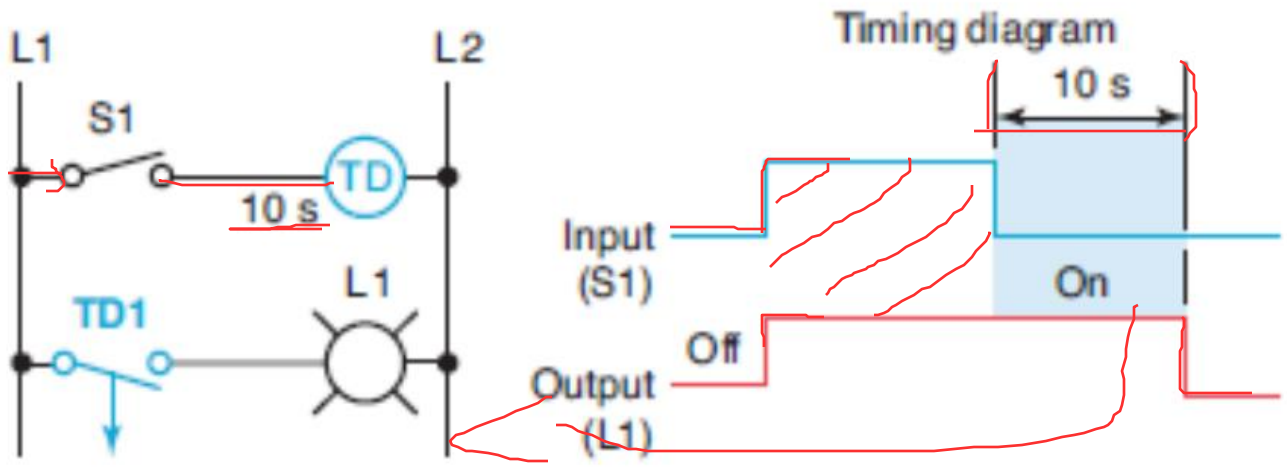
- **On-delay timer circuit that uses a normally open, timed closed (NOTC) contact.**



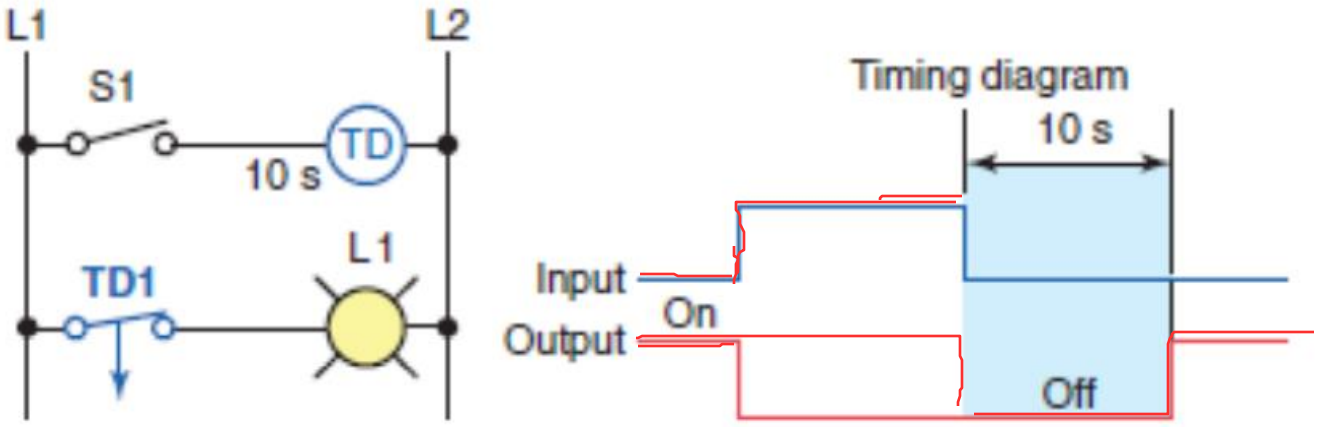
- **On-delay timer circuit that uses a normally closed, timed open (NCTO) contact. Open (NCTO) contact.**



- **Off-delay timer circuit that uses a normally open, timed open (NOTO) contact.**



- **Off-delay timer circuit that uses a normally closed, timed closed (NCTC) contact**



Timer Instructions

TON (Timer On Delay)

Counts time-based intervals when the instruction is true.

تقوم بحساب الفترات الزمنية عندما تكون نتيجة تنفيذ التعليمة صحيحة

TOF (Timer Off Delay)

Counts time-based intervals when the instruction is false.

RTO (Retentive Timer On)

Counts time-based intervals when the instruction is true and retains the accumulated value when the instruction goes false or when

RES (Reset)

Counts time-based intervals when the instruction is false.

MECHATRONICS SPECTRUM TEAM

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Timer In Quantities associated with the timer instruction: strictions

• The preset time represents the time duration for the timing circuit. For example, if a time delay of 10 s is required, the timer will have a preset of 10 s.

• The accumulated time represents the amount of time that has elapsed from the moment the timing coil became energized.

• Every timer has a time base.

o Once the timing rung has continuity, the timer counts in time-based intervals and times until the preset value and accumulated value are equal or, depending on the type of controller, up to the maximum time interval of the timer.

o The intervals that the timers time out at are generally referred to as the time bases of the timer.

o Timers can be programmed with several different time bases: 1 s, 0.1 s, and 0.01 s are typical time bases.

o If a programmer entered 0.1 for the time base and 50 for the number of delay increments, the timer would have a 5-s delay .

o The smaller the time base selected, the better the accuracy of the timer

MECHATRONICS SPECTRUM TEAM

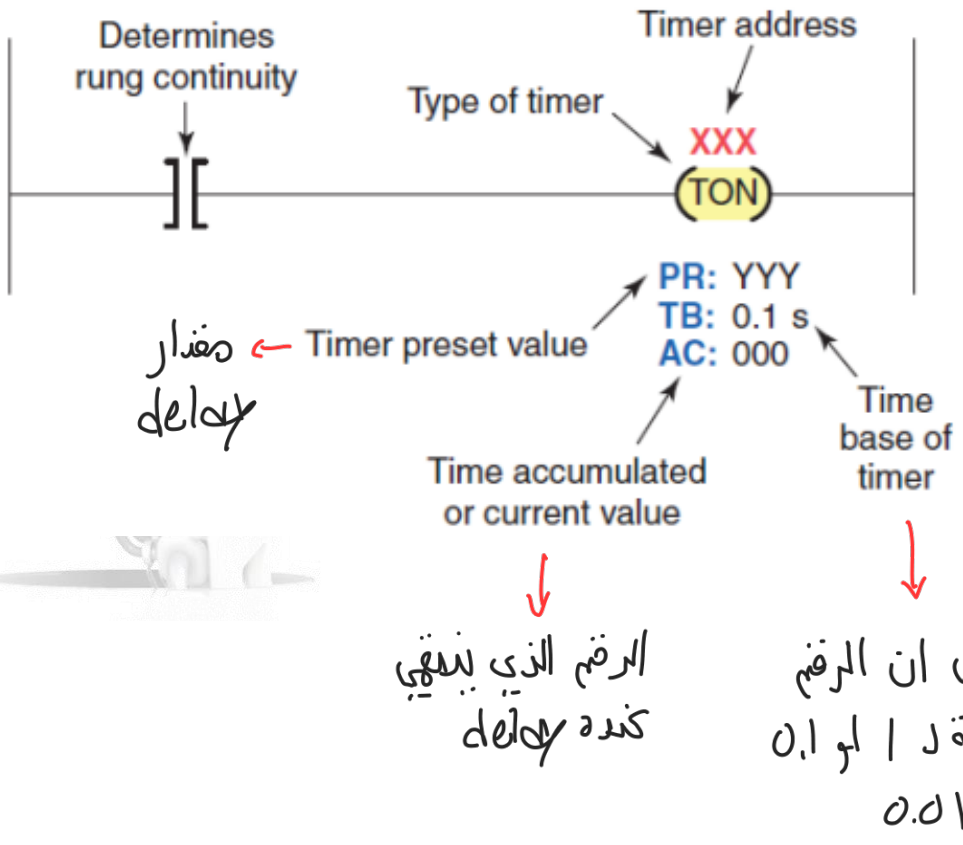
Coil-formatted timer instruction.

- The timer is assigned an address and is identified as a timer.
- Also included as part of the timer instruction is the time base of

the timer, the timer's preset value or time-delay period, and the

accumulated value or current time-delay period for the timer

MECHATRONICS



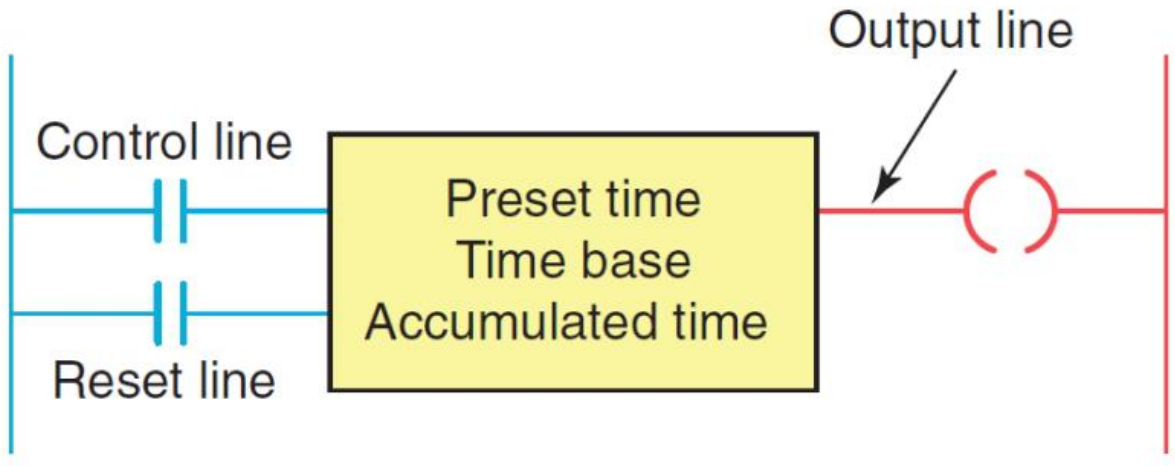
• When the timer rung has logic continuity, the timer begins counting

time-based intervals and times until the accumulated value equals the preset value.

• When the accumulated time equals the present time, the output is energized, and the timed output contact associated with the output is closed. The timed contact can be used as many times as you wish throughout the program as an NO or NC contact

Block-formatted timer instruction

• The timer block has two input conditions associated with it, namely, the control and reset



• The control line controls the actual timing operation of the timer. Whenever this line is true or power is supplied to this input, the timer will time. Removal of power from the control line input halts the further timing of the timer.

• The reset line resets the timer's accumulated value to zero.

On-Delay Timer Instruction

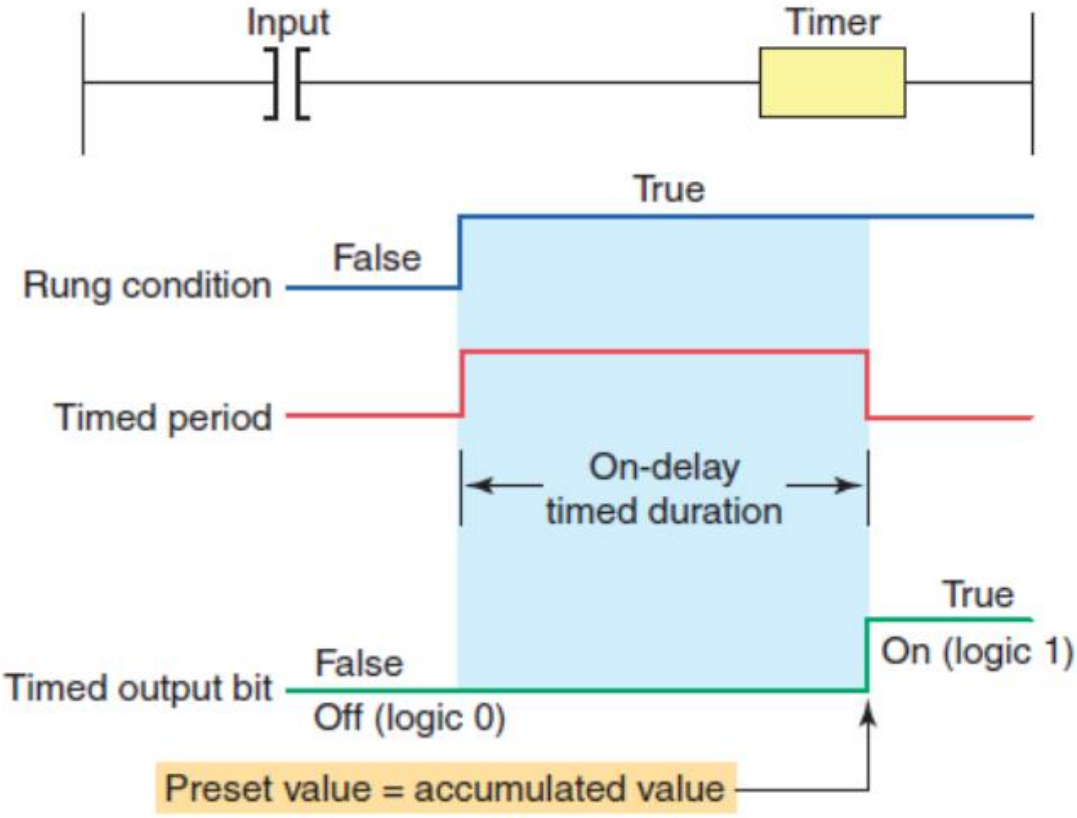
Principle of operation of an on-delay timer.

• The on-delay timer operates such that when the rung containing the timer is true, the timer time-out period commences.

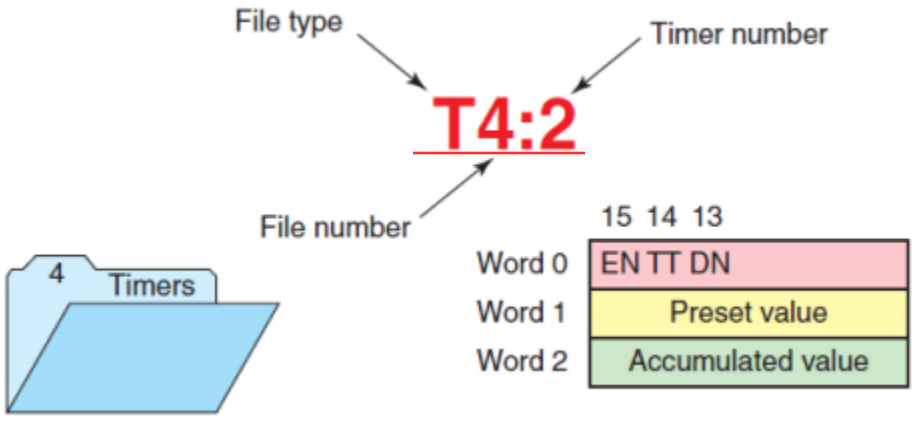
•يعمل جهاز ضبط وقت التأخير على النحو التالي عندما تكون الدرجة المحتوية على المؤقت قيمتها صحيحة ، تبدأ فترة مهلة المؤقت.

• At the end of the timer time-out period, an output is made true.

•في نهاية فترة المهلة الموقت ، يتم جعل قيمة المخرج صحيح



Timer File Example: SLC 500 timer file



-The preset value (PRE) word is the set point of the timer, that is, the value up to which the timer will time. The preset word has a range of 0 through 32,767 and is stored in binary form. The preset will not store a negative number.

-The accumulated value (ACC) word is the value that increments as the timer is timing

1 Enable (EN) bit

The enable bit is true (has a status of 1) whenever the timer instruction is true. When the timer instruction is false, the enable bit is false (has a status of 0)

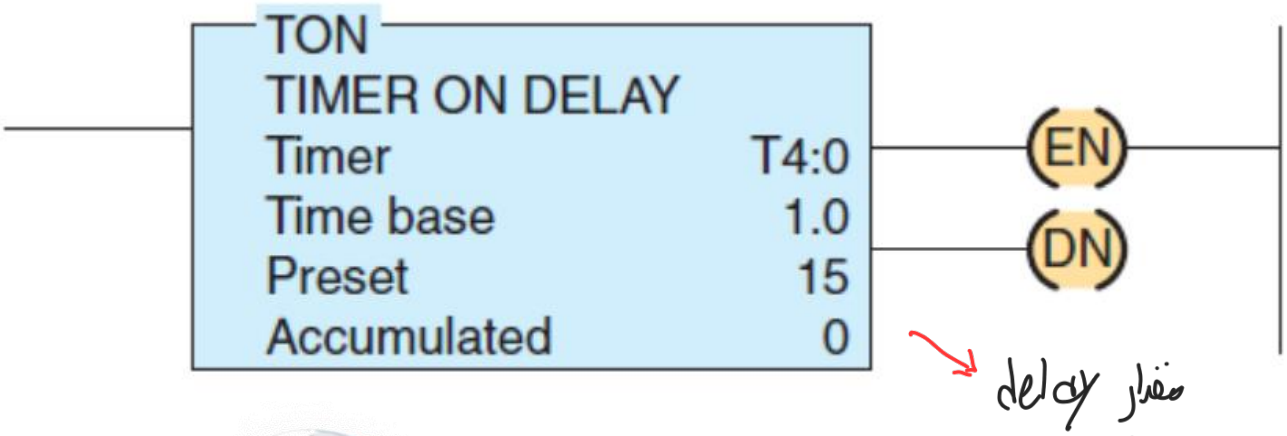
2 Timer-timing (TT) bit

The timer-timing bit is true whenever the accumulated value of the timer is changing, which means the timer is timing. When the timer is not timing, the accumulated value is not changing, so the timer-timing bit is false

3 Done (DN) bit

The done bit changes state whenever the accumulated value reaches the preset value. Its state depends on the type of timer being used

An example of the on-delay timer



1 Timer number

This number must come from the timer file. In the example shown, the timer number is T4:0, which represents timer file 4, timer 0 in that file. The timer address must be unique for this timer and may not be used for any other timer

2 Time base

The time base (which is always expressed in seconds) may be either 1.0 s or 0.01 s. In the example shown, the time base is 1.0 s

MECHATRONICS SPECTRUM TEAM

3 Preset value

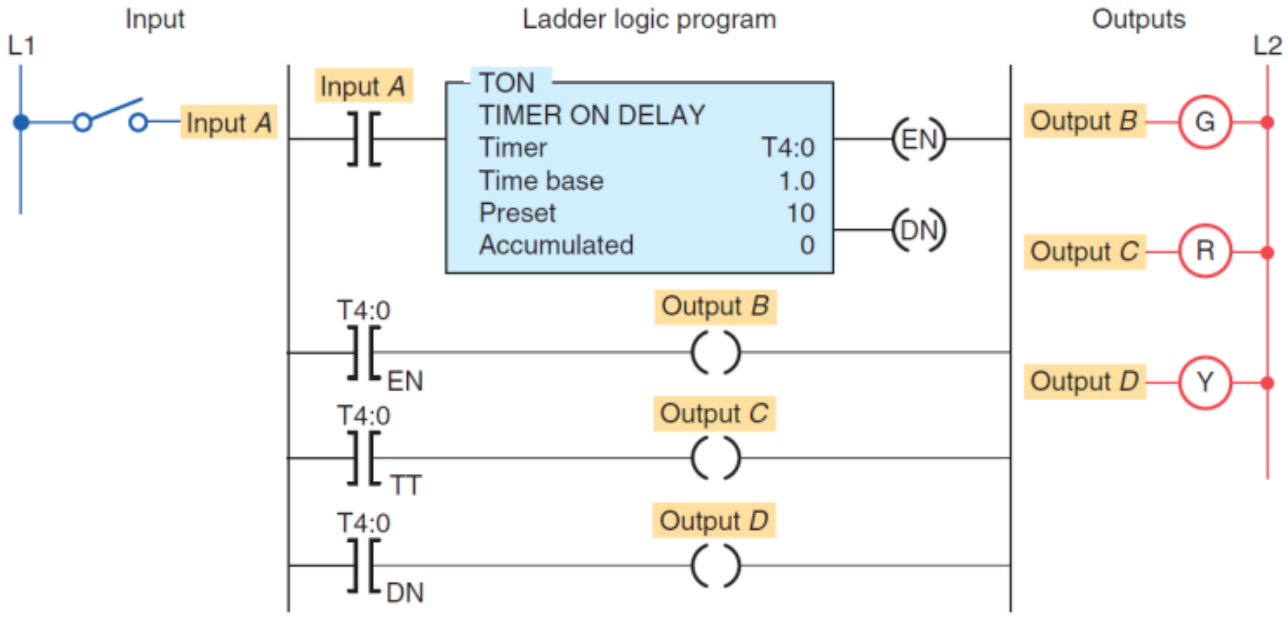
In the example shown, the preset value is 15. The timer preset value can range from 0 through 32,767

4 Accumulated value

In the example shown, the accumulated value is 0.

The timer's accumulated value normally is entered as 0, although it is possible to enter a value from 0 through 32,767. Regardless of the value that is preloaded, the timer value will become 0 whenever the timer is reset

PLC on-delay timer program



The operation of the program can be summarized as follows:

• The timer is activated by input switch A.

• The preset time for this timer is 10 s, at which time output D will be energized.

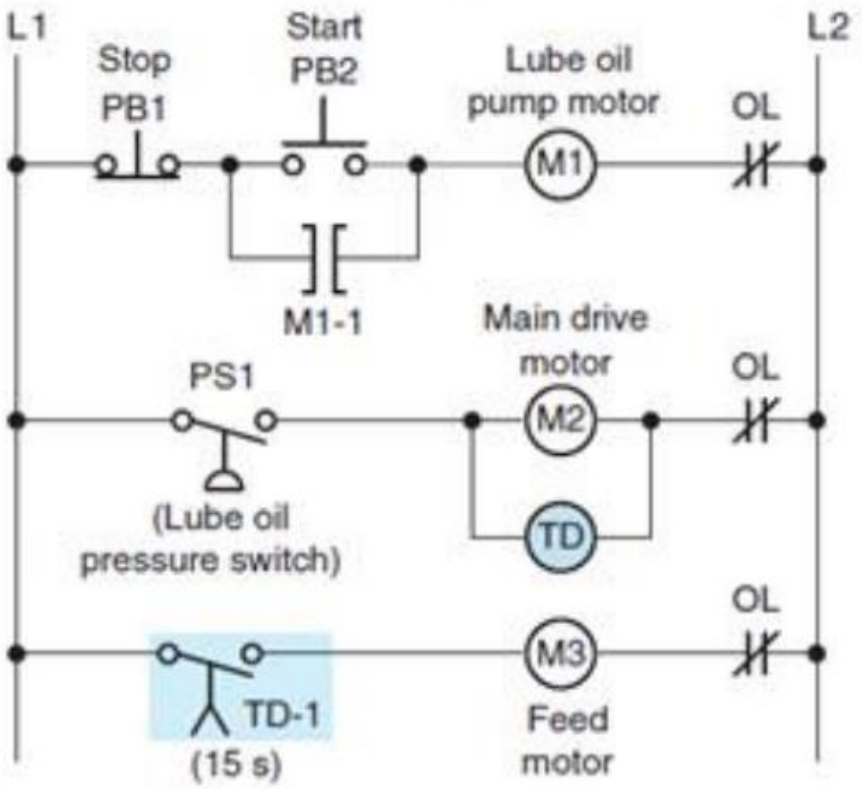
• When input switch is A is closed, the timer becomes true and the timer begins counting and counts until the accumulated time equals the preset value; the output D is then energized.

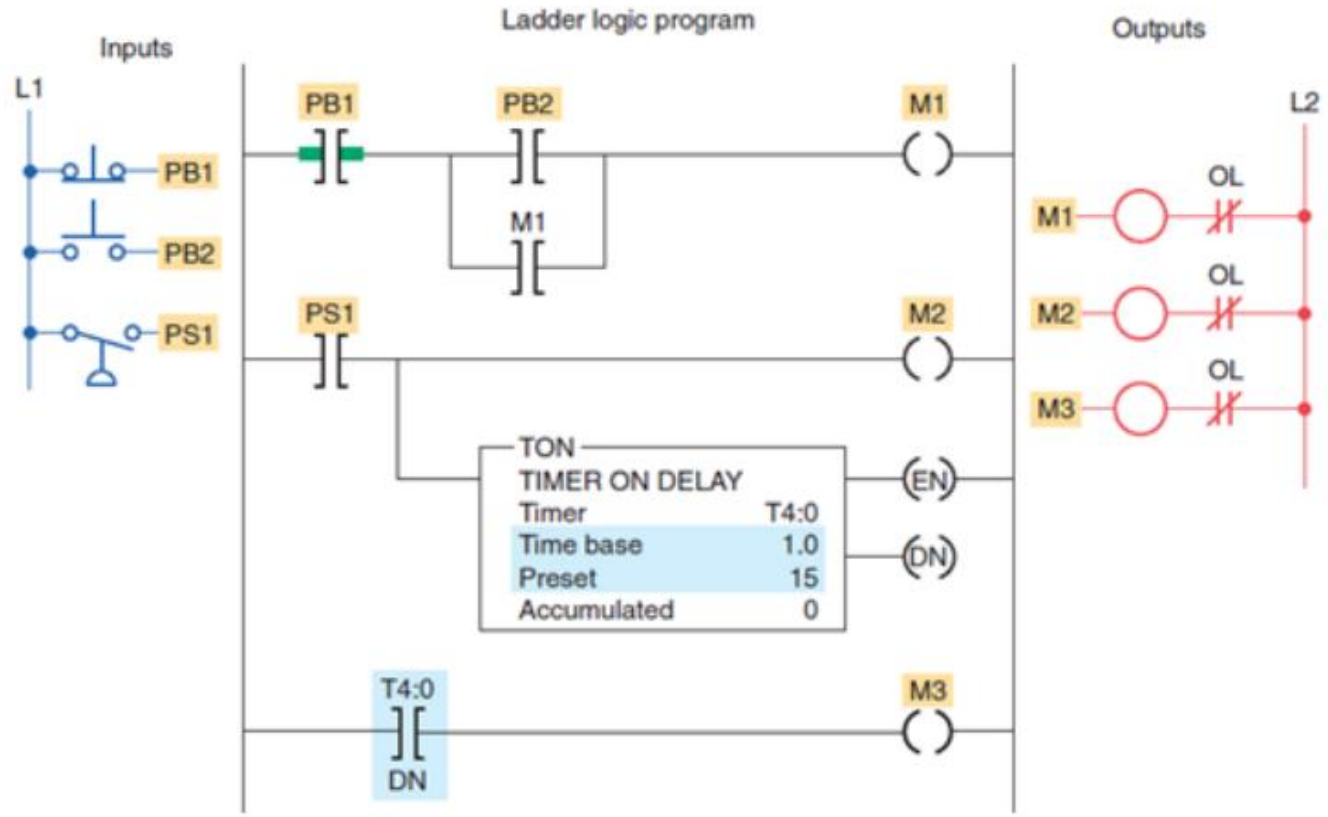
• If the switch is opened before the timer is timed out, the accumulated time is automatically reset to 0.

• This timer configuration is termed non-retentive because any loss of continuity to the timer causes the timer instruction to reset.

• This timing operation is that of an on-delay timer because output D is switched on 10 s after the switch has been actuated from the off to the on position.

Hardwired relay circuit



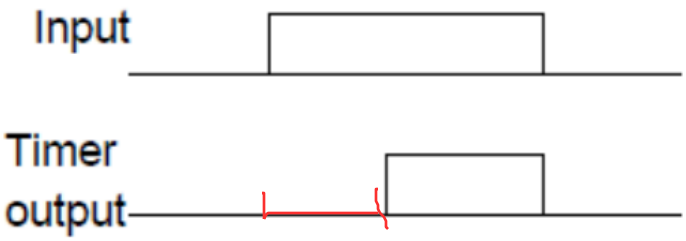


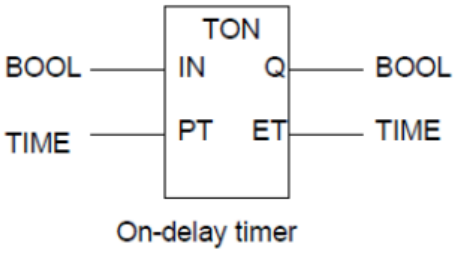
Siemens Timer

PLC semens timer

- T37
- T38
- T39
- T40
- T41

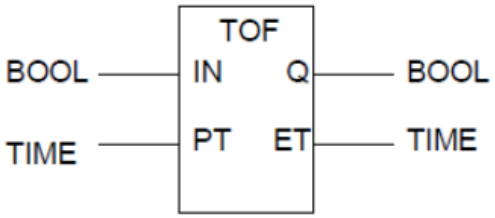
1 TON



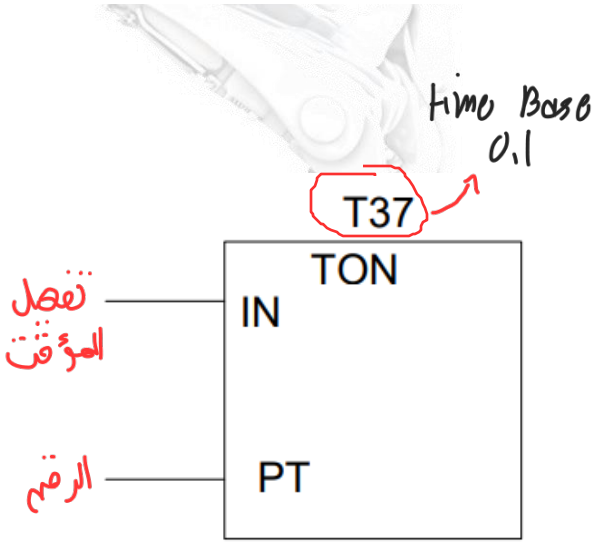


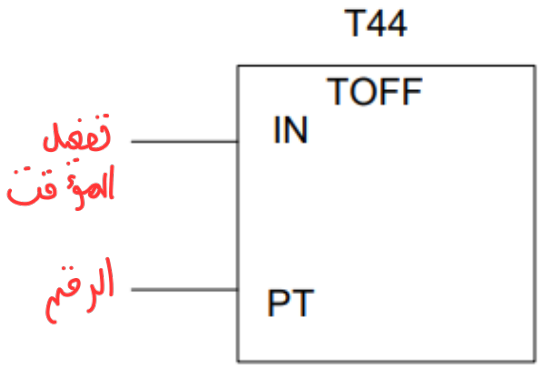
On-delay timer

2 TOFF



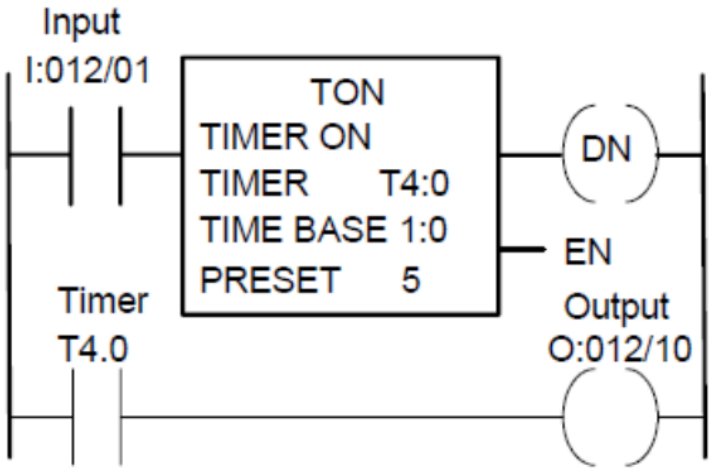
Off-delay timer

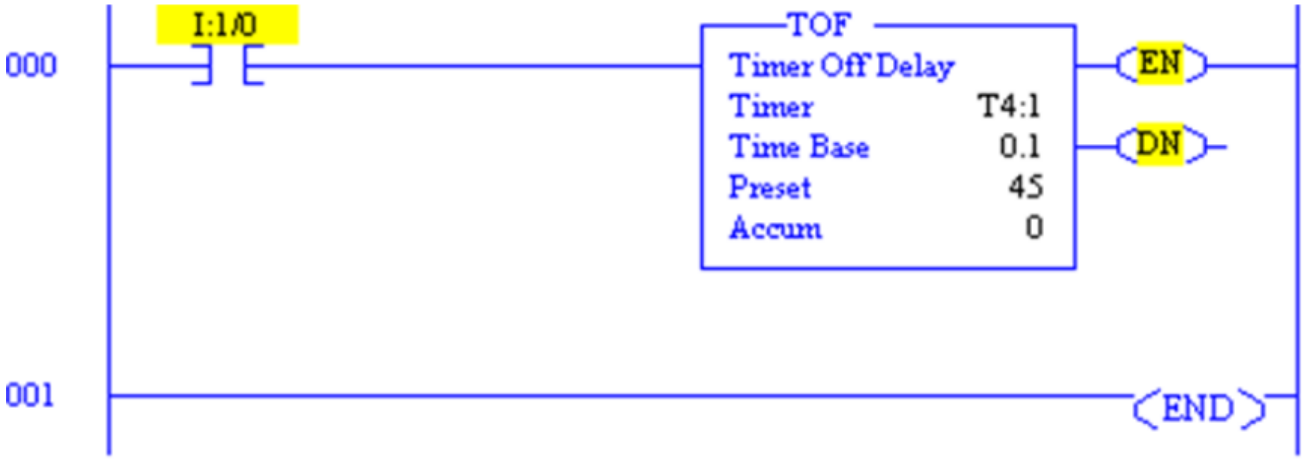




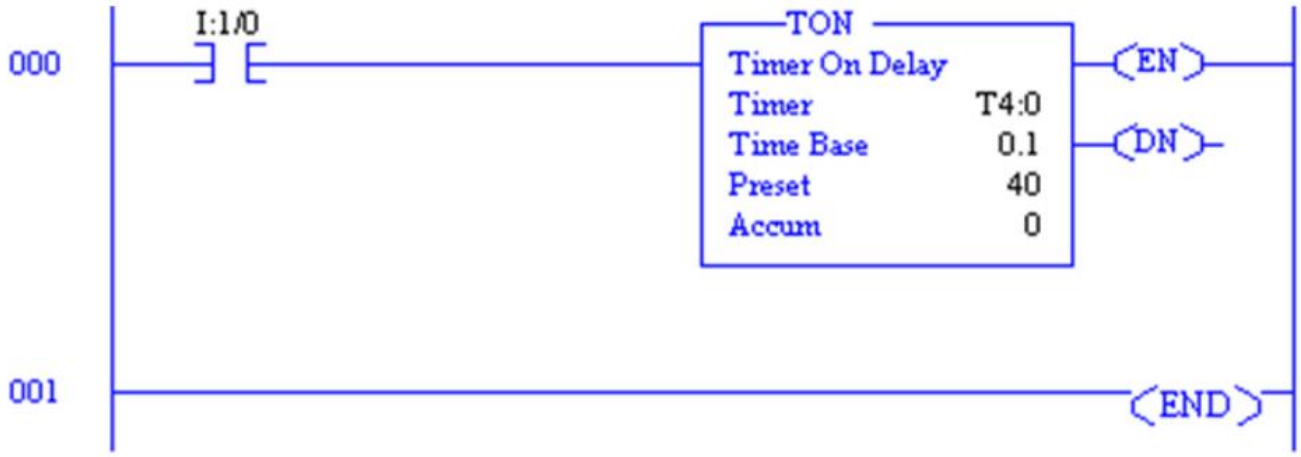
Allen-Bradley Timer

1 TON

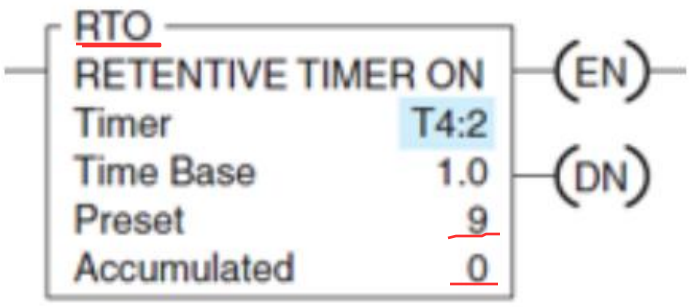




2 TOF

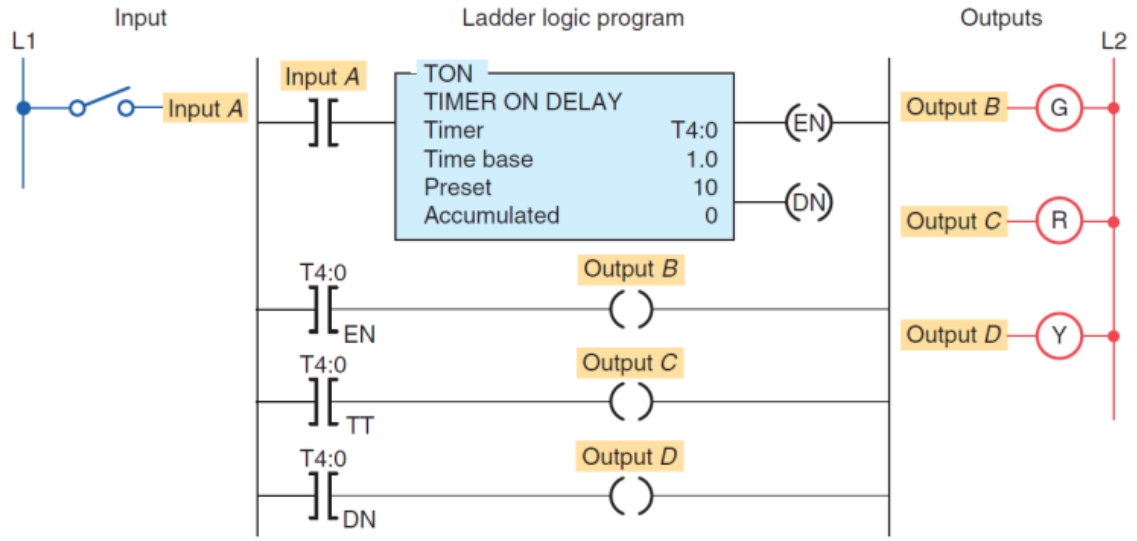


3 Retentive or Accumulating timers (RTO)



هذا النوع من المؤقتات يحتفظ بقيمة الزمن في حال فصل التغذية عن ملف التشغيل للمؤقت

Examples



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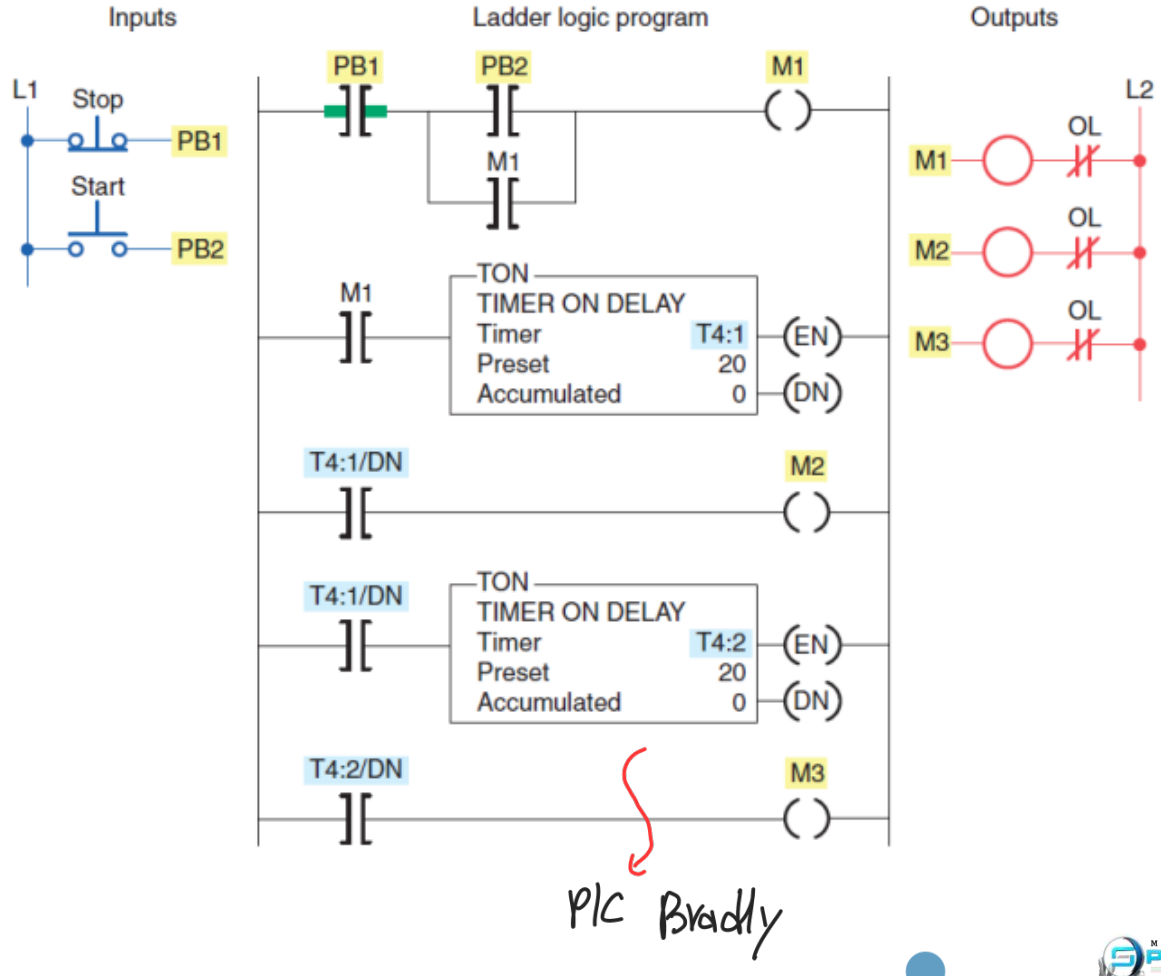

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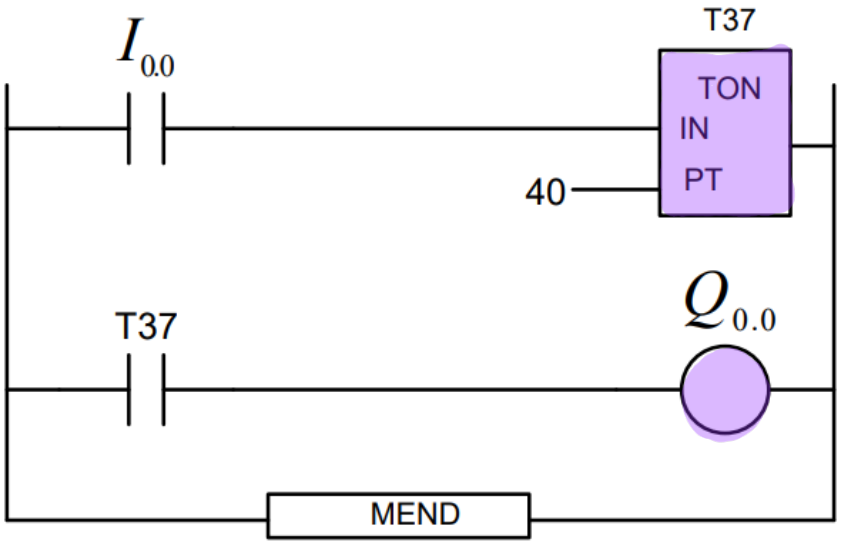
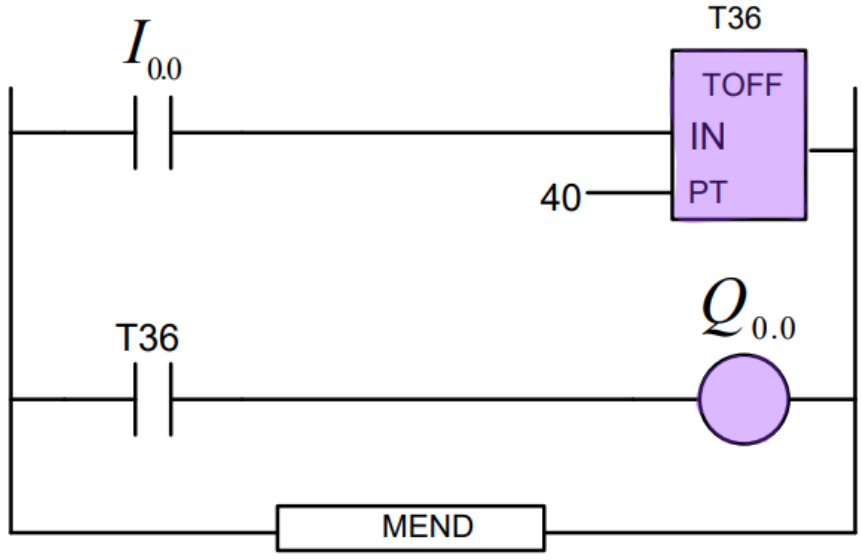
PLC semins




```
LD      I0.0
TON     #40.T37
LD      T37
=       Q0.0
MEND
```




```
LD      I0.0
TOFF    T36,# 40
LD      T36
=       Q0.0
MEND
```

Control Three Mixing Devices in a Processing Line

Problem Description

-Control three mixing devices in a process line. Two mixing devices are running in the parallel to each other. When any one of the mixing process stops, individual stop switches turn the third mixing device ON and the device operated by the stop button is stopped. Implement automation of this system in PLC using Ladder Diagram programming language.

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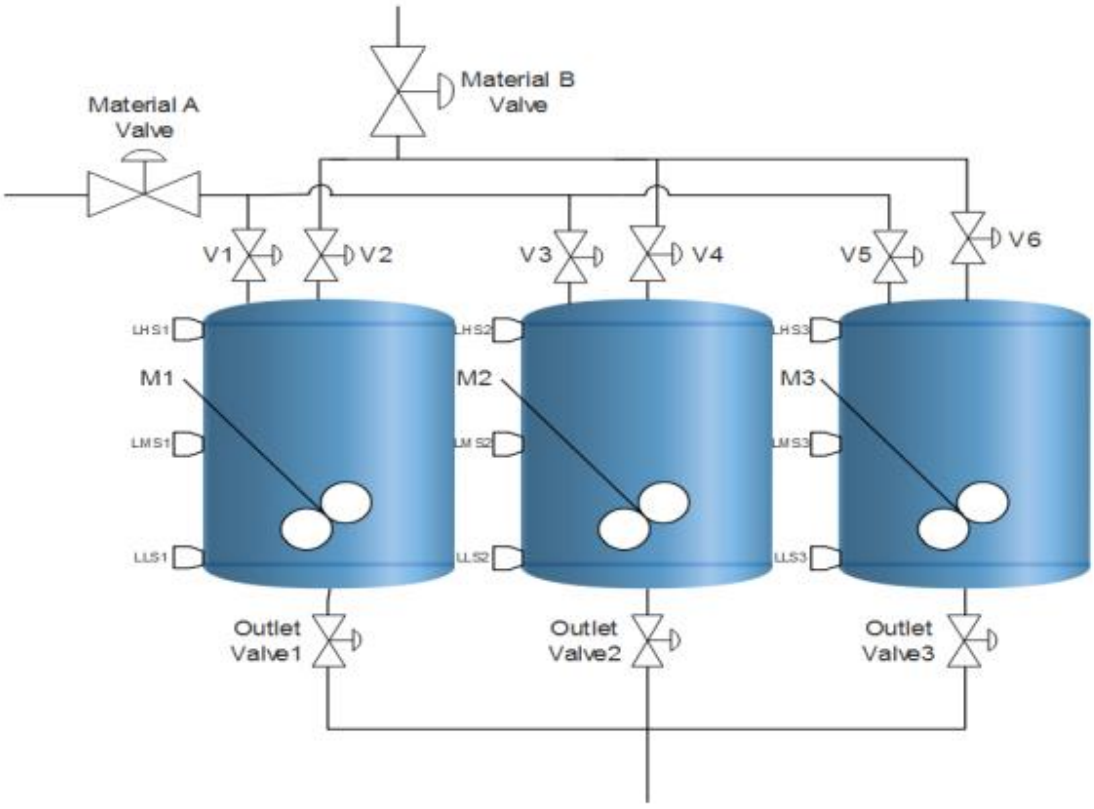


Problem Solution

- Use limit switches, control valves and mixing motor separately for all three mixing tanks.

- The only common input to all these tanks is Material feeding by two valves. Use separate timers for each tank for mixing.

- First two tanks are tagged as primary mixing tanks and the third one is said to be secondary mixing tank.

- Use switches of stop buttons of both Primary mixing processes as an input to start the tank third to enter the process sequence.

Extra Inputs and Outputs for Standby Process

I:3/0 = Individual Stop for tank 1 (Start Tank 3) (Input)

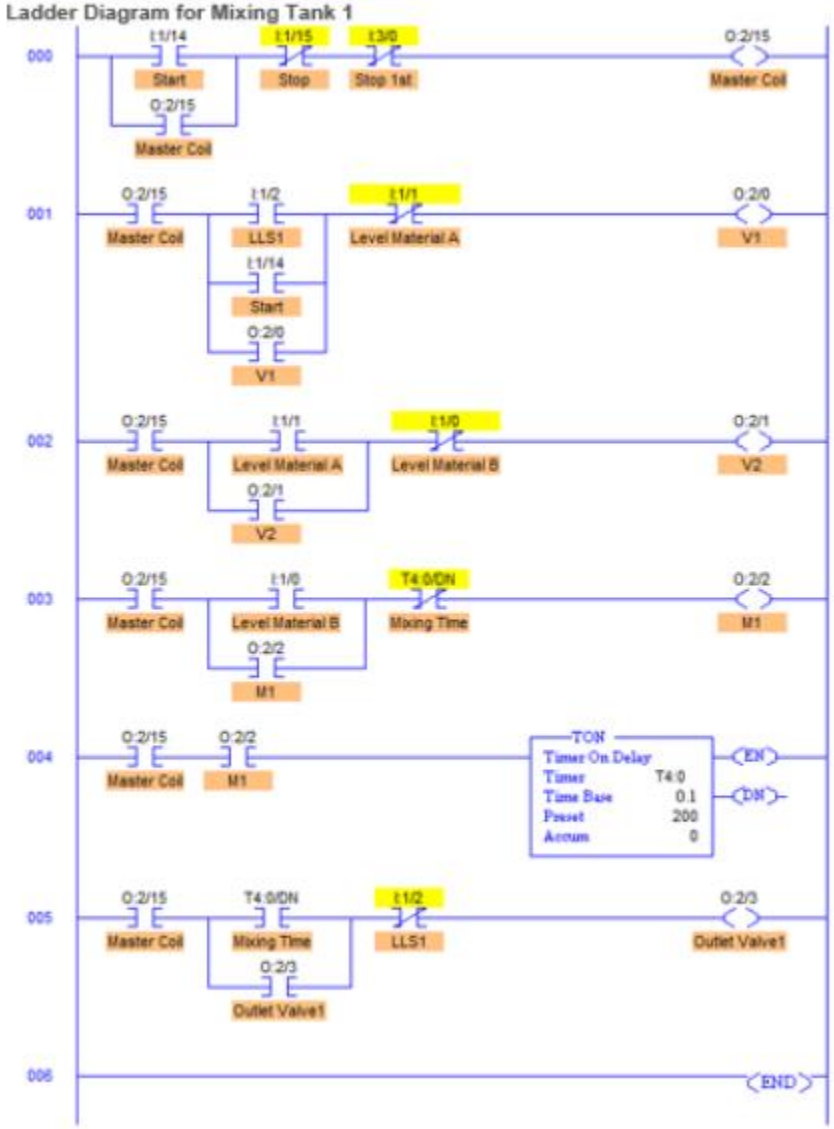
I:3/1 = Individual Stop for tank 2 (Start Tank 3) (input)

- In Mixing Tank 3, when any of the Primary tanks Tank1 or Tank 2 fails, operation of mixing tank 3 takes place

- Input of this Mixing tank 3 is Individual stop buttons of Tank 1 and Tank 2 which must be pressed in case of failure in either of the primary mixing tanks.

```
List of Inputs and Outputs
 I:1/14                    = Start                                    (Input)
 I:1/15                    = Stop                                     (Input)
 O:2/15                    = Master Coil                              (Output)
 I:1/0, I:1/5, I:1/7       = Level of Material B Switches             (Input)
 I:1/1, I:1/4, I:1/8       = Level of Material A switches             (Input)
 I:1/2, I:1/3, I:1/6       = Low Level Switch (empty tank)            (Input)
 O:2/0, O:2/4, O:2/8       = Inlet Valves to feed Material A          (Output)
 O:2/1, O:2/5, O:2/10      = Inlet Valves to feed Material B          (Output)
 O:2/2, O:2/6, O:2/9       = Agitator Motors M1, M2 and M3            (Output)
 O:2/3, O:2/7, O:2/11      = Outlet Valves for product outlet         (Output)
 T4:0, T4:1, T4:2          = Timers to mix materials                  (Timer)
```

Ladder Diagram for Mixing Tank 1



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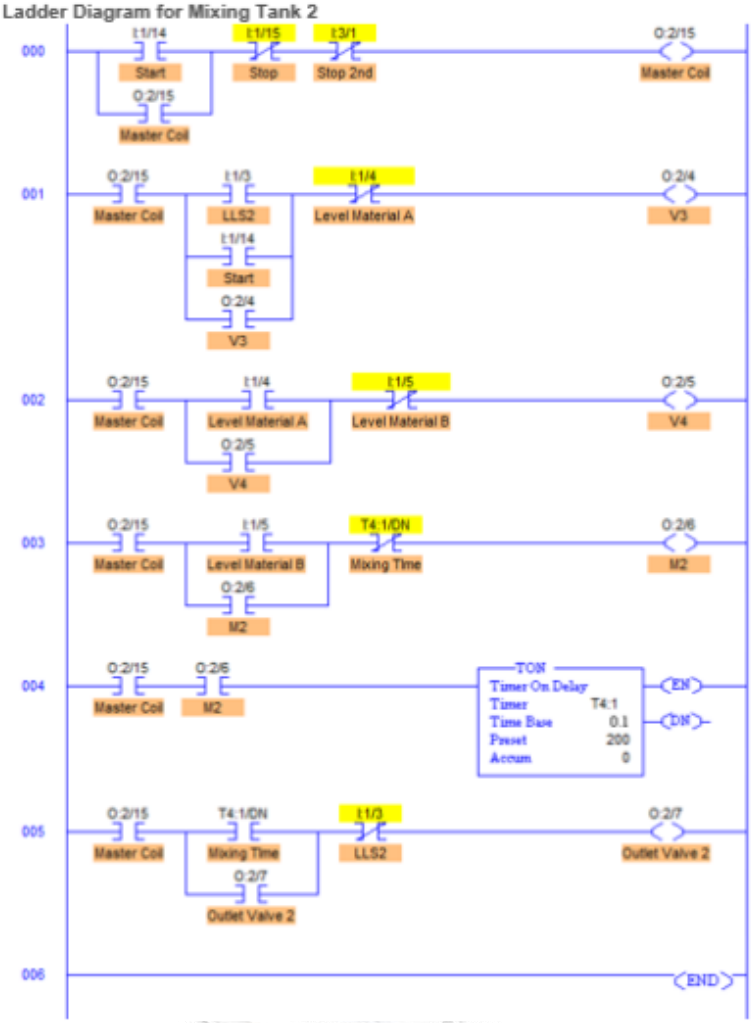
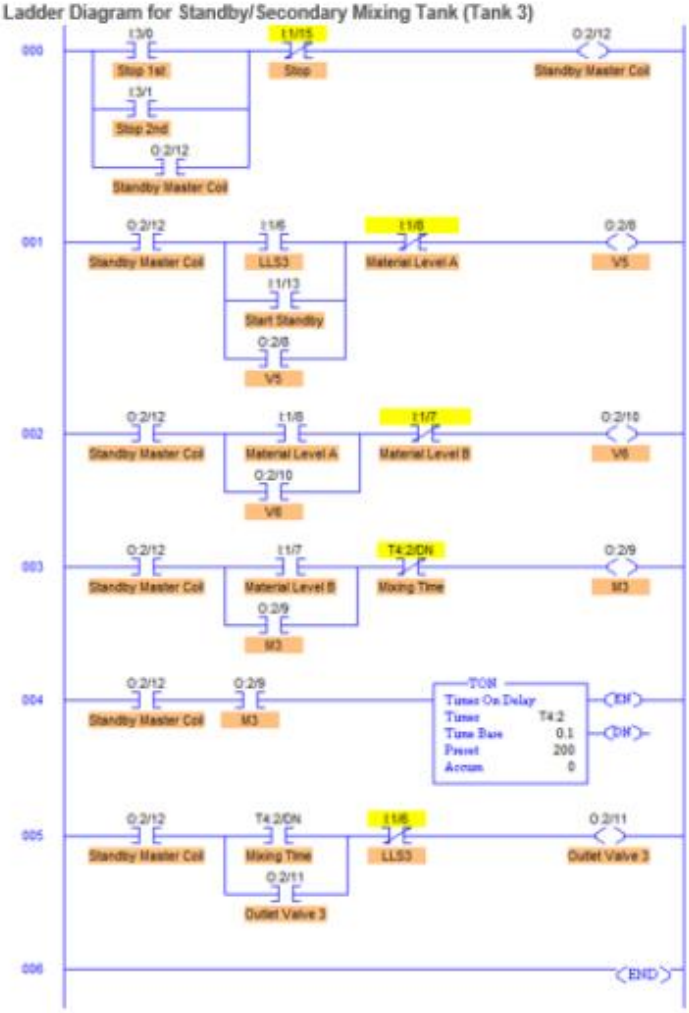
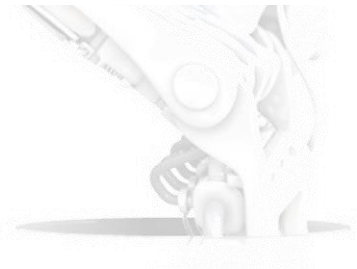
Ladder Diagram for Mixing Tank 2



Ladder Diagram for Standby/Secondary Mixing Tank (Tank 3)





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Chapter 7

Counters

Introduction

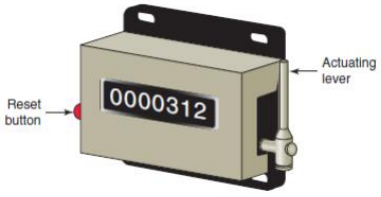
• PLC counters have programming formats which are similar to timer formats.

• Transitions on counter input rung causes the counter to count up (or down).

• Counter reset is accomplished via the (Reset) instruction

The difference between counters and Timers

• Counters are similar to timers except that they do not operate on an internal clock but are dependent on external or program sources for counting.



Mechanical counter

Electronic counters

MECHATRONICS SPECTRUM TEAM
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Methods used to represent a counter within a PLC's ladder logic

Main specification

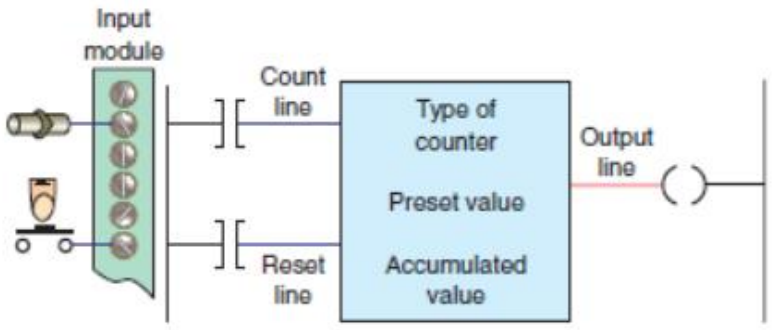
1.Counter type

2.Counter address

3.Counter preset

4.value

5.Accumulated count



MECHATRONICS SPECTRUM TEAM

Counter Applications

- Straight counting in a process

– The counter output goes on after the set count is received by repetitive pulses to the counter input.

- A process where a timed interval is started when a count reaches a preset value.

- A process where a count of events is to start after a fixed time interval

Up Counter (CTUP)

• The CTUP is an instruction that counts false-to-true rung transitions.

- Rung transitions can be caused by events occurring in the program (from internal logic or by external devices) such as parts traveling past a detector or actuating a limit switch.

• When rung conditions for a CTU instruction have made a false-to-true transition, the accumulated value is incremented by one count, provided that the rung containing the CTU instruction is evaluated between these transitions.

- The ability of the counter to detect false–to–true transitions depends on the speed (frequency) of the incoming signal

- The accumulated value is retained when the rung conditions again become false.

• The accumulated count is retained until cleared by a reset (RESET) instruction.

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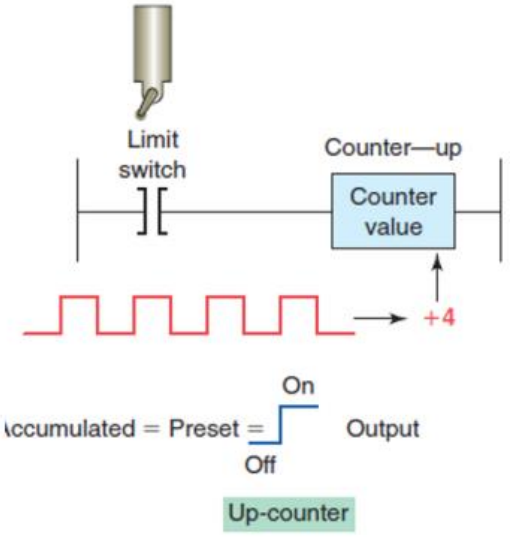
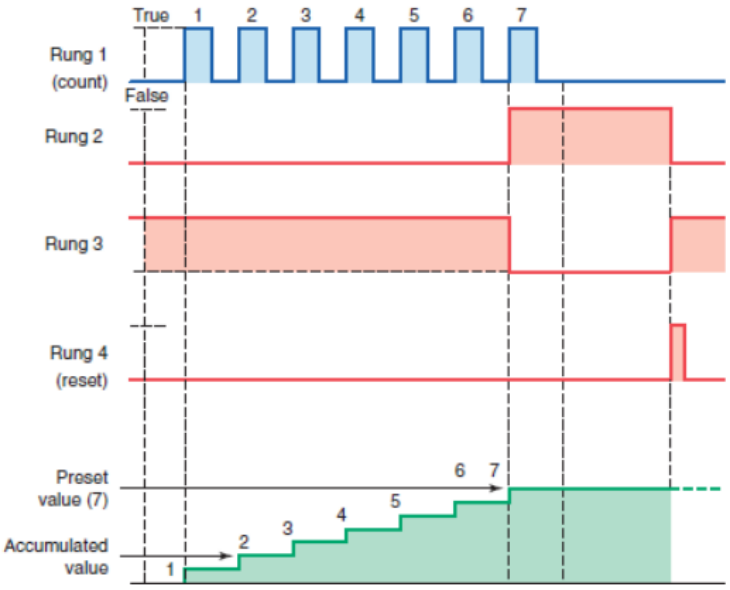
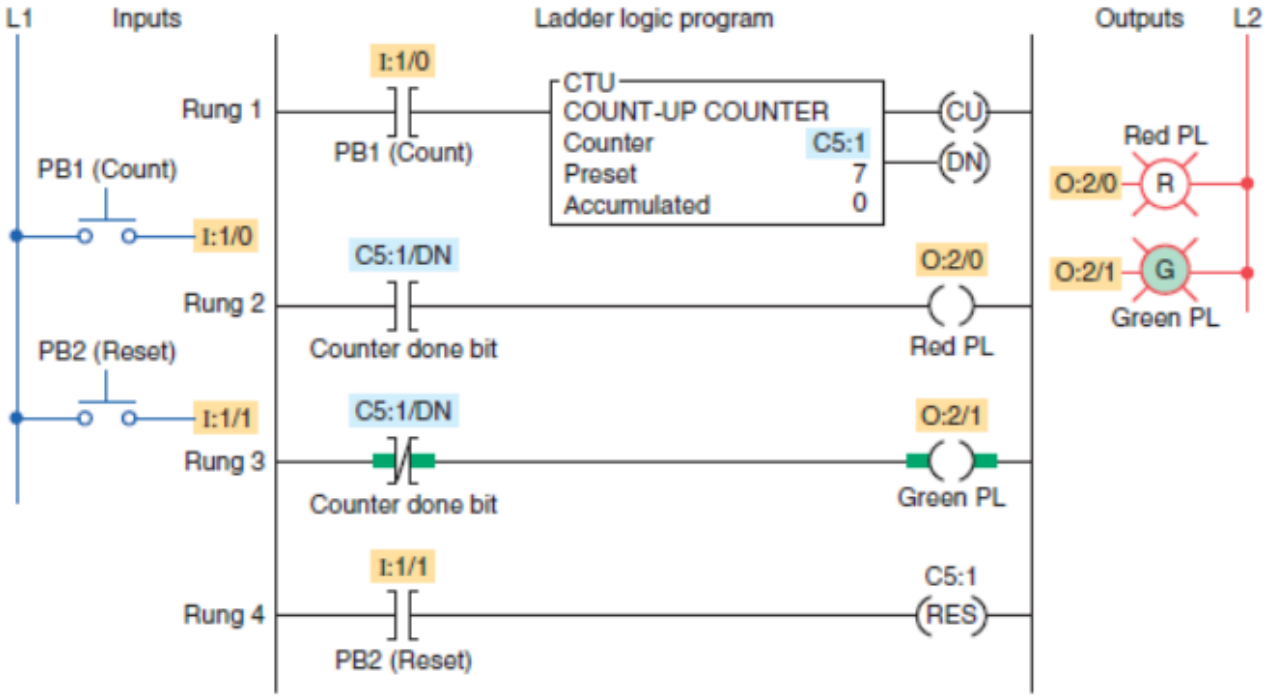

Limit switch
Counter—up
Counter value
+4
On
Accumulated = Preset =
Output
Off
Up-counter


True
1
2
3
4
5
6
7
Rung 1 (count)
False
Rung 2
Rung 3
Rung 4 (reset)
Preset value (7)
Accumulated value


L1
Inputs
Ladder logic program
Outputs
L2
Rung 1
I:1/0
PB1 (Count)
CTU
COUNT-UP COUNTER
Counter C5:1
Preset 7
Accumulated 0
(CU)
(DN)
Red PL
O:2/0
R
O:2/1
G
Green PL
Rung 2
C5:1/DN
Counter done bit
O:2/0
Red PL
PB2 (Reset)
I:1/1
Rung 3
C5:1/DN
Counter done bit
O:2/1
Green PL
Rung 4
I:1/1
PB2 (Reset)
C5:1
(RES)

• Operating pushbutton PB1 provides the off-to-on transition pulses that are counted by the counter.

• The preset value of the counter is set for 7.

• Each false-to-true transition of rung 1 increases the counter's accumulated value by 1.

• After 7 pulses, or counts, when the preset counter value equals the accumulated counter value, output DN is energized.

• As a result, rung 2 becomes true and energizes output O:2/0 to switch the red pilot light on.

• At the same time, rung 3 becomes false and deenergizes output O:2/1 to switch the green pilot light off.

• The counter is reset by closing pushbutton PB2, which makes rung 4 true and resets the accumulated count to zero.

• Counting can resume when rung 4 goes false again

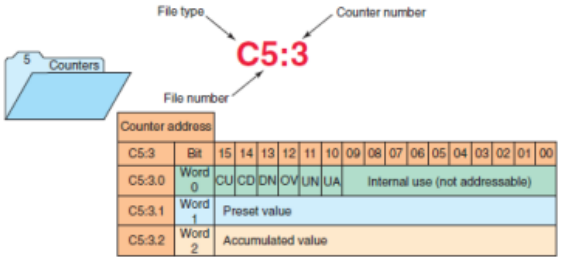
MECHATRONICS SPECTRUM TEAM

Counter files

Counter Table

	/CU	/CD	/DN	/OV	/UN	/UA	.PRE	.ACC
C5:0	0	0	0	0	0	0	0	0
C5:1	0	0	0	0	0	0	0	0
C5:2	0	0	0	0	0	0	0	0
C5:3	0	0	0	0	0	0	50	0
C5:4	0	0	0	0	0	0	0	0
C5:5	0	0	0	0	0	0	0	0

Address C5:3 Table: C5: Counter



- C5 5 Counter file 5
- :3 5 counter element 3 (0–255 counter elements per file)
- C5:3/DN is the address for the done bit of the counter.
- C5:3/CU is the address for the count-up enable bit of the counter.
- C5:3/CD is the address for the count-down enable bit of the counter.
- C5:3/OV is the address for the overflow bit of the counter.
- C5:3/UN is the address for the underflow bit of the counter.

T E A M

control words

Count-Up (CU) Enable Bit —The count-up enable bit is used with the count-up counter and is true whenever the count-up counter instruction is true. If the count-up counter instruction is false, the CU bit is false.

Count-Down (CD) Enable Bit —The count-down enable bit is used with the count-down counter and is true whenever the count-down counter instruction is true. If the count-down counter instruction is false, the CD bit is false

Done (DN) Bit —The done bit is true whenever the accumulated value is equal to or greater than the preset value of the counter, for either the count-up or the count-down counter.

Overflow (OV) Bit —The overflow bit is true whenever the counter counts past its maximum value, which is 32,767. On the next count, the counter will wrap around to 32,768 and will continue counting from there toward 0 on successive false-to-true transitions of the count-up counter.

Underflow (UN) Bit —The underfill ow bit will go true when the counter counts below 32,768. The counter will wrap around to 32,767 and continue counting down toward 0 on successive false-to-true rung transitions of the count-down counter.

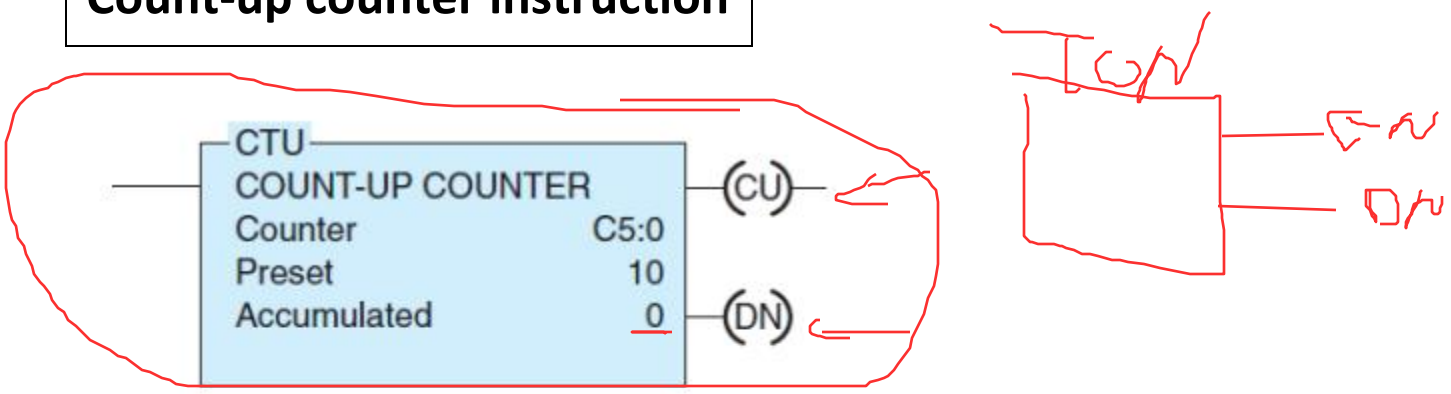
Update Accumulator (UA) Bit —The update accumulator bit is used only in conjunction with an external HSC (high-speed counter). Counter Number —This number must come from the counter file. In the example shown, the counter number is C5:0, which represents counter fi le 5, counter 0 in that fi le. The address for this counter should not be used for any other count-up counter.

Counter Number —This number must come from the counter file. In the example shown, the counter numbers C5:0, which represents counter fi le 5, counter 0 in that fi le. The address for this counter should not be used for any other count-up counter.

Preset Value —The preset value in the example shown is 10. Accumulated Value —The accumulated value retained until cleared by a reset (RESET) instruction.

Accumulated Value —The accumulated value retained until cleared by a reset (RESET) instruction

Count-up counter instruction



C5:0/CU — Counter enable bit

C5:0/DN — Counter done bit

C5:0/OV — Overflow status bit

C5:0 (RES) — The reset instruction resets the counter's accumulated value back to zero.

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Examples

- Accumulated count is reset only by the (RESET) instruction.

- The counter will increment the accumulator value even after the preset is reached

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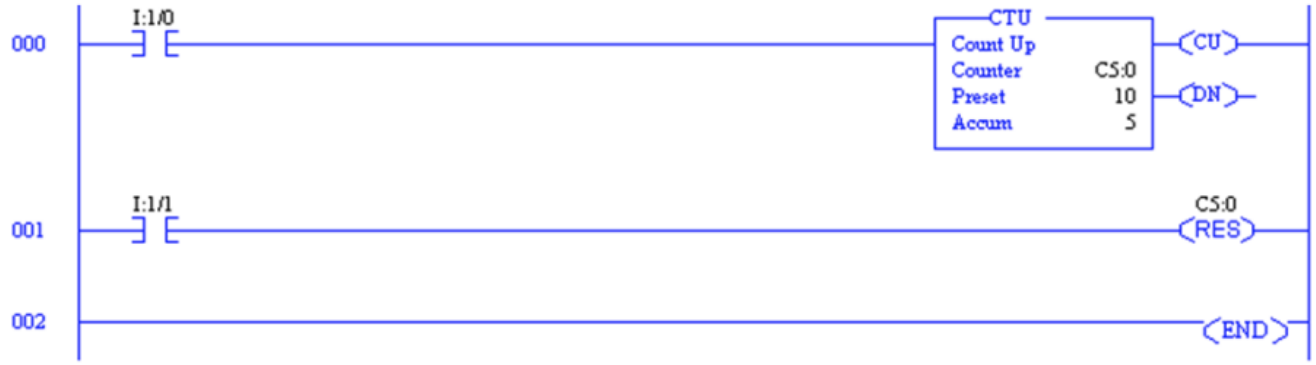
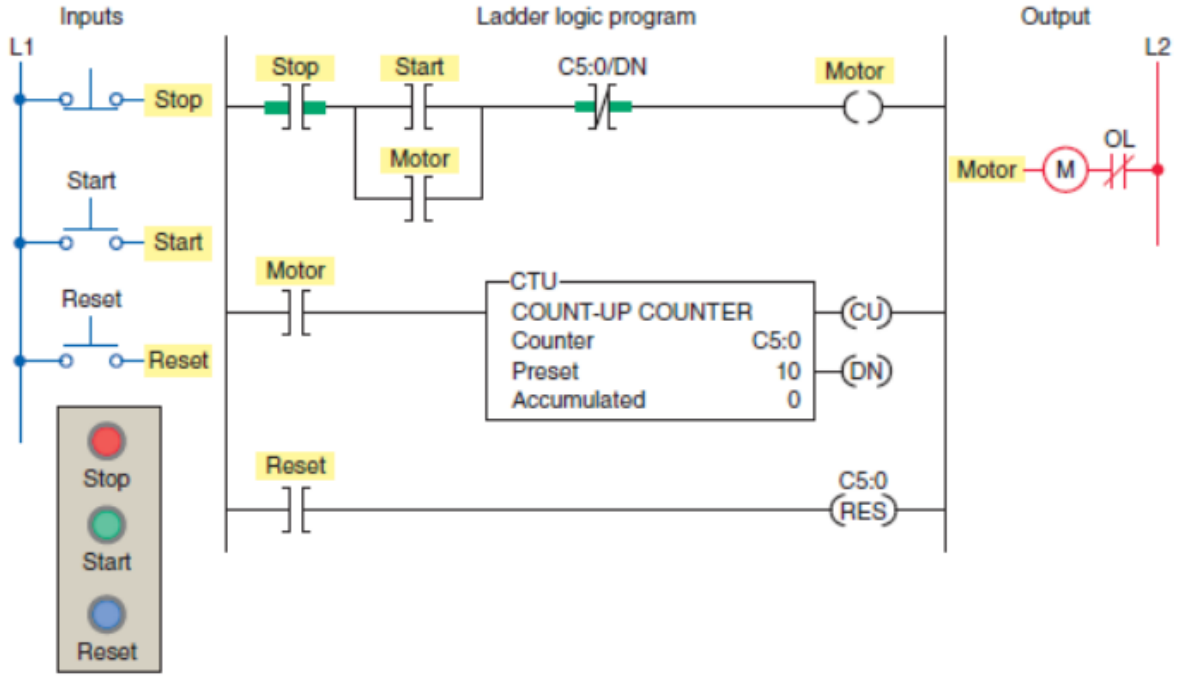


PLC counter program used to stop a motor from running after 10 operations.



- Up-counter C5:0 counts the number of off/on operations of the motor.
- The preset value of the counter is set to 10.

• A counter done bit examine-off instruction is programmed in series with the motor output instruction.

• A motor output examine-on instruction is used to increment the accumulated value of the counter for each off/on operation

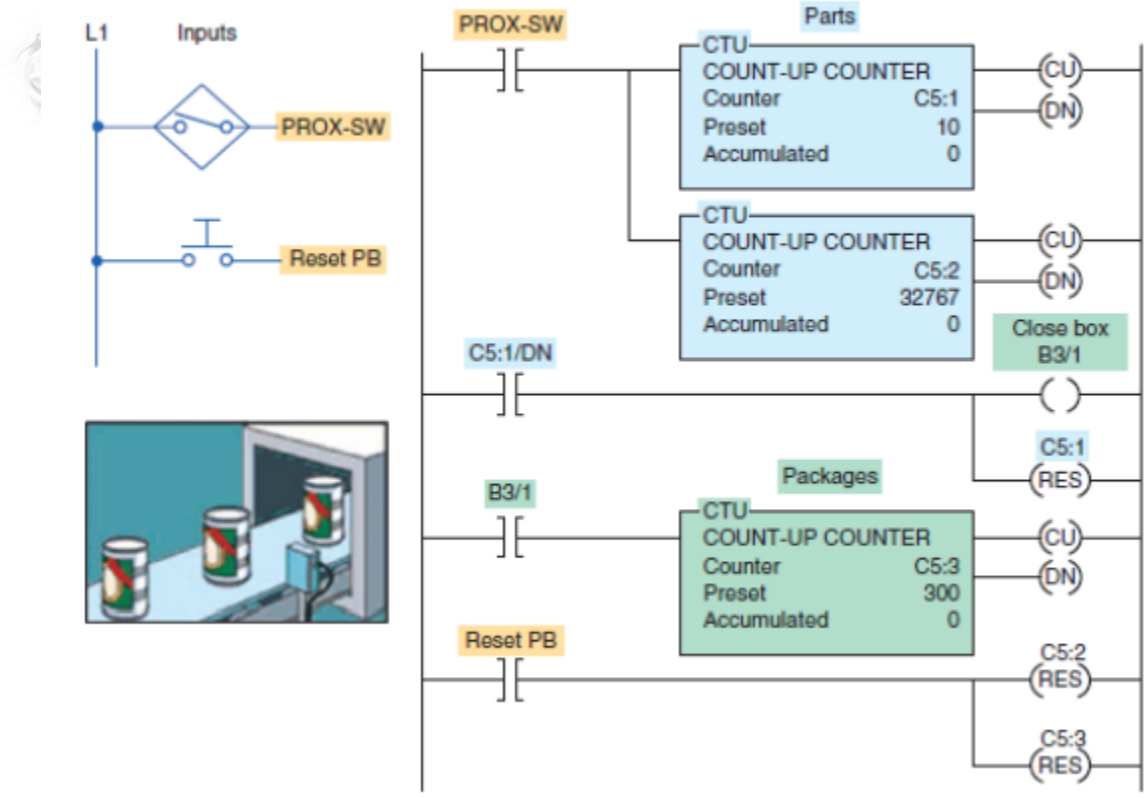
• After the count of 10 is reached the counter done bit examine-off instruction goes false preventing the motor from being started.

• Closure of the reset pushbutton resets the accumulated count to zero.

MECHATRONICS SPECTRUM TEAM

Can-counting program

Ladder logic program



The operation of the program can be summarized as follows:

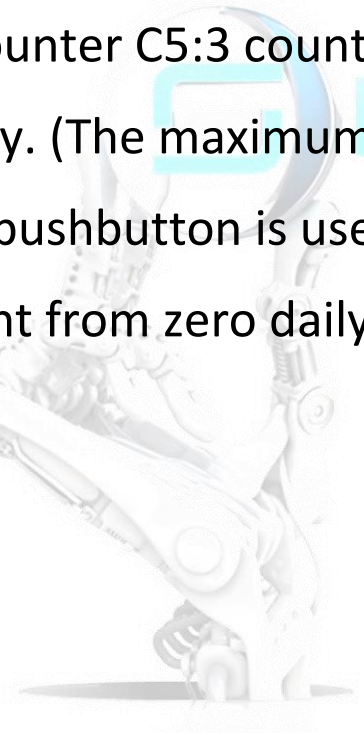
• Counter C5:2 counts the total number of cans coming off an assembly line for final packaging.

• Each package must contain 10 parts.

• When 10 cans are detected, counter C5:1 sets bit B3/1 to initiate the box closing sequence.

• Counter C5:3 counts the total number of packages filled in a day. (The maximum number of packages per day is 300.)

• A pushbutton is used to restart the total part and package count from zero daily.



Down Counter (CTD)

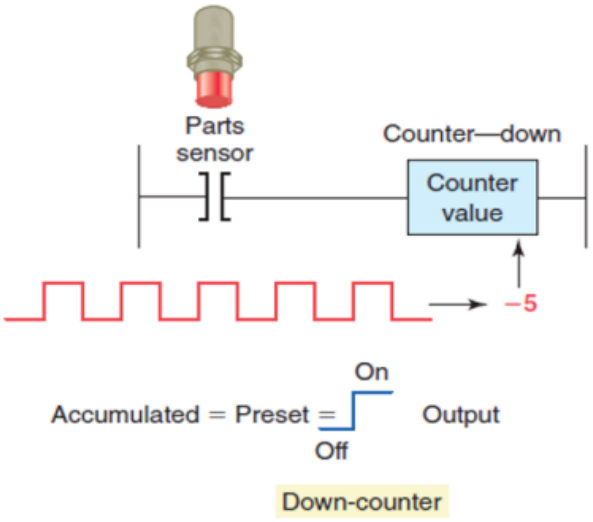
• The CTD is an instruction that counts false-to-true rung transitions.

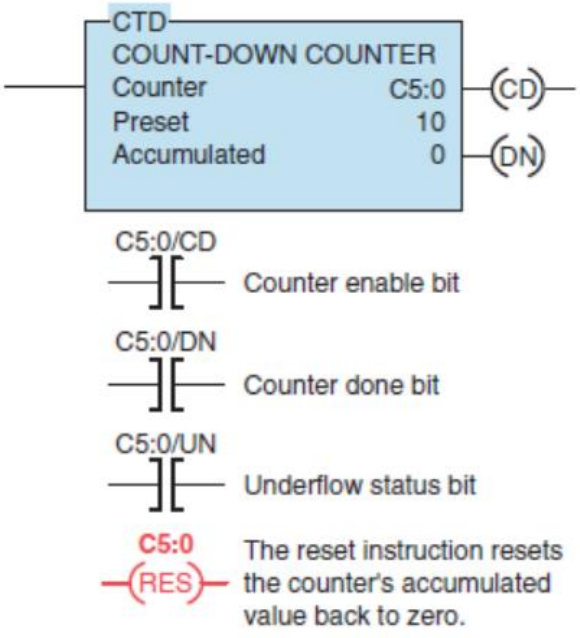
– Rung transitions can be caused by events occurring in the program such as parts traveling past a detector or actuating a limit switch.

• When rung conditions for a CTD instruction have made a false-to-true transition, the accumulated value is decremented by one count, provided that the rung containing the CTD instruction is evaluated between these transitions.

-The accumulated counts are retained when the rung conditions again become false.

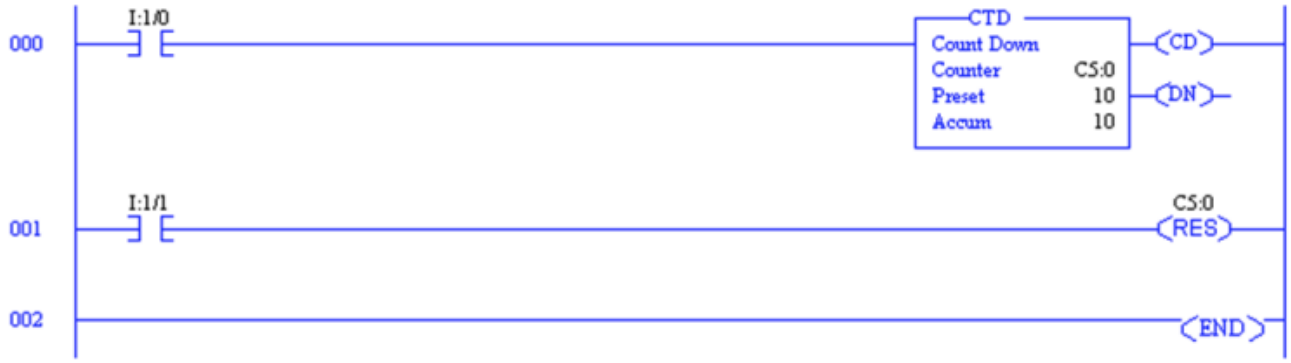
• The accumulated count is retained until cleared by a reset (RESET) instruction.



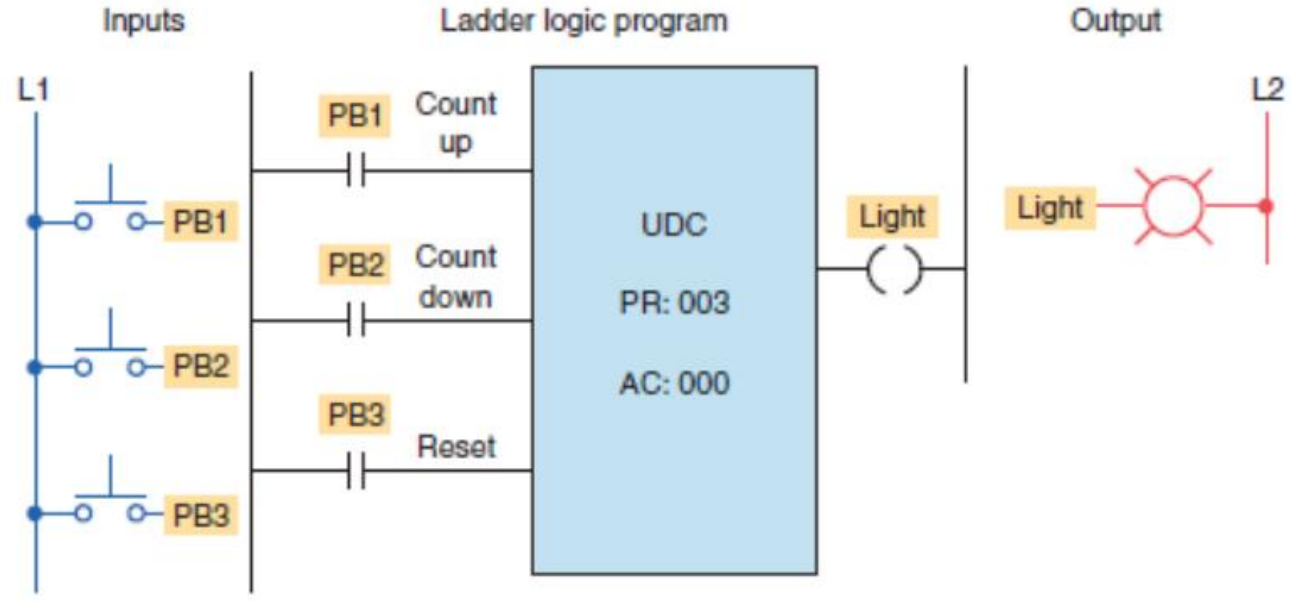


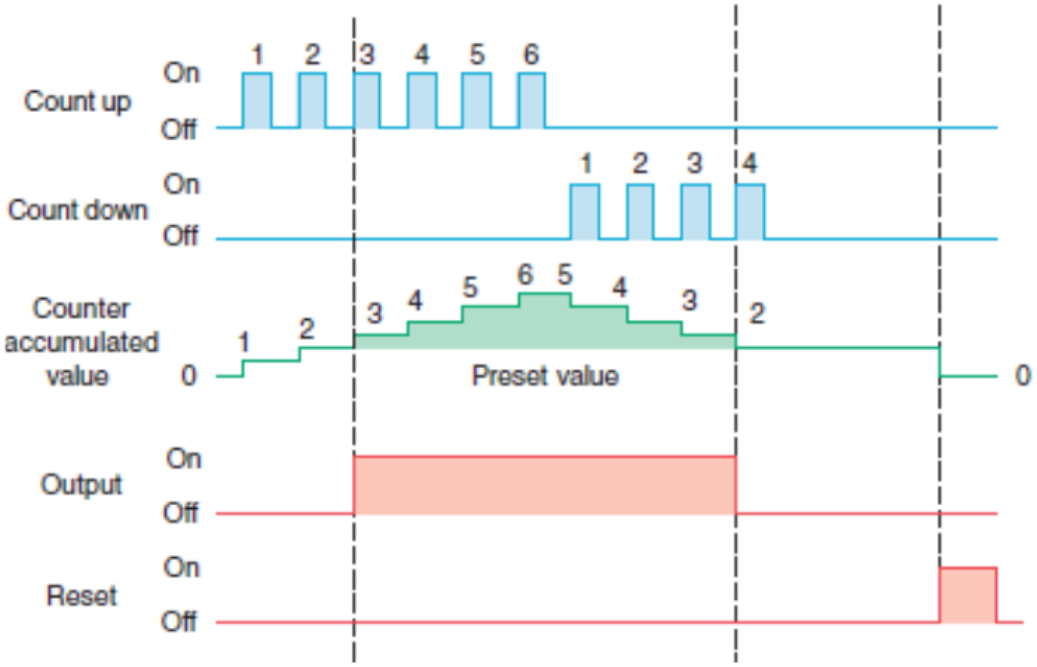
Down Counter Example

- Accumulated count is reset only by the (RES) instruction
- The counter will decrement the accumulator value even after a 0 count is reached.



Counter-UP-Down





MECHATRONICS SPECTRUM TEAM

• Separate count-up and count-down inputs are provided.

• Assuming the preset value of the counter is 3 and the accumulated count is 0, pulsing the count-up input (PB1) three times will switch the output light from off to on.

• This particular PLC counter keeps track of the number of counts received above the preset value. As a result, three additional pulses of the count-up input (PB1) produce an accumulated value of 6 but no change in the output.

• If the count-down input (PB2) is now pulsed four times, the accumulated count is reduced to 2 (6 2 4). As a result, the accumulated count drops below the preset count and the output light switches from on to off.

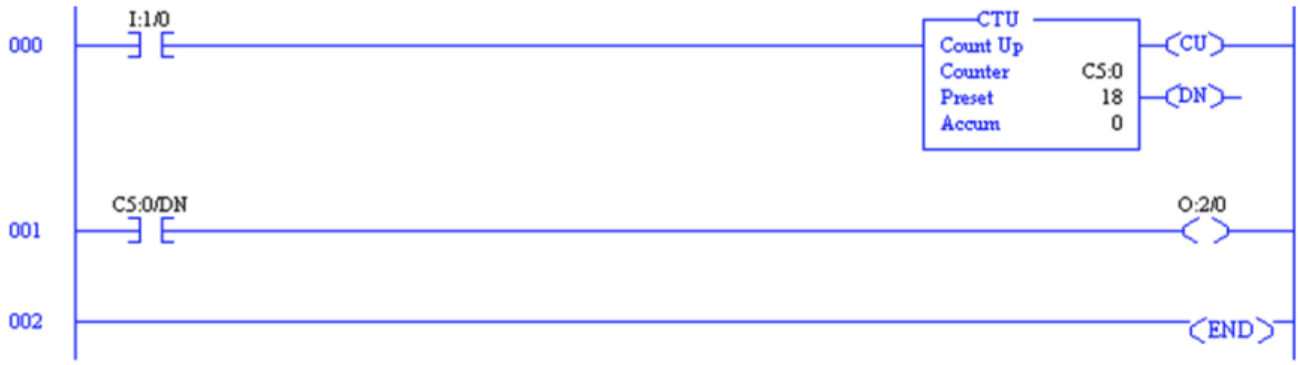
• Pulsing the reset input (PB3) at any time will reset the accumulated count to 0 and turn the output light off

بسواعدنا نبنيها

Counters Examples

Straight Counting in a Process

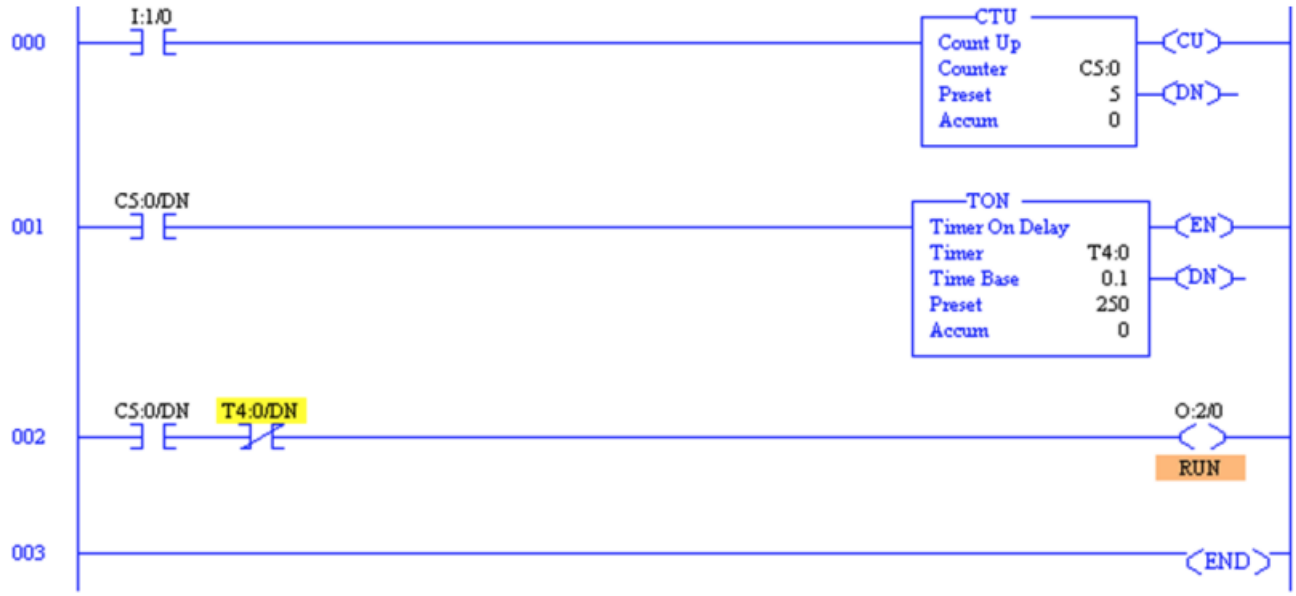
- After a certain number of counts occur, the output goes on.
- The output can be used to energize an indicator.
- The output status could also be utilized in the ladder diagram logic in the form of a contact.
- Either counter will function (count) if its input rung transitions from false-to-true.
- After the count input receives 18 pulses, the O:2/0 output will energize



MECHATRONICS SPECTRUM TEAM

Time a Process after a Count has been Reached

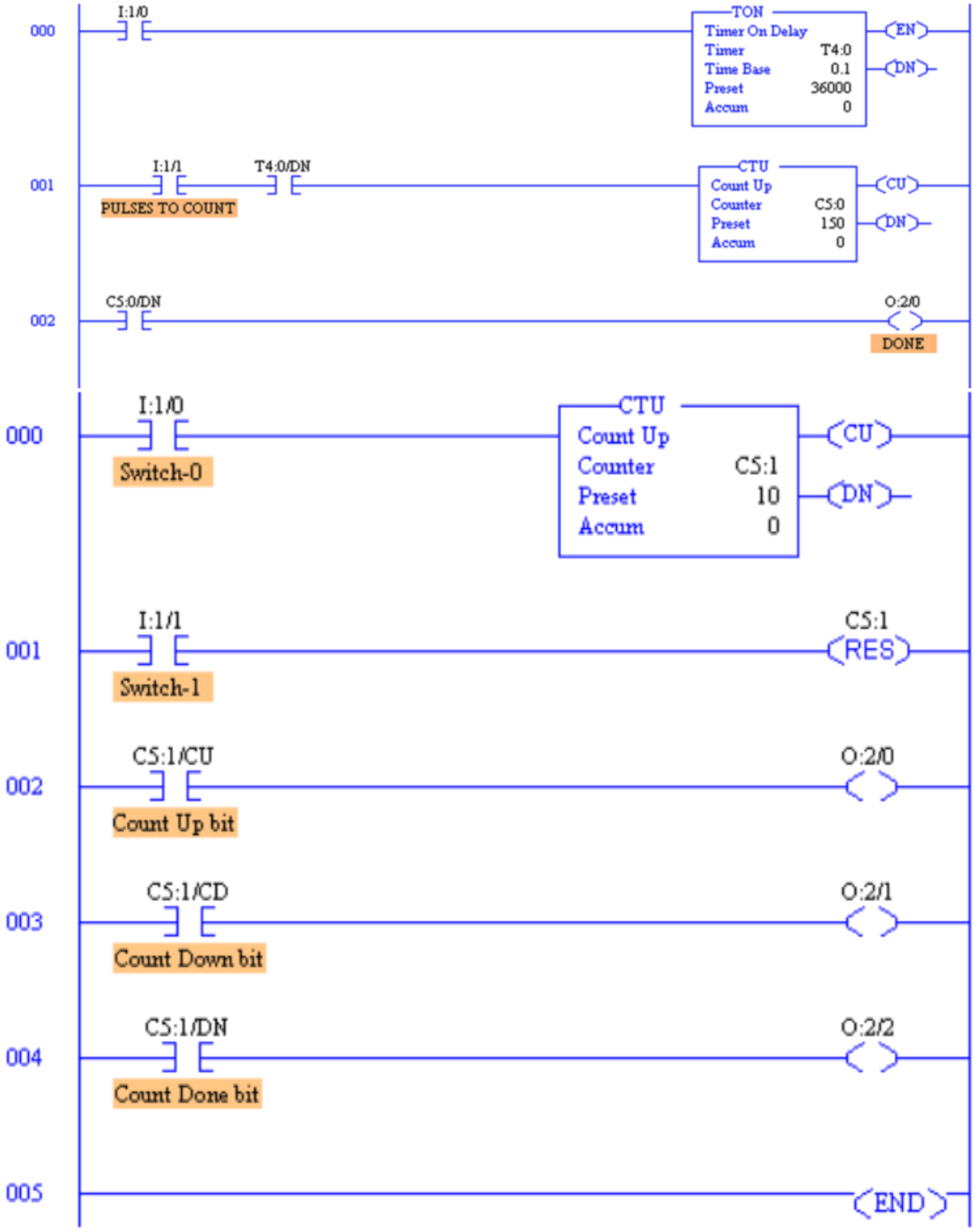
After a count of 5 from a sensor, a paint spray is to run for 25 seconds.



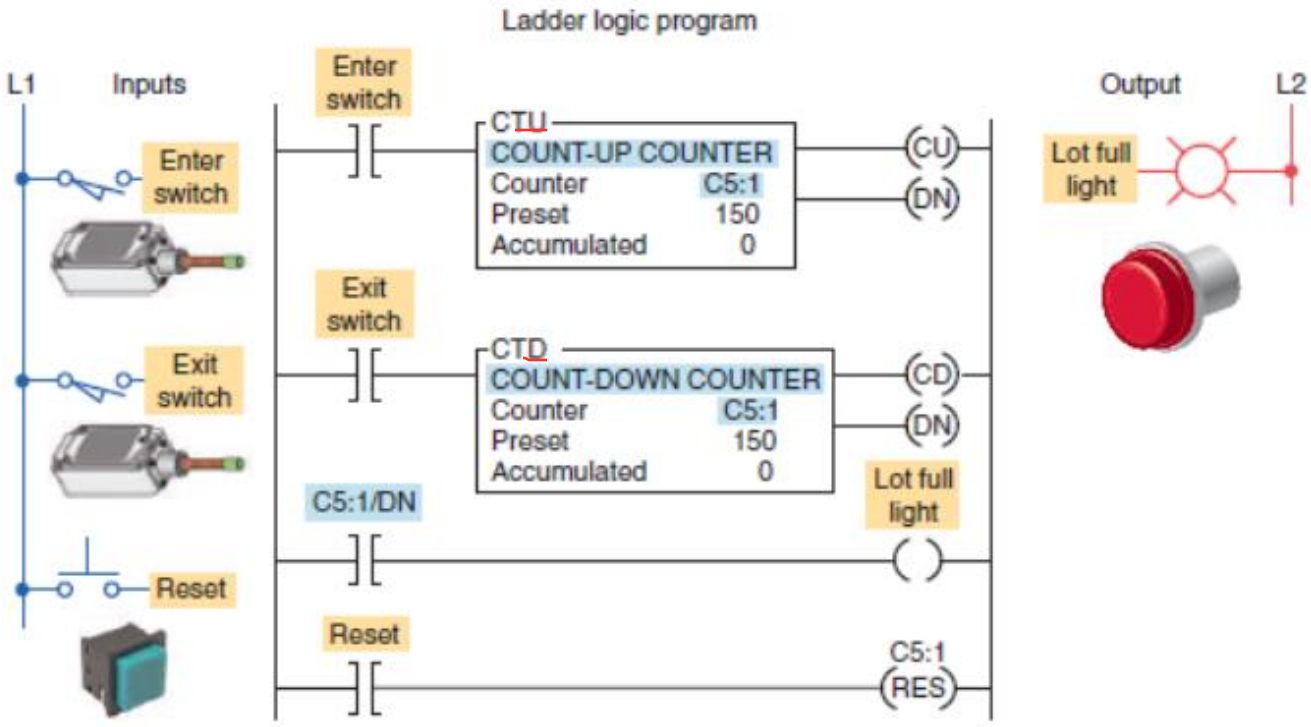
Delay of the Start of the Counting Process

In this process we do not wish to start counting until one hour after the process starts.

- A timer output contact in the timer run line closes after the time period.
- The closure then enables the counter to start counting input pulses.
- After a count of 150, the output comes on.


000
I:1/0
TON
Timer On Delay
Timer T4:0
Time Base 0.1
Preset 36000
Accum 0
EN
DN
001
I:1/1
PULSES TO COUNT
T4:0/DN
CTU
Count Up
Counter C5:0
Preset 150
Accum 0
CU
DN
002
C5:0/DN
O:2/0
DONE
000
I:1/0
Switch-0
CTU
Count Up
Counter C5:1
Preset 10
Accum 0
CU
DN
001
I:1/1
Switch-1
C5:1
RES
002
C5:1/CU
Count Up bit
O:2/0
003
C5:1/CD
Count Down bit
O:2/1
004
C5:1/DN
Count Done bit
O:2/2
005
END

Parking garage counter



• As a car enters, the enter switch triggers the upcounter output instruction and increments the accumulated count by 1.

• As a car leaves, the exit switch triggers the downcounter output instruction and decrements the accumulated count by 1.

• Because both the up- and down-counters have the same address, C5:1, the accumulated value will be the same in both instructions as well as the preset.

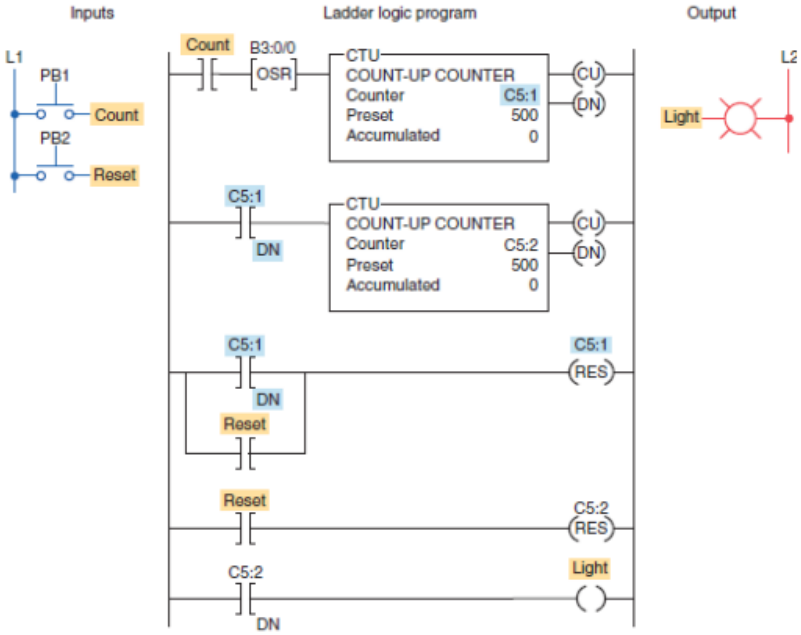
• Whenever the accumulated value of 150 equals the preset value of 150, the counter output is energized by the done bit to light up the Lot Full sign.

• A reset button has been provided to reset the accumulated count

Cascading Counters

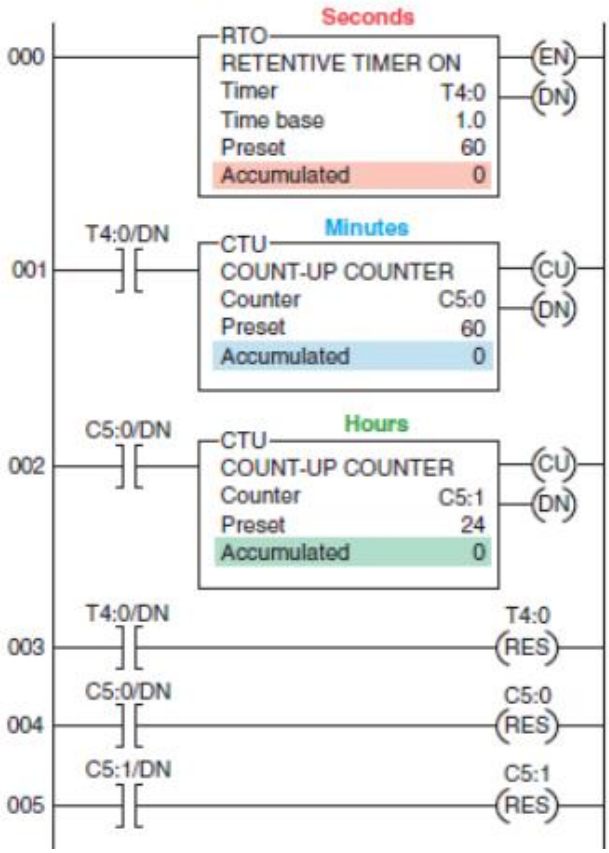
The operation of the program can be summarized as follows:

- The output of the first counter is programmed into the input of the second counter.
- The status bits of both counters are programmed in series to produce an output.
- These two counters allow twice as many counts to be measured.



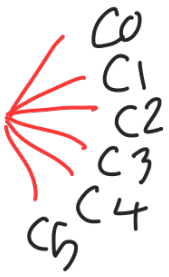
24-hour clock program

كحل لسابقة
بستخدام PLC

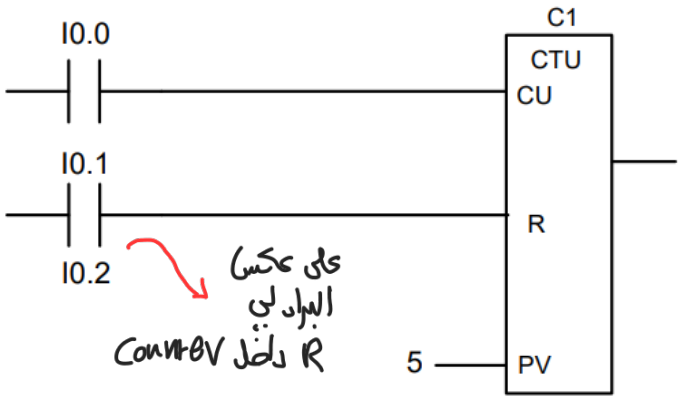


Siemens Up Counter

PLC semins Counter

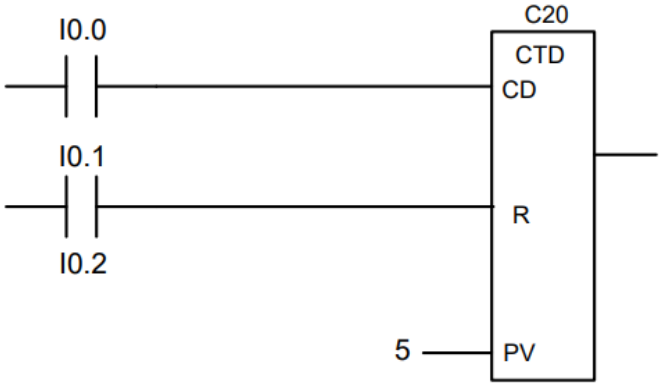


1 Counter up



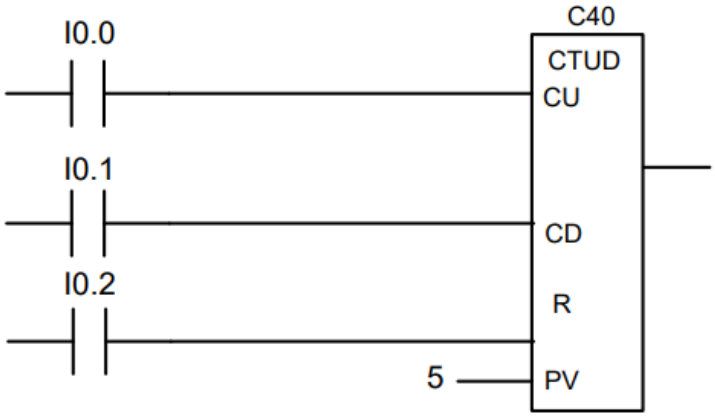

```
LD   I0.0
LD   I0.1
CTU  C1, 5
```

2 Counter Down




```
LD   I0.0
LD   I0.1
CTD  C20, 5
```

3 Counter Up Down




```
LD   I0.0
LD   I0.1
LD   I0.2
CTUD C40, 5
```

Chapter 8

Timer and Counter Examples

problem

Write a logic ladder diagram to implement the following process using a Siemens s7-200 PLC.

1. Start push-Button is used to start the process and a stop push-Button is used to stop the process.

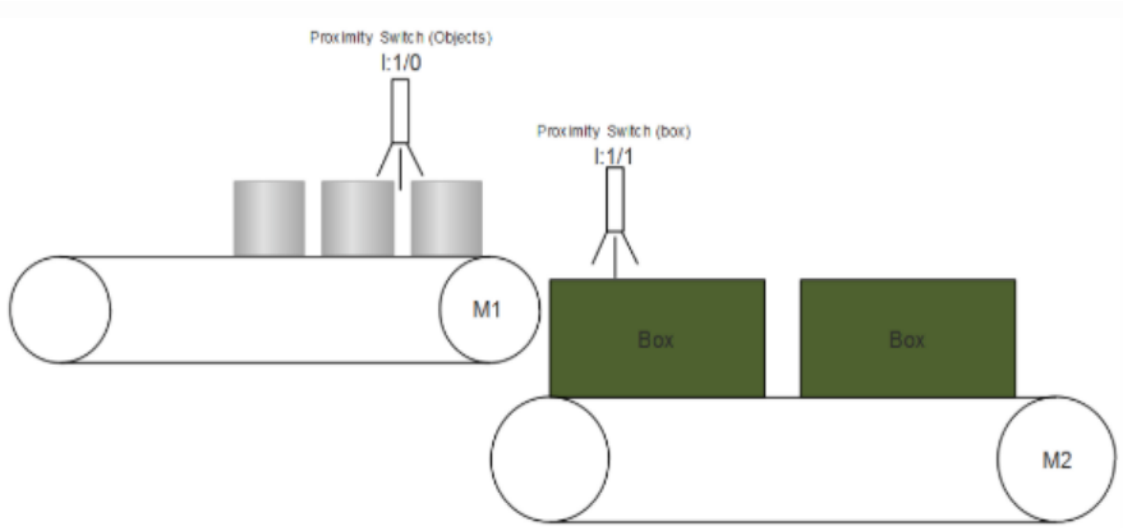
2. A proximity switch is used to detect moving objects on the conveyor belt 1.

3. Another proximity switch is used to detect an empty box on conveyor belt 2.

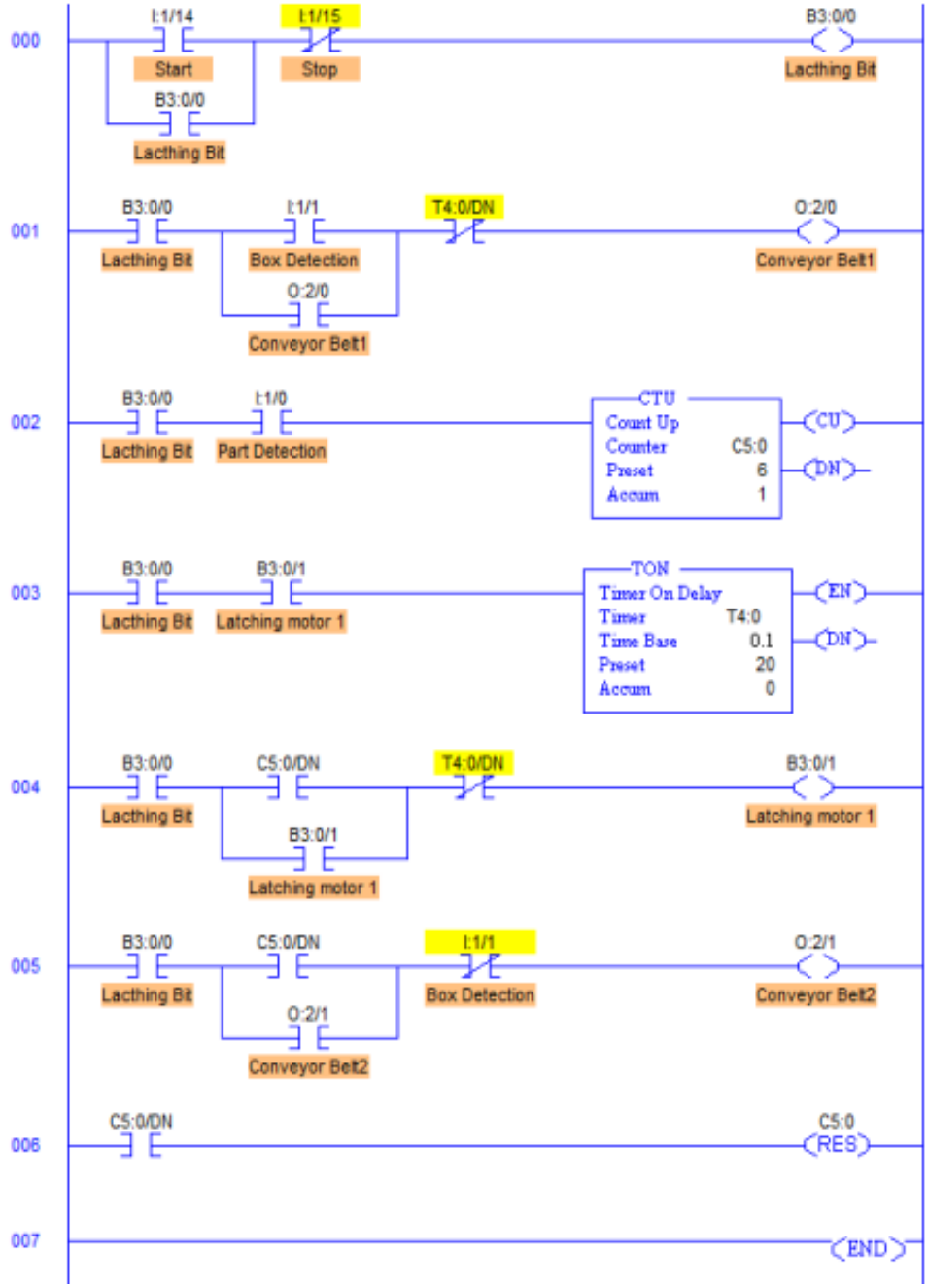
4. 24 objects should be placed in every box.

5.When number of objects to be packed are detected timer is activated the time between the moving of empty boxes is set to be 60 sec. When time is over, it moves the conveyor bell 2 until next empty box is detected

MECHATRONICS SPECTRUM TEAM




```
List of Inputs and Outputs
 I:1/14      = Start                          (Input)
 I:1/15      = Stop                           (Input)
 B3:0/0      = Latching Bit                   (Bit)
 B3:0/1      = Latching Motor 1 bit           (Bit)
 I:1/0       = Part detection                 (Input)
 I:1:1       = Box detection                  (Input)
 O:2/0       = Conveyor Belt 1                (Output)
 O:2/1       = Conveyo2 Belt 2                (Output)
 C5:0 = Part counter                          (Counter)
 T4:0 = Timer to stop conveyor                (Timer)
```


000
I:1/14
Start
I:1/15
Stop
B3:0/0
Lacthing Bit
B3:0/0
Lacthing Bit
B3:0/0
Lacthing Bit
001
B3:0/0
Lacthing Bit
I:1/1
Box Detection
T4:0/DN
O:2/0
Conveyor Belt1
O:2/0
Conveyor Belt1
002
B3:0/0
Lacthing Bit
I:1/0
Part Detection
CTU
Count Up
Counter C5:0
Preset 6
Accum 1
CU
DN
003
B3:0/0
Lacthing Bit
B3:0/1
Latching motor 1
TON
Timer On Delay
Timer T4:0
Time Base 0.1
Preset 20
Accum 0
EN
DN
004
B3:0/0
Lacthing Bit
C5:0/DN
T4:0/DN
B3:0/1
Latching motor 1
B3:0/1
Latching motor 1
005
B3:0/0
Lacthing Bit
C5:0/DN
I:1/1
Box Detection
O:2/1
Conveyor Belt2
O:2/1
Conveyor Belt2
006
C5:0/DN
C5:0
RES
007
END

Program Description

- RUNG000 is for master start and stop the process.

- RUNG001 is for controlling Conveyor Belt 1 motor M1 which is operated by Box Detection proximity switch. When the process is started and empty box is detected by Proximity having address of I:1/1, power supply to the motor of Conveyor Belt 1 is given. This drives the Motor M1 moving objects on the conveyor belt 1.

- Every time a part is detected by the I:1/1 Part detector, Counter C5:0 is incremented by 1. Counter has a preset value of 6. Here that means when total 5 parts are detected, counter done bit C5:0/DN is set to 1.

- When C5:0/DN bit is set, timer T4:0 is activated for 2secs which is an assumption that the part will take approximately 2secs to fall into the box after it is detected by the Pat Detector proximity switch I:1/0.

- Counter is immediately reset when C5:0/DN goes true. So one shot is generated by C5:0/DN bit. It latches the Latching Motor1 bit before it goes false.

- Conveyor Belt 2 motor M2 (O:2/1) is operated when counter value reaches to 6 that is when total 5 parts are detected. And it is stopped when an empty box is detected again.

- And the process is repeated.

- Important thing to note here is that Counter Preset is set 6 that is because when the counter is reset, the value is the accumulator is set to 1 because of latching bit.

MECHATRONICS SPECTRUM TEAM

بسواعدنا نبنيها

PLC Program to Drive Motors Simultaneously with Interlocking

Problem Description

Two Motors are running in a sequence one by one for a particular time. If the start button is pressed Motors run in sequence such that 1st Motor stays ON for 5secs and then 2nd Motor is turned ON and stays ON for 5secs. And the cycle is repeated until it is interrupted. While motors are running in the sequence, if one motor is running and the button of other motor is pressed, then the running Motor should stop and the other motor should run. Implement this logic in PLC using Ladder Diagram programming language.

Problem Solution

• Define addresses of motors.

• Use pushbuttons to start and stop the sequential operation.

• Use TON timer to generate a particular time delay.

• Use the enable bit of first timer to energize 2nd motor coil and activate

second timer.

• No need to use Reset contact, program will reset the timers

MECHATRONICS SPECTRUM TEAM

- List of Inputs and Outputs
- I:1/0 = Motor 0 Start (Input)
- I:1/1 = Motor 1 Stop (Input)
- I:1/2 = Master Start (Input)
- I:1/3 = Master Stop (Input)
- O:2/2 = Master Coil (Output)
- O:2/0 = Motor 0 (Output)
- T4:0 = Motor 0 Timer (Timer)
- O:2/1 = Motor 1 (Output)
- T4:1 = Motor 1 Timer (Timer)

Program Description

• RUNG000 is Latching coil for Master Start and Stop.

• RUNG001-RUNG002 and RUNG003-RUNG004 are arrangements to operate Motor0 O:2/0 and Motor1 O:2/1 respectively.

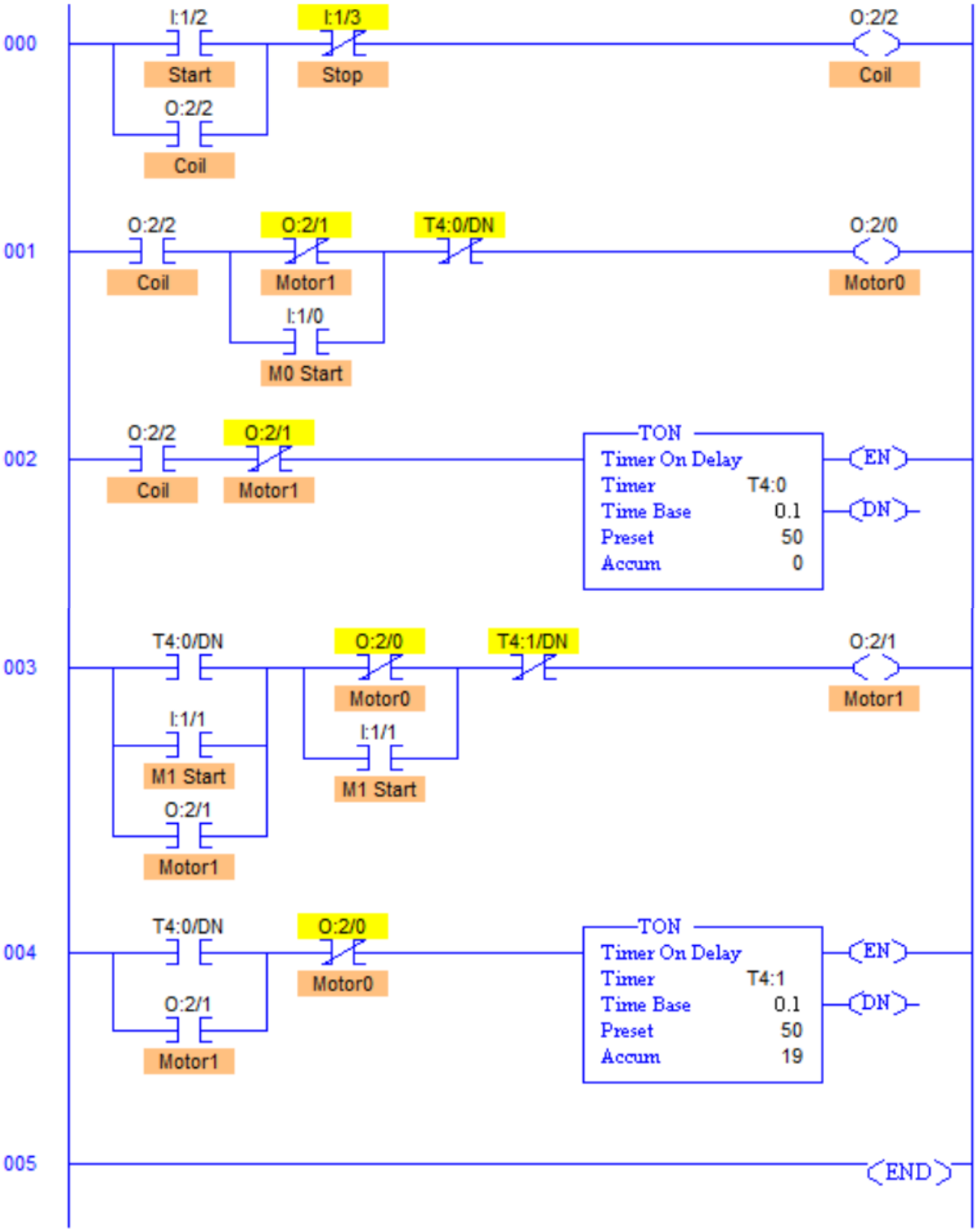
• When Start button is pressed momentarily, it latches the coil starting the energizing the output O:2/0, hence turning Motor0 ON.

• Timer T4:0 is used to run Motor0 for 5secs (Less because of simulation purpose).

• Similarly T4:1 is used to run Motor1 for 5secs.

• While Motor0 is running, and M1 Start I:1/1 is pressed, it stops the Motor0 (O:2/0 de-energizes) and Motor1 (O:2/1 energizes) starts and timer t4:1 is activated. And vice-versa.

• XIO contact of each motor's output is used to turn other motor off by de-energizing other motor's output which cuts off the current supply in the field.


000
I:1/2
Start
I:1/3
Stop
O:2/2
Coil
O:2/2
Coil
O:2/2
Coil
001
O:2/2
Coil
O:2/1
Motor1
T4:0/DN
O:2/0
Motor0
I:1/0
M0 Start
002
O:2/2
Coil
O:2/1
Motor1
TON
Timer On Delay
Timer T4:0
Time Base 0.1
Preset 50
Accum 0
EN
DN
003
T4:0/DN
I:1/1
M1 Start
O:2/1
Motor1
O:2/0
Motor0
I:1/1
M1 Start
T4:1/DN
O:2/1
Motor1
004
T4:0/DN
O:2/1
Motor1
O:2/0
Motor0
TON
Timer On Delay
Timer T4:1
Time Base 0.1
Preset 50
Accum 19
EN
DN
005
END

Control of a set of traffic lights in one direction

Operation of the program:

• Transition from red light to green light to amber light is accomplished by the interconnection of the three TON timer instructions.

• The input to timer T4:0 is controlled by the T4:2 done bit.

• The input to timer T4:1 is controlled by the T4:0 done bit.

• The input rung to timer T4:2 is controlled by the T4:1 done bit.

• The timed sequence of the lights is:

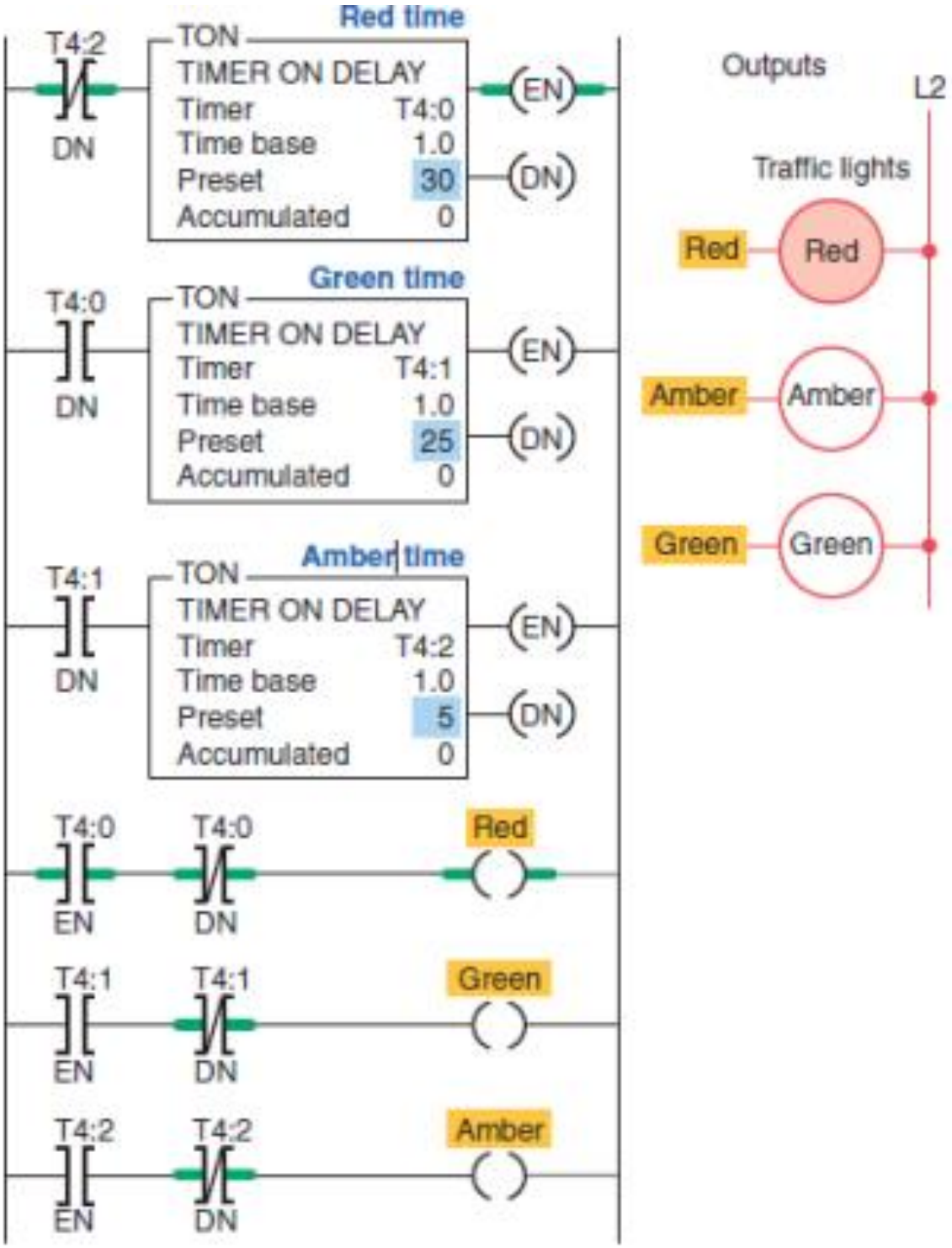
• Red—30 s on

•Green—25 s on

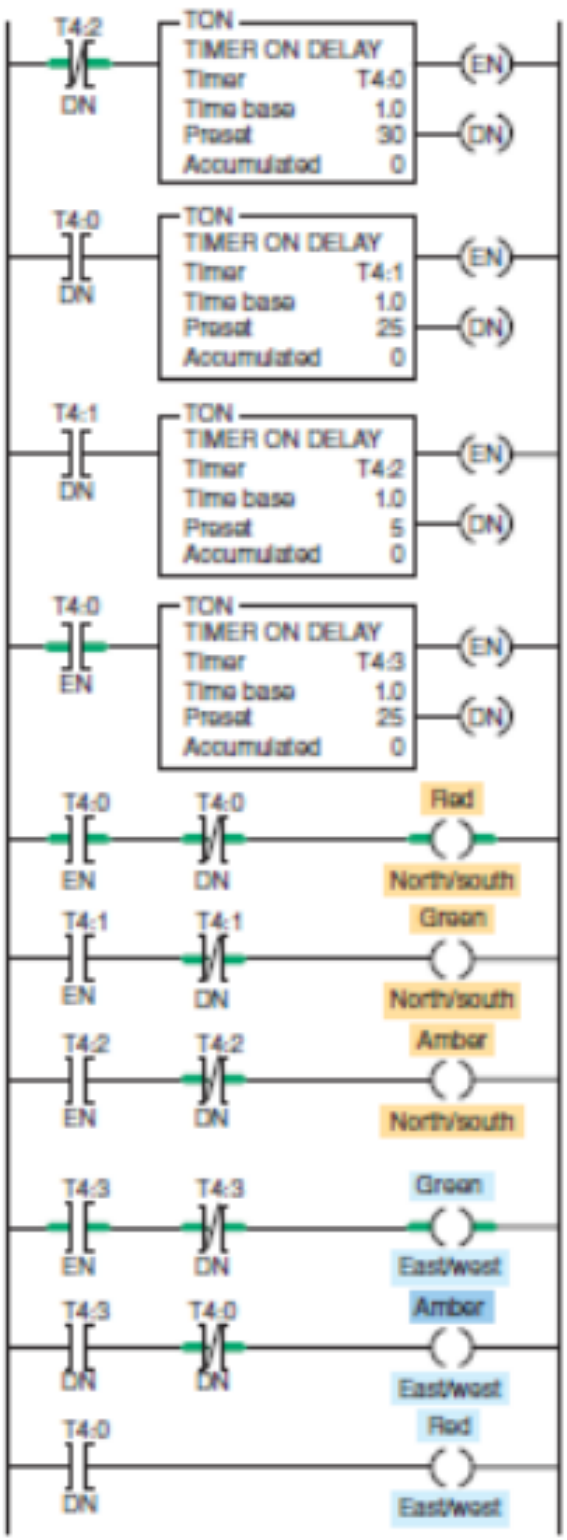
• Amber—5 s on

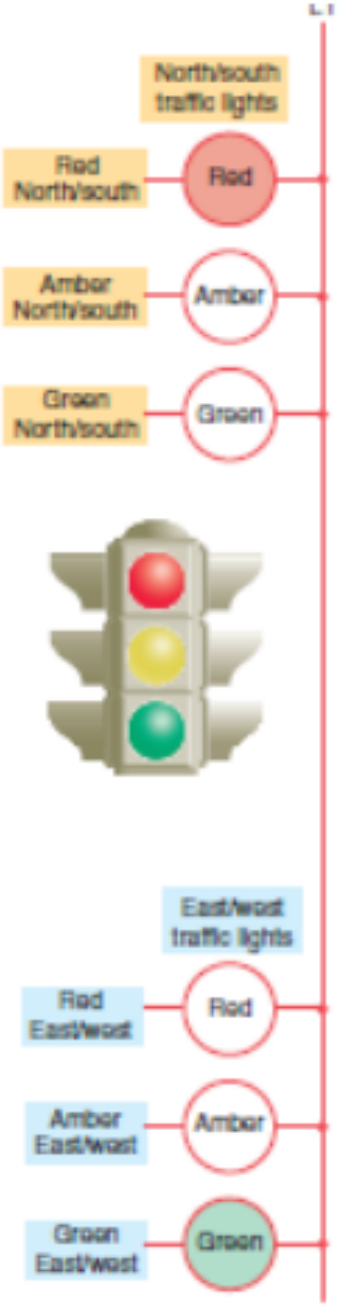
The sequence then repeats itself

MECHATRONICS SPECTRUM TEAM


Red time
T4:2
DN
TON
TIMER ON DELAY
Timer T4:0
Time base 1.0
Preset 30
Accumulated 0
EN
DN
Green time
T4:0
DN
TON
TIMER ON DELAY
Timer T4:1
Time base 1.0
Preset 25
Accumulated 0
EN
DN
Amber time
T4:1
DN
TON
TIMER ON DELAY
Timer T4:2
Time base 1.0
Preset 5
Accumulated 0
EN
DN
T4:0
EN
T4:0
DN
Red
T4:1
EN
T4:1
DN
Green
T4:2
EN
T4:2
DN
Amber
Outputs
L2
Traffic lights
Red
Red
Amber
Amber
Green
Green

Control of a set of traffic lights in two direction





ONICS
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PLC Program to Maintain the Capacity of a Particular Classroom

Problem Description

A classroom has a capacity of maximum 120 students. There are two doors, one for Entry and the other for Exit. When number of students in the classroom is less than 120, Entry door has a Green light on it which remains ON. When number of students in the classroom is 120 or more than that, Red light goes ON turning OFF the Green light which indicates that the classroom has reached its maximum capacity and is full.

Problem Solution

• Considering the availability of two separate doors for Entry and Exit, two separate Proximity Switches can be used to detect entry and exit of students.

• One proximity switch is mounted at the Entry door and the other at the Exit door.

• Both the switches will generate two different outputs which can be then fed to PLC to operate the lights according to the Ladder Logic Program written in its memory.

• Counter must be used to count the number of students entering and exiting.

• Comparator must also be used to compare the count value with the given maximum capacity of 120

I:1/0 = Proximity Switch to detect Entry of a student. (Input)

I:1/1 = Proximity Switch to detect Exit of a student. (Input)

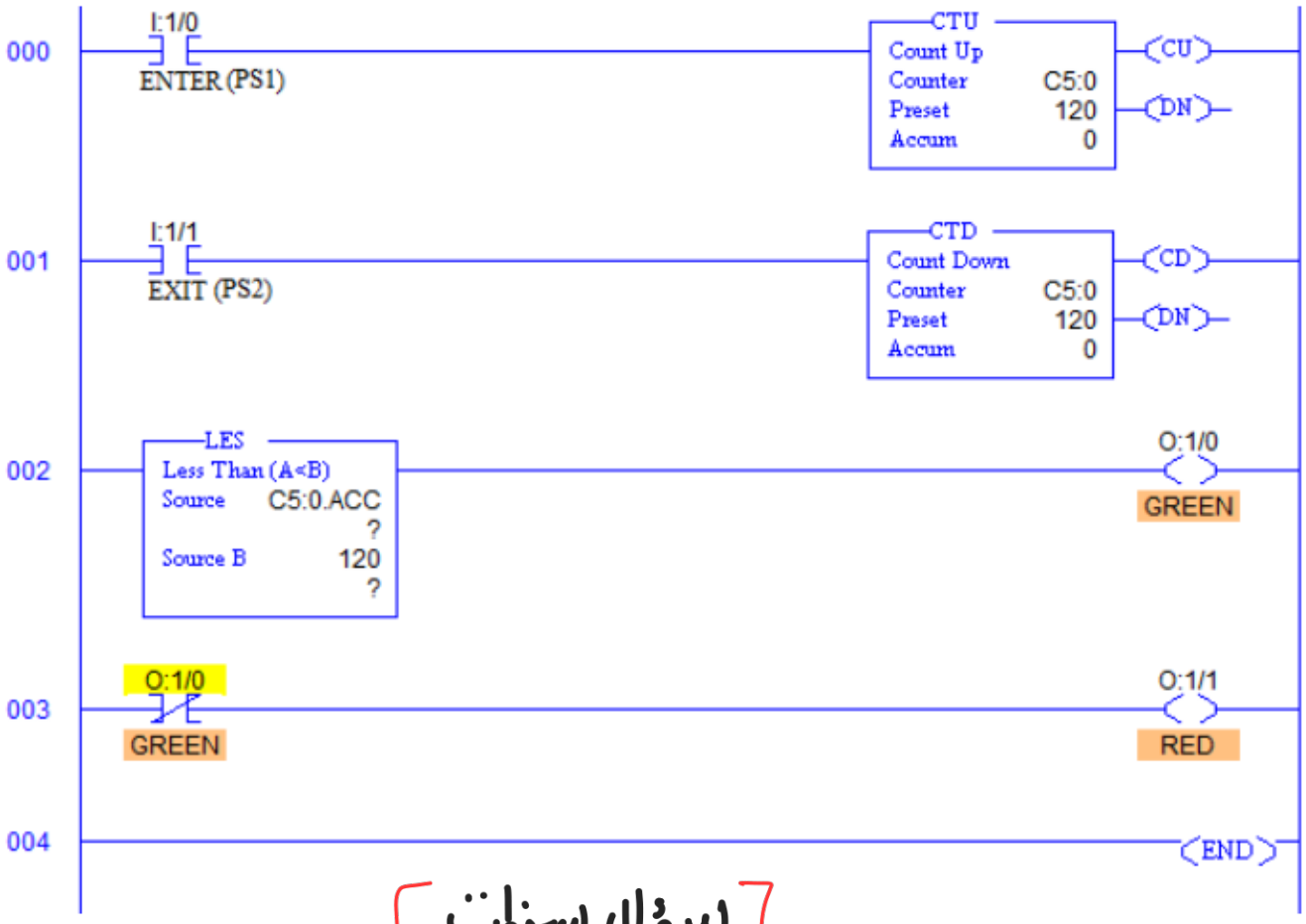
O:1/1 = Red Light to indicate availability in the classroom (Output)

O:1/0 = Light to indicate classroom's maximum capacity. (Output)

C5:0 = Counter to count the number of students entering. (Counter)

C5:0 = Counter to count the number of students exiting. (Counter)

LES = Comparator to compare the counter value



سؤال سنوات

PLC Program to Drive Motor in Forward and Reverse Direction

Problem Solution

• For any three-phase AC motor, reversing can be accomplished by reversing any two leads. For single-phase motor, reversing start lead with respect to the main leads. And for DC motor, reversing the field leads with respect to the armature leads.

• There are certain ways to reverse the motor one is by using Reversing Contactors.

• Reversing Contactor is used to reverse Three-Phase AC motors.

Single-phase motors are not widely used for reversing operation. In fact, these are not even available widely with reversing capability. As here we have a three-phase AC motor, reversing any two leads will drive the motor in reverse direction. Contactor is an electrical switch used for switching an electrical power circuit.

Two magnetic contactors are used, one for forward connections and the other for reverse connections. Only Push Button switches are used to control the direction of this three-phase AC motor.

MECHATRONICS SPECTRUM TEAM

- We have the input commands to these Push Button switches which are then internally processed by PLC and then there is the output terminal which activates the corresponding relay to energize the relevant magnetic contactors.

- Configure forward and reverse wiring of the motor with contactors such

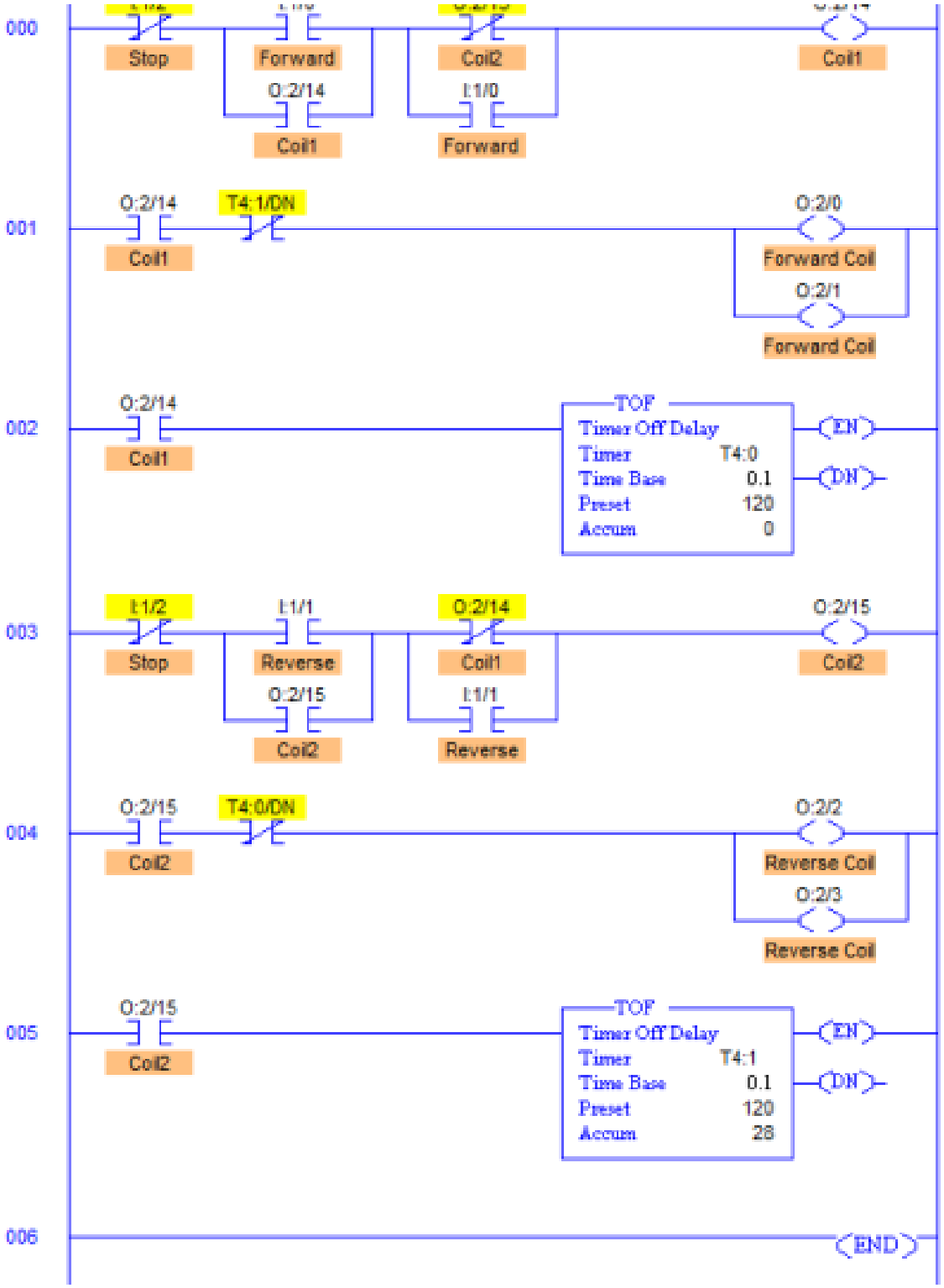
that forward contactor is connected directly in the normal direct phasing of the motor terminal and reverse contactor is connected with two of the motor terminals in the opposite phase.

- When it is switched to reverse direction, forward rotation does not stop

instantaneously hence we have to determine what time it takes to

completely stop one particular direction. Then provide time delay of a second or two and activate the other contactor.

I:1/0	= Forward Start	(Input)
I:1/1	= Reverse Start	(Input)
I:1/2	= Stop	(Input)
O:2/14	= Latched coil 1 for forward direction	(Output)
O:2/0 & O:2/1	= Forward Contactor	(Output)
T4:1	= Delay before forward direction	(Timer)
O:2/15	= Latched coil 2 for reverse direction	(Output)
O:2/2 & O:2/3	= Reverse Contactor	(Output)
T4:0	= Delay before reverse direction	(Timer)



[سؤال سنوات]

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MECHATRONICS SPECTRUM TEAM

PLC Program to Implement an Automatic Car-Wash Process

Problem Description

When a Car enters the hall, a certain sequence is to be followed automatically. Steps are,

1) Soaping.

2) Washing.

3) Rinsing.

4) Drying.

Implement this process sequence in PLC using Ladder Diagram programming language.

Problem Solution

• To detect the car automatically, load cells can be used, or any other sensor such as Infrared Sensor can also be used.

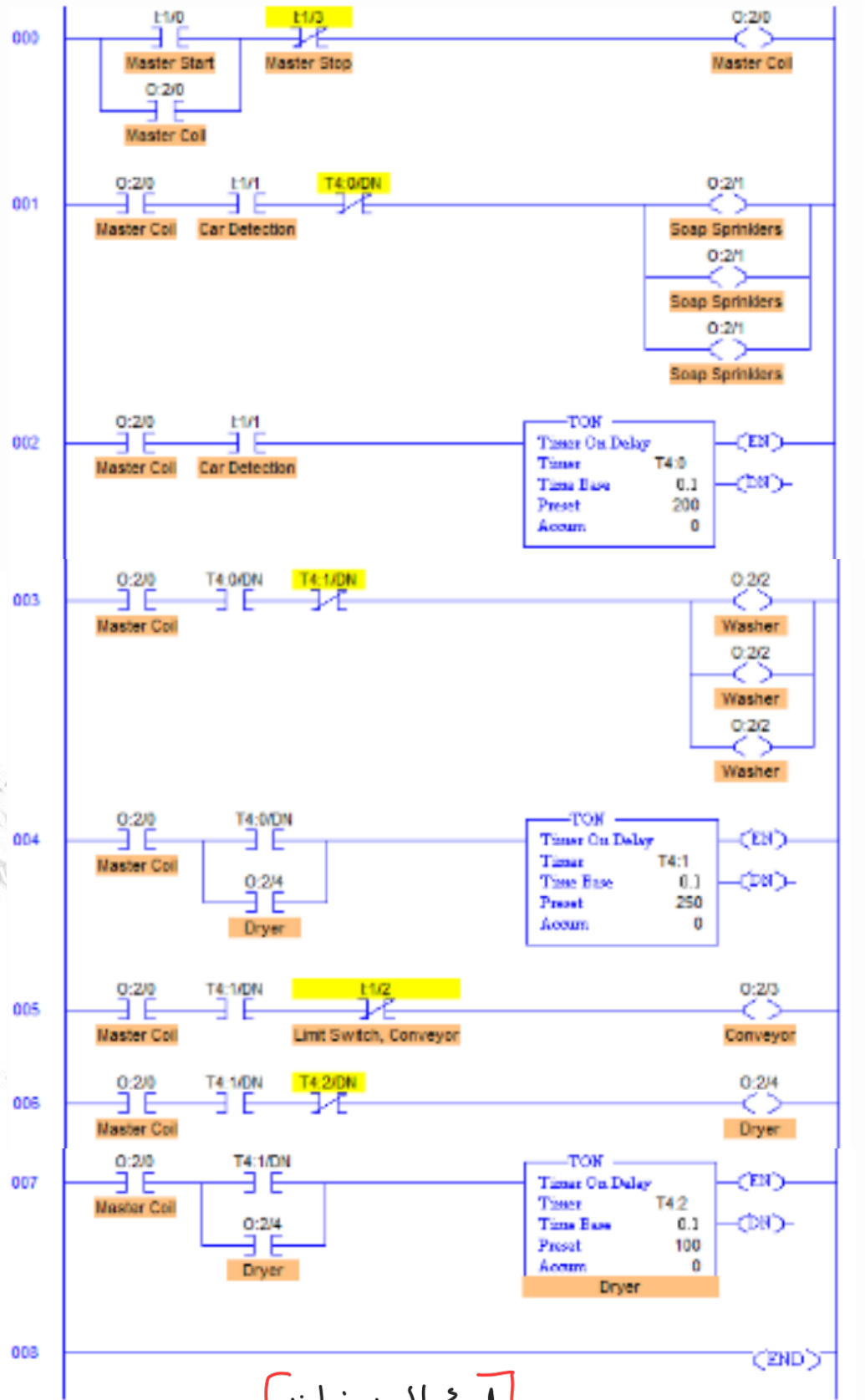
• Soaping, Washing, Rinsing and Drying are performed for a particular time, hence to generate time delay for these outputs become mandatory.

• To operate this process, for soaping, washing, and drying, four different timers are used.

• IR sensor detects everything whatever restricts the signal but in load cell, particular Low Level and High Level can be set to detect heavily weighted cars only. Load Cell can be here more effective here than IR sensors.

MECHATRONICS SPECTRUM TEAM

```
List of Inputs and Outputs
 I:1/0  = Master Start                    (Input)
 I:1/1  =  Car Detection                  (Input)
 I:1/2  = Limit Switch, Conveyor          (Input)
 I:1/3  =  Master Stop                    (Input)
 O:2/0  = Master Coil                     (Output)
 O:2/1  = Soap Sprinkler                  (Output)
 O:2/2  = Washer                          (Output)
 O:2/3  = Conveyor                        (Output)
 O:2/4  = Dryer                           (Output)
 T4:0   = Soaping Time                    (Timer)
 T4:1   = Washing Time                    (Timer)
 T4:2   = Drying Time                     (Timer)
```



[سؤال سنوات]

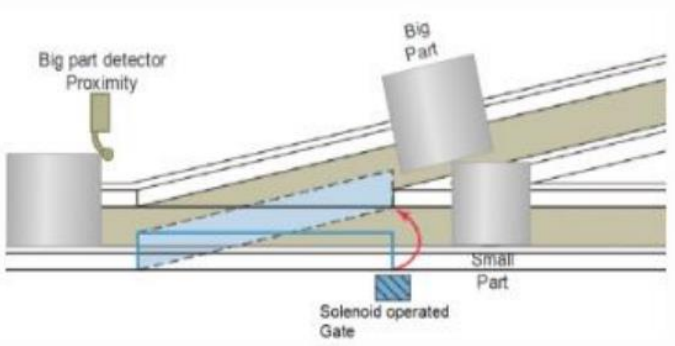
بسواعدنا نبنيها
MECHATRONICS SPECTRUM TEAM

PLC Program to Separate Big and Small Parts

Problem Description

Two different-sized particles are being moved on the conveyor belt. To pack these particles, two different boxes are used separately. These products must not be packed in the same box. The separation of these particles is to be controlled. Implement automation to perform this operation in PLC using Ladder Diagram programming language.

MECHATRONICS



Problem Solution

- Small parts do not need to be detected as it has a straight conveyor path.
- Big parts are to be diverted to the adjacent conveyor which is connected with the main conveyor as shown in diagram above.
- Detection of big particles has to be done in order to divert the path of such particles.
- To detect big parts, a Proximity Limit switch is used.
- The conveyor has a fixed speed; hence timer is to be set to energize Gate Solenoid which diverts big parts to the adjacent conveyor.

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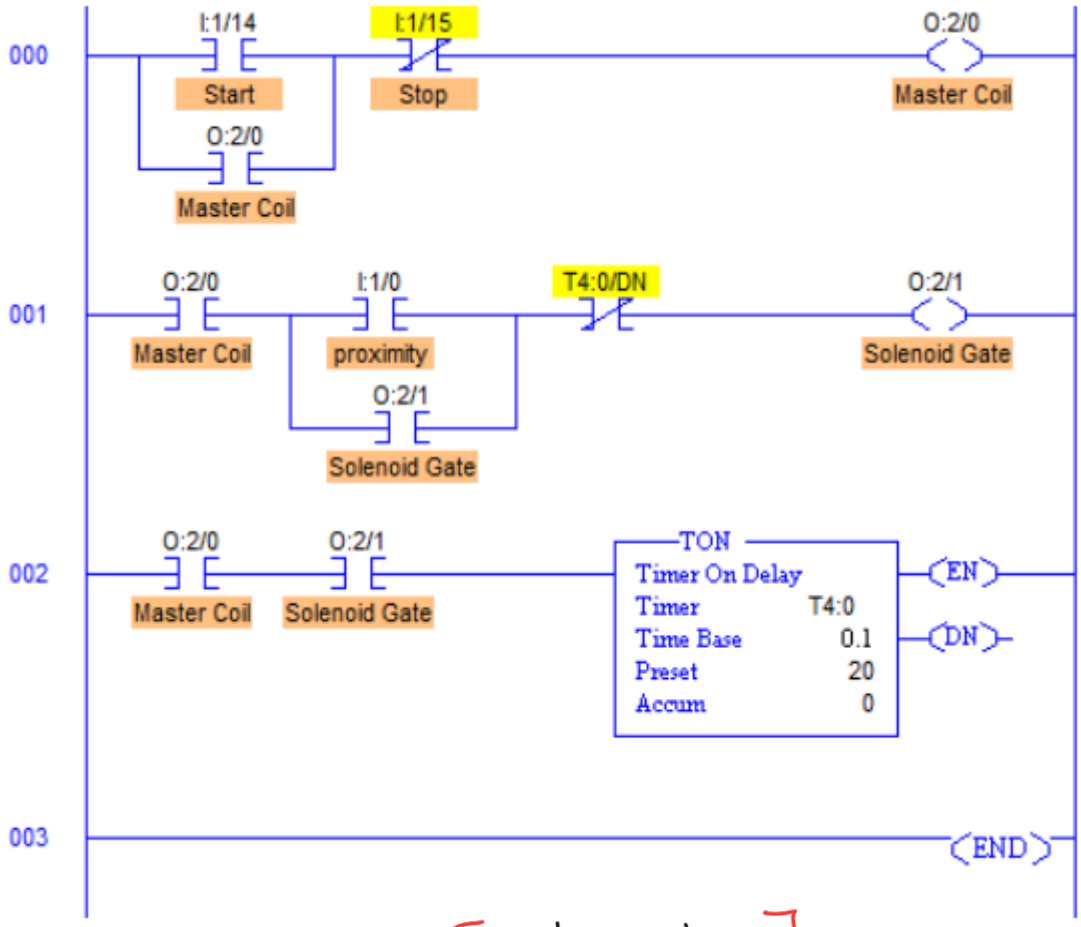
MECHATRONICS SPECTRUM TEAM

```
List of Inputs and Outputs
 I:1/14  = Start                                  (Input)
 I:1/15  = Stop                                   (Input)
 O:2/0   = Master Coil                            (Output)
 I:1/0   = Proximity to detect big parts          (Input)
 O:2/1   = Gate solenoid coil                     (Output)
 T4:0    = Timer to operate gate                  (Timer)
```



[سؤال سنوات]

بسواعدنا نبنيها
MECHATRONICS SPECTRUM TEAM

Program Description

• RUNG000 is to Master Start and Stop the separation process.

• RUNG001 and RUNG002 operate solenoid coil.

• When the Proximity switch detects a big part, it energizes and latches Solenoid Coil which is connected to O:2/1.

• Let us assume it takes approximately 2secs for a big part to transfer from the main conveyor to the adjacent one.

• When the Solenoid coil is energized, Timer T4:0 is started.

• The solenoid coil remains energized for 2secs that is when the part is being diverted to the adjacent conveyor.

• After 2secs, when a big part is diverted to the adjacent conveyor, XIO of T4:0/DN de-energizes O:2/1, and the Gate is returned back to its main position to let Small parts move on the main conveyor

MECHATRONICS SPECTRUM

PLC Program to Drive Motors Simultaneously with Interlocking

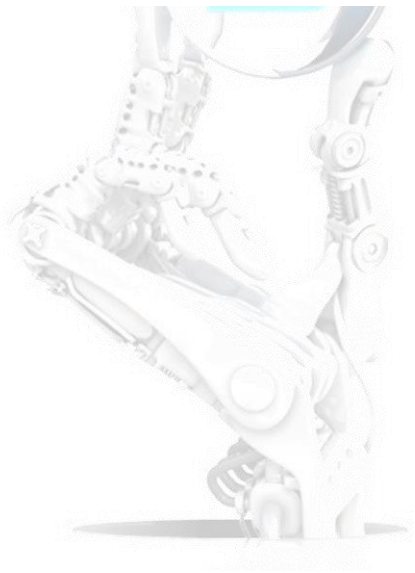
Problem Description

Two Motors are running in a sequence one by one for a particular time. If the start button is pressed Motors run in sequence such that 1st Motor stays ON for 5secs and then 2nd Motor is turned ON and stays ON for 5secs. And the cycle is repeated until it is interrupted. While motors are running in the sequence, if one motor is running and the button of other motor is pressed, then the running Motor should stop and the other motor should run. Implement this logic in PLC using Ladder Diagram programming language.

Problem Solution

- Define addresses of motors.
- Use pushbuttons to start and stop the sequential operation.
- Use TON timer to generate a particular time delay.
- Use the enable bit of first timer to energize 2nd motor coil and activate second timer.
- No need to use Reset contact, program will reset the timers itself after completion of each cycles.
- Use Parallel Start Motor contact to start one motor when other motor is running

I:1/0	= Motor 0 Start	(Input)
I:1/1	= Motor 1 Stop	(Input)
I:1/2	= Master Start	(Input)
I:1/3	= Master Stop	(Input)
O:2/2	= Master Coil	(Output)
O:2/0	= Motor 0	(Output)
T4:0	= Motor 0 Timer	(Timer)
O:2/1	= Motor 1	(Output)
T4:1	= Motor 1 Timer	(Timer)



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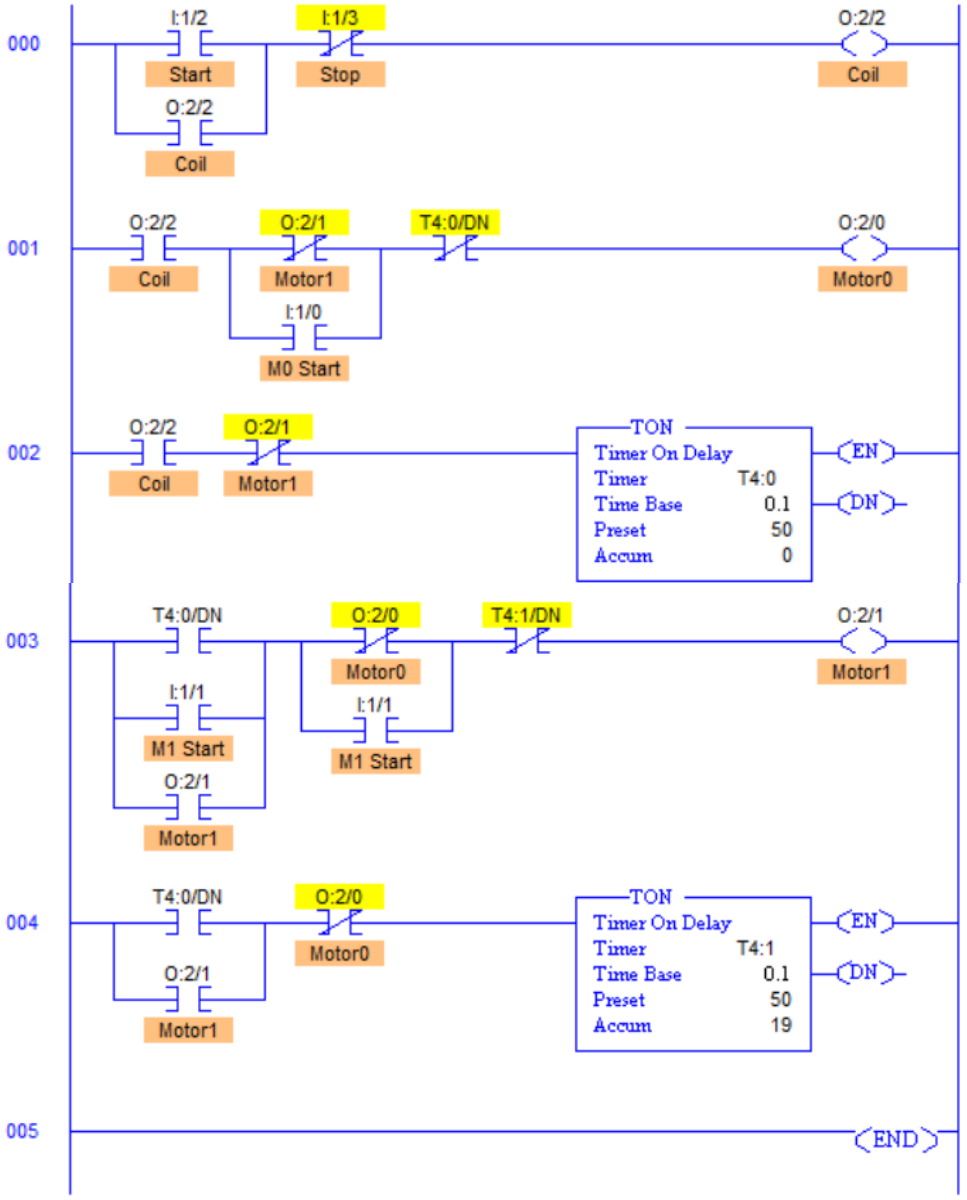
T E A M

بسواعدنا نبنيها
MECHATRONICS SPECTRUM TEAM


000
I:1/2
Start
O:2/2
Coil
I:1/3
Stop
O:2/2
Coil
001
O:2/2
Coil
O:2/1
Motor1
I:1/0
M0 Start
T4:0/DN
O:2/0
Motor0
002
O:2/2
Coil
O:2/1
Motor1
TON
Timer On Delay
Timer T4:0
Time Base 0.1
Preset 50
Accum 0
EN
DN
003
T4:0/DN
I:1/1
M1 Start
O:2/1
Motor1
O:2/0
Motor0
I:1/1
M1 Start
T4:1/DN
O:2/1
Motor1
004
T4:0/DN
O:2/1
Motor1
O:2/0
Motor0
TON
Timer On Delay
Timer T4:1
Time Base 0.1
Preset 50
Accum 19
EN
DN
005
END

PLC Program to Operate 4 Outputs Simultaneously with Time Delay

Problem Description

There is a total four number of outputs that should be run one by one with a particular time delay. Implement this in PLC using Ladder Diagram programming language.

Problem Solution

• Generate Master Start and Stop buttons to activate the sequence and define four outputs.

• Use TON timer to generate a particular time delay, same or different.

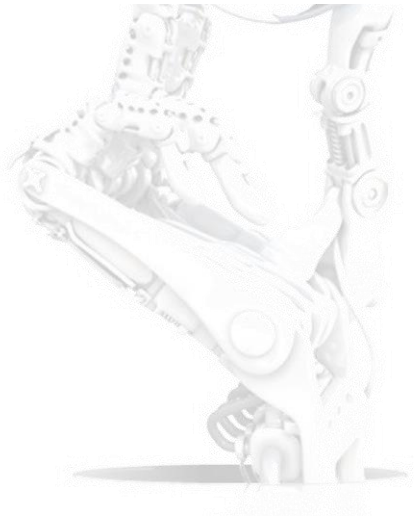
• Use Done bit of first timer to energize other output and activate second timer.

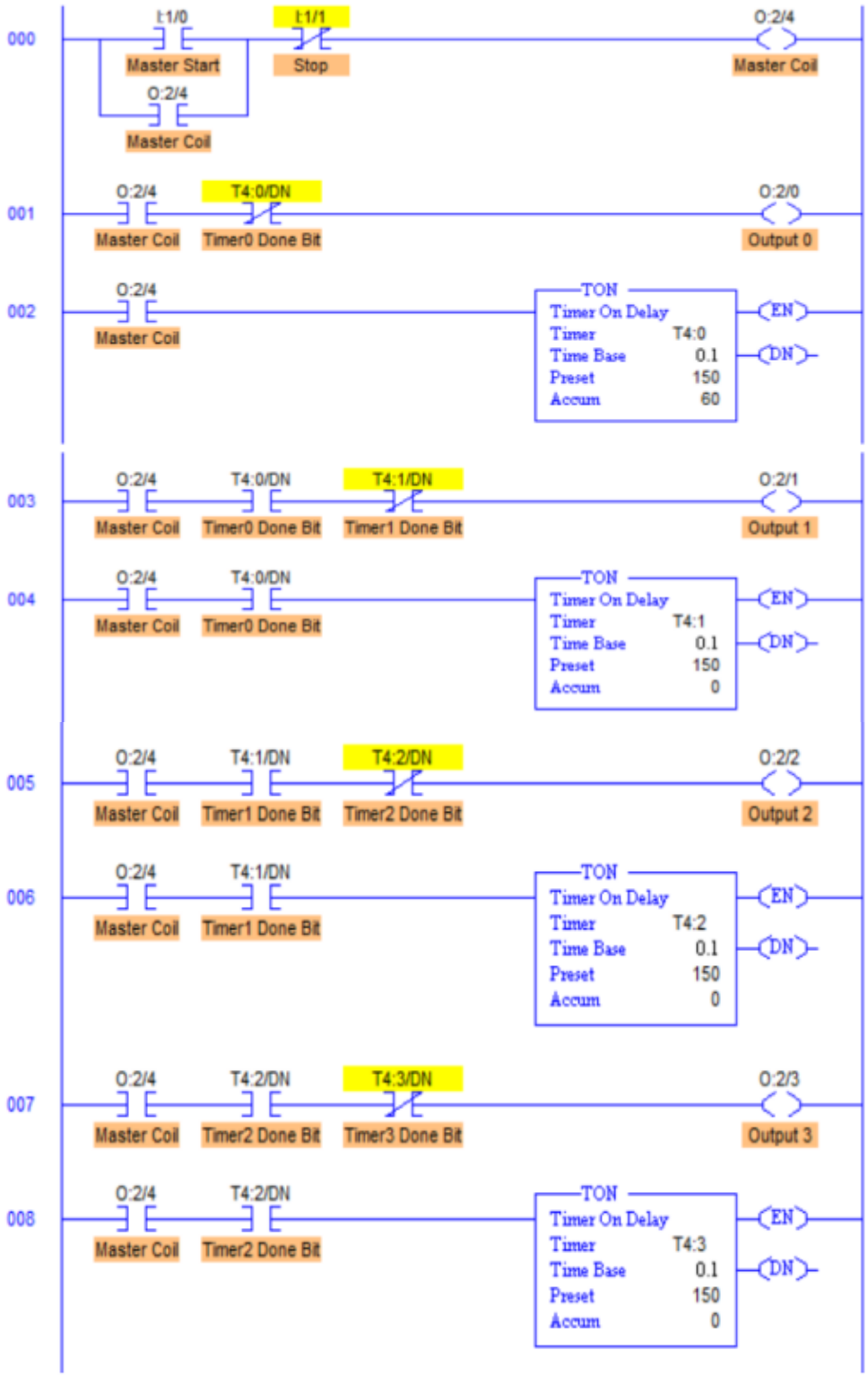
• Repeat this process until the final output is energized and last timer is activated.

• Reset timers after the completion of first cycle if necessary.

• Use LEDs as output to test the program.

I:1/0 = Master Start (Input)
I:1/1 = Master Stop (Input)
O:2/4 = Master Coil (Output)
O:2/0 = Output0 (Output)
T4:0 = Output0 Timer (Timer)
O:2/1 = Output1 (Output)
T4:1 = Output1 Timer (Timer)
O:2/2 = Output2 (Output)
T4:2 = Output2 Timer (Timer)
O:2/3 = Output3 (Output)
T4:2 = Output3 Timer (Timer)




000
I:1/0
Master Start
O:2/4
Master Coil
I:1/1
Stop
O:2/4
Master Coil
001
O:2/4
Master Coil
T4:0/DN
Timer0 Done Bit
O:2/0
Output 0
002
O:2/4
Master Coil
TON
Timer On Delay
Timer T4:0
Time Base 0.1
Preset 150
Accum 60
EN
DN
003
O:2/4
Master Coil
T4:0/DN
Timer0 Done Bit
T4:1/DN
Timer1 Done Bit
O:2/1
Output 1
004
O:2/4
Master Coil
T4:0/DN
Timer0 Done Bit
TON
Timer On Delay
Timer T4:1
Time Base 0.1
Preset 150
Accum 0
EN
DN
005
O:2/4
Master Coil
T4:1/DN
Timer1 Done Bit
T4:2/DN
Timer2 Done Bit
O:2/2
Output 2
006
O:2/4
Master Coil
T4:1/DN
Timer1 Done Bit
TON
Timer On Delay
Timer T4:2
Time Base 0.1
Preset 150
Accum 0
EN
DN
007
O:2/4
Master Coil
T4:2/DN
Timer2 Done Bit
T4:3/DN
Timer3 Done Bit
O:2/3
Output 3
008
O:2/4
Master Coil
T4:2/DN
Timer2 Done Bit
TON
Timer On Delay
Timer T4:3
Time Base 0.1
Preset 150
Accum 0
EN
DN

PLC Program to Operate Screwing of Parts

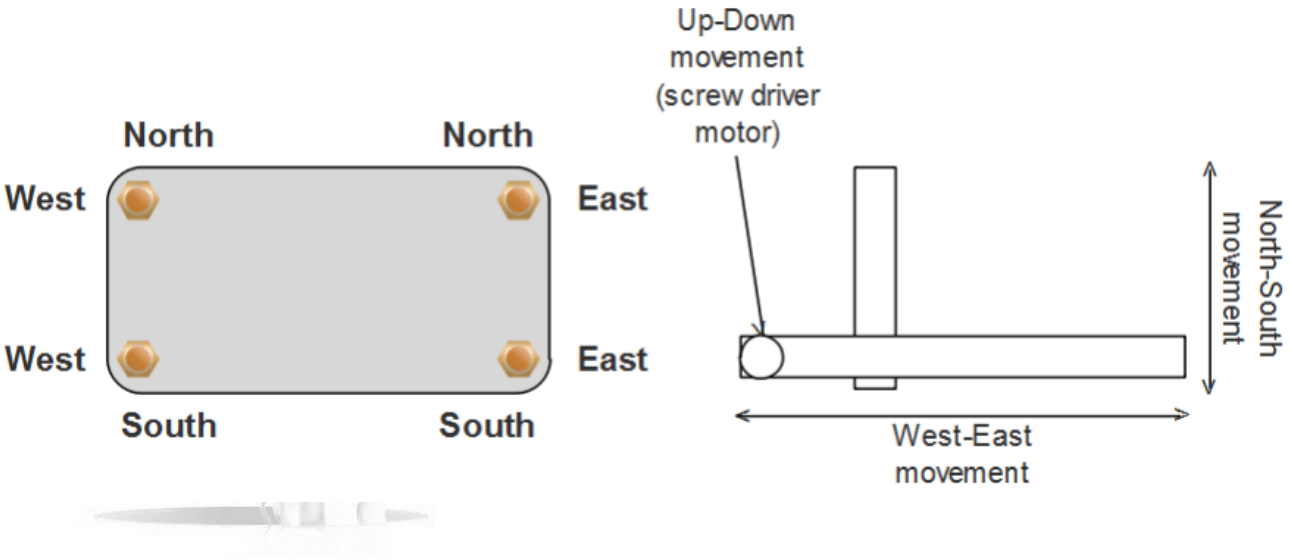
Problem Description

When a plate is placed, screws are to be fit at each corner of the plate.

Use robotic arm to perform this action and implement automation of

this in PLC using Ladder Diagram programming language.

MECHATRONICS SPECTRUM



Problem Solution

•Robotic arm used to perform this screwing action.

•Robotic arm used here is of 3-axis arm.

•Sequence of the arm is to be controlled.

•SQO instruction must be used in order to full fill the requirements.

•Sequence to be followed is as given below:

•1- Arm is initially at South-East position

• 2- Arm lowers and screws.

• 3- Raise arm.

• 4- Move to West-South position

• 5- Arm lowers and screws.

• 6- Raise arm.

• 7- Move to West-North position.

• 8- Arm lowers and screws.

• 9- Raise arm.

•10- Move to North-East position.

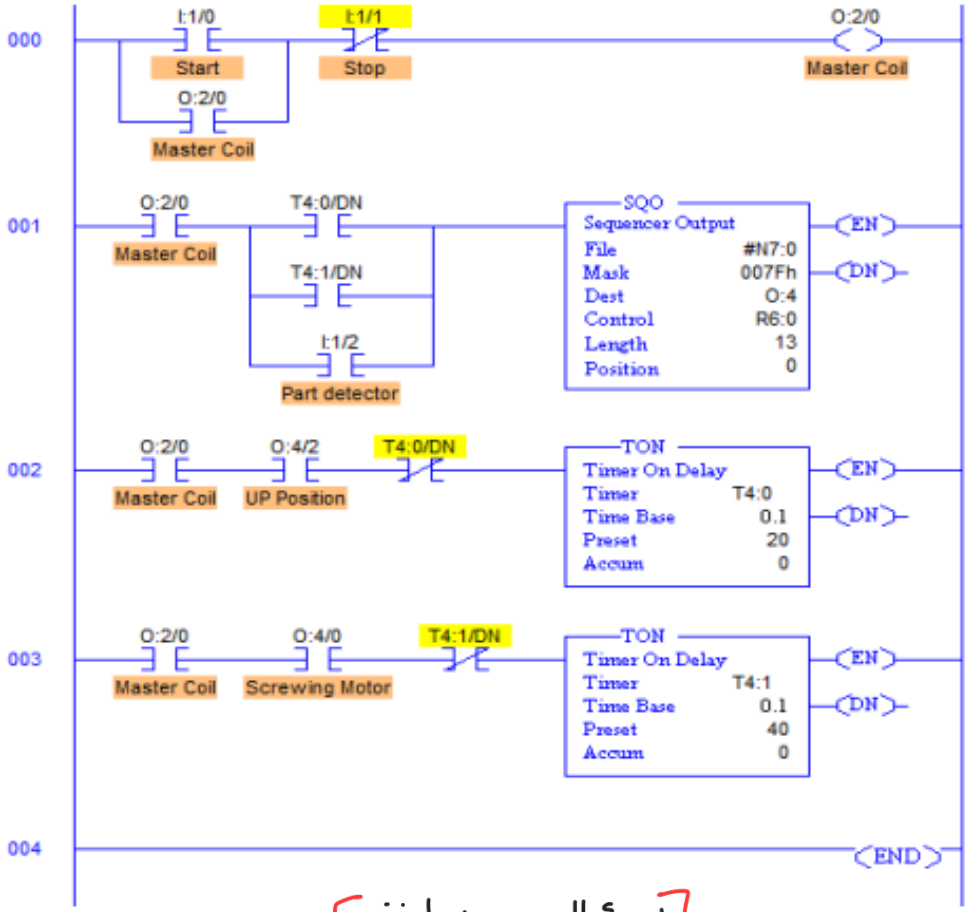
•11- Arm lowers and screws.

•12- Raise arm.

• 13- Move to South-East position

MECHATRONICS SPECTRUM TEAM

I:1/0	= Start PB	(Input)
I:1/1	= Stop PB	(Input)
I:1/2	= Part detector proximity	(Input)
O:2/0	= Master coil	(Output)
O:4/0	= Screwing Motor coil	(Output)
O:4/1	= Down position	(Output)
O:4/2	= Up position	(Output)
O:4/3	= West position	(Output)
O:4/4	= East position	(Output)
O:4/5	= South position	(Output)
O:4/6	= North position	(Output)
N7:0	= File register	(Register)
N7:1 to N7:13	= Sequence storing registers	(Register)
R6:0	= Control Register	(Register)



[لسؤال سنوات]

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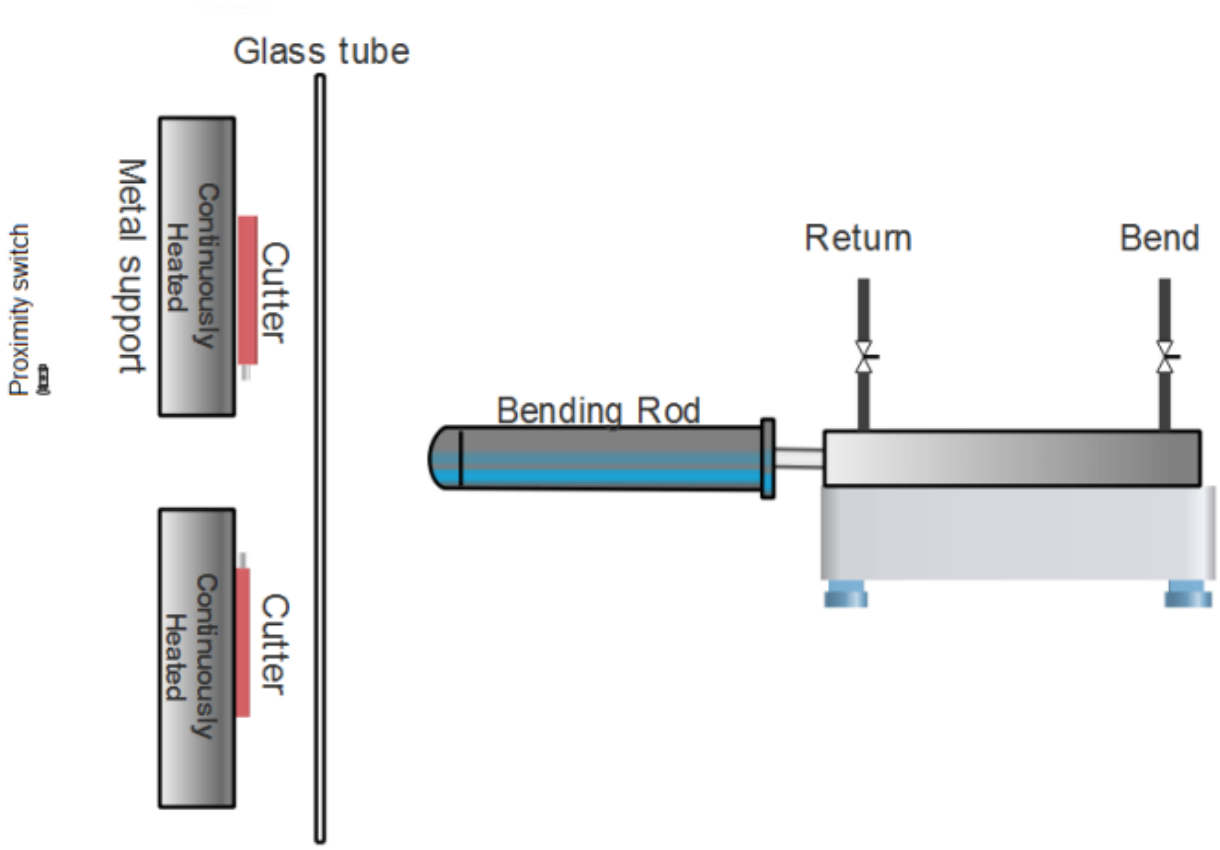
MECHATRONICS SPECTRUM TEAM

PLC Program to Heat and Bend Glass Tubes

Problem Description

Heated glass tubes are passing in a process line having a particular length that is to be bent. To manufacture fluorescent bulbs, these tubes are to be bent in a U-Shape. Automate this process in PLC using Ladder Diagram programming language.



Problem Solution

• Diagram is an example of how bending process is done.

• Pneumatic operated rod moves in right and left direction which forces glass tube to bend through metal support.

• Metal support is continuously heated so that glass tubes do not break while bending.

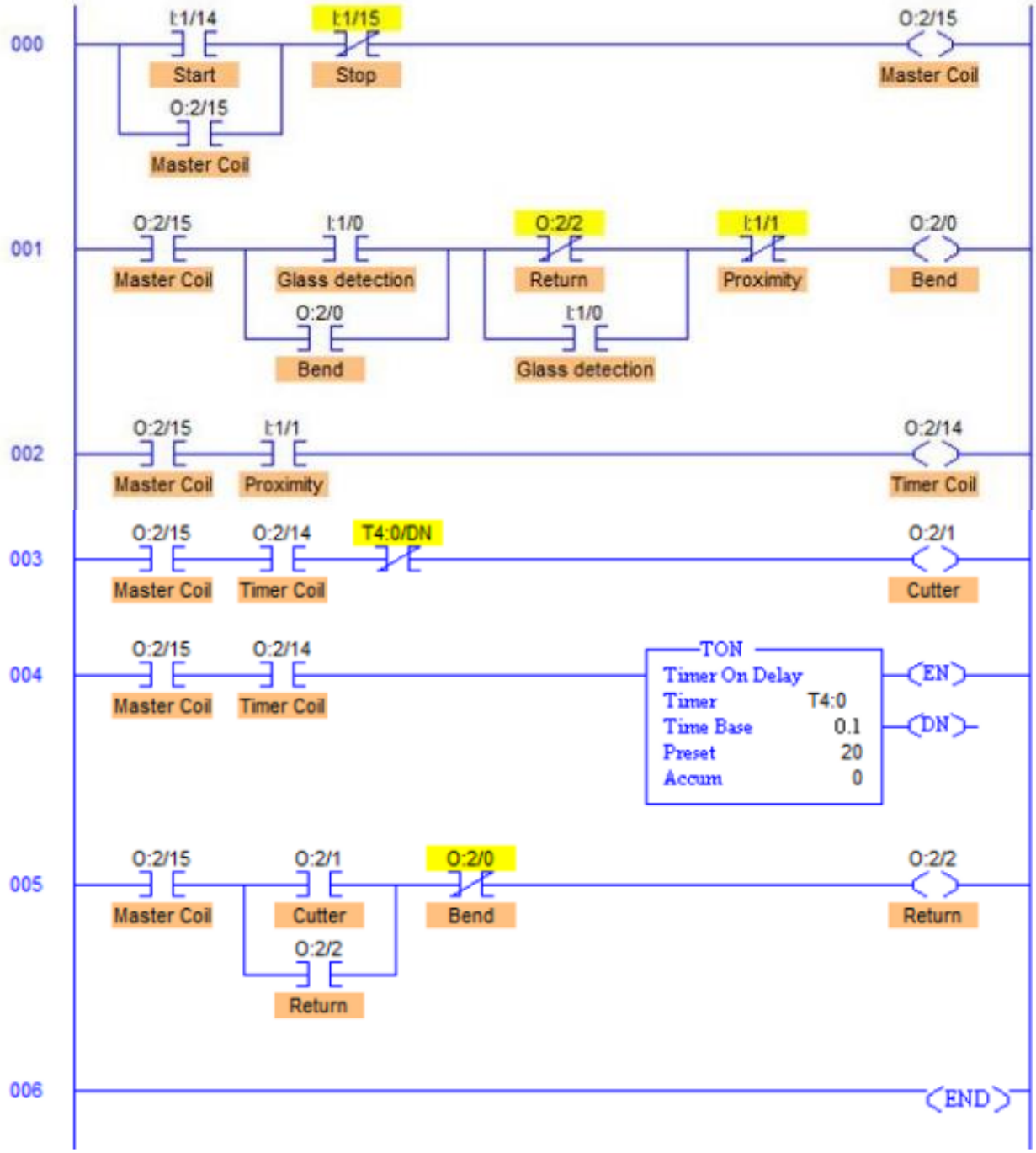
• When bending is done, two cutters are used to cut the bent glass tube.

• Proximity switch detects a length to be cut by the cutter.

• Metal support are mounted close to each other such that only bending rod and glass tube together can pass through it.

• Metal support is not exactly square as shown in the diagram, it has curvy ends for smoother bending operation

I:1/14	= Start	(Input)
I:1/15	= Stop	(Input)
O:2/15	= master Coil	(Output)
I:1/0	= Glass presence detector	(Input)
I:1/1	= Glass length detector (Proximity)	(Input)
O:2/0	= Bend (Air supply)	(Output)
O:2/1	= Cutter	(Output)
O:2/2	= Return (Air supply)	(Output)
O:2/3	= Current supply (To heat metal support)	(Output)
O:2/14	= Timer coil	(Output)
T4:0	= Timer for cutter	(Timer)



[سؤال سنوات]

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MECHATRONICS SPECTRUM TEAM

PLC Program to Store Temperature Data of a Process at Different Time

Problem Description

Temperature of two tanks are being controlled. To display and compare temperature of both the tanks, graphical display is used. Update temperature of these tanks on the display every 5secs for comparison with respect to time. Implement this process in PLC using Ladder Diagram programming language

Problem Solution

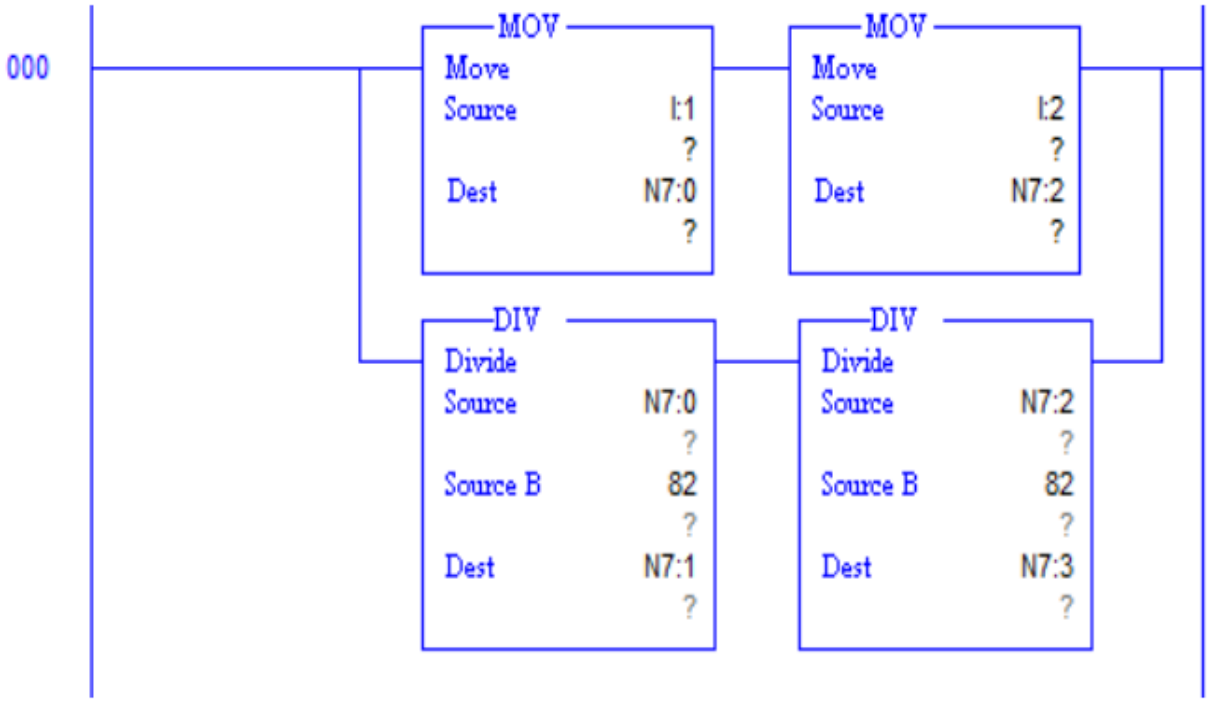
• Using RTD PT100, we can calibrate value 00000 to 65536 into -250-650C degree and divide it into 800 equal parts.

• Use MOV instruction to transfer Input data into the register for further conversion usage.

• Convert outputs of temperature transmitters into equivalent Temperature in Centigrade.

• To do this, divide input by value change per centigrade, we get temperature inside that tank

I:1	= Input from temperature transmitter1	(Input)
I:2	= Input from temperature transmitter2	(Input)
I:3/0	= Start PB	(Input)
I:3/1	= Stop PB	(Input)
O:4/0	= Master coil	(Output)
N7:0 to N7:8	= Registers to store and process data	(Register



- When divided by the total range of RTD PT100

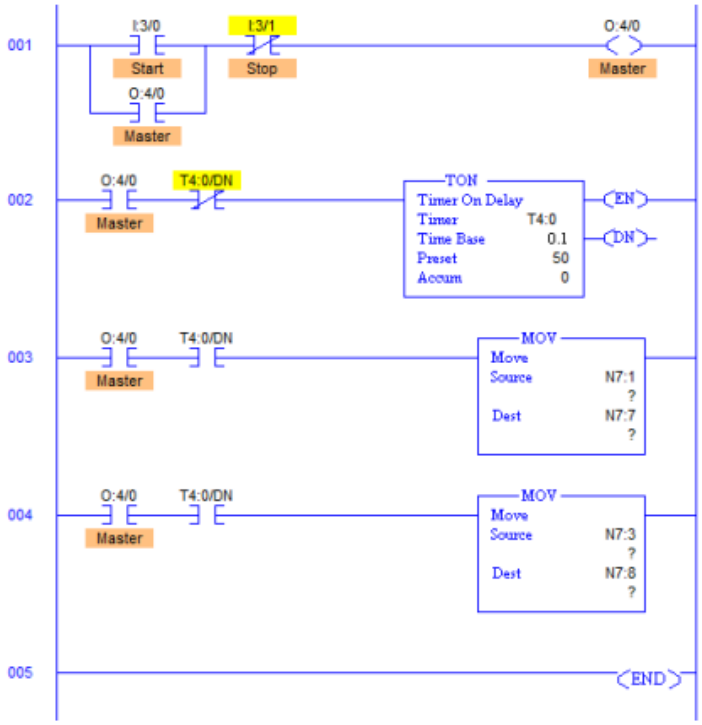
which is -250 to 650 degree centigrade, we get 82 value changes per Centigrade.

- Hence when divided by 82, we get the answer in centigrade.

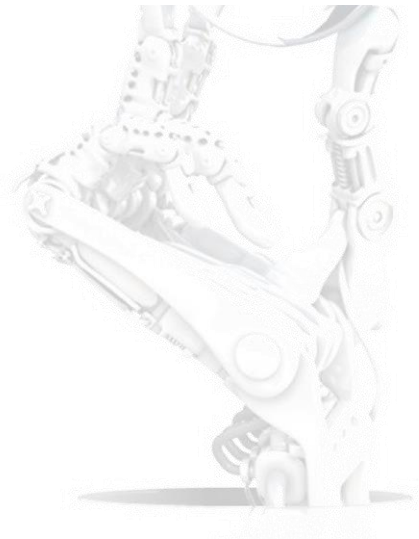
- An important thing to note here is that the input here will always be greater than 23124 which is

the value when the temperature is 32 degree centigrade that is room temperature.

Ladder diagram to update Trend display every 5 secs



- RUNG002, RUNG003 and RUNG004 updates temperature data in corresponding registers every 5secs.
- Timer is in auto reset mode, hence timer will keep running till it is manually stopped by operator.



T E A M

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MECHATRONICS SPECTRUM TEAM

PLC Program to Detect and Remove Burned Chips

Problem Description

Potato chips are made and ready to be packed. But before that, it goes through a conveyor in which a final quality check is done, and burnt chips are detected and removed from the process line.

Implement automation of this process in PLC using the Ladder Diagram programming language.

Problem Solution

• To detect burned chips, light sources, and sensors are used.

• Light source is used so light detectors such as Light Dependent Resistors are used to detect the burned chips.

• Blowers are used to throw burned chips away from the conveyor when detected.

• There are a total number of 8 blowers. Several blowers to be used depending on the width of a conveyor belt.

• Time measurement of an event to take place can be used here to measure what time burned chips take to reach from light source to blowers when detected.

• Set this time as a preset of a timer to operate a particular blower.

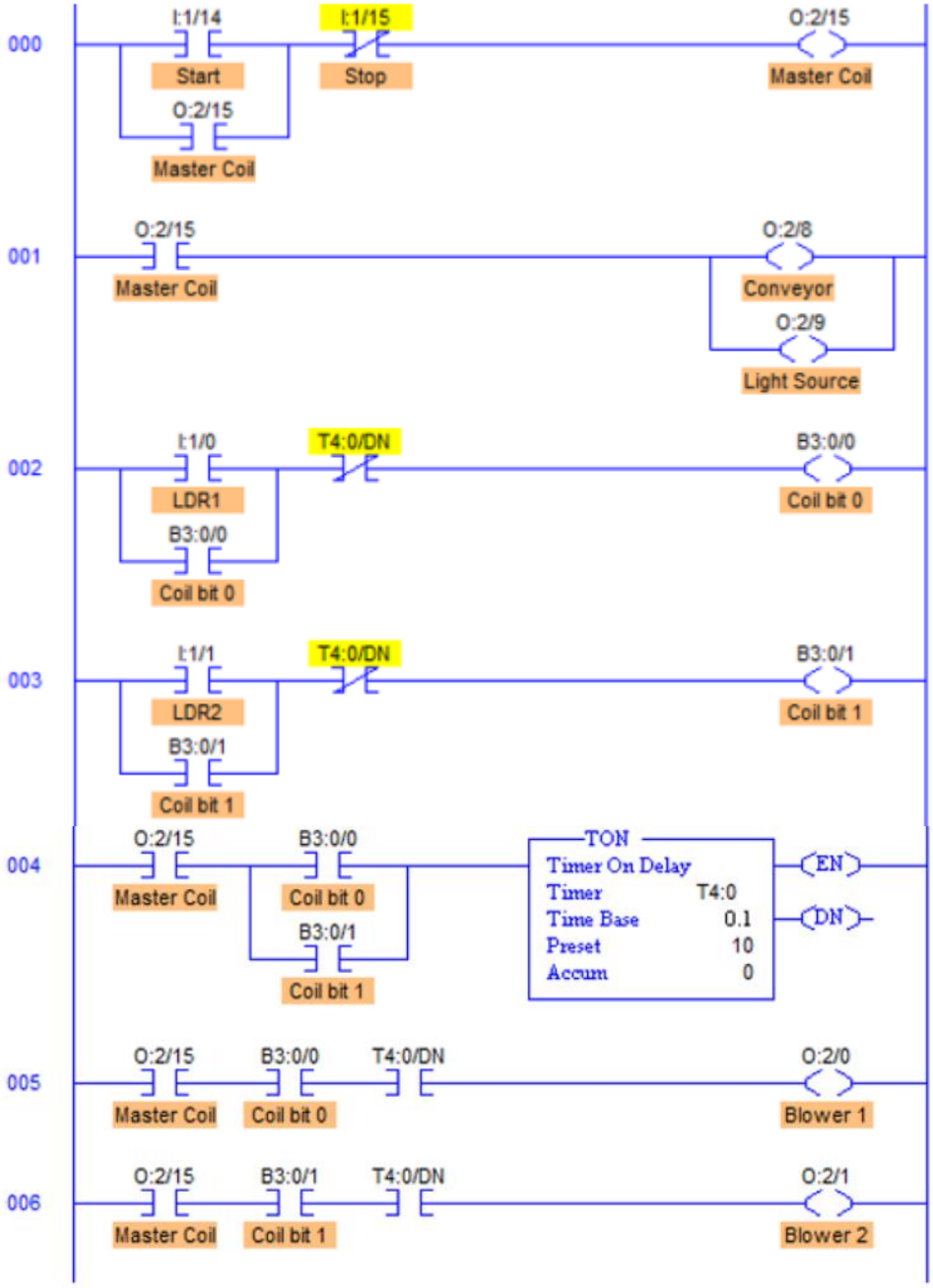
• There are 8 blowers, so 8 light detecting circuits must be used to operate all blowers.

MECHATRONICS SPECTRUM TEAM

• Let us assume we are using Light Dependent Resistor. To use this resistor, the threshold has to be set that is the darkest color to be passed as a good quality product. If chips are darker than the desired level, the light source detects it and activates the corresponding circuit.

• So output of this circuit is normally high and to activate the blower, normally low logic has to be set while programming or we can even invert output from the LDR circuit

I:1/14	= Start	(Input)
I:1/15	= Stop	(Input)
O:2/15	= Master Coil	(Internal)
O:2/8	= Conveyor	(Output)
O:2/9	= Light Source	(Output)
I:1/0 to I:1/7	= Light Dependent Resistors	(Input)
B3:0/0 to B3:0/7	= Bit latching	(Bits)
O:2/0 to O:2/7	= Blowers	(Output)
T4:0	= Timer to activate blower after 1sec	(Timer)


000
I:1/14
Start
I:1/15
Stop
O:2/15
Master Coil
O:2/15
Master Coil
001
O:2/15
Master Coil
O:2/8
Conveyor
O:2/9
Light Source
002
I:1/0
LDR1
T4:0/DN
B3:0/0
Coil bit 0
B3:0/0
Coil bit 0
003
I:1/1
LDR2
T4:0/DN
B3:0/1
Coil bit 1
B3:0/1
Coil bit 1
004
O:2/15
Master Coil
B3:0/0
Coil bit 0
B3:0/1
Coil bit 1
TON
Timer On Delay
Timer T4:0
Time Base 0.1
Preset 10
Accum 0
EN
DN
005
O:2/15
Master Coil
B3:0/0
Coil bit 0
T4:0/DN
O:2/0
Blower 1
006
O:2/15
Master Coil
B3:0/1
Coil bit 1
T4:0/DN
O:2/1
Blower 2

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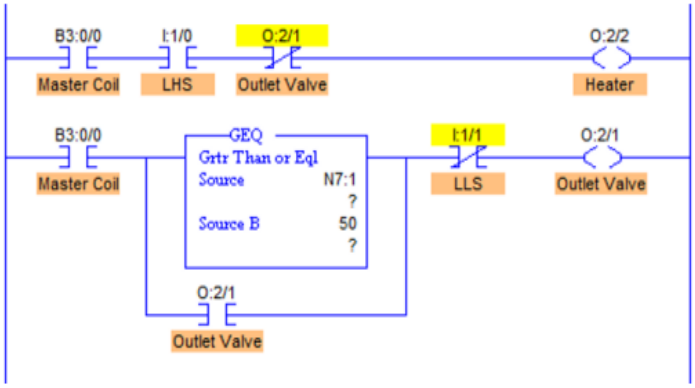

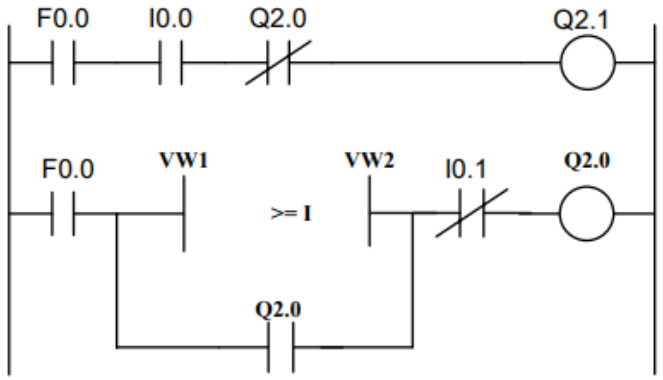

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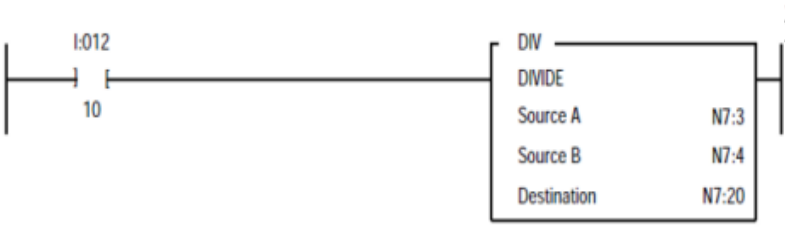

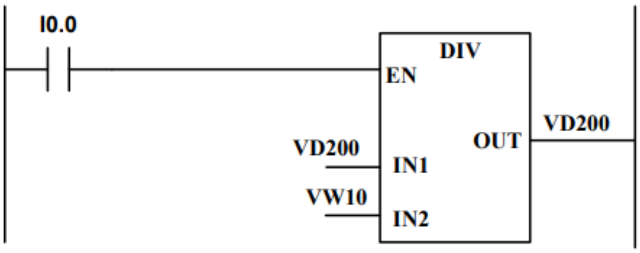

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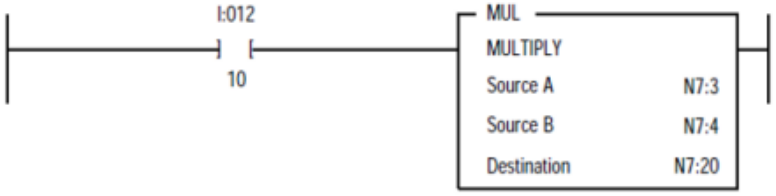

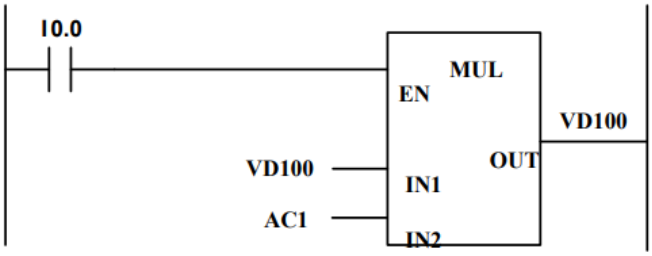

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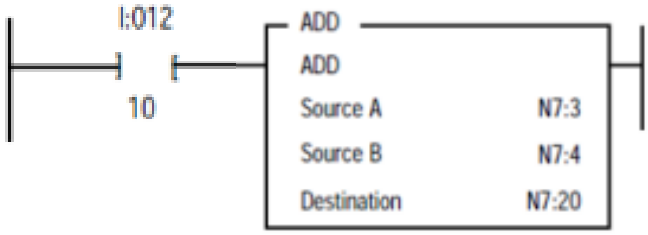

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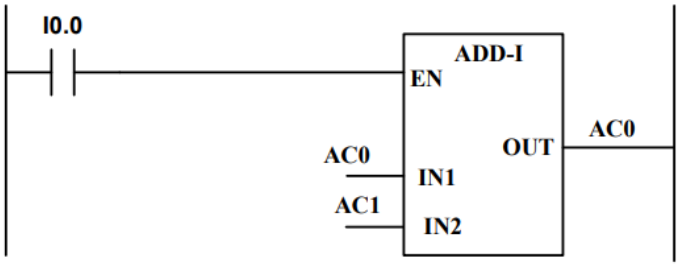
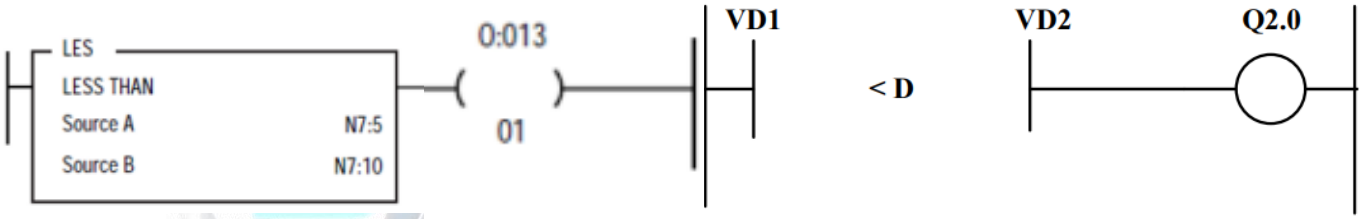


Siemens



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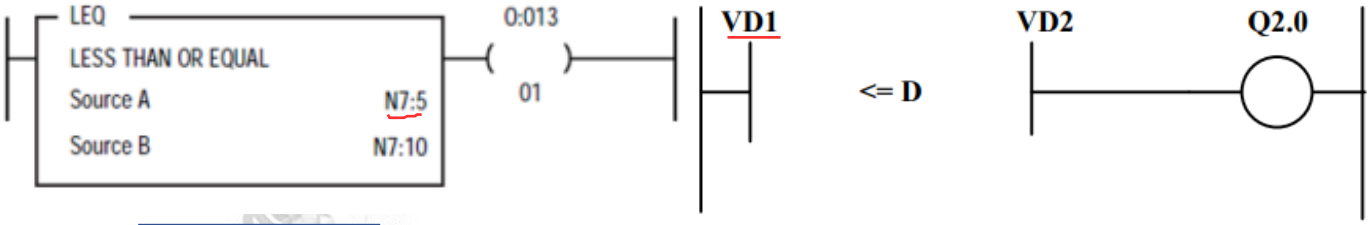

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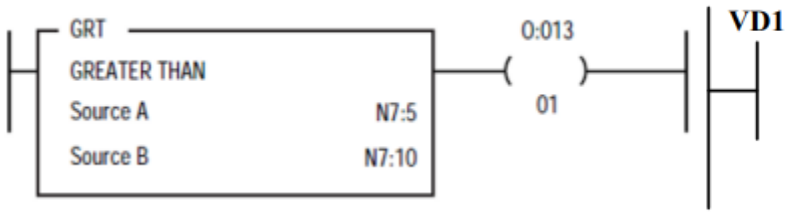
Register N7

Register VD

نراي اختلاف اسماء ورموز Register



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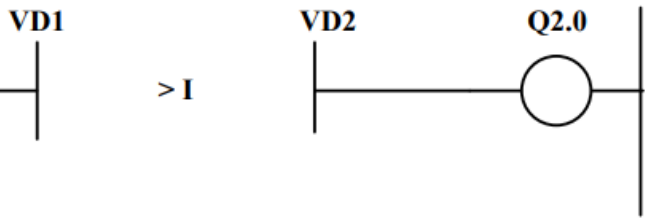

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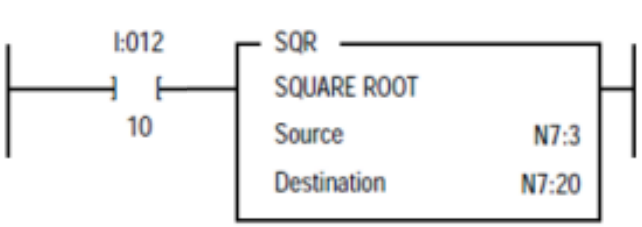

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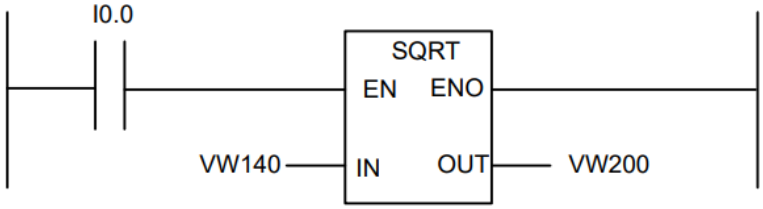

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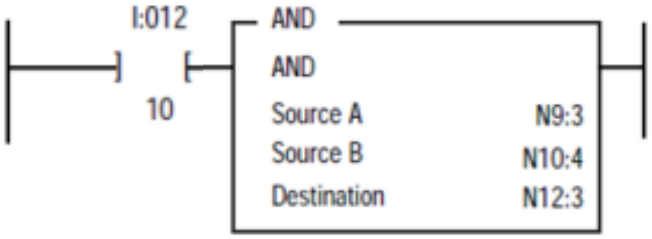

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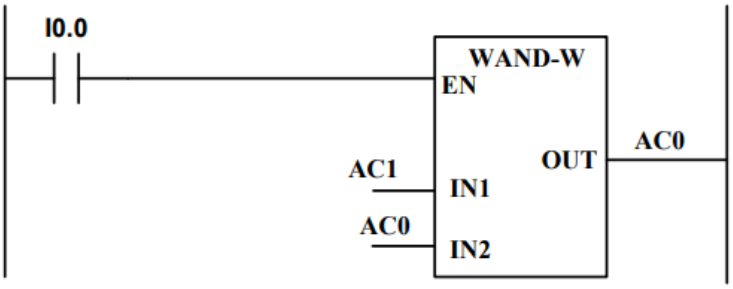

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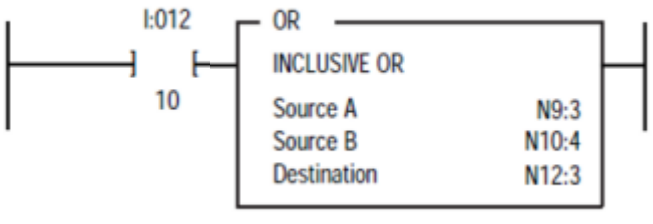

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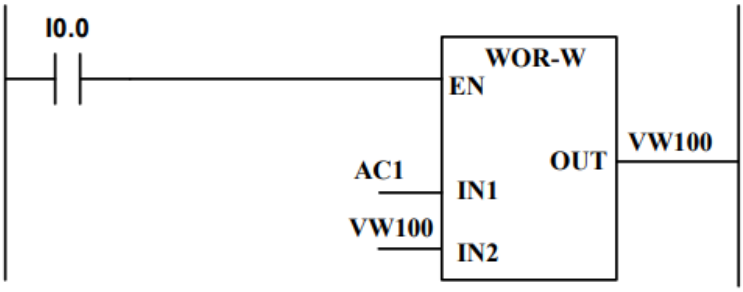

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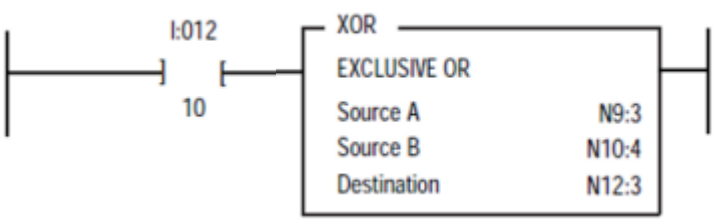

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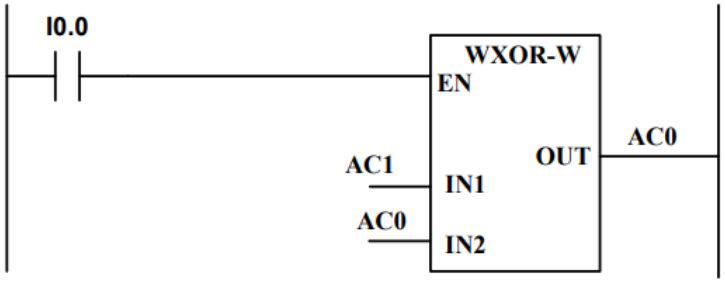

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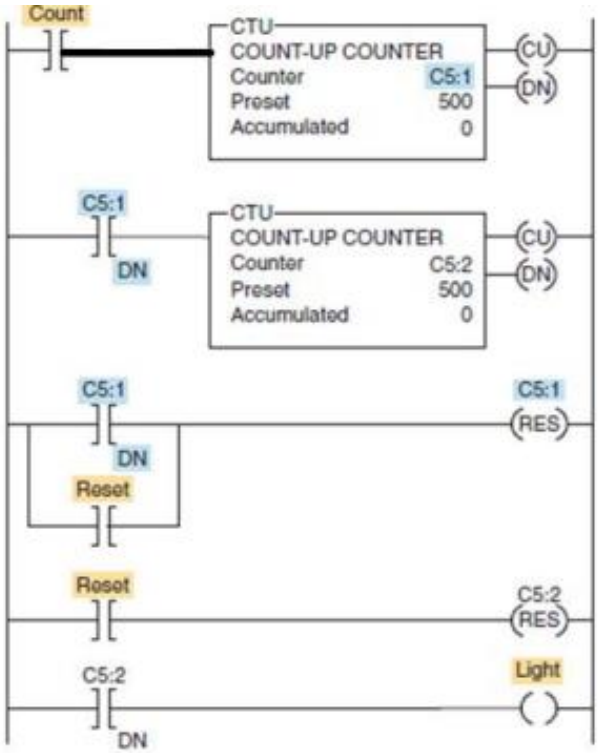

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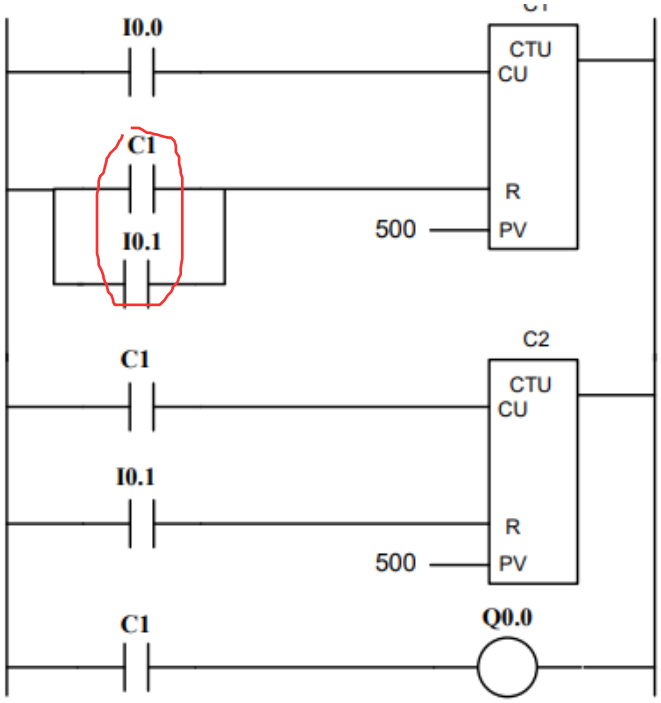

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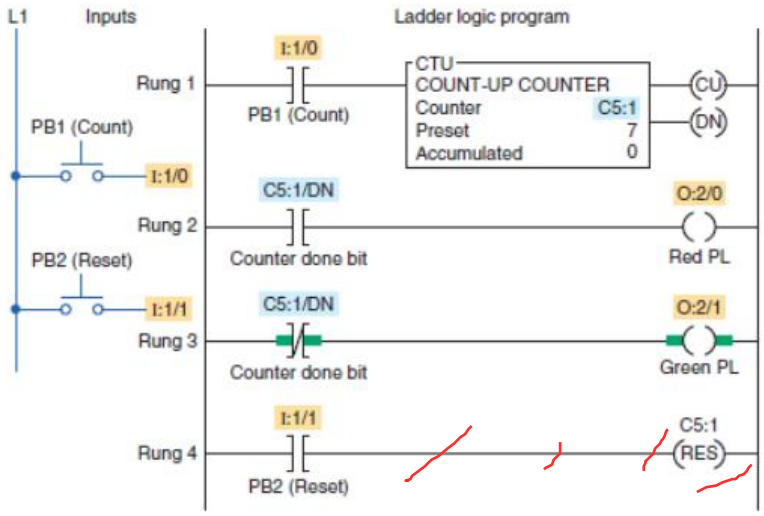

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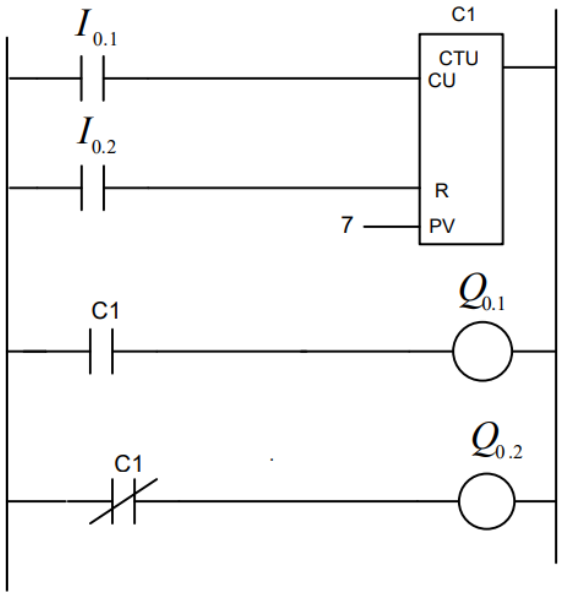

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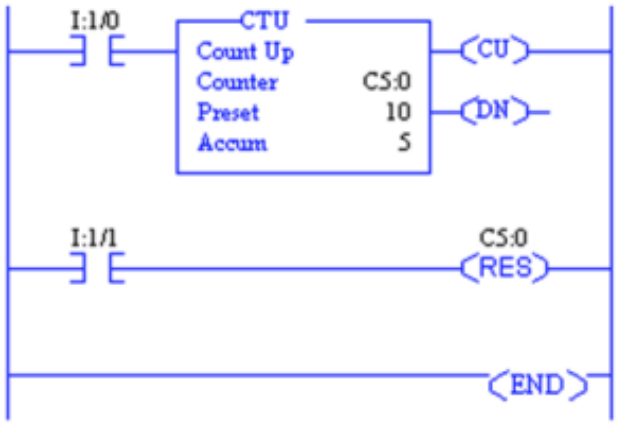

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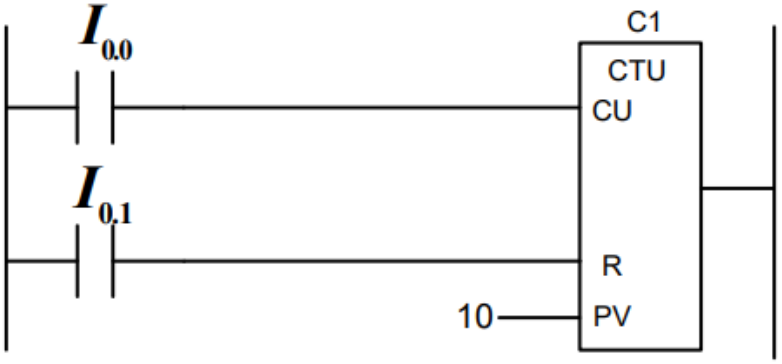

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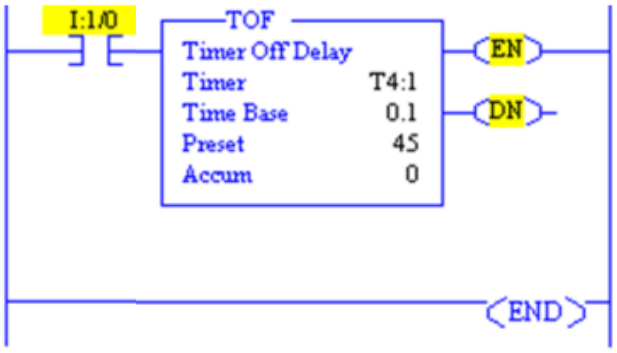

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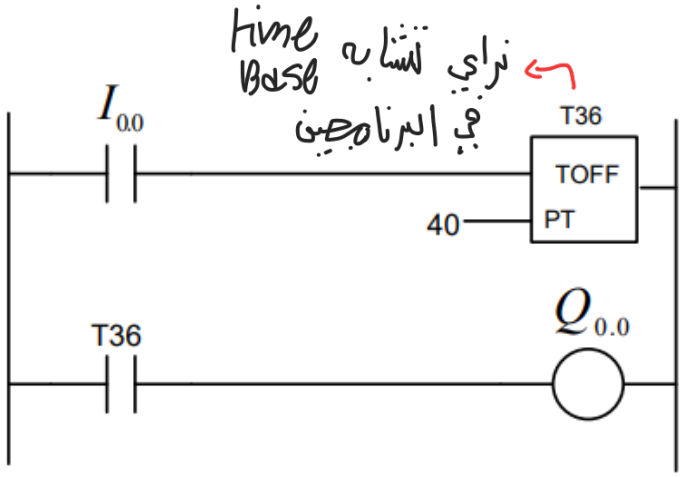

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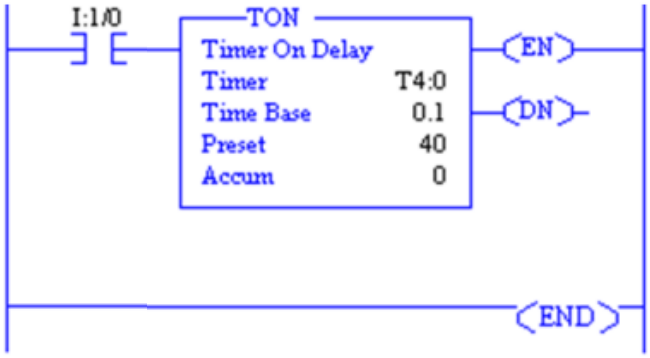

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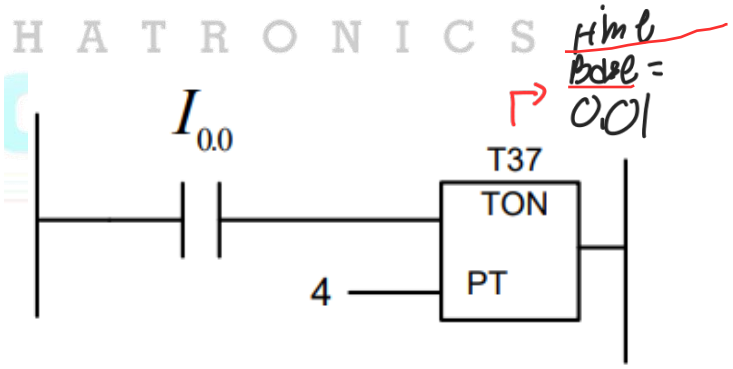

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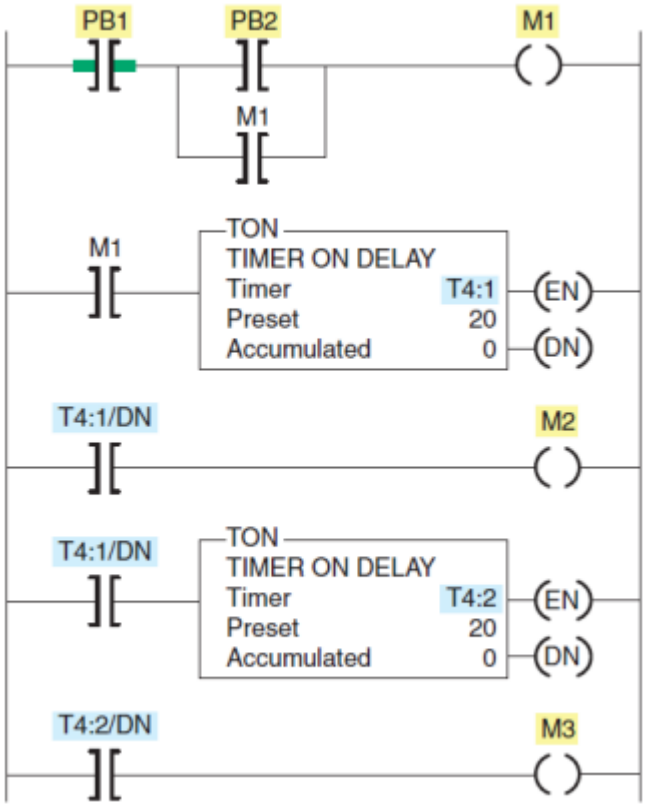

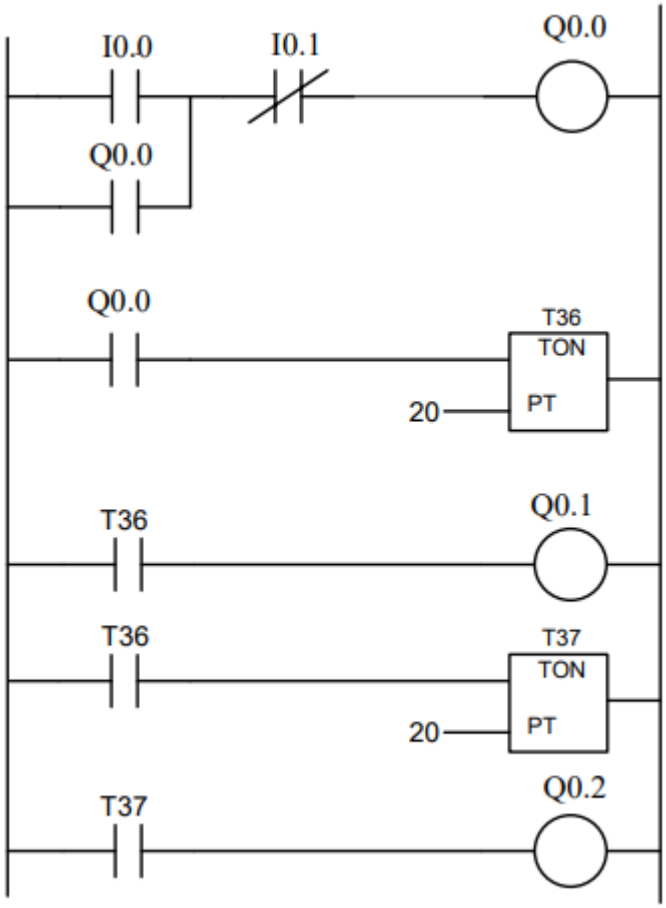

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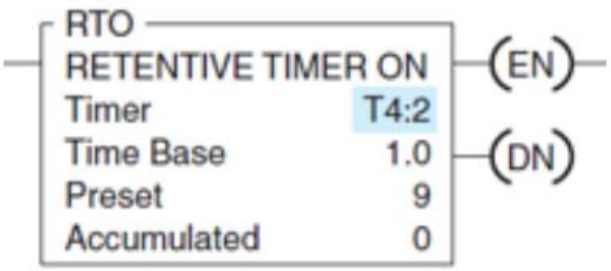

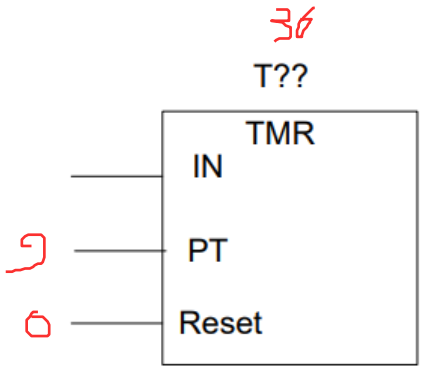

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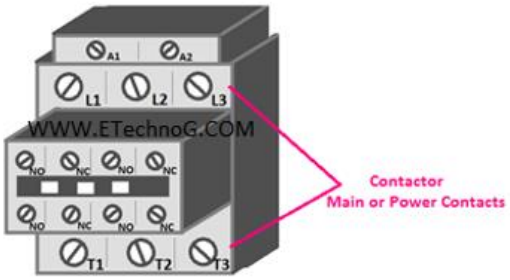
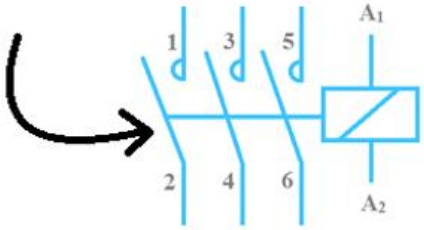
Chapter 9

Contactors

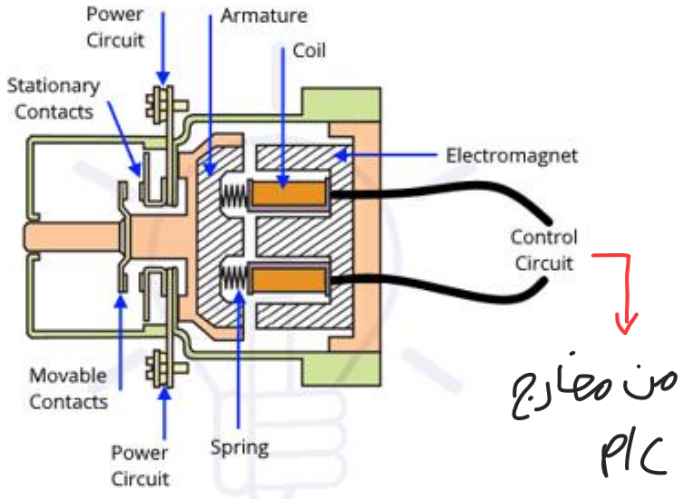
Electrical Contactor Connection and Wiring Diagram





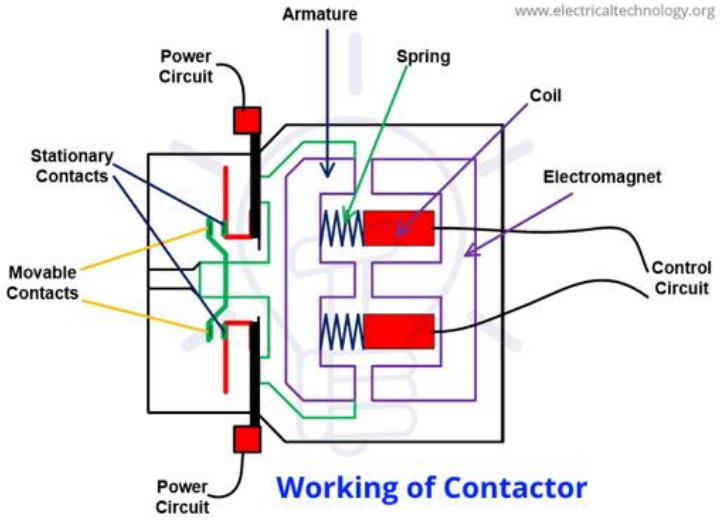
هو عبارة عن
ريليه عفتاح كهروميكانيكي
يمكنه التحكم في احمال
كبيرة و فيه خاصيه الحماية
من ارتفاع التيار

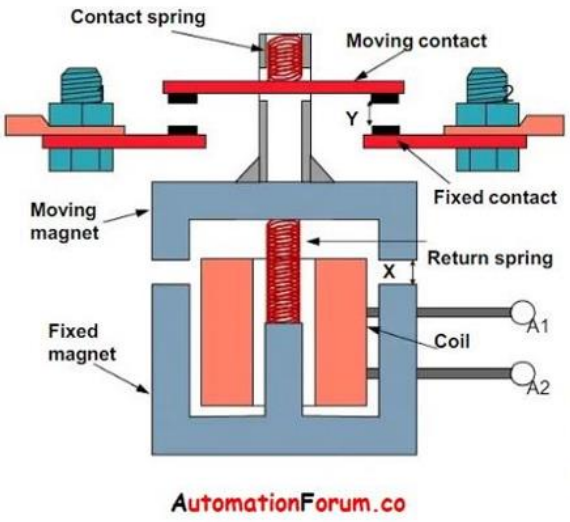
Construction of contactor



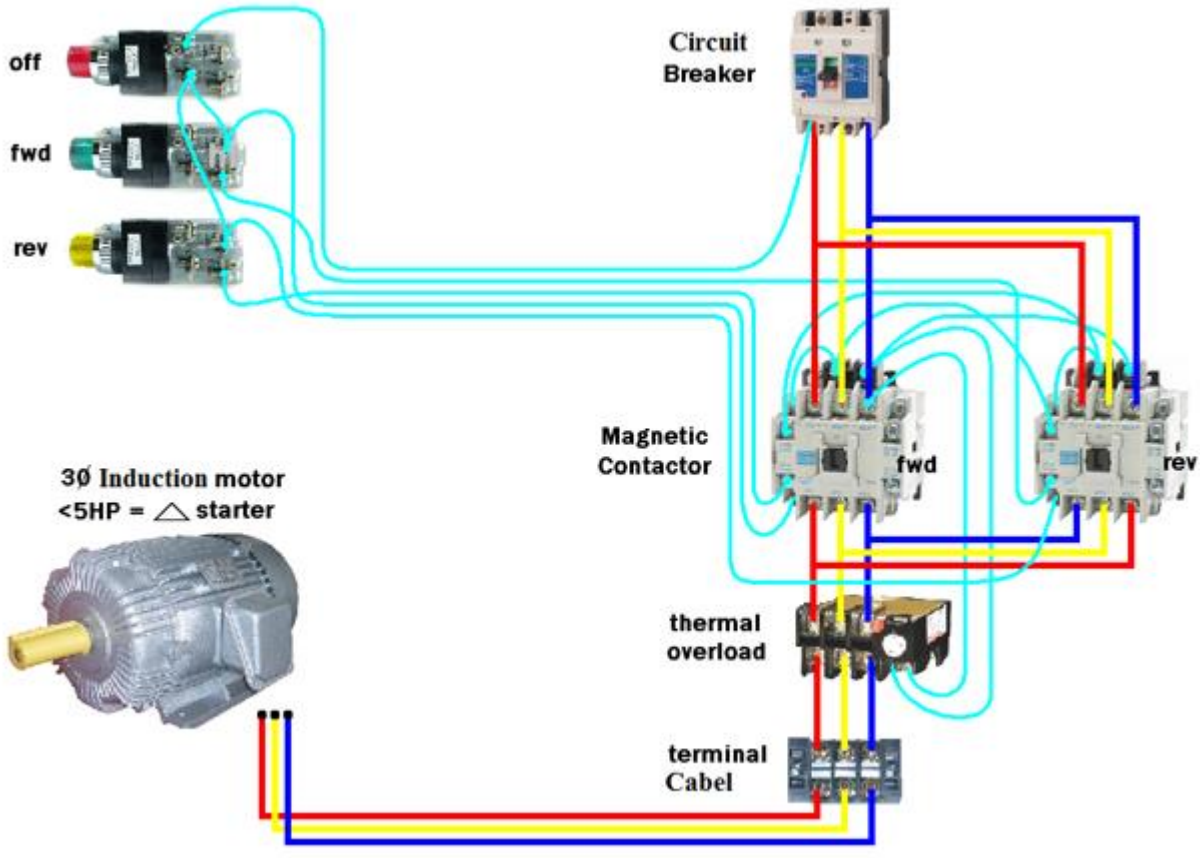
من مخارج
PLC

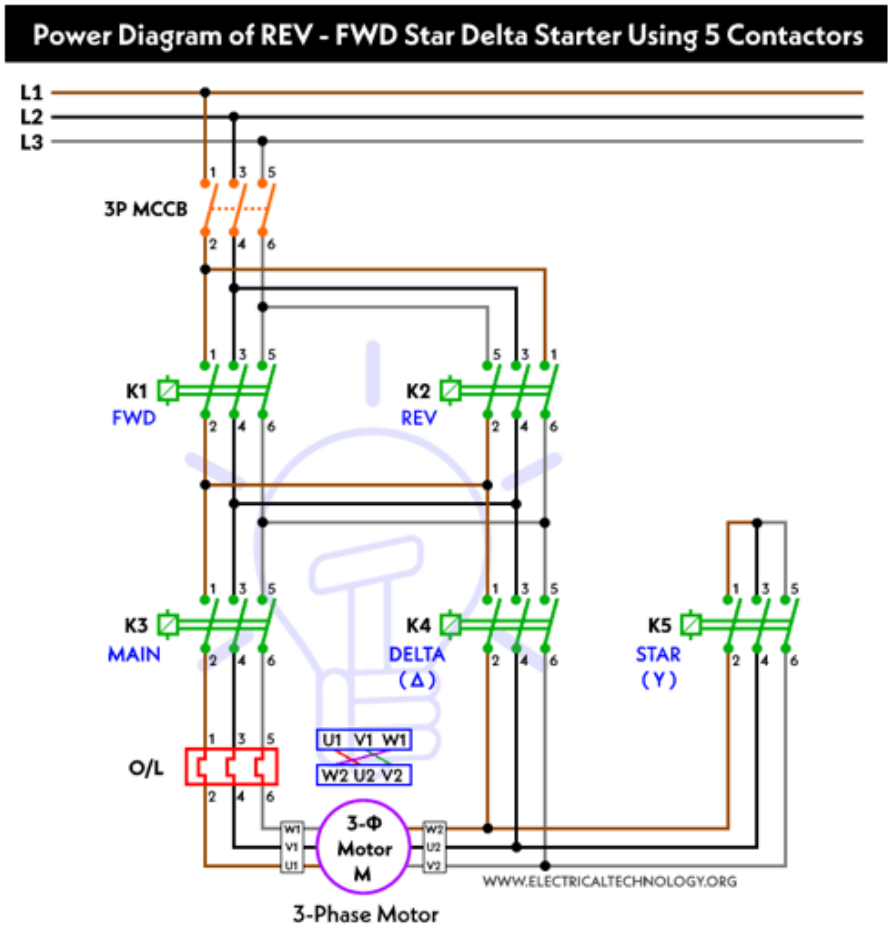
Working of contactor

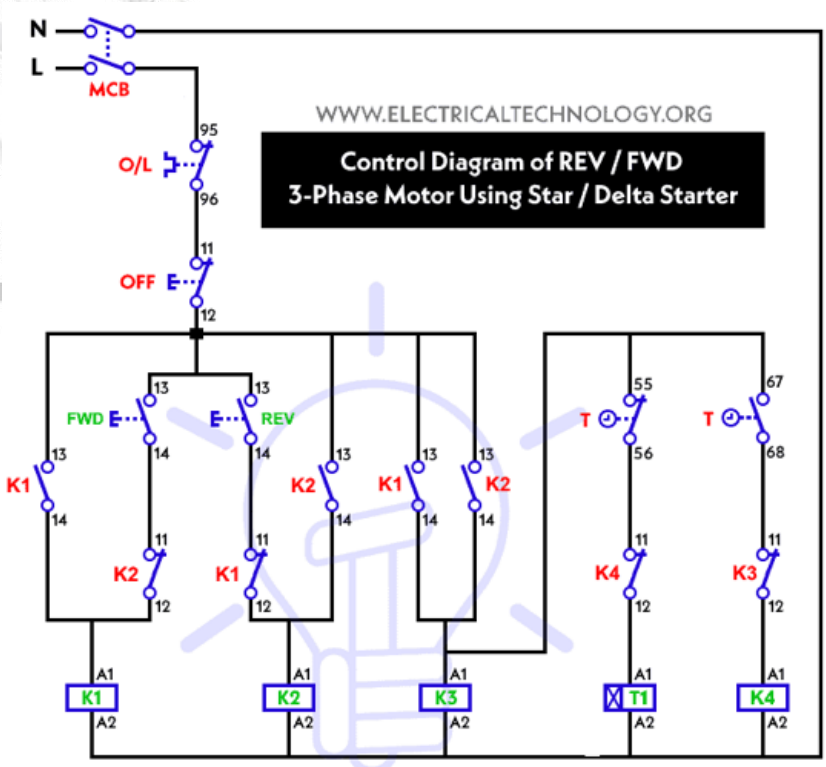




Star-Delta motor control







سؤال لسنوات فاينل ٢٠٢٥ تحويل لـ "ladder diagram"

بسواعدنا نبنيها
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Chapter 10

SCADA

What is SCADA?

• Supervisory Control And Data Acquisition

- Collect measurements and status data from the process
- Remotely get involved in the process
- Centralized system platform
- Based on distributed I/O

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Applicable Processes

• Oil or Gas production facilities

• Pipelines for gas, oils, chemicals or water.

• Railway/Transportation Process

• Nuclear, Gas, Hydro generation plants

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What is controlled by SCADA?

- Tap changers
- Switching devices
- Shunt capacitor/reactor
- Generator set points
- Excitation & power output
- Sequential control

Component of SCADA System

- Field data interface devices:

which interface to field sensing devices and local control switchboxes and valve actuators, such as:

- RTUs : Remote Terminal Units

- PLCs : Programmable Logic Controllers

- A communications system:

Used to transfer data between field data interface devices and control units and the computers in the SCADA central host.

- A central host computer serves:

Servers (sometimes called a SCADA Center, master station, or Master Terminal Unit (MTU)

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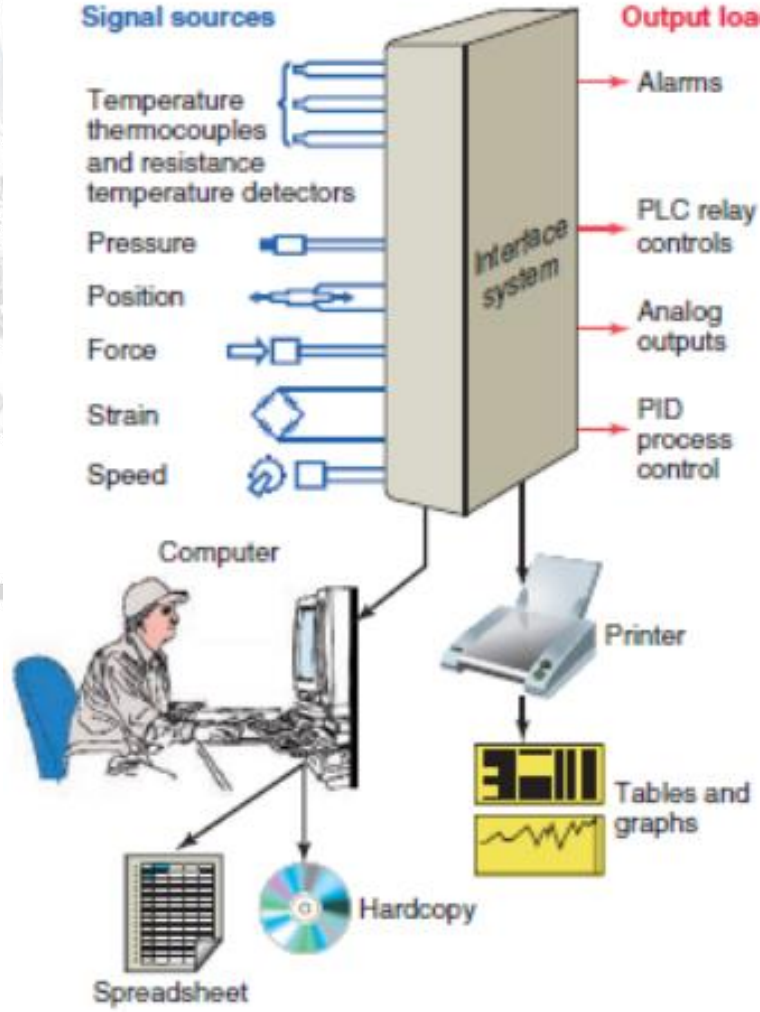
• A collection of standard and/or custom software:

Called Human Machine Interface (HMI) software or Man Machine Interface (MMI) software systems are used to provide the SCADA central host and operator terminal application, support the communications system, and monitor and control remotely located field data interface devices



Human Machine Interface - HMI

Communication between operator and machine

- Input

- Mouse, keyboard, touch screen

- Output

- Screen, audio, print-outs .

Application Servers

Application servers provide the computing platform for the SCADA System, included servers are:

- Real-time database

- Historical database

- Energy Management applications

- State Estimation

- Optimal/Dispatcher Power flow

- Voltage Stability Assessment

- Geographic Information Systems

- Distribution Management

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System Configuration Servers

Allows configuration of the SCADA system environment, typical servers include:

- Data engineering of the SCADA system providing manual data entry into the SCADA topology database .

- Substation Device configuration/Configuration tools and databases/ Remote access tools for configuration

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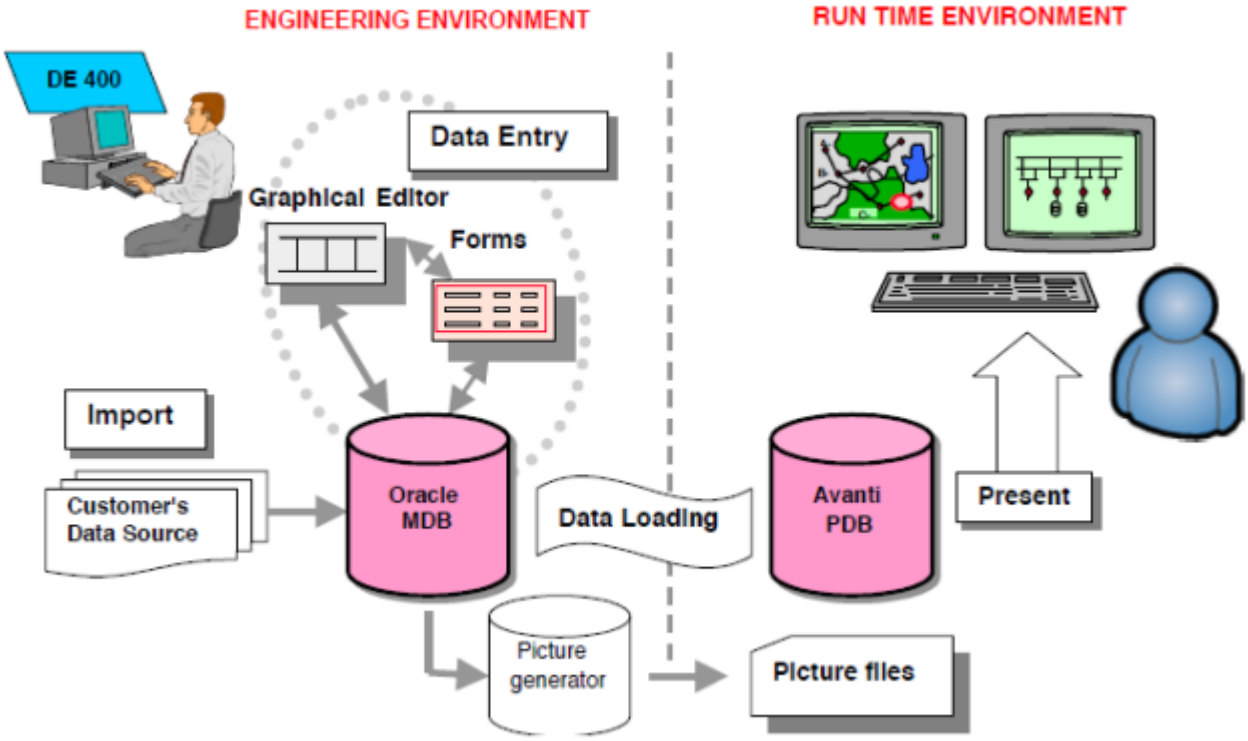
Data Engineering



Communication Servers

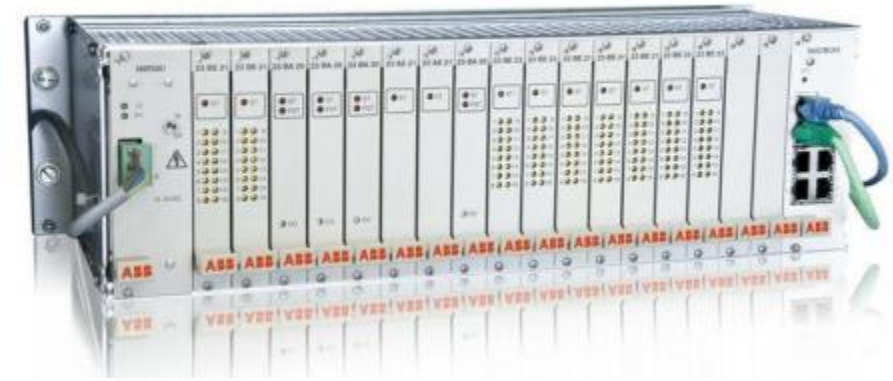
Variety of servers for communication

- Communication to other Control centers

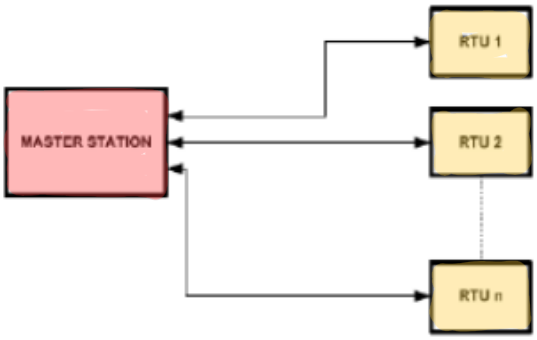
- Communication with office applications

Remote Terminal Unit - RTU

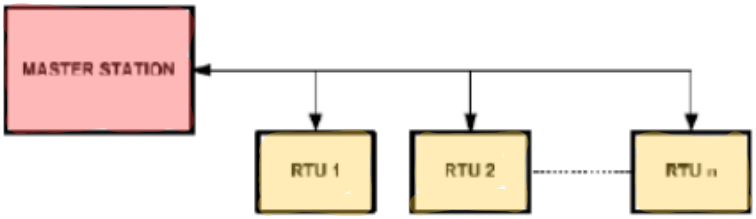
A remote terminal unit (RTU) is a microprocessor controlled electronic device that interfaces objects in the physical world to a distributed control systems or SCADA by transmitting telemetry data to the system, and by using messages from the supervisory system to control connected objects.



Communication

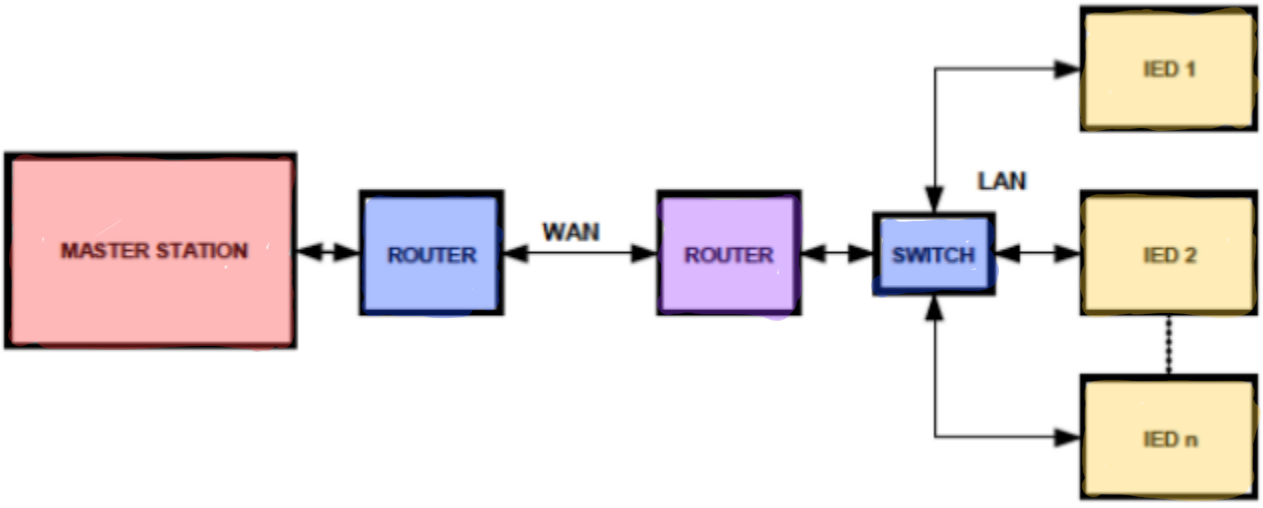


- Radial serial circuit



- Multi-drop circuit

Networked solutions



مصادر ومراجع

1

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في الختام اسأل الله ان يوفقنا واياكم ويهدينا سبيل الرشاد

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